

CHIRAL KOSZUL DUALITY NONCOMMUTATIVE FACTORISATION D -MODULES, ORDERED QUANTUM FIELDS, AND THE GEOMETRY OF DEFECTS

E_1 -algebras in factorisable D -modules on ordered chiral configurations

RAEEZ LORGAT

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Abstract

Let X be a smooth complex curve. The primitive object of this paper is an augmented associative algebra

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1})$$

in factorisable right D -modules. The factorisation coefficient records the complete chiral singularity on X : every principal part, every regular product, and their covariance under coordinate change. The internal E_1 multiplication records ordered fusion along a transverse oriented line. Its analytic realization is a holomorphic–topological factorisation algebra on $X^{\mathrm{an}} \times \mathbb{R}$.

The Ran category carries pointwise tensor product and disjoint-union chiral convolution. Beck–Chevalley base change supplies their interchange, so the relevant factorisation category is duoidal. An ordered chiral algebra is a bimonoid: its pointwise associative multiplication is a morphism of chiral factorisation coalgebras. The ordered bar therefore carries two compatible coproducts, deconcatenation in the interval direction and factorisation in the curve direction.

The geometric carrier is the Grothendieck construction of the associative operad with the chiral configuration category. Its objects are ordered sequences of finite chiral configurations. A two-coloured Boardman–Vogt resolution is indexed by planar levelled forests. Chiral edges resolve collisions in X ; transverse edges resolve associative composition. Codimension-two faces give the chiral relation, the associative relation, and their interchange. Orientation of the two determinant lines makes the corresponding edge differentials anticommute.

The interval bar is the simplicial derived tensor product

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = |[r] \mapsto \mathbf{1} \otimes \bar{A}^{\otimes r} \otimes \mathbf{1}| \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1} \simeq \int_{([0,1], \partial[0,1])} A.$$

Its faces compose adjacent transverse operations. The full chiral OPE remains a morphism of factorisation D -modules and accompanies every bar term. Connected positive-weight algebras and their complete Mittag–Leffler limits satisfy internal associative bar–cobar reconstruction. The independent chiral convolution on $D(\mathrm{Ran} X)$ satisfies the Francis–Gaitsgory pro-nilpotent Chevalley–primitive equivalence. A strong monoidal realization compares the two constructions whenever its essential image is closed under bar and cobar.

The augmentation boundary condition has endomorphism algebra

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

For positive finite-type objects, the double centralizer is the derived augmentation completion. The universal twisting morphism identifies augmentation-complete A -modules with conilpotent comodules over $C = \text{Bar}_X^{\text{ord}}(A)$. Its two-sided form sends the diagonal bimodule to the diagonal bicomodule, transports derived tensor product to derived cotensor product, and identifies Hochschild theory with coHochschild theory. The interval, annulus, boundary dual, and derived centre are four faces of one Koszul–Morita square. Quadratic duality, PBW deformation, curved nonhomogeneous duality, and Verdier duality enter through explicit comparison morphisms to this universal resolution.

Planar transport supplies exchange. An E_2 enhancement, a flat meromorphic connection, or a braided line category produces braid monodromy. Internal Hochschild cochains

$$Z_{\text{ord, ch}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A)$$

form the universal E_2 recipient of central actions. A rigid meromorphic line category with a faithful tensor fibre functor reconstructs a coquasitriangular coordinate bialgebra; its continuous dual gives the corresponding completed quantum algebra. RTT relations identify the rational case with the Yangian. Knizhnik–Zamolodchikov transport and four-dimensional Chern–Simons theory realize the same exchange geometry.

A square-zero BRST derivation extends factorwise to the ordered bar and forms a bicomplex with its bar differential. The resulting spectral sequence compares the bar of BRST cohomology with BRST cohomology of the bar. Cohomological concentration recovers quantum Drinfeld–Sokolov reduction and its principal W -algebra.

The theory has two genus variables. A Calabi–Yau trace on the transverse algebra closes intervals into circles and produces Hochschild chains, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues pointed algebraic curves and produces conformal blocks over Deligne–Mumford moduli. Their simultaneous algebra is the two-edge-coloured master equation

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_X + \frac{1}{2}[\Theta, \Theta]_{\perp} + \hbar_X \Delta_X \Theta + \hbar_{\perp} \Delta_{\perp} \Theta = 0.$$

The two loop operators arise from independent normal directions and supercommute. Curved Koszul elements, determinant-line curvature, BRST anomaly, framing anomaly, and conformal anomaly occupy their native deformation complexes; family index maps and open–closed comparison maps relate them in concrete theories.

The paper proves the structural results above and calculates them on free and square-zero algebras, quantum and Jordan planes, Rees Weyl algebras, quiver and preprojective algebras, finite-group crossed products, enveloping algebras, Heisenberg and affine currents, free ghosts, lattice theories, Virasoro and principal W -algebras, RTT Yangians, KZ transport, prime-form sewing, Siegel theta series, and holomorphic–topological gauge theory. Every scalar appears as the value of a specified trace, index, period, or monodromy operation on a previously constructed object.

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Overture: five geometries of one quantum operation

The diagonal carries collision. The interval carries order. The plane carries exchange. The circle carries trace. The stable curve carries sewing. These five geometric carriers generate five algebraic operations.

On a smooth complex curve X , the singular product is supported on the diagonal:

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta! \mathfrak{g}. \quad (1)$$

On an oriented line, fusion is an associative operation

$$m : A \otimes A \longrightarrow A. \quad (2)$$

On a transverse plane, parallel transport in configuration space produces braid-group representations. On a circle, closure is factorisation homology

$$\int_{S^1} A \simeq \mathrm{HH}_*(A). \quad (3)$$

On the boundary of the moduli space of curves, contraction along stable-graph edges produces sewing.

The common source is

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \quad (4)$$

The factorisable D -module carries the chiral singularity. The E_1 structure carries transverse order. Planar transport adds exchange. A cyclic pairing adds trace. Clutching-compatible state sums add modular sewing.

Theorem 0.1 (Five-carrier theorem). *Let A be as in (4). The following constructions are functorial.*

The chiral coefficient is the system of D -modules on powers of X with factorisation maps and diagonal singularities. The transverse bar is

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1}.$$

Under analytic excision, it is factorisation homology of an interval with augmentation labels at the endpoints. An E_2 lift or flat meromorphic connection supplies braid transport. A cyclic trace supplies the circle object and ribbon contractions. A modular action supplies stable-curve sewing. Each construction is determined by its displayed geometric datum.

Proof. The coefficient system is the definition of a factorisable D -module. The simplicial two-sided bar computes the derived tensor product, and factorisation excision identifies it with interval homology. The little two-disks operad contains exchange paths and therefore braid operations. A cyclic pairing evaluates the two ends of an interval and creates a circle. A modular action assigns contraction maps to separating and nonseparating clutching correspondences. Functoriality follows from functoriality of realization, transport, trace, and clutching. $\square$

The order of dependence is

$$\begin{aligned}
& \text{chiral coefficient} \xrightarrow{E_1\text{-fusion}} \text{ordered chiral algebra}, \\
& \text{ordered chiral algebra} \xrightarrow{E_2\text{-lift or transport}} \text{exchange category}, \\
& \text{ordered chiral algebra} \xrightarrow{\text{cyclic trace}} \text{ribbon amplitudes}, \\
& \text{chiral coefficient} \xrightarrow{\text{modular sewing}} \text{stable-curve amplitudes}.
\end{aligned} \tag{5}$$

The last two arrows coexist in the bicoloured theory and retain independent genus gradings.

The common law

Codimension-one boundary strata carry elementary compositions. Codimension-two corners carry their relations. The oriented boundary of an oriented boundary is zero. Coefficient maps assigned to the faces transform this topological identity into an algebraic master equation.

For the interval, the two ends of a one-dimensional associahedral boundary give

$$(ab)c - a(bc) = 0.$$

For three chiral points, the partial diagonals give the chiral Jacobi identity. For compactified Feynman configurations, renormalised graph weights give the classical or quantum master equation. For stable curves, separating and nonseparating nodes give the bracket and loop terms of the modular equation. Topology supplies the signed incidence relation; the quantum theory supplies the coefficient maps.

Theorem 0.2 (Stokes-to-master theorem). *Let $\overline{C}$ be an oriented stratified space whose codimension-one faces are labelled by degree-one contractions ρ_F of a coefficient complex (E, d_E) . Assume compatibility with restriction to corners and*

$$d_E \rho_F + \rho_F d_E = 0.$$

For

$$D = d_E + \sum_F \rho_F,$$

the component of D^2 supported on a codimension-two stratum equals the signed sum of the composites attached to its incident faces. Hence $D^2 = 0$ exactly when every corner sum is zero.

Proof. Expand D^2 . The internal square is zero and the mixed terms cancel by the chain-map identity. Every remaining term is an ordered pair of incident faces. Grouping these pairs by their common codimension-two stratum gives the oriented cellular boundary relation. $\square$

This theorem organizes the bar differential, chiral Jacobi identity, Yang–Baxter equation, Batalin–Vilkovisky equation, and modular sewing equation. In each case the compactified geometry provides the incidence signs and the field theory provides the operations.

Six realizations

Dirac’s language is distributional: singular support belongs to the observable. The diagonal-supported D -module operation in (1) is its algebraic incarnation.

Feynman's language is compositional: an amplitude is a sum over intermediate states. The ordered bar is the derived complex of those intermediate boundary operations, and its differential fuses adjacent states.

Polyakov's language is anomalous: quantisation produces a determinant line with curvature. A number is obtained by evaluating its characteristic class on a chosen family.

Witten's language is topological: isotopy of line defects produces braid transport, and closure of a line produces a ribbon trace.

Gaiotto's language is categorical: boundary conditions are objects, boundary-changing fields are morphisms, interfaces are bimodules, and junctions are higher morphisms. The algebra (4) is the endomorphism algebra of a boundary condition inside this category.

Costello's language is factorisational: renormalised observables extend across compactified collision strata and satisfy the BV master equation. A theory holomorphic along X and topological along a line realizes (4) analytically.

These six realizations share one geometric syntax and preserve their native object types.

The invariant sentence

A noncommutative chiral quantum field theory on a curve with one ordered topological direction is an augmented E_1 -algebra in factorisable D -modules.

Every subsequent object is obtained from this datum by a named functor, geometric extension, or trace. The geometry specifies the available composition; the coefficient specifies its value; the oriented boundary specifies its equation.

Part I

The noncommutative chiral object

1 The phenomenon and its invariant language

1.1 Two limits of the same experiment

Place local operators on a complex curve X and line defects on an oriented real line. There are two independent short-distance limits. In the first, two points of X approach one another while their transverse positions remain fixed. The product develops a finite principal part along the diagonal. In the second, two transverse intervals become adjacent while their positions on X remain separated. The product is associative and ordered. A theory in which both limits exist therefore carries two operations of different geometric origin:

$$\text{chiral collision on } X, \quad \text{ordered fusion on } \mathbb{R}. \quad (7)$$

The first is encoded by a factorisation D -module. The second is encoded by an E_1 -multiplication internal to the category of such modules.

This distinction is forced by topology. A punctured complex disc has a nontrivial fundamental group, so analytic continuation of insertions can produce monodromy. An ordered chamber

$$\text{Conf}_n^<(\mathbb{R}) = \{t_1 < \dots < t_n\}$$

is contractible, so parallel transport inside one order chamber is trivial. The chiral singularity and the associative order are independent data. Thickening the transverse line to a plane supplies exchange paths and their braid classes.

Definition 1.1 (Ordered chiral algebra). Let X be a smooth complex algebraic curve and let $\text{FactD}(X)$ be the symmetric monoidal stable category of factorisable right D -modules defined in Section 2. An *ordered chiral algebra* on X is an augmented algebra

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).$$

Its multiplication is denoted $m : A \otimes A \rightarrow A$, and its augmentation is denoted $\varepsilon : A \rightarrow \mathbf{1}$.

The word *chiral* belongs to the coefficient category. The word *ordered* belongs to the E_1 -operation. The two adjectives name independent structures.

1.2 The diagonal and the interval

The diagonal is the carrier of the operator product expansion. Let

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2, \quad \Delta : X \hookrightarrow X^2.$$

A chiral singular product is a morphism from an extension off the diagonal to a distribution supported on the diagonal. In the Lie form of Beilinson and Drinfeld it is

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \quad (8)$$

A coordinate expansion of (8) produces all OPE modes. The morphism itself is coordinate-free.

The interval is the carrier of the ordered product. If $I_1 < I_2$ are disjoint subintervals of an oriented interval I , factorisation gives

$$A(I_1) \otimes A(I_2) \longrightarrow A(I).$$

Shrinking I_1 and I_2 inside their order chamber produces the binary multiplication. Three ordered subintervals produce the two composites $(ab)c$ and $a(bc)$; the contractibility of the space of such shrinkings gives their coherent homotopy. After fixing the endpoints and quotienting by affine reparametrisation, the standard real compactification of ordered points is an associahedron.

The two geometries meet on $X \times \mathbb{R}$. A small rectangle $U \times I$ carries a complex of observables. Chiral factorisation varies U in X ; ordered factorisation varies I in $\mathbb{R}$. Their interchange is the geometric origin of an E_1 -algebra in $\text{FactD}(X)$.

1.3 The unique language

A useful language must satisfy four requirements.

First, it must retain the entire singular product. A logarithmic residue detects a simple normal coefficient. A fourth-order Virasoro pole and a double affine pole require higher normal jets. A factorisation D -module contains all of them at once.

Second, it must retain noncommutative fusion before any symmetrisation. Tensor order, line order, and braid exchange are three different structures. The E_1 -multiplication retains the first two; an E_2 enhancement supplies the third.

Third, it must separate algebraic and analytic completion. The bar construction is defined in the stable category. Trace-class sewing and convergence of conformal blocks are additional analytic theorems.

Fourth, it must type every anomaly. Determinant-line curvature, the BRST anomaly class, the BV quantum anomaly, and curved Koszul data live in different complexes. A comparison morphism identifies selected classes in a concrete theory.

These requirements lead to the following square:

$$\begin{array}{ccc} \text{Alg}_{E_1}(\text{FactD}(X)) & \xrightarrow{\text{Bar}_X^{\text{ord}}} & \text{Coalg}_{\text{conil}}(\text{FactD}(X)) \\ \text{an} \downarrow & & \downarrow \text{an} \\ \text{Fact}_{\text{hol,lc}}^{\text{rect}}(X^{\text{an}} \times \mathbb{R}) & \xrightarrow{\int_{([0,1],\partial)}} & \text{Fact}_{\text{hol}}(X^{\text{an}}). \end{array} \tag{9}$$

The upper row is the internal associative bar construction. For positive complete objects, cobar reconstructs it. The lower row is interval factorisation of a rectangular holomorphic–topological theory. On regular holonomic objects, Riemann–Hilbert identifies the vertical arrows. The square expresses one phenomenon in algebraic and analytic language.

1.4 What is genuinely noncommutative

A conventional factorisation algebra on a manifold has symmetric multiplication for disjoint opens because disjoint union is unordered. A codimension-one direction orders the opens and creates noncommutative composition. Boundary conditions, interfaces, and parallel line defects supply such

directions. The resulting multiplication is composition in the defect category, while the chiral OPE remains the diagonal singularity in the coefficient.

Let $\mathcal{C}$ be a factorisation category on X and let b be an object representing a boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b) \quad (10)$$

is an E_1 -algebra in the factorisation coefficient category. Composition of endomorphisms gives the E_1 product. Collision of their insertion points gives the chiral singularity. Formula (10) is the native source of noncommutative chiral algebras.

A simple algebraic model is obtained from a factorisation object V and an associative algebra B . If V is commutative as a factorisation object, then

$$A = B \otimes V$$

is an ordered chiral algebra with multiplication $m_B \otimes m_V$. Matrix-valued free fields and Chan–Paton factors have this form locally. The bar construction sees the noncommutative resolution of B while every term remains a chiral coefficient system on X .

1.5 The bar is transverse

The reduced bar construction of an augmented algebra is generated by ordered words

$$[a_1 | \cdots | a_r],$$

and its differential multiplies adjacent letters. The word order is the order of intervals on $\mathbb{R}$. Each coefficient a_i and every factorisation map of $A^{\otimes r}$ already carries the chiral singularities; the bar differential composes the transverse word.

This observation removes the original ambiguity between collision residues and the associative bar differential. The de Rham differential of a configuration space, the residue differential along a collision divisor, and the simplicial bar differential are three distinct operators. They can coexist in a total complex, but their source and target degrees must be written separately.

Proposition 1.2 (Low-arity bar identity). *Let A be a dg E_1 -algebra in a stable symmetric monoidal category. On homogeneous elements of the augmentation ideal,*

$$b_2[a|b|c] = (-1)^\alpha[(ab)|c] + (-1)^\beta[a|(bc)],$$

with the standard suspension signs, and

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Hence the multiplication part of the bar differential squares to zero exactly by associativity.

Proof. There are two length-two contractions of a word of length three. The suspension orientation gives opposite signs. The two coefficient operations agree by associativity. $\square$

The complete chiral identities are tested elsewhere: by the chiral operad on the formal disc, by factorisation on the Ran space, or by the partition collision complex. Associativity of the transverse product and the Borchers identity of the chiral coefficient are compatible, distinct equations.

1.6 From order to exchange

The distinction between order and exchange is the first physical test of the language. Parallel line defects in a topological line multiply according to their order. A second topological direction supplies an exchange path. The homotopy class of that path is a braid. Thus

$$E_1 \text{ fusion} \xrightarrow{\text{one more topological direction}} E_2 \text{ exchange.}$$

Dunn additivity identifies the second structure with an E_1 -algebra object in E_1 -algebras. The universal E_2 object acting on an E_1 algebra is its Hochschild centre.

This yields a precise physical dictionary:

ordered intervals	fusion of parallel defects,
chiral diagonals	singular OPE on the spectral curve,
second transverse direction	braid exchange,
Hochschild centre	universal bulk or exchange algebra,
Tannaka coend	reconstructed quantum symmetry.

Each arrow in this table is a geometry-determined functor between typed objects.

1.7 The two notions of modularity

A trace on an E_1 algebra closes an interval into a circle. Iterated gluing produces ribbon surfaces. This is the modularity of open topological field theory and Hochschild homology.

A chiral algebra on a varying algebraic curve has a different modularity. Plumbing a node changes the complex curve. The relevant spaces are $\overline{\mathcal{M}}_{g,n}$, and the amplitudes are conformal blocks. The first genus is transverse and ribbon; the second is complex and Deligne–Mumford.

When an ordered chiral algebra carries both sewing structures and their mixed interchange maps, the two contractions supercommute with the product orientation. The resulting master equation has two loop operators. The separation of the two loop parameters is essential in gauge theory: one counts topological colour loops, the other counts handles of the chiral curve.

1.8 Mathematics and physics

The mathematical object is an E_1 algebra in factorisable D -modules. The physical object is a holomorphic–topological factorisation algebra with a boundary or line defect. The comparison is the morphism of factorisation algebras obtained by restricting quantum observables to the defect.

Let $\text{Obs}_{\mathcal{T}}$ be the quantum observables of a renormalised BV theory $\mathcal{T}$ on $X \times \mathbb{R}_{\geq 0}$. A boundary condition gives a factorisation algebra Obs_{∂} on $X \times \mathbb{R}$. When holomorphic dependence admits a factorisable D -module realization and the boundary theory is locally constant in the ordered direction, Theorem 10.5 gives

$$\text{Obs}_{\partial} \in \text{Alg}_{E_1}(\text{FactD}(X)). \quad (11)$$

The renormalized quantum master equation makes the observable differential square to zero and makes the factorisation maps compatible with ultraviolet contractions. A nonzero BV anomaly obstructs this quantum factorisation object.

The perturbative protected sectors of Chern–Simons theory, four-dimensional Chern–Simons theory, suitable holomorphic twists of supersymmetric gauge theories, and open string field theory furnish

instances of this pattern after their boundary or defect observable comparisons are constructed. Their quantum symmetries are determined by their exchange geometries and spectral curves; the interval bar retains the common ordered composition.

1.9 The theorem spine

The paper proves the following sequence of statements.

- (i) Ordered block-Ran factorisation objects are equivalent to E_1 -algebras in $\text{FactD}(X)$.
- (ii) Their ordered bar is interval factorisation homology and has an associahedral cofibrant model.
- (iii) On connected positive-weight objects and their complete Mittag-Leffler limits, the internal associative bar and cobar reconstruct one another. Separately, Ran cardinality is pro-nilpotent for chiral convolution.
- (iv) The augmentation boundary has endomorphism algebra $A^!$, and the double dual is the augmentation completion.
- (v) The Hochschild centre is the universal E_2 algebra acting on an E_1 algebra; a planar enhancement supplies the associated braid operations.
- (vi) Rigid meromorphic line categories reconstruct quantum symmetries by Tannaka coends.
- (vii) BRST derivations extend to the bar and yield a convergent comparison spectral sequence under boundedness and completeness.
- (viii) Cyclic transverse sewing and chiral curve sewing form a bi-modular master equation with two independent loop operators.
- (ix) Under the algebraisation, locality, gauge-fixing, and anomaly-cancellation hypotheses stated in Part V, factorisation observables realize the preceding constructions in perturbative holomorphic-topological field theory.

The remainder of the paper develops the geometry and the exact examples behind these statements. The remainder proves that these subjects are functorial realizations of a single ordered chiral object.

2 Factorisable D -modules and the complete chiral coefficient

2.1 The finite-set presentation

Let X be a smooth complex algebraic curve. Write Fin^{surj} for the category of nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \quad (x_j)_{j \in J} \longmapsto (x_{\alpha(i)})_{i \in I}.$$

The Ran prestack is the colimit

$$\text{Ran}(X) = \operatorname{colim}_{I \in (\text{Fin}^{\text{surj}})^{\text{op}}} X^I.$$

A right D -module on $\text{Ran}(X)$ is represented by a compatible family $\mathcal{F}_I \in D\text{-mod}(X^I)$ with transition morphisms along the diagonals. All functors below are derived.

For a decomposition $I = I_1 \sqcup \cdots \sqcup I_r$, let

$$U(I_1, \dots, I_r) \subset X^I$$

be the open locus on which points belonging to different blocks are disjoint; points in one block may collide. Denote the inclusion by $j_{I_\bullet}$.

Definition 2.1 (Factorisable D -module). A factorisable right D -module $\mathcal{F}$ on X is a compatible Ran family $\{\mathcal{F}_I\}$ together with equivalences

$$j_{I_\bullet}^! \mathcal{F}_I \simeq j_{I_\bullet}^! (\mathcal{F}_{I_1} \boxtimes \cdots \boxtimes \mathcal{F}_{I_r}) \quad (12)$$

for every decomposition of I , compatible with refinement, permutation, and diagonal transition maps.

The equivalence (12) expresses independence on separated supports. The extension of $\mathcal{F}_I$ across partial diagonals carries the singular terms of the operator product. The definition is intrinsic and coordinate-free; propagators enter in analytic realizations.

Let $\text{FactD}(X)$ denote a stable presentable enhancement of this category. Its *coefficient tensor product* is the componentwise $!$ -tensor product

$$(\mathcal{F} \otimes^{\text{pt}} \mathcal{G})_I := \mathcal{F}_I \otimes^! \mathcal{G}_I. \quad (13)$$

The unit is the vacuum factorisation module $\mathbf{1}$.

Proposition 2.2. *Assume that the chosen Ran enhancement is closed under componentwise $!$ -tensor product and componentwise colimits. Then (13) makes $\text{FactD}(X)$ a presentable stable symmetric monoidal category, and tensor product preserves colimits separately.*

Proof. The assertion holds on each X^I . The diagonal compatibilities and the restrictions in (12) are preserved by $!$ -tensor product. The stated closure hypotheses therefore descend the componentwise structure to the factorisation subcategory. $\square$

This is the symmetric monoidal category in which the transverse E_1 multiplication is formed. It must be distinguished from the chiral tensor product on $D(\text{Ran } X)$, introduced below.

2.2 The complete singular coefficient

Consider the binary component $\mathcal{F}_{\{1,2\}}$ on X^2 . Off the diagonal, factorisation identifies it with $\mathcal{F}_{\{1\}} \boxtimes \mathcal{F}_{\{2\}}$. The extension across the diagonal has local cohomology supported on $\Delta(X)$. In a formal coordinate it is expressed by principal parts

$$\sum_{m \geq 0} \frac{c_m(w)}{(z-w)^{m+1}}.$$

Coordinate changes act triangularly on the coefficients and preserve the filtered distribution.

For $D = \Delta(X) \subset X^2$, the pole filtration is

$$F_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}(mD),$$

and

$$\mathrm{gr}_m^F \mathcal{O}_{X^2}(*D) \simeq N_{D/X^2}^{\otimes m}. \quad (14)$$

Thus the associated graded of a pole is a tensorial normal symbol; the filtered object retains the coordinate-dependent mixing of modes.

Principle 2.3 (Complete coefficient principle). *The chiral coefficient of an ordered chiral algebra is the factorisable D -module itself. Residues, normal jets, formal modes, and distributional kernels are realizations of this coefficient.*

Proof. The factorisable D -module is defined on the whole configuration space together with its extensions across every partial diagonal. Restriction to the complement gives the regular product; local cohomology along a diagonal gives supported singular distributions; the pole filtration gives normal symbols; a formal coordinate expands those symbols into modes; de Rham and distributional realization give residues and kernels. Each construction is a functor applied to the same coefficient object. $\square$

For an ordered chiral algebra, the transverse multiplication

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

is a morphism of factorisable D -modules. It therefore respects the whole diagonal filtration and every disjoint-factorisation map. In coordinates it acts on the complete singular expansion, including every pole order and the regular term.

2.3 Two Lie structures and their comparison

Restriction along the operad map $\mathrm{Lie} \rightarrow E_1$ gives the graded commutator

$$[a, b]_{\perp} = m(a, b) - (-1)^{|a||b|} m(b, a). \quad (15)$$

Proposition 2.4 (Internal commutator). *Restriction of operads defines*

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}) \longrightarrow \mathrm{Alg}_{\mathrm{Lie}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}).$$

The bracket (15) satisfies the graded Jacobi identity.

Proof. It is the commutator bracket in an associative algebra object of a symmetric monoidal category. The usual cancellation proof of Jacobi is operadic and therefore applies internally. $\square$

The bracket $[-, -]_{\perp}$ is a Lie operation on the *transverse multiplication*. A Beilinson–Drinfeld chiral Lie bracket is instead a morphism

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

in the chiral pseudo-tensor category. The factorisation coefficient of A may carry such a chiral operation. A morphism of operadic deformation problems compares it with the internal commutator (15). This distinction is essential in boundary field theory: the chiral bracket measures collision on X , while $[-, -]_{\perp}$ measures the discontinuity between two transverse orders.

2.4 Augmentations and boundary conditions

An augmentation $\varepsilon : A \rightarrow \mathbf{1}$ is a morphism of E_1 algebras in $\text{FactD}(X)$. It makes $\mathbf{1}$ a left and right A -module, and the augmentation ideal is

$$\bar{A} = \text{Fib}(A \xrightarrow{\varepsilon} \mathbf{1}).$$

For a right module R and a left module L , the two-sided bar $\text{Bar}(R, A, L)$ computes interval factorisation homology with endpoint labels R and L . The reduced bar is the case $R = L = \mathbf{1}$.

The module formulation is physically primary. A boundary condition is an object whose local endomorphisms form an E_1 algebra. The unit represents the distinguished augmentation boundary state.

2.5 Two monoidal products on the Ran space

There are two different tensor products in the subject.

The coefficient tensor $\otimes^{\text{pt}}$ of (13) is componentwise. Its support satisfies

$$\text{supp}(\mathcal{F} \otimes^{\text{pt}} \mathcal{G}) \subset \text{supp}(\mathcal{F}) \cap \text{supp}(\mathcal{G}).$$

It preserves each cardinality profile. Pro-nilpotence for this tensor product is therefore controlled by augmentation or weight filtrations.

The *chiral tensor product* on $D(\text{Ran } X)$ is defined by the disjoint-union correspondence

$$\text{Ran}(X) \times \text{Ran}(X) \supset (\text{Ran} \times \text{Ran})_{\text{disj}} \longrightarrow \text{Ran}(X).$$

For reduced objects supported on subsets of cardinality at least r and s , respectively, chiral convolution is supported in cardinality at least $r + s$:

$$D(\text{Ran } X)^{\geq r} \otimes^{\text{ch}} D(\text{Ran } X)^{\geq s} \subset D(\text{Ran } X)^{\geq r+s}. \quad (16)$$

Theorem 2.5 (Ran chiral pro-nilpotence). *For the chiral tensor product $\otimes^{\text{ch}}$, the quotient by objects supported in cardinality $> N$ is nilpotent on reduced objects, and the cardinality completion is pro-nilpotent. This statement concerns $D(\text{Ran } X)$ with chiral convolution. The pointwise tensor product of $\text{FactD}(X)$ uses its augmentation or weight filtration instead.*

Proof. Equation (16) implies that an $(N + 1)$ -fold chiral product of reduced objects is supported in cardinality at least $N + 1$ and hence vanishes in the N th quotient. Taking the inverse limit gives the pro-nilpotent category. This is the support argument used in chiral Koszul duality on the Ran space [38]. $\square$

The two products answer different questions. The pointwise tensor permits an E_1 algebra of defects with a fixed chiral coefficient. The chiral tensor assembles disjoint finite subsets and is the ambient product for the Francis–Gaitsgory Lie–commutative-coalgebra equivalence. Any comparison between their bar constructions must name the monoidal functor that relates them.

2.6 Regular holonomic realization

Assume that every $\mathcal{F}_I$ is regular holonomic and that the chosen subcategory is closed under the tensor operations used above. After the standard de Rham shifts, Riemann–Hilbert identifies $\mathcal{F}_I$ with a constructible complex on $X^{I,\text{an}}$. Factorisation equivalences become constructible factorisation equivalences on the disjoint loci.

Proposition 2.6 (Analytic realization). *Under these regular-holonomic closure hypotheses, Riemann–Hilbert gives a symmetric monoidal realization from $(\text{FactD}(X), \otimes^{\text{pt}})$ to constructible factorisation sheaves on X^{an} . It carries internal E_1 algebras, modules, bar constructions, and Hochschild cochains to their constructible realizations whenever it commutes with the required geometric realizations and internal Homs.*

Proof. Riemann–Hilbert identifies regular holonomic D -modules with constructible complexes and carries external tensor products and inverse images along the open configuration embeddings to their analytic counterparts, with the standard de Rham shifts. Hence it transports the factorisation equivalences and the pointwise symmetric monoidal product. Algebra and module structures are operadic diagrams built from these products, so they are preserved. Geometric realization of the simplicial bar and internal Hom are preserved under the additional commutation hypotheses stated in the proposition, which gives the bar and Hochschild comparisons. $\square$

The analytic realization reveals the monodromy of local systems on $\text{Conf}_n(X)$; planar exchange data supplies the transverse braid action.

2.7 Factorisation categories and Morita origin

Let $\mathcal{C}$ be a D -linear factorisation category on X , and let b be a compatible object. Its internal endomorphisms are

$$A_b(U) = \text{RHom}_{\mathcal{C}(U)}(b|_U, b|_U).$$

Composition is ordered, while factorisation in U supplies the chiral coefficient.

Theorem 2.7 (Morita source). *Let $\mathcal{C}$ be a presentable D -linear factorisation category and let b be a dualisable factorisation object. Then $A_b = \text{RHom}_{\mathcal{C}}(b, b)$ is an E_1 algebra in $\text{FactD}(X)$. If b is compact and generates each local category, and these local Morita equivalences commute with factorisation descent, then*

$$\text{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \longrightarrow \text{Mod}_{A_b}(\text{FactD}(X))$$

is a factorisation Morita equivalence.

Proof. Composition gives the E_1 multiplication. Factorisation of b on disjoint opens gives the coefficient factorisation maps. The compact-generator Morita theorem applies locally; the final hypothesis identifies the local equivalences on overlaps and descends them. $\square$

This is the source encountered in gauge theory: a boundary condition or line defect is b ; local endomorphisms form A_b ; transverse composition is noncommutative; singular dependence on the holomorphic insertion point is carried by the factorisable D -module.

2.8 Historical position

Dirac's distributional formalism made the singular support of products of quantum fields unavoidable. Wilson organised short-distance singularities by the operator product expansion. Borchers and Kac encoded their algebraic identities. Beilinson and Drinfeld replaced coordinate series by D -module morphisms on diagonals. Factorisation algebras expressed the same local-to-global law for quantum observables. The present object adds an oriented topological direction and retains its associative order before any exchange or symmetrisation is imposed.

3 The duoidal law: locality as a bialgebra identity

The primitive object admits a second, equivalent description which makes its geometry visible at once. The Ran category carries a pointwise tensor product and a convolution product along disjoint union. One product combines coefficients supported at the same finite subset; the other combines observables supported at disjoint finite subsets. Their compatibility is duoidal. An ordered chiral algebra is a monoid for the first product and a factorisation coalgebra for the second. The assertion that multiplication is a morphism of factorisation objects is therefore a bialgebra identity, but the multiplication and comultiplication belong to different monoidal structures.

3.1 The two products and their interchange

Let

$$\mathcal{D}_X = D(\mathrm{Ran} X).$$

Write $\otimes^{\mathrm{pt}}$ for the pointwise $!$ -tensor product. Write $\otimes^{\mathrm{ch}}$ for chiral convolution along the correspondence

$$\mathrm{Ran} X \times \mathrm{Ran} X \xleftarrow{j} (\mathrm{Ran} X \times \mathrm{Ran} X)_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran} X, \quad F \otimes^{\mathrm{ch}} G := u_! j^! (F \boxtimes G).$$

The diagonal of the Ran space is compatible with disjoint union. The projection formula and Beck–Chevalley base change therefore give a natural interchange morphism

$$(F_1 \otimes^{\mathrm{ch}} F_2) \otimes^{\mathrm{pt}} (G_1 \otimes^{\mathrm{ch}} G_2) \longrightarrow (F_1 \otimes^{\mathrm{pt}} G_1) \otimes^{\mathrm{ch}} (F_2 \otimes^{\mathrm{pt}} G_2), \quad (17)$$

with the middle factors permuted by the symmetry of $\otimes^{\mathrm{pt}}$. In a six-functor realization for which the relevant base-change and projection-formula maps are equivalences, (17) is an equivalence on the disjoint locus. In the general six-functor setting it is the corresponding lax interchange map.

Theorem 3.1 (Duoidal Ran theorem). *Fix a sheaf theory on $\mathrm{Ran} X$ for which pointwise $!$ -tensor, chiral convolution, the projection formula, and disjoint-union base change are defined. Then*

$$(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$$

is a lax duoidal stable category. On the full subcategory on which the base-change morphisms in (17) are equivalences, it is a strong duoidal category.

Proof. Associativity and symmetry of $\otimes^{\mathrm{pt}}$ are inherited componentwise from $D(X^I)$. Associativity of $\otimes^{\mathrm{ch}}$ is the associativity of union of pairwise disjoint finite subsets. Apply the two chiral coproduct correspondences, tensor their outputs pointwise, and then use the diagonal base-change square to

regroup the first and third, and the second and fourth, factors. The resulting natural transformation is (17). The coherence diagrams reduce to the associativity, symmetry, and base-change diagrams of the six-functor formalism. This is precisely the definition of a lax duoidal category; invertibility of the interchange maps gives the strong form. $\square$

The theorem is an instance of the general theory of duoidal ∞ -categories [79]. Its particular content here is geometric: the two products are the diagonal product and disjoint-union convolution on the Ran space.

3.2 Factorisation coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\text{ch}}$. Its coproduct restricts, for disjoint finite subsets S and T , to a map

$$\Delta_{S,T} : F_{S \cup T} \longrightarrow F_S \boxtimes F_T.$$

It is a factorisation coalgebra when this map is an equivalence on every disjointness locus. Francis and Gaitsgory identify factorisation algebras with the corresponding full subcategory of chiral commutative coalgebras [38].

Let

$$\iota : \text{FactD}(X) \hookrightarrow \text{CoCAlg}^{\text{ch}}(\mathcal{D}_X)$$

denote this realization. The pointwise tensor of two factorisation coalgebras is again a factorisation coalgebra: its coproduct is obtained by applying (17) to their two coproducts.

Theorem 3.2 (Ordered chirality as a duoidal bimonoid). *Assume the strong duoidal hypotheses of Theorem 3.1 on the chosen factorisation subcategory. For an object $A \in \text{FactD}(X)$, the following data are equivalent:*

- (i) *an E_1 multiplication on A internal to $\text{FactD}(X)$;*
- (ii) *an E_1 monoid structure for $\otimes^{\text{pt}}$ on ιA such that its multiplication and unit are morphisms of chiral commutative coalgebras;*
- (iii) *a bimonoid in the duoidal category $(\mathcal{D}_X, \otimes^{\text{pt}}, \otimes^{\text{ch}})$ whose chiral coproduct is factorising on disjoint loci.*

The compatibility is the duoidal bialgebra identity

$$\Delta_{\text{ch}} \circ m = (m \otimes^{\text{ch}} m) \circ \chi \circ (\Delta_{\text{ch}} \otimes^{\text{pt}} \Delta_{\text{ch}}), \quad (18)$$

where χ is the interchange morphism.

Proof. An E_1 algebra in the full subcategory of factorisation coalgebras is, by definition, an E_1 multiplication whose structure maps remain in that subcategory. Saying that multiplication is a morphism of coalgebras is exactly (18); saying that the unit is a coalgebra map is the corresponding unit identity. These are the axioms of a bimonoid in a duoidal category. The factorisation condition is the requirement that the chiral coproduct become an equivalence on disjoint loci. $\square$

Equation (18) is the exact algebraic form of locality for the noncommutative theory. Separate two collections of insertions in the chiral curve and fuse them in the ordered direction. One obtains the same operation by fusing inside each separated region and then applying factorisation. The equation is the bimonoid interchange law; cocommutativity and R -matrix relations are additional structures.

3.3 The bar retains factorisation

The simplicial ordered bar is levelwise a chiral commutative coalgebra:

$$B_r(A) = \mathbf{1} \otimes^{\text{pt}} \bar{A}^{\otimes^{\text{pt}} r} \otimes^{\text{pt}} \mathbf{1}.$$

Every face is built from m or the augmentation. By Theorem 3.2, these are coalgebra maps. Every degeneracy is built from the unit and is likewise a coalgebra map.

Theorem 3.3 (Double-coalgebra bar theorem). *Suppose geometric realization in $\mathcal{D}_X$ preserves chiral commutative coalgebras and commutes with the pointwise tensor products appearing in the simplicial bar. Then*

$$C = \text{Bar}_X^{\text{ord}}(A)$$

has two compatible coalgebra structures:

- (i) *the coassociative deconcatenation coproduct of the E_1 bar;*
- (ii) *the commutative chiral factorisation coproduct inherited levelwise from A .*

Deconcatenation is a morphism of chiral factorisation coalgebras. Equivalently, C is a coassociative coalgebra object in chiral commutative coalgebras.

Proof. The simplicial bar is a simplicial object in chiral commutative coalgebras because every simplicial operator is a coalgebra morphism. Geometric realization gives the second coproduct. Deconcatenation is defined by cutting an ordered word. The chiral coproduct acts independently on each letter, so cutting and factorising commute by the interchange law. The resulting square is (18) applied to the iterated pointwise tensor powers of A . $\square$

This theorem explains why the ordered bar is simultaneously an interval object and a Ran-space coefficient. The coassociative coproduct remembers where the interval is cut. The chiral coproduct remembers where the curve is separated. Their compatibility is inherited from locality of the original ordered chiral algebra.

3.4 A Fubini theorem

Let $p : X^{\text{an}} \times \mathbb{R} \rightarrow X^{\text{an}}$ and $q : X^{\text{an}} \times \mathbb{R} \rightarrow \mathbb{R}$ be the projections. Factorisation pushforward along p integrates the locally constant real direction; pushforward along q integrates the chiral direction when the corresponding chiral homology is defined. Product excision gives a Fubini comparison.

Theorem 3.4 (Ordered chiral Fubini). *Let A be an ordered chiral algebra whose analytic realization is rectangularly generated. Assume the relevant factorisation homologies exist and that the pushforwards preserve the geometric realizations of the bar. Then there are natural equivalences*

$$\int_X \text{Bar}_X^{\text{ord}}(A) \simeq \text{Bar}^{\text{ord}}\left(\int_X A\right), \quad (19)$$

$$\int_X \mathrm{HH}_*(A) \simeq \mathrm{HH}_*\left(\int_X A\right). \quad (20)$$

More generally, factorisation homology over a product rectangle may be computed in either order.

Proof. The bar is a geometric realization of the simplicial object $[r] \mapsto \bar{A}^{\otimes_{\mathrm{pt}} r}$. Chiral factorisation homology is a colimit over chiral disks. Under the stated preservation hypotheses, the two colimits commute, giving (19). Hochschild chains are factorisation homology over the transverse circle, or equivalently the cyclic realization of the two-sided bar. Commuting that realization with chiral factorisation homology gives (20). On the analytic product, both assertions are the Fubini theorem for factorisation homology and follow from excision on product-disk covers. $\square$

The first equivalence says that one may integrate the chiral curve before or after resolving ordered fusion. The second says the same after the interval has been closed into a circle. In conformal field theory this is the equality between first forming the open-string trace and then taking chiral coinvariants, or first taking chiral coinvariants and then closing the ordered state space. Analytic convergence of either trace remains a separate theorem.

3.5 Primitives and quantum symmetry

The chiral factorisation coproduct in Theorem 3.3 is a separation map. A quantum-group coproduct acts on tensor products of line representations and is supplied by a tensor-compatible exchange theory. Gauge-theoretic examples compare it with the chiral separation coproduct through the line-operator realization.

The primitive part of the chiral coalgebra records connected support. The primitive part of the coassociative bar coalgebra records indecomposable interval states. Their intersection consists of states that are connected in the curve and indecomposable along the interval. This bidegree is the natural home of a binary collision kernel or a first line-defect interaction. An R -matrix arises only after an exchange functor transports these primitives around a two-dimensional collision.

3.6 Physical meaning

The two products of the duoidal category are two experimental operations. Pointwise tensor places two coefficient systems on the same chiral support and then fuses them in transverse order. Chiral convolution places them on separated supports and then forgets the separation by union. Equation (18) says that a local experiment performed in two separated laboratories is independent of whether fusion is described before or after the laboratories are separated.

This is the geometric content of locality in the ordered theory. It refines associativity by constraining support dependence. Planar exchange paths refine it further to braiding. It is the point at which Beilinson's factorisation language and the noncommutative language of defects become one sentence.

4 The block-ordered chiral configuration category

4.1 What is ordered

Let Ch_X denote a fixed operadic enhancement of the Beilinson–Drinfeld chiral pseudo-tensor structure whose algebras in the chosen derived system of right D -modules are the factorisable D -modules of

Theorem 2.1. The noncommutative chiral operad is

$$\mathbf{OCh}_X := E_1 \otimes \mathbf{Ch}_X, \quad (21)$$

where $\otimes$ is the Boardman–Vogt tensor product of ∞ -operads. An algebra over (21) has a chiral factorisation structure and an associative transverse multiplication, with the interchange relation built into the operad.

The word *ordered* refers to the E_1 inputs. It orders entire chiral input blocks. A typical operation is represented by an ordered sequence

$$(I_1, \dots, I_r), \quad (22)$$

where the finite set I_a labels the chiral insertions carried by the a th transverse interval. Points belonging to different blocks may have the same coordinate on X because their transverse coordinates are distinct. Collisions inside the chiral direction are carried by the D -module extension across diagonals; composition in the transverse direction is carried by E_1 .

4.2 The operadic Grothendieck construction

The block-ordered chiral configuration category $\mathbf{BRan}_X$ is the operadic Grothendieck construction of the associative operad acting on the chiral configuration category. Its objects are pairs

$$([r]; I_1, \dots, I_r),$$

with $[r]$ a finite ordinal and each I_a a finite chiral label set. Its generating morphisms are of three kinds:

- (i) chiral refinements, collisions, and permutations inside one I_a ;
- (ii) adjacent composition of two E_1 slots;
- (iii) insertion or removal of the unit slot.

The relations are the chiral relations, the simplicial identities of the associative nerve, and the interchange squares between the two kinds of operation.

The object space of a fixed block profile (22) is

$$X^{I_1} \times \dots \times X^{I_r}. \quad (23)$$

This product is the geometric carrier for operations; the monoidal structure of $\mathbf{FactD}(X)$ is supplied by the chosen operadic pull–tensor–push correspondence. The coefficient morphism attached to an adjacent block composition is obtained from the E_1 multiplication after the chiral inputs have been placed in the common output profile. In a six-functor model, the required pull–tensor–push correspondence is part of the chosen operadic enhancement. The theorem applies to D -modules equipped with the selected operadic enhancement and its base-change maps.

Definition 4.1 (Block-ordered factorisation D -module). A block-ordered factorisation D -module is an algebra over $\mathbf{OCh}_X = E_1 \otimes \mathbf{Ch}_X$ in the chosen derived D -module system. Equivalently, it is a coCartesian operadic section over $\mathbf{BRan}_X$ whose restriction to each block is chiral factorisable, whose restriction to the ordinal direction is Segal, and whose mixed squares satisfy interchange.

Write $\text{OrdFactD}(X)$ for this category. The definition includes the coefficient correspondences needed to compare the different spaces (23); it is therefore stronger than the datum of a simplicial family of sheaves on their underlying products.

Theorem 4.2 (Block-ordered recognition). *There are natural equivalences*

$$\text{Alg}_{E_1}(\text{FactD}(X)) \simeq \text{Alg}_{E_1 \otimes \text{Ch}_X}(\mathcal{V}_X) \simeq \text{OrdFactD}(X), \quad (24)$$

where $\mathcal{V}_X$ is the chosen finite-set D -module system presenting $\text{FactD}(X)$.

Proof. By definition, $\text{FactD}(X) \simeq \text{Alg}_{\text{Ch}_X}(\mathcal{V}_X)$. The exponential law for ∞ -operads gives

$$\text{Alg}_{E_1}(\text{Alg}_{\text{Ch}_X}(\mathcal{V}_X)) \simeq \text{Alg}_{E_1 \otimes \text{Ch}_X}(\mathcal{V}_X).$$

Unstraightening an algebra over the tensor-product operad gives a coCartesian section over its operadic Grothendieck construction. The restrictions to the two operadic factors give the chiral and Segal conditions; the tensor-product relation gives interchange. Straightening reverses the construction. $\square$

The theorem is the invariant meaning of factorisation over an ordered chiral configuration space. The block picture is a geometric presentation of the tensor-product operad.

4.3 The strict and homotopy-coherent forms

In a strict dg model, an ordered chiral algebra is a factorisable D -module A with maps

$$m : A \otimes^{\text{pt}} A \longrightarrow A, \quad \eta : \mathbf{1} \longrightarrow A,$$

satisfying strict associativity and unitality. In the invariant ∞ -category it is an E_1 algebra. A cofibrant A_∞ model replaces the single multiplication by operations indexed by planar trees. The chiral coefficient is unchanged.

Proposition 4.3 (Interchange). *Let γ be a chiral operation and let m be the transverse multiplication. Whenever the two composites are defined by one product-disk embedding, their images satisfy*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \quad (25)$$

with the symmetry isomorphism that places equal-coloured inputs together.

Proof. Equation (25) is the defining interchange relation of $E_1 \otimes \text{Ch}_X$. In the analytic product-disk model, both sides are the factorisation map associated with the same embedding, evaluated by composing first in opposite coordinate directions. $\square$

The proposition is stronger than the coexistence of two operations and weaker than braiding. It says that holomorphic collision and transverse composition are compatible. A planar enhancement adds paths that exchange transverse factors.

4.4 Analytic chamber model

For $r \geq 1$, let

$$C_r^< = \{(t_1, \dots, t_r) \in \mathbb{R}^r \mid t_1 < \dots < t_r\}.$$

This chamber is contractible. A profile (22) has analytic carrier

$$X^{I_1, \text{an}} \times \dots \times X^{I_r, \text{an}} \times C_r^<. \quad (26)$$

Local constancy in $C_r^<$ erases the distances while retaining the order. The X -coordinates remain holomorphic and may approach any partial diagonal allowed by the coefficient D -module.

Compactifying the configurations of subintervals in one interval modulo translation and positive dilation gives associahedra. Their strata are indexed by planar rooted trees. This compactification resolves the homotopy coherence of the E_1 multiplication.

Proposition 4.4 (Associahedral resolution). *For a locally constant factorisation algebra on an oriented line, cellular chains on compactified ordered interval configurations act by the A_∞ bar operations. Contracting every associahedral cell to its corolla gives the ordinary simplicial bar construction through a quasi-isomorphism of operadic resolutions.*

Proof. Factorisation assigns an operation to every ordered embedding of disjoint intervals. The corresponding embedding spaces are contractible, and their compactified boundary strata are planar nested collisions. Stokes' formula is the sum of edge contractions, hence the Stasheff identity. The cellular chains form a cofibrant resolution of the associative operad, while the simplicial bar is the standard resolution of an algebra over that operad. The comparison is the augmentation of the cofibrant resolution. $\square$

The simplex and the associahedron therefore play different roles. The simplex is the indexing object of the simplicial bar. The associahedron is a geometric cofibrant resolution of associative coherence.

4.5 Mixed collision geometry

A chain model for OCh_X may be built from two-coloured forests. Chiral edges describe nested partial diagonals in X ; transverse edges describe nested intervals in the ordered line. A face contracts one edge of one colour. On a total complex, the two boundary operators anticommute by the product orientation.

Lemma 4.5 (Mixed square-zero identity). *Let d_X be the differential supplied by a chain model of the chiral coefficient and let $b_\perp$ be the E_1 bar differential. If both are morphisms for the interchange structure, then*

$$d_X b_\perp + b_\perp d_X = 0$$

after the standard totalization sign. Hence

$$(d_X + b_\perp)^2 = d_X^2 + b_\perp^2.$$

Proof. The two operators are boundaries in independent factors of the mixed operadic resolution. Exchanging their conormal orientation lines contributes the totalization sign. Compatibility with interchange identifies the two coefficient composites. $\square$

Associativity proves $b_\perp^2 = 0$. The chiral identities prove $d_X^2 = 0$ in the chosen chiral model. The total square is zero when both identities and their mixed interchange hold.

4.6 Modules, interfaces, and marked blocks

A right module is an algebra over the two-coloured operad with ordered algebra inputs followed by one module input. A left module uses the opposite order, and a bimodule has both boundary colours. For a right module R and a left module L , the simplicial two-sided bar is

$$B_r(R, A, L) = R \otimes A^{\otimes r} \otimes L. \quad (27)$$

Its realization is interval factorisation homology with endpoint labels R and L .

Proposition 4.6 (Marked-block recognition). *The category of right, left, and bi-modules over A is equivalent to the corresponding category of marked coCartesian sections over the coloured block-ordered configuration category. Relative tensor product of bimodules is gluing of marked intervals.*

Proof. Apply the straightening argument of Theorem 4.2 to the standard coloured associative operads for modules and bimodules. Excision for the glued interval computes the relative tensor product. $\square$

This is the native geometry of boundary-changing fields and defect junctions.

4.7 Opposite order

Reflection of the transverse line reverses the ordinal factor while leaving X unchanged. It induces

$$\text{rev} : \text{OCh}_X \longrightarrow \text{OCh}_X$$

and sends multiplication to its opposite.

Proposition 4.7 (Geometric opposite). *Under Theorem 4.2, transverse reflection sends A to A^{op} . Its chiral coefficient is unchanged.*

Proof. Reflection reverses the order of the two little intervals in every binary E_1 operation, so the resulting multiplication is $m \circ \tau$. It acts trivially on the curve X and therefore leaves the chiral operadic factor and its coefficient correspondences fixed. Straightening identifies the reflected section with A^{op} . $\square$

This is the canonical inversion available at the E_1 level. An E_2 structure equips this reflection with inverse braiding.

4.8 Three distinct forgetful operations

Three distinct operations occur under the heading of symmetry.

First, forgetting the E_1 multiplication leaves the underlying factorisable D -module. This functor loses all transverse products.

Second, derived commutativisation is the left adjoint

$$\text{Alg}_{E_1}(\text{FactD}(X)) \longrightarrow \text{Alg}_{E_\infty}(\text{FactD}(X))$$

when it exists in the selected presentable category. It imposes coherent commutativity on the transverse product. It is the universal coherent-commutativisation of the transverse algebra.

Third, descent from labelled to unlabelled configurations concerns monodromy of a coefficient local system on $\text{Conf}_n(X)$. It asks for extension from the pure braid group to the full braid group; symmetric exchange further asks that the action factor through Σ_n . This is independent of the existence of a transverse E_1 multiplication.

The separation of these three operations is the first safeguard of noncommutative chiral Koszul duality.

5 The mixed operad of chiral collision and ordered fusion

5.1 The invariant operad

Let Ch_X be an ∞ -operad whose algebras in the chosen coefficient category are factorisable D -modules on X . The ordered chiral operad is defined by the Boardman–Vogt tensor product

$$\text{OrdCh}_X := E_1 \otimes \text{Ch}_X. \quad (28)$$

The tensor product is characterized by interchange: an operation in the E_1 direction commutes with an operation in the chiral direction after the canonical permutation and base-change maps.

Theorem 5.1 (Operadic additivity). *Let $\mathcal{V}$ be a presentable symmetric monoidal ∞ -category for which the indicated algebra categories exist. Then*

$$\text{Alg}_{\text{OrdCh}_X}(\mathcal{V}) \simeq \text{Alg}_{E_1}(\text{Alg}_{\text{Ch}_X}(\mathcal{V})). \quad (29)$$

For the D -module realization of Ch_X , the right-hand side is $\text{Alg}_{E_1}(\text{FactD}(X))$.

Proof. Equation (29) is the universal property of the tensor product of ∞ -operads. It identifies an algebra carrying two compatible operadic actions with an algebra over their tensor product. $\square$

For locally constant operations, Dunn additivity identifies $E_1 \otimes E_n$ with E_{n+1} [28, 54]. The chiral operation on X varies singularly across collision diagonals, so (28) retains a holomorphic singular direction together with an ordered topological direction.

5.2 Three geometric models

Three spaces occur for three different reasons.

The simplex indexes the monadic simplicial bar. The associahedron resolves the coherent multiplication of an E_1 algebra. The Fulton–MacPherson compactification resolves relative chiral configurations near collision diagonals. Together they provide the simplicial, associative-coherent, and collision-geometric resolutions.

Fix an ordered partition $I = I_1 \sqcup \cdots \sqcup I_r$. On that stratum, a local product chart has the form

$$K_r \times \prod_{a=1}^r \text{FM}_{I_a}(X), \quad (30)$$

up to the usual base and normal-bundle factors. When a transverse face merges two blocks, the partition changes and the corresponding Fulton–MacPherson factors are replaced by the factor for

their union. Thus the global resolution is assembled from the charts (30) over the category of ordered partitions; the global resolution is the resulting colimit of the partition-indexed charts.

5.3 Two-coloured levelled forests

Choose a cofibrant dg-operadic presentation

$$\mathbf{Q}_X \longrightarrow \mathbf{Ch}_X$$

whose chiral cells are represented locally by Fulton–MacPherson incidence cells with their screen-chain coefficients. Apply a functorial Boardman–Vogt construction to the tensor-product operad:

$$W(E_1 \otimes \mathbf{Q}_X) \longrightarrow E_1 \otimes \mathbf{Ch}_X. \quad (31)$$

A cellular basis element of this resolution is represented by a rooted levelled forest with two kinds of internal edge:

- a transverse edge, recording composition of consecutive E_1 blocks;
- a chiral edge, recording a nested collision inside a chiral coefficient.

A vertex carries the screen-chain coefficient prescribed by $\mathbf{Q}_X$. The determinant line is

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)). \quad (32)$$

Definition 5.2 (Mixed forest complex). For a finite label set I , let

$$W\mathrm{OrdCh}_X(I) = \bigoplus_F C_*(\mathrm{Screen}(F)) \otimes \mathrm{or}(F), \quad (33)$$

where F ranges over the two-coloured levelled forests in the chosen cellular presentation. The differential is the sum of the internal screen differential and the signed contractions of one transverse or one chiral edge.

This definition is a chain model of the operadic resolution. A smooth compactification carrying these strata, when constructed, realizes the same incidence complex geometrically.

5.4 The boundary identity

Write

$$d = d_{\mathrm{scr}} + d_\perp + d_{\mathrm{ch}}.$$

Theorem 5.3 (Mixed boundary theorem). *With the orientation convention (32),*

$$d_\perp^2 = 0, \quad d_{\mathrm{ch}}^2 = 0, \quad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0,$$

and the screen differential supercommutes with both edge-contraction differentials. Hence $d^2 = 0$.

Proof. A codimension-two forest is obtained either by contracting two transverse edges, two chiral edges, or one edge of each colour. In the first two cases, the two orders of contraction induce opposite boundary orientations in the corresponding one-colour Boardman–Vogt complex. In the mixed case,

the final forest is the same in either order, while exchanging the two one-dimensional determinant factors in (32) changes the sign. The screen differential is compatible with every operadic face map and contributes the standard total-complex signs. $\square$

When endomorphism operations are attached to vertices, the transverse part of this identity becomes the Stasheff relations, the chiral part becomes the identities of the chosen chiral coefficient model, and the mixed part says that transverse multiplication is a morphism of chiral objects.

5.5 Cofibrant comparison

Theorem 5.4 (Mixed Boardman–Vogt resolution). *Assume that chain operads are taken in an admissible monoidal model category, that $\mathbf{Q}_X$ is cofibrant, and that the tensor-product operad and functorial W -construction preserve weak equivalences between the relevant cofibrant objects. Then the augmentation*

$$W(E_1 \otimes \mathbf{Q}_X) \longrightarrow E_1 \otimes \mathbf{Ch}_X \quad (34)$$

is a cofibrant resolution. Its algebras are homotopy-coherent ordered chiral algebras. When rectification holds in the coefficient model category, their localized category is equivalent to $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. The W -construction replaces each internal edge by a length parameter and has contractible length fibres. Under the stated model-categorical hypotheses it is cofibrant and its edge-contraction augmentation is an aritywise weak equivalence. The tensor-product universal property identifies its strict algebras with compatible E_1 and chiral structures. Rectification is the standard change-of-operad equivalence along a weak equivalence between admissible cofibrant operads. $\square$

The hypotheses in this theorem are part of the mathematical statement. The displayed hypotheses supply the cofibrant tensor-product resolution and its interchange cells.

5.6 Incidence and screen geometry

A collision forest has an incidence type and a screen space. The incidence category records which clusters collide and in what order. Its fibre over a fixed root partition has a terminal corolla and hence a contractible nerve. The screen space records relative positions inside each cluster and generally has nontrivial cohomology.

For three points on an affine complex line modulo translation and dilation, the open screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore a proper pushforward of the geometric stratum retains more information than the pushforward of its incidence nerve. Arnold classes, configuration-space periods, and propagator forms belong to the screen coefficient in (33); the screen coefficient system carries them, while the incidence differential carries forest contractions.

5.7 Homotopy transfer and Feynman trees

Let a deformation retract of ordered chiral complexes be given. Homological perturbation transfers the structure by summing over the two-coloured forests: vertices carry original operations, internal edges carry the contracting homotopy, and screen chains carry the chiral coefficient. The resulting operations satisfy the complete forest boundary identity because Theorem 5.3 is the differential of the cofibrant resolution.

This is the precise coincidence between Feynman trees and homotopy transfer. The equality is combinatorial once the propagator and vertex maps have been supplied; the propagator and vertex maps select the configuration-space forms that represent the transferred operations.

5.8 Renormalised Feynman weights

In perturbative field theory, an edge of a graph carries a propagator and a vertex carries an interaction. Renormalisation extends the resulting distribution to a compactified configuration space. Boundary restriction factors through contracted subgraphs.

Proposition 5.5 (Feynman weight morphism). *Let a holomorphic-topological BV theory on $X \times \mathbb{R}$ have renormalised graph weights extending to a geometric realization of the mixed forest model. Assume that restriction to every boundary stratum factorizes by graph contraction and that the renormalised quantum master equation holds. Then the graph weights define a morphism from the mixed cofibrant operad to the endomorphism operad of the boundary observables.*

Proof. Factorisation on a boundary stratum identifies the weight with the composite of the weights of the contracted graph and the quotient graph. Stokes' formula, including the BV boundary and counterterm contributions, is exactly compatibility with the differential of the mixed forest complex. These two identities are the operadic composition and chain-map axioms. $\square$

5.9 Three reversals

Reflection of the transverse line sends an E_1 algebra to its opposite. Reversal of an exchange path, when an E_2 enhancement exists, inverts the braid. Verdier duality reverses $!$ - and $*$ -extensions of the chiral coefficient. These are three different operations. The two-colour grading keeps their domains visible and prevents one from being inferred from another.

6 The ordered chiral bar construction

6.1 The simplicial object

Let A be an augmented ordered chiral algebra and let $\bar{A} = \text{Fib}(A \rightarrow \mathbf{1})$. The reduced simplicial bar object in $\text{FactD}(X)$ is

$$B_r^{\text{ord}}(A) = \bar{A}^{\otimes r}, \quad r \geq 0. \quad (35)$$

The inner face d_i multiplies the i th and $(i+1)$ st factors. The outer faces apply the augmentation. Degeneracies insert the unit in the unreduced model. The ordered chiral bar is the geometric realization

$$\text{Bar}_X^{\text{ord}}(A) := |B_{\bullet}^{\text{ord}}(A)|. \quad (36)$$

Every term of (35) is a factorisable D -module. The differential is a morphism in $\text{FactD}(X)$.

In cochain notation, after desuspension,

$$\text{Bar}_X^{\text{ord}}(A) \simeq \left(\bigoplus_{r \geq 0} (s^{-1} \bar{A})^{\otimes r}, d_A + b_m \right), \quad (37)$$

where b_m is the coderivation induced by the E_1 multiplication. Completion replaces the direct sum by a product.

6.2 The chiral coefficient remains intact

Formula (37) uses the internal differential and the transverse multiplication. Every OPE coefficient belongs to the factorisation D -module $A \otimes A$ and is transported through the multiplication morphism m . The bar face composes adjacent ordered factors, while the chiral coefficient retains the complete Laurent expansion.

A residue complex can be paired with the bar as a separate geometric realization. Such a pairing has the form

$$\text{Bar}_X^{\text{ord}}(A) \longrightarrow \Gamma(\text{FM}_I(X), \mathcal{K}_I \otimes A_I),$$

where $\mathcal{K}_I$ is a chosen complex of forms, currents, or local cohomology classes. The map must specify how each principal part is paired with a normal jet. Its square-zero identity uses both the bar differential and the differential of $\mathcal{K}_I$. The intrinsic bar exists before this choice.

Separation theorem 6.1 (Interval composition and collision residue). *The internal bar differential is the coderivation*

$$b_m : T^c(s^{-1}\bar{A}) \longrightarrow T^c(s^{-1}\bar{A})$$

that lowers transverse word length and applies the full multiplication morphism. Poincaré residue is the geometric map

$$\text{Res}_D : \Omega_Y^p(\log D) \longrightarrow \Omega_D^{p-1}$$

that lowers logarithmic form degree and extracts the first normal symbol. A geometric bar model combines them through a specified tensor-product coefficient map.

Proof. The coderivation b_m is induced by the simplicial face that composes two adjacent interval labels; its source and target are tensor coalgebras and its degree is fixed by suspension. The residue is induced by the exact sequence of logarithmic forms along a divisor; its source and target are de Rham complexes and it lowers form degree. Their tensor product acts on a geometric bar model. The mixed compatibility equation is the vanishing of their graded commutator, which is precisely compatibility of multiplication with the coefficient map along the collision face. $\square$

The displayed sources and targets determine the two operations and their mixed compatibility equation.

6.3 Coalgebra structure

Deconcatenation gives

$$\Delta[a_1 | \cdots | a_r] = \sum_{p=0}^r [a_1 | \cdots | a_p] \otimes [a_{p+1} | \cdots | a_r]. \quad (38)$$

It is coassociative and conilpotent. The bar differential is a coderivation.

Proposition 6.2 (Internal bar coalgebra). *The ordered chiral bar is a conilpotent coassociative coalgebra object in $\text{FactD}(X)$. The coaugmentation is the empty word, and the coradical filtration is the transverse word-length filtration.*

Proof. Deconcatenation is defined componentwise in every symmetric monoidal category. Its coassociativity is the equality of the two ways of cutting a word twice. The multiplication coderivation is local with respect to cuts, so it satisfies the coderivation identity. Iterating the reduced coproduct of a word of length r more than r times gives zero. $\square$

The coalgebra structure records interval splitting. A chiral coproduct of fields is an additional operation. A bialgebra structure consists of that operation together with its compatibility with (38).

6.4 Interval factorisation homology

Let $([0, 1], \partial)$ denote an oriented interval whose endpoints are labelled by the augmentation module $\mathbf{1}$. Excision for factorisation homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1}. \quad (39)$$

The standard two-sided simplicial resolution of the derived tensor product is precisely (35).

Theorem 6.3 (Bar as interval factorisation homology). *There is a canonical equivalence in $\text{FactD}(X)$*

$$\text{Bar}_X^{\text{ord}}(A) \simeq \int_{([0,1],\partial)} A. \quad (40)$$

For a right module R and a left module L ,

$$\text{Bar}(R, A, L) \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Choose a Weiss cover of the interval by an ordered chain of small intervals together with collars at the endpoints. Factorisation excision identifies the colimit of the Čech diagram with the derived relative tensor product. Refining the cover gives the simplicial two-sided bar. The argument takes place in the coefficient category $\text{FactD}(X)$ and is therefore compatible with chiral factorisation. $\square$

The theorem is the physical meaning of the bar construction. It is the amplitude of an interval whose interior carries the boundary operator algebra and whose endpoints carry the vacuum boundary state.

6.5 The associahedral model

The simplicial bar is the derived tensor product associated with a strict E_1 multiplication. A homotopy-coherent multiplication is an algebra over a cofibrant resolution $W\text{Ass} \rightarrow \text{Ass}$. Its bar coderivation contains the transferred operations m_k ,

$$b = \sum_{k \geq 1} b_{m_k}, \quad (41)$$

and $b^2 = 0$ is the family of Stasheff identities.

Let K_r denote the associahedron in the convention in which its cells are planar trees with r leaves. The cellular chains $C_*(K_r)$ form one presentation of $W\text{Ass}(r)$. They resolve the coherent associative operation represented simplicially by the bar.

Theorem 6.4 (Associahedral comparison). *Assume an admissible monoidal dg model of $\text{FactD}(X)$, let A be cofibrant as a $W\text{Ass}$ -algebra, and let A^{str} be a strictification along $W\text{Ass} \rightarrow \text{Ass}$. The operadic bar construction of A has a cellular filtration with terms*

$$\bigoplus_{r \geq 0} C_*(K_r) \otimes (s^{-1}\bar{A})^{\otimes r}, \quad (42)$$

whose differential contains the cellular boundary and the operations m_k . Change of operad and the simplicial two-sided bar give a natural zigzag of weak equivalences

$$\text{Bar}_{W\text{Ass}}(A) \simeq \text{Bar}_{\text{Ass}}(A^{\text{str}}) \simeq \text{Bar}_X^{\text{ord}}(A^{\text{str}}).$$

Proof. Cellular chains of associahedra present the Boardman–Vogt resolution of the associative operad. The change-of-operad adjunction is a Quillen equivalence under the stated admissibility and cofibrancy hypotheses. Bar constructions are homotopy invariant on cofibrant algebras and compute the same derived relative tensor product $\mathbf{1} \otimes_A^L \mathbf{1}$. This yields the zigzag. $\square$

The theorem identifies the simplicial and associahedral homotopy models of the same interval amplitude.

6.6 Low arity

For a strict dg algebra, the normalized reduced bar begins

$$\bar{A} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 2} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 3} \xleftarrow{b} \dots$$

The binary differential is

$$b[a|b] = (-1)^{|a|}[ab].$$

The length-three component of b^2 is the signed associator. In higher length, every term is indexed by two adjacent multiplications; the two orders of contracting the corresponding planar edges occur with opposite bar signs.

For an A_∞ algebra, m_3 fills the homotopy between the two binary composites. At the first two-dimensional associahedral cell, m_4 and composites of m_2, m_3 fill the pentagon. The geometry is exactly the cellular boundary of associahedra.

The chiral coefficient adds an independent filtration. A word $[a|b]$ can carry an arbitrary OPE singularity in the X variables. The bar differential applies the transverse product ab as a morphism of those singular coefficient systems.

6.7 Functoriality

An augmented morphism $f : A \rightarrow B$ of ordered chiral algebras induces

$$\text{Bar}(f) : \text{Bar}_X^{\text{ord}}(A) \longrightarrow \text{Bar}_X^{\text{ord}}(B)$$

by applying f to every bar factor. A homotopy-coherent morphism induces the corresponding morphism of cellular bars.

A square-zero derivation Q of A induces a coderivation Q_{Bar} acting on each factor. Since Q is a

derivation of multiplication,

$$[b_m, Q_{\text{Bar}}] = 0.$$

This identity is the basis of BRST descent.

A monoidal functor $F : \text{FactD}(X) \rightarrow \mathcal{V}$ preserving geometric realizations satisfies

$$F(\text{Bar}_X^{\text{ord}}(A)) \simeq \text{Bar}_{\mathcal{V}}(F(A)). \quad (43)$$

Thus restriction to a formal disc, analytic realization, taking a fibre, and factorisation homology commute with the ordered bar when their continuity hypotheses hold.

6.8 Fubini in the two directions

Let $\mathcal{A}$ be the analytic factorisation algebra on $X^{\text{an}} \times \mathbb{R}$ corresponding to A . For a compact one-manifold M with boundary labels, Fubini for factorisation homology gives

$$\int_{U \times M} \mathcal{A} \simeq \int_U \left(\int_M \mathcal{A} \right). \quad (44)$$

Taking $M = [0, 1]$ with vacuum endpoints yields the analytic realization of (40). Taking $M = S^1$ yields internal Hochschild chains whenever the circle factorisation homology exists. A cyclic trace or trace-class analytic completion is additional data needed only to turn those chains into scalar amplitudes.

Formula (44) is the first exact relation between chiral homology and topological factorisation. It integrates the transverse direction while retaining the chiral direction as a coefficient system.

6.9 Completion

The bar length filtration has finite quotients

$$\text{Bar}_{\leq N}(A) = \bigoplus_{r=0}^N (s^{-1}\bar{A})^{\otimes r}.$$

The completed bar is

$$\widehat{\text{Bar}}_X^{\text{ord}}(A) = \prod_{r \geq 0} (s^{-1}\bar{A})^{\otimes r}. \quad (45)$$

The coderivation is continuous when every component has finite output in a fixed total filtration degree. This holds automatically for strict E_1 multiplication and for filtered A_∞ operations that raise a complete positive filtration.

If conformal weight is unbounded, one can first restrict to a finite weight window and then complete. The two inverse limits commute under the Mittag-Leffler condition described in [Theorem 7.3](#).

Proposition 6.5 (Cohomology of completion). *Let C_N be the tower of finite bar-length and finite-weight truncations with surjective transition maps on cohomology. Then*

$$H^*(\varprojlim_N C_N) \simeq \varprojlim_N H^*(C_N).$$

Proof. The Milnor exact sequence has a $\varprojlim^1_N$ term. Surjectivity gives the Mittag-Leffler condition

and annihilates that term. □

Completion is the inverse-limit theorem for the finite weight system.

6.10 The bar of a boundary endomorphism algebra

Let $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ be the endomorphism algebra of a compact boundary condition. The two-sided bar

$$\mathrm{Bar}(b, A_b, b)$$

resolves the identity kernel of the full subcategory generated by b . Under the Morita equivalence of Theorem 2.7, it is the factorisation analogue of the diagonal bimodule resolution.

Applying internal Hom to this resolution computes Hochschild cochains. Applying a trace computes Hochschild chains. Applying a second boundary label computes defect-changing operators. Thus the ordered bar is the common resolution behind the centre, the trace, and the module category.

6.11 The perturbative reading

In a perturbative boundary field theory, a bar word is a sequence of boundary insertions ordered along a collar. The bar differential fuses adjacent insertions. A Feynman propagator on X controls the chiral singularity of that fusion. Counterterms extend the coefficient distribution to the collision diagonal. The two operations are separate but compatible.

For a free Gaussian theory, Wick contraction gives an explicit morphism from the bar to the configuration-space de Rham complex. For an interacting renormalisable theory, Costello's renormalisation constructs the corresponding factorisation map order by order in $\hbar$. The quantum master equation is precisely the condition that the resulting differential squares to zero.

This is the exact sense in which a bar word is an interval amplitude. The bar differential is composition of boundary operations; the propagator supplies the coefficient of that composition.

7 Ordered chiral bar–cobar reconstruction

7.1 The internal adjunction

Put

$$\mathcal{C}_X := (\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1}).$$

Assume throughout this section that $\mathcal{C}_X$ admits the geometric realizations and totalizations used below, and that tensor product preserves geometric realizations separately. These hypotheses hold in the standard presentable derived enhancements after restricting to the complete subcategories on which the indicated limits exist.

For a coaugmented conilpotent coassociative coalgebra C with coaugmentation coideal $\bar{C}$, define

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \left(\bigoplus_{r \geq 0} (s\bar{C})^{\otimes r}, d_C + d_{\Delta} \right), \quad (46)$$

where d_{Δ} is the derivation induced by the reduced coproduct. A complete cobar uses the product over tensor lengths and the continuous extension of the same derivation.

A degree-one morphism $\tau : C \rightarrow A$ is a twisting morphism when

$$d\tau + m(\tau \otimes \tau)\bar{\Delta} = 0. \quad (47)$$

The universal twisting morphisms give natural equivalences of mapping spaces

$$\mathrm{Map}_{\mathrm{Alg}_{E_1}}(\mathrm{Cobar}_X^{\mathrm{ord}} C, A) \simeq \mathrm{Tw}(C, A) \simeq \mathrm{Map}_{\mathrm{Coalg}_{\mathrm{coAss}}} (C, \mathrm{Bar}_X^{\mathrm{ord}} A).$$

Hence

$$\mathrm{Cobar}_X^{\mathrm{ord}} : \mathrm{Coalg}_{\mathrm{coAss}}^{\mathrm{conil}}(\mathcal{C}_X) \rightleftarrows \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathcal{C}_X) : \mathrm{Bar}_X^{\mathrm{ord}} \quad (48)$$

is an adjunction. The unit and counit become equivalences under connectivity, conilpotence, and completeness hypotheses.

7.2 The connected positive-weight theorem

Definition 7.1 (Admissible positive object). An augmented algebra A in $\mathcal{C}_X$ is admissibly positive if it has a complete weight decomposition

$$A \simeq \mathbf{1} \oplus \prod_{w \geq 1} A_w, \quad A_p \otimes A_q \longrightarrow A_{p+q},$$

with each finite weight truncation dualisable and bounded below after every component evaluation $\mathrm{ev}_I : \mathrm{FactD}(X) \rightarrow D\text{-mod}(X^I)$. The evaluations are required to be conservative and to commute with the bar and cobar constructions. A coalgebra is admissibly positive when its coaugmentation coideal has the analogous positive, complete, conilpotent weight grading.

The positivity condition makes every fixed weight of a bar or cobar contain only finitely many tensor lengths. It is the internal analogue of a connected weight-graded augmented dg algebra.

Theorem 7.2 (Positive ordered bar–cobar reconstruction). *For an admissibly positive augmented ordered chiral algebra A , the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A \quad (49)$$

is an equivalence. For an admissibly positive conilpotent coalgebra C , the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C) \quad (50)$$

is an equivalence.

Proof. Filter the two composites by tensor length and then take a fixed total weight. Positivity makes the resulting filtration finite. After applying ev_I , the associated graded is the ordinary connected associative bar–cobar resolution in complexes; its augmentation has the standard extra-degeneracy contraction. Thus each evaluated unit and counit is a quasi-isomorphism in every finite weight. Conservativity of the evaluations gives the equivalence in $\mathcal{C}_X$, and completeness reconstructs the full object from its finite weight truncations. $\square$

Corollary 7.3 (Completed reconstruction). *Let $A \simeq \varprojlim_N A_{\leq N}$ be an inverse limit of admissibly positive finite truncations, and suppose the transition maps are Mittag-Leffler on cohomology and*

tensor product commutes with the inverse limit. Then the continuous counit

$$\widehat{\text{Cobar}}_X^{\text{ord}} \widehat{\text{Bar}}_X^{\text{ord}}(A) \longrightarrow A$$

is an equivalence. The analogous assertion holds for a complete conilpotent coalgebra.

Proof. The theorem applies at every finite stage. The Milnor exact sequence identifies cohomology of the inverse limit with the inverse limit of cohomology because the $\varprojlim^1$ term vanishes under the Mittag-Leffler hypothesis. $\square$

This is the associative bar–cobar theorem internal to the coefficient category. Its convergence is controlled by weight or augmentation completion for the pointwise tensor product.

7.3 The separate Ran theorem

The Ran space carries another monoidal structure, the chiral tensor product $\otimes^{\text{ch}}$. By Theorem 2.5, its reduced cardinality quotients are nilpotent. Francis and Gaitsgory prove the enhanced bar–cobar equivalence there for operads satisfying their reduced or pro-nilpotent hypotheses. The reduced associative operad gives the outer E_1 case, while their chiral Lie construction gives the Lie–commutative–coalgebra equivalence [38].

Theorem 7.4 (Enhanced Ran bar–cobar theorem). *In the pro-nilpotent monoidal category $(D(\text{Ran } X), \otimes^{\text{ch}})$, enhanced bar and cobar are inverse for the reduced associative operad, and more generally for the operads covered by the Francis–Gaitsgory enhanced theorem. The corresponding Lie case identifies chiral Lie algebras with the appropriate conilpotent commutative coalgebras. A restriction to a factorisation subcategory is valid when that subcategory is closed under the enhanced constructions and the unit and counit remain factorisation equivalences.*

Proof. The cardinality filtration is pro-nilpotent by Theorem 2.5. Francis–Gaitsgory’s enhanced Koszul-duality theorem applies to reduced operads satisfying their pro-nilpotent hypotheses and identifies the enhanced unit and counit with equivalences [38]. The reduced associative operad and the chiral Lie operad lie in this class. Closure of a factorisation subcategory under the two functors and equivalence of the restricted unit and counit make the ambient equivalence restrict to that subcategory. $\square$

Theorem 7.4 uses chiral convolution on the Ran space. Theorem 7.2 uses pointwise tensor product and resolves the transverse E_1 multiplication. A strong monoidal comparison functor relates their bars and cobars.

7.4 The augmentation boundary

The unit is a left A -module through the augmentation. Define

$$A^! := \text{RHom}_A(\mathbf{1}, \mathbf{1}). \quad (51)$$

Composition gives $A^!$ an E_1 multiplication. When the bar terms are dualisable, or when continuous internal Hom is used, the bar resolution gives

$$A^! \simeq \text{RHom}_{\mathcal{C}_X}(\text{Bar}_X^{\text{ord}}(A), \mathbf{1}). \quad (52)$$

Theorem 7.5 (Boundary Koszul duality). *Let A be an admissibly positive augmented ordered chiral algebra and let $A^!$ be defined by (51). The adjunction*

$$- \otimes_{A^!}^L \mathbf{1} \dashv \mathrm{RHom}_A(\mathbf{1}, -)$$

restricts to an equivalence between the thick subcategory of A -modules generated by $\mathbf{1}$ and the thick subcategory of right $A^!$ -modules generated by $A^!$. If $\mathbf{1}$ is a compact generator of the selected A -module category, the equivalence extends to its presentable completion.

Proof. The unit of the adjunction is an equivalence on the free rank-one $A^!$ -module, and the counit is an equivalence on $\mathbf{1}$. Exactness and preservation of retracts extend these equivalences to the two thick closures. Compact generation extends the result by Ind-completion. $\square$

Thus $A^!$ is the algebra of local operators on the augmentation boundary condition. It retains the complete endomorphism algebra of the augmentation boundary condition.

7.5 The double dual and completion

There is a canonical map

$$A \longrightarrow A^{!!} := \mathrm{RHom}_{A^!}(\mathbf{1}, \mathbf{1}). \quad (53)$$

Theorem 7.6 (Double dual as augmentation completion). *Let A be admissibly positive, with dualisable finite weight pieces. Then (53) identifies $A^{!!}$ with the derived completion of A for the augmentation ideal. In particular, if A is complete for its positive weight filtration, then $A \simeq A^{!!}$.*

Proof. At every finite weight, the bar resolution is finite and dualisable. Dualising it once computes $A^!$; dualising the dual bar computes the finite augmentation quotient of A . The maps are compatible with the weight tower. Taking the inverse limit gives the derived augmentation completion, and completeness gives the final equivalence. $\square$

The completion is part of the statement. The weight completion is the inverse limit of the finite truncations and may contain infinite compatible sequences beyond the direct sum.

7.6 Quadratic comparison

Suppose that A has a connected quadratic presentation

$$A = T_{\mathcal{C}_X}(V)/(R)$$

with dualisable positive weight pieces. The quadratic coalgebra

$$A^i = C(s^{-1}V, s^{-2}R) \subset T^c(s^{-1}V)$$

has a canonical twisting morphism and a comparison

$$q_A : A^i \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}}(A). \quad (54)$$

Theorem 7.7 (Quadratic recognition). *Assume that component evaluation is conservative and preserves the quadratic twisting constructions. The following conditions are equivalent:*

- (i) q_A is an equivalence;
- (ii) $\text{Cobar}(A^i) \rightarrow A$ is an equivalence;
- (iii) the left quadratic twisted tensor product is acyclic;
- (iv) the right quadratic twisted tensor product is acyclic;
- (v) the evaluated bar homology is concentrated on the quadratic diagonal in every finite weight.

Proof. After component evaluation this is the fundamental theorem for Koszul twisting morphisms. Conservativity lifts the equivalences to $\mathcal{C}_X$. Finite positive weights justify the passage between the quadratic coalgebra and the full bar. $\square$

Quadratic Koszulness concerns the comparison (54). It is stronger than the universal reconstruction of Theorem 7.2.

7.7 PBW and nonhomogeneous relations

Let $F_\bullet A$ be a complete exhaustive multiplicative filtration and suppose $\text{gr}_F A \cong A_0$. Filter the bar by total PBW degree.

Theorem 7.8 (Filtered PBW comparison). *Assume that the filtered comparison is complete and separated, its spectral sequence converges strongly in every finite chiral-cardinality and conformal-weight window, and A_0 is quadratic Koszul. Then the nonhomogeneous Koszul coalgebra determined by the quadratic, linear, and constant parts of the relations maps equivalently to $\text{Bar}_X^{\text{ord}}(A)$.*

Proof. The associated graded of the comparison is the quadratic equivalence for A_0 . Strong convergence lifts it through the filtration. Linear terms become part of the coalgebra differential, and constant terms become its curvature functional. $\square$

An inhomogeneous algebra is computed by its quadratic-linear-constant dual and the induced differential and curvature.

7.8 Curved coalgebras

A curved coalgebra has a coderivation d_C and a curvature functional $h : C \rightarrow \mathbf{1}[2]$ satisfying

$$d_C^2 = (h \otimes \text{id} - \text{id} \otimes h)\bar{\Delta}, \quad h d_C = 0.$$

Its curved cobar has a square-zero total differential because the curvature functional supplies the constant part of the cobar derivation. This curvature is an algebraic feature of an inhomogeneous presentation. A family index map may subsequently send this algebraic element to a two-form on a moduli space.

Curved modules use the coderived or contraderived category when coacyclic or contraacyclic complexes form the relevant null system [60]. The module theory and completion determine the choice.

7.9 Verdier duality

Let $\mathbb{D}_{\text{Ran}}$ denote Verdier duality. If the finite bar stages are dualisable and duality commutes with the completion, then

$$A_{\text{Verdier}}^{\vee} := \mathbb{D}_{\text{Ran}} \text{Bar}_X^{\text{ord}}(A) \quad (55)$$

acquires an algebra structure by dualising deconcatenation.

A continuous perfect pairing compares the Verdier algebra (55) with the boundary endomorphism algebra (51):

$$\text{Bar}_X^{\text{ord}}(A) \otimes A^! \longrightarrow \mathbf{1}.$$

Proposition 7.9 (Three dual constructions). *For an augmented ordered chiral algebra, the following are distinct until an explicit equivalence is proved:*

- (i) *the covariant reconstruction $\text{Cobar Bar}(A) \rightarrow A$;*
- (ii) *the boundary algebra $A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$;*
- (iii) *the Verdier dual algebra $\mathbb{D}_{\text{Ran}} \text{Bar}(A)$, when defined.*

Proof. The first construction is covariant and lands in augmented algebras. The second is contravariant through the module variable and lands in endomorphism algebras of one boundary object. The third is contravariant through sheaf duality and is defined only under Verdier dualisability and continuity. Their variances, domains, and targets therefore differ. An equivalence between any two is additional comparison data. $\square$

This separation assigns opposite order, inverse braiding, level reflection, and Verdier duality to their respective comparison maps.

7.10 The complete deformation complex

The deformation complex of the transverse E_1 multiplication is the shifted internal Hochschild complex

$$\text{Def}_{E_1}(A) = \text{HH}_{\text{FactD}(X)}^{\bullet}(A, A)[1]. \quad (56)$$

The chiral coefficient has its own deformation complex. On a formal disc it is the chiral operadic complex governing vertex or nonlocal-vertex operations. The deformation theory of the full ordered chiral object is controlled by a cofibrant resolution of the tensor-product operad $E_1 \otimes \text{Fact}_X$.

Proposition 7.10 (Bigraded deformation complex). *Choose an admissible cofibrant resolution of $E_1 \otimes \text{Fact}_X$. The convolution deformation complex of an ordered chiral algebra has a complete filtration by the number of transverse and chiral vertices. Its associated graded is spanned by two-coloured trees; the horizontal axis restricts to the Hochschild deformation complex of the E_1 multiplication, the vertical axis restricts to the deformation complex of the chiral coefficient, and the mixed components impose the interchange law.*

Proof. The cellular Boardman–Vogt resolution is filtered by the two edge colours. Taking convolution with the endomorphism operad sends its cellular boundary to the deformation differential. Forests containing only transverse edges give Hochschild cochains; forests containing only chiral edges give coefficient deformations; forests of both colours give the mixed compatibility terms. $\square$

A Morita equivalence identifies the Hochschild axis. It identifies the full ordered chiral deformation problem only when it also identifies the chiral coefficient and the mixed interchange maps.

8 The Koszul–Morita square and the universal boundary state

The bar coalgebra is simultaneously a resolution of the augmentation and the coalgebraic form of the universal boundary state. Once this point is made explicit, the interval, the diagonal bimodule, the annulus, Hochschild theory, and the derived centre become different shadows of one two-sided construction.

Throughout this section let

$$\mathcal{C}_X = (\text{FactD}(X), \otimes^{\text{pt}}, \mathbf{1}),$$

let A be an admissibly positive augmented E_1 algebra in $\mathcal{C}_X$, and put

$$C = \text{Bar}_X^{\text{ord}}(A).$$

Let $\tau : C \rightarrow A$ be the universal twisting morphism. All tensor and cotensor products below are derived in the complete subcategories selected by the positive weight filtration.

8.1 One-sided Koszul transforms

For a left A -module M , define the twisted left C -comodule

$$\mathbb{K}_A(M) = (C \otimes M, d_C + d_M + d_\tau), \quad (57)$$

where d_τ applies the reduced coproduct to C , evaluates one factor by τ , and lets the result act on M . For a conilpotent left C -comodule N , define

$$\mathbb{L}_A(N) = (A \otimes N, d_A + d_N + d_\tau). \quad (58)$$

The Maurer–Cartan equation for τ is exactly the statement that both displayed differentials square to zero.

Theorem 8.1 (Module–comodule Koszul equivalence). *The functors $\mathbb{K}_A$ and $\mathbb{L}_A$ are inverse equivalences between the derived category of augmentation-complete left A -modules and the coderived category of conilpotent left C -comodules. The same statement holds for right modules and right comodules.*

Proof. Filter both composites by the coradical degree of C . In each fixed positive weight, the filtration is finite. The associated graded twisting differential vanishes, and the unit and counit reduce to the ordinary bar–cobar contractions. The extra degeneracy of the simplicial bar gives the contraction at each finite weight. Conservativity of component evaluation and completeness lift the equivalence to $\mathcal{C}_X$. When the coalgebra differential is curved or infinite products are essential, the same filtered contraction has coacyclic cone and therefore gives the asserted coderived equivalence. $\square$

The theorem is the module form of ordered chiral Koszul duality. It is stronger than an equality of Ext groups and weaker than a Morita equivalence between all A -modules and all modules over $A^!$. It identifies precisely the sector generated from the augmentation boundary.

8.2 The two-sided transform

For an A -bimodule M , set

$$\mathbb{K}_A^{\text{bi}}(M) = C \otimes_{\tau} M \otimes_{\tau} C. \quad (59)$$

The two outer factors carry the left and right C -coactions. The differential has four terms: the differentials of C and M , the left twisting action, and the right twisting action. The two twisting terms anticommute because the left and right A -actions commute.

Conversely, for a conilpotent C -bicomodule N , set

$$\mathbb{L}_A^{\text{bi}}(N) = A \otimes_{\tau} N \otimes_{\tau} A. \quad (60)$$

Theorem 8.2 (Bimodule–bicomodule equivalence). *The functors $\mathbb{K}_A^{\text{bi}}$ and $\mathbb{L}_A^{\text{bi}}$ are inverse equivalences between augmentation-complete A -bimodules and conilpotent C -bicomodules in the corresponding derived or coderived categories.*

Proof. Apply Theorem 8.1 first to the left action and then to the right action. The two transforms commute because their twisting coderivations act on different outer coalgebra factors and the bimodule identity makes their mixed commutator zero. Equivalently, filter (59) by the sum of the two coradical degrees and use the two-sided extra-degeneracy contraction on the associated graded. $\square$

8.3 The diagonal bicomodule

The diagonal A -bimodule is A itself. The diagonal C -bicomodule is the coalgebra C equipped with the two coactions

$$C \xrightarrow{\Delta} C \otimes C, \quad C \xrightarrow{\Delta} C \otimes C,$$

one read as a left coaction and the other as a right coaction. Denote it by C_{Δ} .

Theorem 8.3 (Diagonal correspondence). *There is a canonical equivalence of C -bicomodules*

$$\mathbb{K}_A^{\text{bi}}(A) \simeq C_{\Delta}. \quad (61)$$

Proof. The two-sided transform of A is

$$C \otimes_{\tau} A \otimes_{\tau} C.$$

The bar–cobar counit contracts the middle A successively against the left and right twisting cochains. Filter by the total coradical degree of the two outer factors. On the associated graded, the twisting terms vanish and the contraction identifies the complex with one copy of C . Under this identification the two surviving coactions are the two legs of Δ_C . Finite positive weights and completeness lift the associated-graded equivalence. $\square$

The theorem gives a precise mathematical form to the phrase “universal boundary state.” The diagonal bimodule is the identity kernel for tensoring over A ; its Koszul transform is the identity kernel for cotensoring over C .

Corollary 8.4 (Composition unit). *For every conilpotent C -bicomodule N ,*

$$C_{\Delta} \square_C^L N \simeq N, \quad N \square_C^L C_{\Delta} \simeq N.$$

Proof. The diagonal bicomodule is the regular C -bicomodule. The standard cobar resolution of a conilpotent comodule has an extra codegeneracy after cotensoring with C_Δ , contracting it onto the original comodule. Equivalently, transport the identity $A \otimes_A^L M \simeq M$ through Theorems 8.3 and 8.5. $\square$

8.4 Tensor becomes cotensor

Let M be an A – B bimodule and N a B – D bimodule, with all algebras admissibly positive and all modules complete. Relative tensor product composes defects. The coalgebraic operation dual to it is derived cotensor product.

Theorem 8.5 (Tensor–cotensor gluing). *Let C_A, C_B, C_D be the ordered bar coalgebras. Then there is a canonical equivalence*

$$\mathbb{K}^{\text{bi}}(M \otimes_B^L N) \simeq \mathbb{K}^{\text{bi}}(M) \square_{C_B}^L \mathbb{K}^{\text{bi}}(N). \quad (62)$$

Proof. Choose two-sided bar resolutions for the relative tensor product and cobar resolutions for the cotensor product. After inserting the universal twisting morphisms, both total complexes are indexed by the same strings of B -operations between M and N . The Eilenberg–Zilber comparison identifies the two bisimplicial totalizations. In fixed positive weight the comparison is finite and is a quasi-isomorphism by the standard two-sided Koszul contraction. Completion gives (62). $\square$

This theorem is the algebraic form of cutting and regluing an interval. Tensoring sums over intermediate open states. Cotensoring imposes equality of the two coalgebraic boundary states.

8.5 Hochschild and coHochschild theories

Let

$$C^e = C \otimes C^{\text{cop}}.$$

For a C -bicomodule N , define

$$\text{coHH}_*(C, N) := C_\Delta \square_{C^e}^L N,$$

and

$$\text{coHH}^*(C, N) := \text{RHom}_{C^e}(C_\Delta, N)$$

in the selected coderived category.

Theorem 8.6 (Chiral Hochschild–coHochschild duality). *For every complete A -bimodule M there are natural equivalences*

$$\text{HH}_*(A, M) \simeq \text{coHH}_*(C, \mathbb{K}_A^{\text{bi}}(M)), \quad (63)$$

$$\text{HH}^*(A, M) \simeq \text{coHH}^*(C, \mathbb{K}_A^{\text{bi}}(M)). \quad (64)$$

In particular,

$$\text{HH}_*(A) \simeq C_\Delta \square_{C^e}^L C_\Delta,$$

$$\text{HH}^*(A) \simeq \text{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Proof. Hochschild homology is the derived tensor product of the diagonal bimodule with M over A^e . Transport this expression through Theorem 8.2. By Theorem 8.3, the diagonal bimodule maps to

C_Δ , and by Theorem 8.5, derived tensor becomes derived cotensor. This proves (63). Derived Hom is preserved by an equivalence of derived categories, which proves (64). $\square$

The annulus is therefore the self-cotensor of the universal boundary state. The derived centre is its coalgebraic self-extension. The bimodule–bicomodule equivalence identifies the two descriptions.

8.6 The Koszul–Morita square

The preceding theorems assemble into a commutative square of functors:

$$\begin{array}{ccc}
 D(A^e\text{-mod})_{\text{comp}} & \xrightarrow[\sim]{\mathbb{K}_A^{\text{bi}}} & D^{\text{co}}(C^e\text{-comod})_{\text{conil}} \\
 A \otimes_{A^e}^L (-) \downarrow & & \downarrow C_\Delta \square_{C^e}^L (-) \\
 D(\mathcal{C}_X) & \xlongequal{\quad\quad\quad} & D(\mathcal{C}_X).
 \end{array} \tag{65}$$

The corresponding square for derived Hom computes the centre. This is the Koszul–Morita square. Its horizontal maps change algebraic presentation. Its vertical maps close an interval into a circle. The square says that duality commutes with closure.

Corollary 8.7 (Centre as the closed endomorphism of the boundary state). *The ordered chiral centre has the equivalent descriptions*

$$Z_{\text{ord, ch}}(A) \simeq \text{RHom}_{A^e}(A, A) \simeq \text{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Its E_2 operations are transported by the bimodule–bicomodule equivalence to the brace operations of the diagonal bicomodule.

Proof. The first description is the definition of Hochschild cochains as endomorphisms of the diagonal bimodule. The second follows from Theorems 8.2 and 8.3: an equivalence of enriched derived categories preserves derived endomorphism objects. Brace operations are natural under this equivalence because both are induced by composition of kernels, so the E_2 structures agree. $\square$

8.7 Calabi–Yau transport

Suppose the perfect A -module category carries a d -Calabi–Yau trace. Equivalently, the diagonal bimodule has a nondegenerate cyclic pairing of degree d compatible with composition. Transporting the pairing through Theorem 8.2 gives a cyclic copairing on C_Δ .

Theorem 8.8 (Transport of the open trace). *Assume that the bimodule–bicomodule equivalence lifts to cyclic objects and commutes with geometric realization of the cyclic bar constructions. Then a perfect d -Calabi–Yau structure on the augmentation-generated A -module category induces a nondegenerate degree- d cyclic structure on the diagonal bicomodule C_Δ . Under the resulting S^1 -equivariant Hochschild–coHochschild equivalence, the Connes operator and the pair-of-pants product correspond; when the framed circle action is defined, so do the induced Batalin–Vilkovisky operators.*

Proof. A Calabi–Yau trace is a cyclic natural transformation on the Morita category of perfect modules. Theorem 8.5 transports composition. By the assumed lift to cyclic objects, the cyclic bar construction, its rotation, and its gluing maps are carried to their coHochschild counterparts. The Connes and

pair-of-pants operations therefore agree under the equivalence. The same argument applies to the BV operator when the framed circle action has been supplied. $\square$

This is the exact bridge from ordered Koszul duality to ribbon geometry. It creates the open topological surfaces generated by the transverse trace; modular geometry of varying complex curves is supplied by the independent chiral sewing system.

8.8 Physical interpretation

Let a factorization-homology realization identify a boundary condition b with the endomorphism algebra $A = \mathrm{RHom}(b, b)$. Under interval excision, the bar coalgebra C is the state object of an interval ending on the augmentation boundary. An A – B bimodule is an interface. Relative tensor product composes interfaces. The Koszul transform turns the same interface into a bicomodule of boundary states, and cotensor product performs the same gluing in coalgebraic variables.

Closing an interface into an annulus gives Hochschild homology. Inserting a local bulk operation gives Hochschild cohomology. Theorem 8.6 says that both amplitudes can be computed before or after ordered chiral Koszul duality. The operation is the same; only the variables have changed.

The square (65) is therefore the precise mathematical form of open–closed complementarity on the augmentation-generated sector. It is a categorical equivalence. Applying a specified trace produces its numerical shadows.

9 Exchange begins beyond the ordered line

9.1 The planar origin of braiding

For every n , the chamber

$$\mathrm{Conf}_n^<(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is convex. It remembers the order of factors but has trivial fundamental groupoid.

Theorem 9.1 (Planar origin of braiding). *Ordered configurations on an oriented line decompose into contractible chambers. Thickening the line to a plane connects these chambers by exchange paths, whose fundamental group is the braid group. Hence an E_2 enhancement, a flat connection on planar configuration space, or equivalent exchange data supplies the braid action of an ordered chiral algebra.*

Proof. Parallel transport inside a contractible chamber is canonically trivial. A braid generator is represented by a path interchanging points in a plane or another two-dimensional configuration space. $\square$

Noncommutativity and braiding are distinct structures. The associative multiplication governs chamberwise fusion; the exchange local system governs transport between chambers.

9.2 The E_2 enhancement

Replacing the transverse line by an oriented plane gives a forgetful functor

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

An E_2 lift supplies exchange paths. Three-disk isotopy gives the braid relation. Passing to module categories produces the pentagon and hexagon coherences. Analytically, the lift is a factorization algebra holomorphic on X and locally constant in two real transverse directions.

9.3 Internal Hochschild cochains

For an E_1 algebra A in $\text{FactD}(X)$, define

$$Z_{\text{ord, ch}}(A) := \text{RHom}_{A \otimes A^{\text{op}}}(A, A). \quad (66)$$

The diagonal bar resolution gives the cup product, braces, and Gerstenhaber operations internally in the coefficient category.

Theorem 9.2 (Universal E_2 centre). *Assume that the diagonal bimodule admits a homotopically correct bar resolution in $\text{FactD}(X)$. Then $Z_{\text{ord, ch}}(A)$ is an E_2 algebra. For every E_2 algebra B acting on A through the Swiss-cheese operad, there is a contractibly unique E_2 morphism*

$$B \longrightarrow Z_{\text{ord, ch}}(A)$$

that induces the given action.

Proof. The higher Deligne theorem equips endomorphisms of the unit object A in the monoidal category of A – A bimodules with an E_2 structure. The Swiss-cheese universal property identifies a central action with a morphism to these endomorphisms. The mapping space of such factorizations is contractible by the universal property of the endomorphism object. $\square$

The theorem identifies the centre as the universal recipient of every central bulk action. A bulk-to-centre quasi-isomorphism identifies a physical bulk with this recipient.

9.4 Bimodules and categorical centres

The category $\text{Bimod}_A(\text{FactD}(X))$ is monoidal under relative tensor product, with unit the diagonal bimodule A .

Proposition 9.3 (Hochschild cochains as unit endomorphisms). *There is a natural equivalence*

$$\text{End}_{\text{Bimod}_A}(A) \simeq Z_{\text{ord, ch}}(A),$$

and the E_2 structure on the right is the endomorphism E_2 structure of the unit in the monoidal bimodule category. The Drinfeld centre $Z(\text{Bimod}_A)$ is braided and acts on this unit endomorphism algebra. If the selected line category satisfies monoidal affineness, the action functor gives an equivalence

$$Z(\text{Bimod}_A) \simeq \text{Mod}_{Z_{\text{ord, ch}}(A)}.$$

Proof. The unit of the monoidal category of A – A bimodules is the diagonal bimodule A . Its derived endomorphisms are, by definition, $\text{RHom}_{A \otimes A^{\text{op}}}(A, A)$. The higher endomorphism operations of a monoidal unit are the Deligne E_2 operations. A central bimodule endofunctor acts on the unit and hence on its endomorphisms, which gives the braided action. Under monoidal affineness, the unit is

a compact generator and the Barr–Beck comparison identifies central objects with modules over its endomorphism E_2 algebra. $\square$

The bimodule category, or another explicitly monoidal line-defect category, carries the categorical centre. Additional convolution data may equip selected left modules with a monoidal product.

9.5 Meromorphic exchange

Suppose line defects are parametrized by points of a spectral curve. A meromorphic exchange is a family

$$R_{L,M}(z, w) : L_z \otimes M_w \longrightarrow M_w \otimes L_z \quad (67)$$

defined away from the collision divisor. Isotopy of three lines gives

$$R_{12}(z_1, z_2)R_{13}(z_1, z_3)R_{23}(z_2, z_3) = R_{23}(z_2, z_3)R_{13}(z_1, z_3)R_{12}(z_1, z_2). \quad (68)$$

The poles are chiral singularities on the spectral curve; the exchange path is topological in the transverse plane.

9.6 Pure-braid discrepancy

Let M_n be a complex with a B_n action and let $P_n = \ker(B_n \rightarrow \Sigma_n)$. Its derived pure-braid invariants and coinvariants are

$$M_n^{hP_n} = \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n), \quad (M_n)_{hP_n} = \mathbb{C} \otimes_{\mathbb{C}[P_n]}^L M_n,$$

with residual Σ_n actions.

Definition 9.4 (Pure-braid discrepancy). Writing $q : B_n \rightarrow \Sigma_n$ and q^* for inflation, define

$$\mathfrak{C}_n^{\mathrm{inv}}(M) := \mathrm{Cofib}(q^* M_n^{hP_n} \longrightarrow M_n), \quad (69)$$

$$\mathfrak{C}_n^{\mathrm{coinv}}(M) := \mathrm{Fib}(M_n \longrightarrow q^*(M_n)_{hP_n}). \quad (70)$$

The two objects are different derived measures of what is lost when pure-braid transport is removed. They are defined only after a braid action has been supplied.

Proposition 9.5 (Relative group model). *The homology of (70) is computed by the reduced group-bar complex of P_n with coefficients in M_n . The cohomology of (69) is computed by reduced group cochains. The residual Σ_n action is induced by conjugation in B_n .*

Proof. Homotopy coinvariants are the derived tensor product

$$(M_n)_{hP_n} \simeq \mathbb{C} \otimes_{\mathbb{C}[P_n]}^{\mathbb{L}} M_n,$$

computed by the normalized group-bar resolution. The fibre of the augmentation from M_n to this complex is its reduced version. Homotopy invariants are

$$M_n^{hP_n} \simeq \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n),$$

computed by normalized group cochains, and the cofiber of the constants is the reduced cochain complex. Since P_n is normal in B_n and $B_n/P_n \cong \Sigma_n$, conjugation descends to the residual action. $\square$

A constructed map from collision strata to braid cells may filter the group-bar complex by a partition complex.

9.7 Meromorphic Tannaka reconstruction

Let K be the function field of the spectral curve. Let $\mathcal{L}$ be a small rigid K -linear tensor category with finite-dimensional Hom spaces. A fibre functor may carry either strong monoidal or only quasi-monoidal comparison maps.

Assume first that

$$F : \mathcal{L} \longrightarrow \text{Vect}_K$$

is faithful, exact, and strong monoidal, and that the coend

$$\mathcal{H} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L) \quad (71)$$

exists. A meromorphic braiding on $\mathcal{L}$ is understood over the complement of the collision divisor.

Theorem 9.6 (Meromorphic Tannaka reconstruction). *The strong monoidal pair $(\mathcal{L}, F)$ reconstructs a bialgebra $\mathcal{H}$ over K ; rigidity supplies an antipode whenever the coend lies in the rigid finite or continuous-dual class. A braiding reconstructs a coquasitriangular structure. If the fibre functor is only quasi-monoidal, the same coend carries a coquasi-bialgebra structure whose coassociator is the transported tensor constraint. When the relevant dual exists, these statements dualize to quasitriangular Hopf or quasi-Hopf algebras, respectively.*

Proof. The coend represents natural endomorphisms of the fibre functor. The strong tensor constraints define a coassociative coproduct and counit; rigidity defines the antipode. The braiding gives the universal R -form, and the hexagon identities give coquasitriangularity. For a quasi-monoidal fibre functor, the defect of strict coassociativity is exactly the transported associator, which satisfies the pentagon and hence defines a coquasi-bialgebra. $\square$

A presentation as a Yangian, quantum loop algebra, elliptic quantum group, or Hall double is an additional recognition theorem.

9.8 Morita invariance

Derived Morita equivalent E_1 algebras have equivalent bimodule categories and equivalent Hochschild cochains as E_2 algebras.

Theorem 9.7 (Centre transport through a Morita context). *Let A and B be ordered chiral algebras joined by an invertible A – B bimodule in the Morita $(\infty, 2)$ -category of complete factorisation algebras. Then*

$$Z_{\text{ord, ch}}(A) \simeq Z_{\text{ord, ch}}(B)$$

as E_2 algebras, and the induced monoidal equivalence of bimodule categories identifies their Drinfeld centres.

For $B = A^!$, the conclusion applies only when the boundary Koszul equivalence of Theorem 7.5 extends to such an invertible bimodule Morita context. The conclusion uses the invertible bimodule Morita context in addition to the augmentation-generated equivalence.

Proof. An invertible A – B bimodule induces an equivalence between the Morita endomorphism categories of the identity A - and B -modules. Hochschild cochains are the endomorphisms of these identity kernels, with their E_2 multiplication supplied by composition in the Morita $(\infty, 2)$ -category. Conjugation by the invertible bimodule identifies the two endomorphism objects and their E_2 operations. The same conjugation identifies the bimodule categories, and functoriality of the Drinfeld centre gives the monoidal-centre equivalence. $\square$

The theorem is the precise algebraic content of two boundary presentations having the same universal central bulk. It is the categorical content of two boundary presentations sharing one universal central bulk; a trace produces any numerical complementarity statement.

9.9 Topological enhancement

The centre is E_2 in the two transverse directions and remains holomorphic on the complex curve. Suppose both real translations underlying the holomorphic coordinate become null-homotopic, coherently with all E_2 operations.

Proposition 9.8 (Dimension of the topological centre). *Under this coherent topologization, the analytic centre is locally constant on the four-real-dimensional manifold $X^{\text{an}} \times \mathbb{R}^2$. It therefore determines an E_4 algebra.*

Proof. The null-homotopies make the factorization algebra locally constant in the two real directions of X . It was already locally constant in two transverse directions. The recognition theorem for locally constant factorization algebras gives an E_4 algebra; equivalently, Dunn additivity combines the two E_2 structures. $\square$

Topologizing the whole complex curve gives an E_4 algebra. Topologizing one additional real direction gives the intermediate E_3 structure.

9.10 The bulk-to-centre map

Let a bulk factorization algebra Obs_{bulk} act on the boundary algebra A . Bringing a bulk observable to the boundary gives a central action and hence a map

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord, ch}}(A). \quad (72)$$

Criterion 9.9 (Algebraically complete boundary). *The boundary condition reconstructs the chosen local bulk observables exactly when (72) is an equivalence in the relevant factorization category. The homotopy fibre of this map is the complex of bulk observables whose induced universal central boundary action is null-homotopic.*

Proof. Reconstruction means that restriction to the boundary identifies the selected bulk factorization algebra with the universal centre. This is exactly the assertion that β is an equivalence. Applying the fibre functor in the stable category gives the displayed homotopy fibre, and its universal property identifies its elements with bulk observables equipped with a null-homotopy of their central action. $\square$

A bulk factor with unit boundary action lies in this homotopy fibre. A generation or nondegeneracy theorem removes that fibre and yields reconstruction.

10 Holomorphic–topological recognition

10.1 Product-generated local observables

Let $M = X^{\text{an}} \times \mathbb{R}$. A product disk is an open set $U \times I$, with $U \subset X^{\text{an}}$ a complex disc and $I \subset \mathbb{R}$ an oriented interval. Let $\text{Disk}_{X \times \mathbb{R}}^{\text{rect}}$ be the symmetric monoidal category generated by finite disjoint unions of product disks and their embeddings. The recognition theorem applies to the local class generated from product disks by Weiss descent and excision.

Definition 10.1 (Rectangular holomorphic–topological factorization algebra). A rectangular holomorphic–topological factorization algebra is a factorization algebra $\mathcal{F}$ on $X^{\text{an}} \times \mathbb{R}$ such that:

- (i) $\mathcal{F}$ is the factorization left Kan extension of its restriction to $\text{Disk}_{X \times \mathbb{R}}^{\text{rect}}$;
- (ii) for fixed I , the dependence on U is the analytic realization of a factorisable right D -module;
- (iii) for fixed U , the dependence on I is locally constant and orientation preserving.

Write $\text{Fact}_{\text{hol,lc}}^{\text{rect}}(X \times \mathbb{R})$ for this category.

The first condition extends product-disk data to all opens. The second gives chiral regularity. The third gives the little-interval action. All three conditions are metric-independent.

10.2 The exponential law

The local operation category of product disks is the tensor product of the chiral disk category with the little-interval operad:

$$\text{Disk}_X^{\text{ch}} \otimes E_1.$$

The tensor-product universal property gives an exponential law for algebras.

Theorem 10.2 (Holomorphic–topological recognition). *There is a natural equivalence*

$$\text{Alg}_{E_1}(\text{FactD}(X)) \simeq \text{Fact}_{\text{hol,lc}}^{\text{rect}}(X \times \mathbb{R}). \quad (73)$$

On the analytic side, the statement is restricted to coefficient systems in the image of the chosen Riemann–Hilbert realization. On the algebraic side, it is an equivalence of local factorization structures generated by product disks.

Proof. An object on the left is an algebra over E_1 in algebras over the chiral disk category. By the tensor-product universal property,

$$\text{Alg}_{E_1}(\text{Alg}_{\text{Disk}_X^{\text{ch}}}(\mathcal{V})) \simeq \text{Alg}_{E_1 \otimes \text{Disk}_X^{\text{ch}}}(\mathcal{V}).$$

The latter is exactly a factorization algebra on the rectangular disk category that is chiral in U and locally constant in I . Factorization left Kan extension gives the right-hand side of (73). Restriction back to product disks is inverse because rectangular generation is part of its definition. Riemann–Hilbert identifies the regular-holonomic algebraic coefficient with its analytic constructible realization. $\square$

The theorem identifies the local algebraic structure of every theory satisfying its hypotheses. A Lagrangian realization adds fields, gauge symmetry, a BV pairing, and quantization data.

10.3 The two short-distance limits

Let A correspond to $\mathcal{F}$. For disjoint intervals $I_1 < I_2 \subset I$ and discs $U_1, U_2 \subset U$, factorization gives

$$\mathcal{F}(U_1 \times I_1) \otimes \mathcal{F}(U_2 \times I_2) \longrightarrow \mathcal{F}(U \times I). \quad (74)$$

Approaching the diagonal in $U_1 \times U_2$ probes the singular chiral coefficient. Bringing I_1 and I_2 together inside their order chamber probes the E_1 multiplication. The two operations satisfy interchange because they are the two factors of one product-disk operation.

Proposition 10.3 (Interchange of short-distance limits). *Let m be the transverse multiplication and let γ be a chiral factorization map. On every product-disjoint locus where both composites are defined,*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \quad (75)$$

with the canonical permutation of coefficient factors.

Proof. Both sides are the image of one embedding of four product disks into a larger product disk. They are equal by functoriality of the factorization algebra. $\square$

This interchange is stronger than the coexistence of two products and weaker than braiding. Braiding requires motion in a second transverse direction.

10.4 Boundary and line-defect origins

A boundary condition b in a holomorphic-topological theory has a factorisable endomorphism object

$$A_b = \mathrm{RHom}(b, b).$$

Composition in the normal or topological direction is E_1 multiplication; collision of insertion points on X is the chiral coefficient. Boundary-changing operators are bimodules.

Parallel line defects in a theory on $X \times \mathbb{R}^2$ give the same structure after their transverse positions are restricted to an oriented line. Allowing the lines to move in the full plane may promote fusion to an E_2 structure and supply exchange. The E_1 fusion exists before that promotion.

Principle 10.4 (Defect interpretation). *The E_1 product is composition of boundary operators or line defects. The factorisable D -module records holomorphic collision. An exchange operator is codimension-two transport in a transverse plane.*

Proof. Gluing two consecutive boundary intervals composes their boundary operators and gives the little-interval product. Moving insertion points on X toward a diagonal produces the chiral D -module singularity. Exchanging two defects requires a path around the collision locus in a two-dimensional transverse configuration space; parallel transport along that path is the exchange operator. $\square$

10.5 Classical and quantum observables

Let $(\mathcal{E}, Q, \{-, -\})$ be the local fields of a classical BV theory. A local interaction I satisfies

$$QI + \frac{1}{2}\{I, I\} = 0. \quad (76)$$

The classical observables have differential $Q + \{I, -\}$. A perturbative quantization is represented by $I_\hbar = I + \hbar I_1 + \dots$ satisfying the renormalized quantum master equation

$$QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar \Delta I_\hbar = 0. \quad (77)$$

The equation makes the renormalized observable differential square to zero and makes graphwise factorization compatible with contraction of ultraviolet subgraphs.

Theorem 10.5 (BV realization of an ordered chiral algebra). *Let a perturbatively renormalized BV theory on $X \times \mathbb{R}$ satisfy the quantum master equation. Assume that its quantum observables are rectangularly generated, locally constant in the real direction, and admit a factorisable D -module realization in the holomorphic direction. Then they determine*

$$\text{Obs}_{X \times \mathbb{R}}^q \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).$$

A weak equivalence of renormalized theories that induces an objectwise quasi-isomorphism of these factorization observables induces an equivalence of ordered chiral algebras.

Proof. The quantum master equation supplies the square-zero differential and compatibility of renormalized factorization maps. The three additional hypotheses place the observable factorization algebra in the right-hand side of Theorem 10.2. Applying that equivalence gives the ordered chiral algebra. Objectwise quasi-isomorphisms become equivalences after localization. $\square$

A Lagrangian reconstruction requires the additional data of fields, gauge symmetry, a BV pairing, a real form, counterterms, state spaces, and integration cycles.

10.6 Interval evolution

With augmentation boundary conditions at both ends of an interval, excision gives

$$\int_{([0,1], \partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1} \simeq \text{Bar}_X^{\text{ord}}(A). \quad (78)$$

The ordered bar is therefore the complex of interval amplitudes between the two chosen boundaries. Cutting the interval gives deconcatenation.

Corollary 10.6 (Orientation reversal). *Reflection of the interval identifies the underlying bar objects of A and A^{op} by reversing every word and inserting the Koszul sign of the reversal permutation.*

Proof. Reflection exchanges the two endpoints and reverses the order of every subinterval. On normalized bar chains it sends $[a_1 | \dots | a_r]$ to the reversed word. Moving the suspended homogeneous factors through the reversal permutation gives the Koszul sign. The face maps become the face maps for opposite multiplication, so the map is a chain isomorphism. $\square$

The corollary produces opposite multiplication. Inverse braiding arises after a planar exchange structure has been supplied.

Part II

The chiral coefficient and its geometric resolutions

11 The diagonal as the primitive chiral coefficient

11.1 From singular products to geometry

Two quantum fields are placed at distinct points, multiplied there, and extended toward the diagonal. The singular part of this extension is the operator product expansion. The regular part is its renormalized continuation. Both are carried by the diagonal.

Let X be a smooth complex curve and let

$$j_2 : X^2 \setminus \Delta(X) \hookrightarrow X^2, \quad \Delta : X \hookrightarrow X^2$$

be the open complement and the diagonal embedding. A chiral operation on a right D_X -module $\mathfrak{g}$ is a morphism

$$\mu : j_{2,*}j_2^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}. \quad (79)$$

The source allows arbitrary finite poles along the diagonal. The target records that the singular product is supported where the two points meet. Antisymmetry and Jacobi make $\mathfrak{g}$ a chiral Lie algebra in the sense of Beilinson and Drinfeld [5].

Equation (79) is the common language of several classical constructions. In a formal coordinate it is the singular part of a vertex operator. In distribution theory it is a distribution supported on the diagonal. In perturbative quantum field theory it is a local counterterm or contact operation. In factorization theory it is the first nontrivial structure map. In representation theory its horizontal sections produce differential equations for conformal blocks. Across these categories, diagonal support is the invariant geometric condition.

Principle 11.1 (Diagonal principle). *The invariant datum of a chiral product is the morphism supported on the diagonal. Laurent coefficients, residues, modes, propagators, and contact terms are presentations of that morphism after auxiliary choices.*

Proof. The source $j_{2,*}j_2^*(\mathfrak{g} \boxtimes \mathfrak{g})$ is the sheaf of sections on the complement of the diagonal with finite principal parts along it, and $\Delta_!\mathfrak{g}$ is intrinsically supported on the diagonal. A coordinate trivializes the formal normal line and expands the morphism in its powers; a contour, propagator, or jet pairs with those powers. Coordinate changes act on the expansion while preserving the underlying D -module morphism. Hence the supported morphism is invariant and the listed coefficients are its realizations. $\square$

The principle has a sharp consequence. The complete operator product is carried by the full normal filtration. The logarithmic form $d \log(z - w)$ detects its first normal coefficient. A double pole,

a fourth-order pole, and a regular normally ordered term belong to different pieces of the normal filtration. The complete distribution is primary; its coefficients are derived from it.

11.2 The historical line

Dirac's delta distribution made support a mathematical operation. Wilson's operator product expansion made short distance a hierarchy of local fields. Borchers compressed the consistency of that hierarchy into the vertex identity. Beilinson and Drinfeld replaced formal coordinates by D -modules and diagonals. Fulton and MacPherson resolved simultaneous collisions. Segal, Tsuchiya–Ueno–Yamada, and the theory of conformal blocks organized sewing. Getzler and Kapranov encoded stable sewing by modular operads. Drinfeld and Kohno identified braid monodromy with quantum groups. Costello and Gwilliam expressed perturbative quantum field theory as a factorization algebra of observables. The chiral coefficient branch of this paper is the exact overlap of these constructions; the transverse E_1 multiplication is the independent ordered structure developed in Part I.

The overlap becomes visible only when the order of the objects is respected:

$$\text{diagonal operation} \xrightarrow{\text{Chev}^{\text{ch}}} \text{factorization coalgebra on } \text{Ran}(X) \xrightarrow{\text{Prim}} \text{chiral Lie algebra} \quad (80)$$

The same local operation has three independent enlargements:

$$\begin{array}{ccccc} & & \text{labelled configurations} & & \\ & & \downarrow \text{monodromy} & & \\ \text{formal disc} & \xrightarrow{\text{coordinate expression}} & \text{chiral operation} & \xrightarrow{+ \text{cyclic sewing data}} & \text{stable curves} \\ & \nwarrow \text{localization} \quad \nearrow \text{Ran globalization} & \downarrow & & \downarrow \text{master equation} \\ & & \text{factorization coalgebra} & & \text{modular graph operations} \end{array} \quad (81)$$

Each arrow in these diagrams is a typed functor, comparison map, or monodromy construction.

11.3 The central theorem

The first theorem identifies the geometry that lies between the diagonal operation and the Ran coalgebra.

Theorem 11.2 (Geometric collision theorem). *Let X be a smooth algebraic curve over a field of characteristic zero and let $\mathfrak{g}$ be a chiral Lie algebra on X . For every nonempty finite set I there are three canonically related complexes.*

(i) *The partition collision complex*

$$C_I^{\text{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} \Delta_{\pi,!} \left(\bigotimes_{B \in \pi}^{\text{unord}} \mathfrak{g}[1] \right),$$

with differential given by merging two blocks and applying the suspended chiral bracket.

(ii) *The corolla-normalized forest complex $W_I^{\text{FM}}(\mathfrak{g})$, obtained by resolving each block of a partition by the normalized chains of the rooted-tree category and coupling this vertical resolution to the collision differential.*

(iii) *The labelled component $\mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g})$ of the chiral Chevalley coalgebra.*

The square of the differential on $C_I^{\mathrm{coll}}(\mathfrak{g})$ is the suspended chiral Jacobiator. There are canonical quasi-isomorphisms

$$W_I^{\mathrm{FM}}(\mathfrak{g}) \xrightarrow{\sim} C_I^{\mathrm{coll}}(\mathfrak{g}) \xrightarrow{\cong} \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \quad (82)$$

The first map applies the augmentation from each rooted-tree nerve to its terminal corolla. The second identifies the partition decomposition with the cofree commutative coalgebra in the chiral monoidal category.

Proof. The square formula follows from the classification of length-two intervals in the partition lattice: disjoint mergers cancel by the two suspension orientations, and three-block mergers sum to the suspended chiral Jacobiator. The rooted-tree category resolving a fixed collision block has a terminal corolla, so its normalized nerve augments quasi-isomorphically to the ground field. Tensoring these augmentations blockwise and filtering by nerve degree proves the first quasi-isomorphism. The labelled cofree commutative coalgebra on $\mathfrak{g}[1]$ is indexed by surjections from I to an unlabelled set of primitive factors, equivalently by partitions of I ; its Chevalley coderivation merges two factors by the chiral bracket. This identifies the second map and completes the proof. The detailed sign and factorization verifications appear in Sections 13, 14 and 17. $\square$

The theorem is developed in Sections 13, 14 and 17. Its force is conceptual. The partition lattice carries the algebraic differential. The Fulton–MacPherson boundary supplies a geometric resolution of the incidence pattern. The Ran space globalizes the result into a factorization coalgebra. Screen configuration spaces and differential forms are genuine additional coefficients; their cohomology remains as an additional coefficient system.

11.4 The formal disc and the full operator product

Choose a formal coordinate z on a disc. A weakly translation-equivariant chiral algebra becomes a vertex algebra, and the morphism (79) becomes the full operator product

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}. \quad (83)$$

A general coordinate change acts triangularly on the modes $a_{(n)}b$ and preserves the filtered singular distribution. The quotient of pole order $n + 1$ by pole order n is a tensor power of the normal line to the diagonal. This is the geometric reason that modes transform triangularly.

The algebraic chiral operad $\mathcal{P}^{\mathrm{ch}}$ makes the statement exact. If V is a vector superspace with translation operator, an odd element $X \in W_1^{\mathrm{ch}}(\mathrm{IIV})$ determines a nonunital vertex algebra exactly when

$$[X, X] = 0. \quad (84)$$

The associated coderivation obeys

$$d_X^2 = \frac{1}{2}d_{[X, X]}. \quad (85)$$

Thus the full vertex identity is the coordinate expression of the chiral Jacobi identity. Arnold’s relation contributes to the geometry of configuration-space forms, while the chiral Jacobi identity supplies the Borchers coefficients and the normal filtration supplies every pole order.

11.5 Five exact separations

The common language relates several structurally distinct objects through typed maps.

First, the chiral Chevalley coalgebra exists for the chiral Lie object in the Ran category. A quadratic or Verdier-dual algebra appears after the relevant presentation, dualizability, and continuity data have been supplied.

Second, braid monodromy gives a braided category or a local-system representation. Tannaka reconstruction with a strong or quasi-monoidal fibre functor produces a bialgebra/Hopf or coquasi/quasi-Hopf object; a further recognition theorem produces a named quantum group.

Third, projective curvature on moduli is the curvature of a connection or the class of an Atiyah algebra. Tensoring by an inverse anomaly line cancels it when that line exists.

Fourth, the derived center is the universal recipient of compatible algebraic actions. A bulk theory determines a map to the center; an open–closed quasi-isomorphism identifies it with the physical bulk.

Fifth, numerical invariants are traces of structures. Central charge, level, rank, Hodge classes, theta coefficients, and entropy coefficients inhabit different functors and are compared through their specified trace and index maps.

These typed comparisons preserve the full mathematical and physical content.

11.6 Mathematical conventions

All derived categories are stable and $\mathbb{C}$ -linear. Right D -modules are used on algebraic curves. For a finite set I , X^I is the labelled Cartesian power, $\text{Part}(I)$ is its partition lattice, and $\Delta_\pi : X^\pi \hookrightarrow X^I$ is the partial diagonal associated with π . Tensor products indexed by unordered finite sets are taken in the symmetric monoidal derived category; the suspension $[1]$ carries the Koszul signs. The symbol $\text{FM}_I(X)$ denotes the algebraic Fulton–MacPherson compactification. Real oriented compactifications are used only in the analytic and field-theoretic sections, where Stokes’ theorem is explicit.

For a vertex algebra, modes follow the convention

$$Y(a, z)b = \sum_{n \in \mathbb{Z}} (a_{(n)}b)z^{-n-1}.$$

For a simple Lie algebra $\mathfrak{g}$, the invariant form and the dual Coxeter number are normalized so that the KZ parameter is $\hbar = (k + h^\vee)^{-1}$; other conventions change only the displayed exponent defining q .

12 Chiral operations on a curve

12.1 The chiral tensor product

Let X be a smooth algebraic curve. The ordinary tensor product of right D_X -modules describes fields that occupy the same point. A chiral product must first separate the points. For two right D_X -modules M and N , write

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2.$$

The binary chiral tensor product is represented by $j_*j^*(M \boxtimes N)$, followed by a pushforward to the diagonal when an operation is evaluated. The distinction between j_* and $\Delta_!$ is the distinction between a meromorphic product away from collision and a distribution supported at collision.

For a finite set I , the open configuration locus is

$$U_I = X^I \setminus \bigcup_{i \neq j} \Delta_{ij}, \quad j_I : U_I \hookrightarrow X^I.$$

An I -ary chiral operation with values in a right D_X -module M is a morphism

$$\mu_I : j_{I,*} j_I^* (\boxtimes_{i \in I} M_i) \longrightarrow \Delta_{I,!} M, \quad (86)$$

where $\Delta_I : X \hookrightarrow X^I$ is the small diagonal. Partial compositions are defined by allowing a subset of points to collide first, applying one operation, and then colliding the resulting point with the remaining points. The associativity of these partial compositions is a base-change theorem for the corresponding diagrams of open embeddings and diagonals.

Definition 12.1 (Chiral Lie algebra). A chiral Lie algebra on X is a right D_X -module $\mathfrak{g}$ with an antisymmetric operation

$$\mu : j_{*,*} j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_{,!} \mathfrak{g}$$

whose three partial composites satisfy the chiral Jacobi identity.

The word ‘‘Lie’’ refers to the symmetry and Jacobi law of the diagonal-supported operation; its coefficients are D -modules with arbitrary poles and derivatives. On a formal disc, the chiral Jacobi identity becomes the vertex identity.

12.2 The Jacobi identity as a relation among partial diagonals

Let $I = \{1, 2, 3\}$. There are three pair diagonals and one small diagonal. The composite that first collides 1 with 2 and then collides the result with 3 is a morphism supported on Δ_{123} . The two other parenthesizations have the same target. After the symmetry signs are inserted, their sum is the Jacobiator

$$J_\mu = \mu \circ_{12} \mu + \mu \circ_{23} \mu + \mu \circ_{31} \mu. \quad (87)$$

The chiral Jacobi identity is $J_\mu = 0$.

The support geometry already predicts the form of the relation. Two disjoint collisions commute. Three points colliding into one admit three intermediate pair collisions. Every square of a collision differential therefore splits into disjoint-pair terms and triple-collision terms. The former cancel by the Koszul sign; the latter are the Jacobiator. This observation is made precise in Section 13.

12.3 Units and the vacuum

A unital chiral algebra contains a morphism from the dualizing module ω_X that plays the role of the vacuum. The unit axiom says that a field inserted next to the vacuum extends regularly across the diagonal and returns the field. On the formal disc this becomes

$$Y(\mathbf{1}, z) = \text{id}, \quad Y(a, z)\mathbf{1} \in a + zV[[z]].$$

The nonunital operation and its Jacobi identity contain the singular and regular binary product. The vacuum is additional nullary data. This separation is useful because chiral Koszul duality is naturally formulated for nonunital or augmented objects, while vertex operator algebras are ordinarily presented with a vacuum.

12.4 Chiral algebras and factorization algebras

A factorization algebra on X assigns an object to each finite configuration and identifies the object on disjoint unions with the tensor product of the objects on the pieces. A chiral algebra supplies the infinitesimal collision law from which this factorization structure is generated. Beilinson and Drinfeld construct the chiral enveloping algebra and the corresponding factorization algebra; Francis and Gaitsgory place the relation in a derived categorical setting [5, 38].

The relation may be read in two directions. Starting from a chiral Lie algebra, the chiral Chevalley construction forms a commutative factorization coalgebra. Starting from a conilpotent factorization coalgebra, primitives recover the chiral Lie algebra. The equivalence is a Koszul duality internal to the chiral monoidal category and applies to the pro-nilpotent D -module objects of that category.

Historical note 12.2. Beilinson and Drinfeld's use of right D -modules incorporates the transformation law of fields into the definition. A coordinate gives a calculational presentation of the intrinsic D -module operation. This is the decisive step that separates a chiral algebra from a list of formal Laurent series.

12.5 Distributions and contact terms

The map (79) has an analytic analogue. A distribution on X^2 supported on the diagonal is a finite sum of normal derivatives of the delta distribution with coefficients along the diagonal. In a coordinate z ,

$$T(z, w) = \sum_{m=0}^N A_m(w) \partial_z^m \delta(z - w). \quad (88)$$

Cauchy's formula identifies the same coefficients with pole terms in a meromorphic kernel. The formal-distribution identity

$$\partial_w^m \delta(z - w) = \iota_{z,w} \frac{(-1)^m m!}{(z - w)^{m+1}} - \iota_{w,z} \frac{(-1)^m m!}{(z - w)^{m+1}}$$

relates the two expansions. Thus a chiral operation may be read either as a principal part or as a contact distribution. The equality is between two presentations of one diagonal-supported object.

In perturbative field theory, local counterterms are also distributions supported on diagonals. Renormalization extends distributions from configuration space to the compactification. Two extensions differ by a local distribution supported on collision strata. Chiral algebra encodes the allowed local extensions and their consistency relations.

12.6 The complete principal-part carrier

Let $D \subset Y$ be a smooth divisor. Poincaré residue is a map

$$\text{Res}_D : \Omega_Y^p(\log D) \longrightarrow \Omega_D^{p-1}.$$

It detects the coefficient of dt/t in a logarithmic form. The successive meromorphic coefficients are carried by the full normal-symbol filtration described below.

Theorem 12.3 (Principal-part carrier). *For every $m \geq 1$, the exact sequence*

$$0 \longrightarrow \mathcal{O}_Y((m-1)D) \longrightarrow \mathcal{O}_Y(mD) \longrightarrow N_{D/Y}^{\otimes m} \longrightarrow 0$$

identifies the order- m symbol of a meromorphic function with a section of $N_{D/Y}^{\otimes m}$. Poincaré residue is the order-one logarithmic quotient. The direct system $\{\mathcal{O}_Y(mD)\}_{m \geq 0}$, together with its normal-symbol quotients, carries the complete finite principal part.

Proof. In a normal coordinate t , the order- m symbol is the class of t^{-m} . Under $t' = ut + O(t^2)$ its leading coefficient transforms by u^m , which is the transition function of $N_{D/Y}^{\otimes m}$. The logarithmic generator dt/t gives the first quotient, and the pole filtration supplies every higher quotient functorially. $\square$

Section 15 assembles these quotients into the collision Rees geometry.

12.7 The affine-line presentation

On $X = \mathbb{A}^1$ with coordinate z , translation-invariant sections of $j_*\mathcal{O}_{X^2 \setminus \Delta}$ are Laurent polynomials in $z_1 - z_2$. A chiral operation can therefore be written as a finite sum

$$\{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b, \quad (89)$$

combined with the regular product. The relation

$$\{\partial a_\lambda b\} = -\lambda \{a_\lambda b\}, \quad \{a_\lambda \partial b\} = (\partial + \lambda) \{a_\lambda b\}$$

is the coordinate form of D -module covariance. The Jacobi identity becomes

$$\{a_\lambda \{b_\mu c\}\} - (-1)^{p(a)p(b)} \{b_\mu \{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}. \quad (90)$$

The binomial coefficients of the Borchers identity arise by expanding the translated variable $\lambda + \mu$. The Arnold relation controls differential forms on configuration space, while equation (90) produces the binomial coefficients through expansion of $\lambda + \mu$.

12.8 Factorization as locality

Suppose $U_1, \dots, U_n \subset X$ are pairwise disjoint discs contained in a larger disc V . A factorization algebra supplies a product

$$\mathcal{F}(U_1) \otimes \cdots \otimes \mathcal{F}(U_n) \longrightarrow \mathcal{F}(V).$$

On the disjoint-disc locus, these products form a local system. When discs collide, the chiral operation describes the singular extension. This is the precise relation between OPE and locality: the OPE is the boundary value of factorization.

The factorization axiom is stronger than associativity at a point. It compares all decompositions by disjoint subdiscs and is compatible with isotopy. On a complex curve the holomorphic dependence of the products adds the D -module structure. The result is a geometric object that can be transported, integrated, and sewn.

12.9 The physical reading

In Euclidean quantum field theory, a local observable supported in a small disc is an element of a cochain complex. Disjoint observables multiply. The product depends smoothly or holomorphically on their positions. A quantization satisfying locality therefore determines a factorization algebra [14, 15].

If one direction is holomorphic and the other directions are topological, restriction to a holomorphic boundary produces a chiral algebra. The local operation (79) is then the renormalized collision of boundary observables.

The mathematical and physical statements agree once a comparison map has been supplied between the algebraic D -module and the analytic factorization algebra. The comparison map identifies the algebraic and analytic realizations. Its homotopy fibre records the metric, reality, positivity, and global state-space data carried by the physical theory.

13 The partition collision complex

13.1 Partitions and partial diagonals

Let I be a finite nonempty set. A partition π of I is a collection of disjoint nonempty blocks whose union is I . The refinement order is written $\pi \leq \rho$ when every block of π is contained in a block of ρ . The minimal partition has singleton blocks; the maximal partition has one block.

To π associate the partial diagonal

$$\Delta_\pi : X^\pi \hookrightarrow X^I,$$

where the coordinates indexed by a block are required to coincide. If $\pi < \rho$ is a cover relation, then ρ is obtained from π by merging exactly two blocks. Every saturated chain in $\text{Part}(I)$ is an order of pairwise collisions.

Let $s\mathfrak{g} = \mathfrak{g}[1]$. For a partition π , set

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!} \left(\bigotimes_{B \in \pi}^{\text{unord}} s\mathfrak{g} \right). \quad (91)$$

The unordered tensor product is the coinvariant form of the tensor product in the symmetric monoidal derived category. Suspension supplies the Koszul sign: it turns the antisymmetric chiral bracket into a symmetric degree-one contraction, exactly as in the Chevalley complex of an ordinary Lie algebra.

Definition 13.1 (Partition collision complex). The I -labelled collision object is

$$C_I^{\text{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} C_\pi(\mathfrak{g}). \quad (92)$$

For a cover $\pi < \rho$ that merges blocks B_1 and B_2 , the component

$$d_{\pi,\rho} : C_\pi(\mathfrak{g}) \longrightarrow C_\rho(\mathfrak{g})$$

applies the suspended chiral bracket to the factors indexed by B_1, B_2 and acts identically on the remaining factors. The total differential is the internal differential plus the signed sum over covers.

The sign is determined by moving the two suspended factors next to one another, applying the degree-one bracket, and returning the merged factor to the position determined by the orientation line of the cover. Because the tensor product is unordered, the definition is independent of a chosen linear order on the blocks.

13.2 The square of the collision differential

Every length-two interval $[\pi, \sigma]$ in the partition lattice has one of two forms.

If σ is obtained by merging two disjoint pairs of blocks, there are exactly two intermediate partitions. The coefficient operations act on disjoint tensor factors and commute before suspension. Interchanging the two degree-one contractions changes the sign, so the two paths cancel.

If σ is obtained by merging three blocks into one, there are exactly three intermediate partitions. The three composites are the three terms of the suspended Jacobiator.

Theorem 13.2 (Square of the collision differential). *Let μ be an antisymmetric chiral binary operation on a right D_X -module $\mathfrak{g}$. The collision operator defined from μ satisfies*

$$d_{\text{coll}}^2 = \frac{1}{2}d_{[\mu, \mu]}. \quad (93)$$

Consequently $C_I^{\text{coll}}(\mathfrak{g})$ is a complex for every finite I if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal differential squares to zero and its mixed commutator with d_{coll} vanishes precisely when μ is a chain map. The remaining square is a sum over length-two intervals in $\text{Part}(I)$. Disjoint mergers cancel in pairs by the Koszul sign. Triple mergers sum to the three suspended partial composites of μ . This sum is $\frac{1}{2}[\mu, \mu]$ in the convolution Lie algebra of chiral operations. These two cases exhaust the length-two intervals. $\square$

Corollary 13.3 (Selection by collisions). *An antisymmetric collection of diagonal-supported binary operations defines a chiral Lie algebra exactly when the corresponding partition collision operators form differentials in all arities.*

Proof. The arity-three summand of d_{coll}^2 is the suspended Jacobiator by Theorem 13.2; its vanishing gives the chiral Jacobi identity. Antisymmetry is part of the input, and the chain-map condition is the mixed component of the same square. Conversely, these identities make every length-two interval vanish, so all arities are complexes. $\square$

This is the correct geometric selection statement. It uses the complete chiral operation. The coefficient Jacobi identity and any central cocycle compatibility are evaluated in the chiral operation, while differential-form relations belong to a later geometric realization.

13.3 Arity two and three

For $I = \{1, 2\}$ there are two partitions. The complex is

$$\Delta_{1|2,!}(sg \boxtimes sg) \xrightarrow{s\mu} \Delta_{12,!}(sg).$$

The map is the collision operation.

For $I = \{1, 2, 3\}$, the Hasse diagram is

$$\begin{array}{ccccc}
 & & 1|2|3 & & \\
 & \swarrow & \downarrow & \searrow & \\
 12|3 & & 13|2 & & 23|1 \\
 & \searrow & \downarrow & \swarrow & \\
 & & 123 & &
 \end{array} \tag{94}$$

The square from the top to the bottom is the sum of three composites. In a coordinate presentation these are the three expansions of the vertex Jacobi identity. In the D -module presentation they are three maps with the same source and target, supported on the small diagonal.

13.4 The cofree commutative coalgebra

Let $\mathcal{C}$ be a symmetric monoidal stable category. The cofree conilpotent commutative coalgebra on an object V has underlying object

$$\mathrm{coSym}(V) = \bigoplus_{n \geq 1} (V^{\otimes n})_{\Sigma_n}.$$

In the chiral monoidal category on the Ran space, the labelled component of $\mathrm{coSym}(\mathfrak{sg})$ decomposes by the ways in which labels are identified. Those ways are the partitions of I . The summand associated with π is exactly (91).

Proposition 13.4 (Chevalley identification). *For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is canonically the I -component of its chiral Chevalley coalgebra:*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \cong \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{95}$$

Under this identification the collision differential is the Chevalley coderivation induced by the chiral bracket.

Proof. The cofree commutative coalgebra is the sum over finite collections of primitive factors. Pulling its Ran component back to X^I records a surjection from I to the set of primitive factors. A surjection modulo relabeling of its target is a partition of I . The Chevalley coderivation merges two primitive factors by the bracket, hence agrees with the cover differential of the partition lattice. $\square$

13.5 The factorization coproduct

Suppose $I = I_1 \sqcup I_2$. Restriction to the open locus where the points of I_1 remain disjoint from the points of I_2 gives a coproduct

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \longrightarrow C_{I_1}^{\mathrm{coll}}(\mathfrak{g}) \boxtimes C_{I_2}^{\mathrm{coll}}(\mathfrak{g}). \tag{96}$$

A partition contributes to this restriction exactly when every block lies entirely in I_1 and I_2 . The coproduct is therefore deconcatenation at the level of primitive blocks. Coassociativity follows from the associativity of disjoint union.

Proposition 13.5. *The family $\{C_I^{\mathrm{coll}}(\mathfrak{g})\}_I$, with the coproducts (96), is a conilpotent commutative factorization coalgebra. Its primitive part is $\mathfrak{g}[1]$.*

Proof. On the disjointness locus, a partition splits uniquely into its restrictions to I_1 and I_2 exactly when each block lies in one side. Iterating this restriction is independent of parenthesization, which gives coassociativity; exchanging the sides gives cocommutativity. The collision differential merges blocks and therefore commutes with restriction. The coradical filtration by the number of primitive blocks is finite on every labelled component. Its one-block quotient is $\mathfrak{g}[1]$, which is the primitive part. $\square$

Conilpotence is measured by the number of blocks. Repeated reduced coproducts eventually reach singleton partitions. This filtration is intrinsic and is defined intrinsically in the coalgebra.

13.6 Incidence geometry and Möbius inversion

The Euler characteristic of the collision complex is governed by the Möbius function of the partition lattice. For the interval from the discrete partition to the one-block partition,

$$\mu_{\text{Part}(I)}(\hat{0}, \hat{1}) = (-1)^{|I|-1}(|I| - 1)!.$$

This integer is also the top reduced Euler characteristic of the proper partition complex. It explains the factorial multiplicities that appear in Lie operad components and in the top cohomology of configuration spaces. The equality is combinatorial; the representations carried by the two sides require their respective coefficient systems.

The proper partition complex is homotopy equivalent to a wedge of $(|I| - 1)!$ spheres of dimension $|I| - 3$. Its top homology is the multilinear part of the Lie operad, up to the sign representation. This is the classical topological shadow of the Lie–commutative Koszul duality that becomes chiral on the Ran space.

13.7 Modules and marked points

Let M be a chiral module over $\mathfrak{g}$. Add a distinguished label $*$ whose distinguished block carries the module input. A partition of $I \sqcup \{*\}$ with a distinguished block containing $*$ indexes a summand in the module collision complex. Merging an ordinary block with the distinguished block applies the module action; merging two ordinary blocks applies the bracket. The same length-two interval analysis proves that the square is the sum of the Lie Jacobiator and the module identity.

This pointed complex is the geometric origin of derived coinvariants and conformal blocks with insertions. It is also the correct place to compare BRST reduction, because the BRST differential acts on both the algebra and the marked module.

13.8 Screen data adjoining the partition complex

The partition complex records which groups of points collide and in what incidence pattern. The relative shape of a collision is carried by the adjoining screen configuration space. For three points on a complex line, the ratio of their relative positions survives after translation and dilation. That ratio belongs to the screen configuration space of the Fulton–MacPherson boundary. Differential forms on screens produce Arnold and Orlik–Solomon classes. They enter through the geometric resolution of the next section.

14 The Fulton–MacPherson forest resolution

14.1 Compactifying collisions

The configuration space $\text{Conf}_I(X)$ is the complement of the diagonals in X^I . Its Fulton–MacPherson compactification $\text{FM}_I(X)$ is obtained by successively blowing up partial diagonals in an order compatible with inclusion [40]. The result is smooth, and the boundary is a normal-crossing divisor. A boundary point records the colliding labels together with their limiting relative configuration at every scale.

Boundary strata are indexed by nested collections of subsets, equivalently by rooted forests whose leaves are the labels in I . An internal vertex represents a cluster. An edge represents one scale of collapse. Roots represent the final clusters that remain distinct in X .

Let F be such a forest. The root components determine a partition $q(F)$ of I . The corresponding stratum maps to the partial diagonal $\Delta_{q(F)}(X^{q(F)})$. Its fiber is a product of screen configuration spaces. On a curve, a screen for a cluster S is the compactification of configurations of $|S|$ points on a one-dimensional tangent space modulo translation and dilation.

14.2 Tree categories over collision blocks

For a nonempty finite set B , let $\text{Tree}(B)$ be the category of rooted trees with leaves labelled by B and whose internal vertices have valence at least two. Morphisms contract internal edges. The corolla with leaf set B is terminal. Write

$$K(B) = N_*(N \text{Tree}(B)), \quad (97)$$

for the normalized chain complex of its nerve, placed in nonpositive cohomological degrees. The augmentation and the terminal-corolla inclusion are chain maps

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C}, \quad \mathbb{C} \xrightarrow{\iota_B} K(B), \quad \epsilon_B \iota_B = \text{id}. \quad (98)$$

Because $\text{Tree}(B)$ has a terminal object, ϵ_B is a quasi-isomorphism.

For a partition π of I , define

$$K(\pi) = \bigotimes_{B \in \pi}^{\text{unord}} K(B). \quad (99)$$

An element of $K(\pi)$ is a chain of contractions in a forest whose root components are the blocks of π . This is the incidence information carried by the face category of the Fulton–MacPherson boundary; the screen cohomology enters as a separate geometric coefficient system.

14.3 The corolla-normalized forest resolution

Set

$$W_I^{\text{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \text{Part}(I)} C_\pi(\mathfrak{g}) \otimes K(\pi). \quad (100)$$

Its differential has three terms. The internal term is induced by the differential of $\mathfrak{g}$. The vertical term is the normalized nerve differential on the factors $K(B)$. For a cover $\pi < \rho$ merging blocks B, C , the horizontal term applies the suspended chiral bracket to the B, C coefficient factors and applies

the chain map

$$K(B) \otimes K(C) \xrightarrow{\epsilon_B \otimes \epsilon_C} \mathbb{C} \xrightarrow{\iota_{B \cup C}} K(B \cup C) \quad (101)$$

while leaving the other tree factors unchanged. Thus a collision of two root components is represented by the terminal corolla of the merged block; all nonterminal tree data remains in the vertical resolution.

The maps (101) are symmetric and strictly associative because they factor through the augmentations. They commute with the vertical differential. Consequently the square of the horizontal differential is exactly the square of the partition collision differential, with a common terminal-corolla factor.

There is an augmentation

$$\epsilon_I : W_I^{\text{FM}}(\mathfrak{g}) \longrightarrow C_I^{\text{coll}}(\mathfrak{g}) \quad (102)$$

obtained by applying ϵ_B to every tree factor.

Theorem 14.1 (Forest resolution). *For every chiral Lie algebra $\mathfrak{g}$ and finite set I , the differential on (100) squares to zero and the augmentation (102) is a quasi-isomorphism. The construction is natural under relabelling and compatible with disjoint union.*

Proof. The vertical differential squares to zero. Its mixed commutator with the horizontal differential vanishes because the augmentations and corolla inclusions in (101) are chain maps. Strict associativity of the corolla-normalized merge identifies the horizontal square with the Jacobiator computed in Theorem 13.2; it therefore vanishes for a chiral Lie algebra.

Filter by the total nerve degree of the tree factors. The filtration is finite for fixed I . Its first spectral sequence computes the homology of every $K(B)$. Since each ϵ_B is a quasi-isomorphism, the first page is precisely $C_I^{\text{coll}}(\mathfrak{g})$, and the induced differential is its collision differential. The spectral-sequence comparison proves that ϵ_I is a quasi-isomorphism. Relabelling preserves terminal corollas, and disjoint union tensors the tree complexes, giving naturality and factorization compatibility. $\square$

Corollary 14.2. *The corolla-normalized forest complex resolves the chiral Chevalley coalgebra:*

$$W_I^{\text{FM}}(\mathfrak{g}) \xrightarrow{\sim} \text{coChev}_I^{\text{ch}}(\mathfrak{g}).$$

Proof. Compose the forest-resolution quasi-isomorphism of Theorem 14.1 with the Chevalley identification of Proposition 13.4. $\square$

The word “geometric” refers here to the tree categories labelling the faces of the compactification. The actual screen fibres contribute a second coefficient system through their de Rham or constructible cohomology.

14.4 Screen spaces as geometric coefficients

Let $p_I : \text{FM}_I(X) \rightarrow X^I$ be the proper blowdown. The fibre over a partial diagonal contains compactified screen configurations. For $|I| = 3$, translation and dilation leave one relative moduli coordinate. The screen cohomology therefore contributes to the derived pushforward $Rp_{I,*} \mathbb{C}_{\text{FM}_I(X)}$.

Separation theorem 14.3 (Incidence resolution and screen pushforward). *The terminal corolla contracts the nerve of the forest incidence category. The geometric pushforward has the stratification spectral sequence*

$$E_1^{p,q} = \bigoplus_{\text{codim } S=p} H_c^{p+q}(S; \text{or}_S),$$

whose coefficient groups include the cohomology of the compactified screens. A quasi-isomorphism from a de Rham or constructible Fulton–MacPherson model to the collision complex is specified by a map that sends these screen coefficients to the incidence resolution and proves acyclicity of its cone.

Proof. Contractibility of the incidence nerve follows from its terminal corolla. The displayed spectral sequence is the standard spectral sequence of the boundary-codimension filtration. Each stratum is locally a root-diagonal base times compactified screens, so its coefficient group contains the screen cohomology by Kunneth. A derived equivalence with the incidence complex is therefore exactly a chain map whose cone kills the additional screen rows. $\square$

The distinction separates two independent structures:

incidence of collision strata	geometry inside a screen
partition lattice	configuration space
edge contraction	de Rham differential
chiral Jacobi	Arnold–Orlik–Solomon relations
Chevalley coderivation	propagator and period classes

14.5 The stratification spectral sequence

Filter $\mathrm{FM}_I(X)$ by the codimension of its boundary strata. For a constructible complex $\mathcal{F}$ that is locally constant along strata, the standard spectral sequence of a filtered space has first page

$$E_1^{p,q} = \bigoplus_{\mathrm{codim} S=p} H_c^{p+q}(S; \mathcal{F}|_S \otimes \mathrm{or}_S) \implies H_c^{p+q}(\mathrm{FM}_I(X); \mathcal{F}). \quad (103)$$

The differential is the signed sum of incidence maps between adjacent strata.

For coefficients pulled back from the root diagonal, a stratum splits locally into a base and a product of screens. The Kunneth theorem expresses its cohomology as

$$H_c^*(S; \mathcal{F}|_S) \cong H_c^*(\Delta_{q(F)}; \mathcal{F}_{q(F)}) \otimes \bigotimes_{v \in V_{\mathrm{int}}(F)} H_c^*(\mathrm{Screen}(v)). \quad (104)$$

After barycentric replacement of the face incidence category and corolla normalization, the degree-zero screen classes map to the forest incidence resolution. Positive-degree screen classes record the topology of relative configurations.

Theorem 14.4 (Incidence–screen separation). *For a coefficient system constant along Fulton–MacPherson strata, the boundary-codimension spectral sequence separates the collision incidence maps from the cohomology of screen configuration spaces. After replacing the face poset by its normalized nerve and applying the terminal-corolla contractions blockwise, the row obtained from degree-zero screen classes maps quasi-isomorphically to the forest resolution W_I^{FM} .*

Proof. The local product description of every stratum gives the tensor factorization (104). The first spectral-sequence differential is the signed incidence map tensored with restriction on screen cohomology. In screen degree zero, every connected compactified screen contributes its unit class. The remaining complex is the normalized nerve of the forest face category with the corolla-normalized collision maps of (101). The blockwise terminal-corolla augmentations are quasi-isomorphisms, so this row is quasi-isomorphic to W_I^{FM} . $\square$

This theorem is the precise meeting point of the chiral Chevalley complex and configuration-space forms. The spectral-sequence differential couples the two structures through an explicit comparison map.

14.6 Arnold forms on screens

On an affine coordinate chart, set

$$\eta_{ij} = d \log(z_i - z_j).$$

On $\text{Conf}_3(\mathbb{C})$ the Arnold relation is

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \quad (105)$$

It follows by the elementary substitution $x = z_1 - z_2$, $y = z_2 - z_3$. The Orlik–Solomon algebra of the braid arrangement is generated by the classes of the η_{ij} with these three-term relations [4].

The Arnold relation expresses flatness of the universal logarithmic connection once the coefficients satisfy infinitesimal braid relations. The full chiral Jacobi identity is carried by the coefficient operation on the root diagonal. In the stratification spectral sequence, (105) belongs to the screen coefficient system; the chiral Jacobi identity belongs to the coefficient operation on the root diagonal. A square-zero total differential combines both sources in their assigned bidegrees.

14.7 Real oriented compactification and Stokes’ theorem

For perturbative integrals one uses the real oriented blowup of the diagonals. The boundary hypersurfaces carry outward normal orientations. If α is a compactly supported form of top degree minus one, Stokes’ theorem gives

$$\int_{\text{FM}_I^{\text{or}}(M)} d\alpha = \sum_{H \subset \partial \text{FM}_I^{\text{or}}(M)} \int_H \alpha.$$

Applying Stokes twice yields cancellation of codimension-two corners because each corner occurs with opposite induced orientations from its two parent hypersurfaces. This is the analytic form of the incidence differential squaring to zero.

When forms carry algebraic coefficients, the total square contains three distinct terms: the de Rham square, the incidence square, and the coefficient Jacobiator. A correct proof specifies the bidegrees and the mixed commutators. The coefficient identities enter through the supplied algebraic operations, and boundary-of-boundary supplies their incidence signs.

14.8 Renormalization forests

The same forest combinatorics appears in the BPHZ forest formula. A divergent subgraph determines a collision subconfiguration; nested divergent subgraphs determine a forest; subtraction is performed recursively from smaller clusters to larger clusters. Fulton–MacPherson compactification turns this recursion into geometry. Counterterms are supported on boundary strata, and locality is the compatibility of their restrictions under further collisions.

This parallel is exact at the level of incidence categories. A particular quantum field theory supplies a propagator, a density, a gauge fixing, and a renormalization prescription. Those analytic data form the screen coefficient system. The universal forest resolution supplies the combinatorics on which they live.

15 Pole filtrations and normal geometry

15.1 Principal parts along a divisor

Let Y be a smooth variety and $D \subset Y$ a smooth divisor. For $m \geq 0$ there is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \longrightarrow \mathcal{O}_D((m+1)D) \longrightarrow 0. \quad (106)$$

Since $\mathcal{O}_D(D) \cong N_{D/Y}$, the associated graded of the pole filtration is

$$\mathrm{gr}_{m+1}^F \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes(m+1)}. \quad (107)$$

If t is a local defining equation for D , the class of $t^{-(m+1)}$ is a local generator of this line. Under $t' = ut + O(t^2)$ it transforms by $u^{-(m+1)}$ modulo lower pole order. This is precisely the tensorial transformation law of the normal symbol.

For a normal-crossing divisor $D = \bigcup_{a \in A} D_a$, use the multi-filtration

$$F_m \mathcal{O}_Y(*D) = \mathcal{O}_Y\left(\sum_{a \in A} m_a D_a\right).$$

On a stratum $D_J = \bigcap_{a \in J} D_a$, the multi-associated graded is

$$\mathrm{gr}_m^F \mathcal{O}_Y(*D)|_{D_J} \cong \bigotimes_{a \in J} N_{D_a/Y}^{\otimes m_a} \otimes \mathcal{O}_{D_J}(*D_J^{\mathrm{res}}). \quad (108)$$

The residual divisor records further boundary components meeting the stratum.

15.2 The full operator product

Let D_{12} be the pair-collision divisor in $\mathrm{FM}_2(X) = X^2$. In a coordinate $t = z_1 - z_2$, the singular operator product is

$$\sum_{m \geq 1} A_m(z_2) t^{-m}.$$

The coefficient A_m belongs intrinsically to the m th normal symbol. For a vertex algebra, $A_m = a_{(m-1)}b$. Therefore

$$a_{(r)}b \longleftrightarrow \mathrm{gr}_{r+1}^F \mathcal{O}(*D_{12}) \cong N_{D_{12}/X^2}^{\otimes(r+1)}. \quad (109)$$

The correspondence is coordinate-free at the associated-graded level. A coordinate chooses a trivialization of the normal line and thereby names the coefficient as a mode.

The zeroth regular coefficient belongs to F_0 and is the normally ordered product; its higher regular Taylor coefficients are obtained from it by translation and normal differentiation. Thus the complete filtered chiral operation contains both the singular modes and the regular multiplication. Keeping only the associated graded forgets extensions between pole orders; these extensions encode the triangular mixing under coordinate changes.

15.3 Nested collisions

Let T be a rooted collision tree indexing a boundary stratum $D_T \subset \text{FM}_I(X)$. Each internal edge e corresponds to one normal direction, hence to a line N_e . Locally the normal bundle splits

$$N_{D_T/\text{FM}_I(X)} \cong \bigoplus_{e \in E_{\text{int}}(T)} N_e. \quad (110)$$

The multi-associated graded of a meromorphic coefficient is therefore

$$\bigotimes_{e \in E_{\text{int}}(T)} N_e^{\otimes m_e}. \quad (111)$$

Assigning a positive integer m_e to every internal edge is the geometric version of assigning a pole order to every intermediate OPE channel.

Tree substitution contracts a subtree and replaces it by one incoming branch. Over an arbitrary curve the root point and its tangent line must remain part of the colour: Fulton–MacPherson spaces form a coloured operadic family over root points and tangent lines; a translation or framing trivializes those colours. The intrinsic object is therefore a coloured collision coefficient system over X , whose colours are root points together with their tangent lines.

Theorem 15.1 (Normal-cone collision operad). *The multi-normal symbols of nested collision strata form a coloured operad over X . For a collision tree T , its coefficient line is*

$$\bigotimes_{e \in E_{\text{int}}(T)} N_e^{\otimes m_e},$$

with one positive pole order m_e on every internal edge. Substitution is induced by the canonical morphisms of Rees algebras for nested regular embeddings. It is associative and independent of local coordinates.

Proof. A nested collision stratum is an iterated regular embedding. The associated graded of its formal neighborhood is the symmetric algebra of the conormal bundle, and (110) identifies the independent conormal directions with internal edges. For a chain of nested regular embeddings, deformation to the normal cone is functorial: composing the corresponding Rees morphisms either before or after grouping a subtree gives the same morphism. Tensoring the resulting normal lines gives the stated substitution law and its associativity. The construction uses only ideals of diagonals and their normal bundles, hence is coordinate-free. $\square$

A strict filtered identification of formal neighborhoods requires choices of compatible local coordinates or tubular neighborhoods. Different choices act by filtration-positive automorphisms. The associated-graded coloured operad is canonical; the filtered operation is canonical as a chiral distribution, while every Laurent series is one coordinate representative.

15.4 The collision Rees space

The pole filtration can be assembled into a Rees algebra

$$\mathcal{R}_D(\mathcal{O}_Y(*D)) = \bigoplus_{m \geq 0} F_m \mathcal{O}_Y(*D) u^m. \quad (112)$$

For $u \neq 0$ its localization recovers the meromorphic coefficient algebra. At $u = 0$ it becomes the normal-symbol algebra. Locally, the deformation is the familiar equation

$$tv = u,$$

where v represents the inverse normal coordinate. The generic fiber contains the full pole; the special fiber records its normal cone.

For the full boundary of $\mathrm{FM}_I(X)$ one obtains a multi-Rees space. Collision scales become independent deformation parameters. The special fiber is a union of normal-cone pieces indexed by forests. This geometry explains why classical limits are controlled by trees and why quantum corrections mix different pole orders.

15.5 The chiral and classical operads

The algebraic chiral operad is filtered by the order of poles along diagonals. Bakalov–De Sole–Heluani–Kac construct the corresponding classical operad and a canonical comparison from chiral symbols to classical operations [6, 7]. Geometrically, a chiral product passes to its normal symbols; when a filtration on the state space is compatible with the products, the regular multiplication becomes commutative and the leading singular symbols define the Poisson vertex bracket.

Theorem 15.2 (Chiral-to-classical symbol map). *On a translation-invariant formal disc, the pole filtration of the chiral operad has a canonical morphism of operads*

$$\mathrm{gr}^F \mathcal{P}^{\mathrm{ch}} \longrightarrow \mathcal{P}^{\mathrm{cl}}.$$

Under the normal-symbol identification (109), this is the symbol morphism of the geometric pole filtration. If a filtered vertex algebra is compatible with its filtration, the image of its chiral Maurer–Cartan operation is the Poisson vertex algebra operation on the associated graded.

Proof. Pole order is additive under operadic substitution, so the principal symbol of a composite depends only on the principal symbols of its factors. This defines a morphism of associated-graded operads. The normal exact sequence identifies each principal part with its normal tensor, giving the geometric description. For a compatible filtration on V , the commutator of regular products and every lower pole contribution drops one filtration step; the regular principal symbol is commutative and the leading singular symbol is a Lie conformal bracket satisfying Leibniz. These are the defining operations of $\mathcal{P}^{\mathrm{cl}}$, and the Maurer–Cartan equation passes to the associated graded. This is the operadic filtration theorem of Bakalov–De Sole–Heluani–Kac [6, 7]. $\square$

The Li filtration is the standard compatible filtration. On the flat quantization locus, its symbol map identifies the associated graded vertex algebra with the prescribed Poisson vertex algebra.

15.6 Residues, jets, and differential operators

Higher coefficients may be extracted by pairing principal parts with jets. If t is a coordinate normal to D , then

$$A_m = \frac{1}{2\pi i} \oint t^{m-1} \left(\sum_{r \geq 1} A_r t^{-r} \right) dt. \quad (113)$$

The factor t^{m-1} is a choice of jet. Intrinsically, the pairing is between $N_D^{\otimes m}$ and $N_D^{\vee \otimes m}$. The contour integral is the analytic realization of this pairing. Ordinary residue is $m = 1$.

In a D -module, normal derivatives of delta distributions supply the dual description. The identity between principal parts and delta derivatives is a local Fourier transform in the normal direction. This is why the lambda bracket encodes all singular modes polynomially in λ .

15.7 Coordinate changes and the stress tensor

A projective connection is required to make the stress tensor transform tensorially. Under $z \mapsto w(z)$,

$$T(z) = \left(\frac{dw}{dz} \right)^2 T(w) + \frac{c}{12} \{w, z\}, \quad (114)$$

where $\{w, z\}$ is the Schwarzian derivative. The inhomogeneous term is the central extension of the action of vector fields, detected by the fourth-order pole in the TT operator product.

The pole filtration makes the distinction visible. The central term belongs to the fourth normal symbol. The transformation law mixes it with lower terms through the choice of projective connection. The invariant object is the chiral operation of the Virasoro algebra together with its coordinate-change action.

15.8 Pole order and homotopy depth

The maximal pole order of a chosen strong generating set is a local analytic invariant. The length of a minimal resolution, the nonvanishing range of transferred higher operations, and the genus at which a trace appears are homological or modular invariants. They form independent coordinates of the theory and are related only by family-specific comparison theorems. A first-order system combines simple poles with rich representation theory, while a Virasoro field combines a fourth-order self-OPE with a deformation complex determined by all of its structure operations.

This separation replaces scalar “depth classifications” by canonical filtrations: pole order, coradical degree, collision codimension, graph genus, and cohomological degree. Each filtration has a geometric origin and a convergent spectral sequence under its own boundedness hypotheses.

16 The formal disc and vertex algebras

16.1 Translation-equivariant chiral operations

Let V be a vector superspace with an even endomorphism ∂ . On the affine line, translation-invariant rational functions on configurations are generated by the differences $z_i - z_j$ and their inverses. The algebraic chiral operad has n -ary component

$$\mathcal{P}_V^{\text{ch}}(n) = \text{Hom}_{\mathcal{D}_n^T} \left(V^{\otimes n} \otimes \mathbb{C}[z_i - z_j, (z_i - z_j)^{-1}], V[\lambda_1, \dots, \lambda_n] / \langle \partial + \sum_i \lambda_i \rangle \right), \quad (115)$$

where $\mathcal{D}_n^T$ is the algebra of translation-invariant differential operators. The variables λ_i record normal derivatives. The relation in the target expresses simultaneous translation covariance.

The operad (115) is the formal-coordinate presentation of the Beilinson–Drinfeld chiral endomorphism operad. It contains the regular multiplication and all finite principal parts. Composition expands rational functions in the domains corresponding to nested collisions.

16.2 The universal Lie superalgebra of an operad

For a linear operad $\mathcal{P}$, set

$$W(\mathcal{P}) = \bigoplus_{k \geq -1} W_k(\mathcal{P}), \quad W_k(\mathcal{P}) = \mathcal{P}(k+1)^{\Sigma_{k+1}}.$$

Operadic insertion defines a pre-Lie product and hence a Lie superbracket. For $\mathcal{P} = \mathcal{P}_{\Pi V}^{\text{ch}}$, write $W^{\text{ch}}(V)$. An odd binary operation is an element $X \in W_1^{\text{ch}}(\Pi V)$.

The following theorem is the algebraic selection principle.

Theorem 16.1 (Bakalov–De Sole–Heluani–Kac). *An odd element $X \in W_1^{\text{ch}}(\Pi V)$ satisfies*

$$[X, X] = 0 \tag{116}$$

if and only if X defines a nonunital vertex algebra structure on V with translation operator ∂ . The differential $\text{ad } X$ on $W^{\text{ch}}(V)$ is the vertex-algebra deformation differential.

The theorem includes all coefficients of the integral lambda bracket. Skewsymmetry, the normally ordered product, the noncommutative Wick formula, and the Borchers identity are different coefficient forms of the single equation (116) [6].

16.3 The coderivation identity

Let $\mathcal{C}$ be a conilpotent dg cooperad and let $\alpha : \mathcal{C} \rightarrow \mathcal{P}^{\text{ch}}$ be an operadic twisting morphism, for example the canonical twisting morphism from the bar cooperad. An operation X then determines, through α , a coderivation d_X of the cofree $\mathcal{C}$ -coalgebra $\text{cofree}_{\mathcal{C}}(sV)$. The commutator of coderivations corresponds to the convolution bracket.

Proposition 16.2 (Square of the vertex coderivation). *In every such bar-coalgebra model, an odd chiral operation X satisfies*

$$d_X^2 = \frac{1}{2}[d_X, d_X] = \frac{1}{2}d_{[X, X]}. \tag{117}$$

Consequently d_X is a differential exactly when the image of X is Maurer–Cartan in the chiral convolution Lie algebra; for the binary operation of Theorem 16.1, this is exactly the vertex-algebra condition.

Proof. Coderivations of a cofree conilpotent coalgebra are determined by their corestriction to co-generators. The corestriction of $[d_X, d_Y]$ is the signed sum of single insertions represented by the convolution pre-Lie product; antisymmetrizing gives $[X, Y]$. Setting $Y = X$ gives the formula. $\square$

Combining Theorems 15.1, 16.1 and 16.2 gives the geometric statement: the full pole-filtered collision operation produces a square-zero coderivation exactly when the vertex identities hold. The geometry represents the coefficients; the operadic bracket supplies the identity.

16.4 The Borchers identity

For homogeneous $a, b, c \in V$, the coefficient form of the vertex Jacobi identity is

$$\sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)} b_{(k+j)} c - (-1)^n (-1)^{p(a)p(b)} b_{(n+k-j)} a_{(m+j)} c) = \sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c. \quad (118)$$

The binomial coefficients arise from translating one formal variable relative to another. Equation (118) is therefore invisible to a complex that retains only a simple residue and discards jets. The normal-symbol filtration retains precisely the powers needed to recover the translated expansions.

At the level of distributions, the same identity is the equality of three expansions of a rational distribution on the complement of the three diagonals. The chiral operad records the expansions and their common meromorphic origin.

16.5 Vacuum and conformal structure

A vacuum vector $\mathbf{1}$ and the creation axiom extend the nonunital structure to a vertex algebra. A conformal vector ω supplies operators $L_n = \omega_{(n+1)}$ with Virasoro commutation relations and $L_{-1} = \partial$. The central charge is the coefficient of the fourth-order pole in

$$Y(\omega, z)Y(\omega, w) \sim \frac{c/2}{(z-w)^4} + \frac{2Y(\omega, w)}{(z-w)^2} + \frac{\partial_w Y(\omega, w)}{z-w}.$$

The vacuum, translation, and conformal vector are distinct structures. Chiral Koszul duality applies to the underlying chiral Lie object; conformal blocks and modular anomalies use the conformal structure.

16.6 The classical limit

Let $F_\bullet V$ be a filtration compatible with all vertex products, for example the Li filtration. The associated graded is commutative with a derivation and a lambda bracket. The regular normally ordered product becomes the commutative product; the first nontrivial singular symbols become the Poisson vertex bracket.

Theorem 16.3 (Classical symbol operad). *There is a canonical morphism from the pole-associated graded of the chiral operad to the classical operad $\mathcal{P}^{\text{cl}}$. An odd Maurer–Cartan element in $W(\mathcal{P}_{\text{IV}}^{\text{cl}})$ is a Poisson vertex algebra structure. For a vertex algebra equipped with a compatible filtration, the image of the principal chiral symbol is the Maurer–Cartan element of the associated Poisson vertex algebra.*

Proof. The symbol morphism is Theorem 15.2. The universal Lie superalgebra construction sends operad morphisms to Lie-superalgebra morphisms and therefore preserves Maurer–Cartan equations. The classical operad governs Poisson vertex algebras, so its odd square-zero binary operations are precisely commutative differential products with compatible Lie conformal brackets. Applying the symbol map to a filtered vertex Maurer–Cartan element gives the associated Poisson vertex operation [6, 7]. $\square$

The theorem makes quantization a lifting problem in the chiral deformation complex. Obstructions lie in the second cohomology of the classical differential; choices of lifts lie in the first. This is ordinary Maurer–Cartan obstruction theory applied to the filtered operad.

16.7 Deformations

Fix a vertex operation X . The complex

$$(W^{\text{ch}}(\Pi V), d_X = [X, -]) \quad (119)$$

controls formal deformations. If

$$X_{\hbar} = X + \hbar X_1 + \hbar^2 X_2 + \cdots,$$

then the coefficient of $\hbar$ in $[X_{\hbar}, X_{\hbar}] = 0$ says $d_X X_1 = 0$. Gauge-equivalent first-order deformations differ by a coboundary. The second-order obstruction is the class of $\frac{1}{2}[X_1, X_1]$ in H^2 .

This cohomology is the appropriate home of local deformation classes. Hochschild cohomology of a mode algebra and cohomology of the moduli space of curves arise from separate complexes; explicit comparison maps relate them to chiral deformation cohomology in the settings where those maps are constructed.

16.8 Homotopy chiral structures

Let $\mathcal{P}_{\infty} \rightarrow \mathcal{P}^{\text{ch}}$ be a cofibrant resolution and let $\mathcal{C}_{\infty}$ be a conilpotent cooperadic model carrying its canonical twisting morphism to $\mathcal{P}_{\infty}$. A homotopy chiral algebra is an algebra over $\mathcal{P}_{\infty}$, equivalently a Maurer–Cartan element in the completed convolution L_{∞} -algebra

$$\text{Hom}_{\Sigma}(\overline{\mathcal{C}_{\infty}}, \text{End}_V).$$

The higher operations are indexed by cells in a resolution of collision composition. Their identities are the components of the Maurer–Cartan equation. This is the correct setting for transferred structures and for BRST models before cohomology concentration.

A strict vertex algebra is a homotopy chiral algebra whose higher generators vanish in a chosen model. Homotopy transfer preserves the quasi-isomorphism type and generally produces a homotopy-coherent model. Therefore a transferred higher operation is a property of the contraction as well as of the quasi-isomorphism class; its cohomology class is the invariant obstruction.

16.9 The meaning of “chiral Koszul”

The adjective “chiral” refers to the monoidal category and the diagonal operation. The phrase “chiral Koszul duality” is reserved in this paper for the Chevalley–primitives equivalence on the Ran space. A vertex algebra is a formal-coordinate manifestation of a chiral algebra. A quadratic presentation of a vertex algebra may admit a smaller quadratic dual, but that is an additional Koszul property, an additional presentation-level Koszul property.

17 The chiral Chevalley coalgebra on the Ran space

17.1 The Ran space

The Ran space of X is the prestack of nonempty finite subsets of X :

$$\mathrm{Ran}(X) = \operatorname{colim}_{I \in \mathrm{Fin}^{\mathrm{surj}, \mathrm{op}}} X^I.$$

A point of X^I maps to the subset of its coordinate values; coincident coordinates are allowed in the colimit. The transition maps are partial diagonals. Thus the Ran space remembers every finite collision simultaneously.

The category $D(\mathrm{Ran} X)$ carries two monoidal structures. The star tensor product is induced by union of finite subsets along arbitrary unions of finite subsets. The chiral tensor product is induced by union restricted to disjoint subsets. Chiral Lie algebras are Lie algebra objects for the chiral tensor product. Factorization objects are those whose restriction to disjoint configurations is the tensor product of their one-point pieces.

The Ran prestack is the colimit of Cartesian powers along diagonal maps, and its D -module category is controlled by the Cartesian powers X^I and their diagonal transition maps. Every formula in the preceding sections is therefore a labelled chart of a Ran object.

17.2 Chevalley chains

Let $\mathfrak{g}$ be a chiral Lie algebra. Its chiral Chevalley chains are the cofree commutative coalgebra on $\mathfrak{g}[1]$ with the coderivation induced by the chiral bracket:

$$\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) = \left(\mathrm{coSym}^{\mathrm{ch}}(\mathfrak{g}[1]), d_{\mathfrak{g}} + d_{\mu} \right). \quad (120)$$

Pullback to X^I gives the partition decomposition of [Theorem 13.4](#). The factorization coproduct is induced by the decomposition of a finite set into disjoint subsets.

The primitive filtration is the filtration by the number of factors. Its associated graded is the cofree commutative coalgebra with only the internal differential. This filtration is complete in the conilpotent direction: every element has finite primitive length in a labelled finite component.

Proposition 17.1 (Geometric realization of Chevalley chains). *The forest incidence objects $W_I^{\mathrm{FM}}(\mathfrak{g})$ assemble, under disjoint union of labels and diagonal pullback, into a simplicial resolution of $\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$ on $\mathrm{Ran}(X)$.*

Proof. Let $\iota_I : C_I^{\mathrm{coll}}(\mathfrak{g}) \rightarrow W_I^{\mathrm{FM}}(\mathfrak{g})$ insert the terminal corolla in every block. Then $\epsilon_I \iota_I = \mathrm{id}$. For a surjection of label sets $\phi : I \twoheadrightarrow J$, let T_{ϕ} be the canonical Ran transition on the partition collision complexes and define the transition on the resolution by

$$\tilde{T}_{\phi} = \iota_J T_{\phi} \epsilon_I.$$

Because $\epsilon \iota = \mathrm{id}$, the maps $\tilde{T}_{\phi}$ are strictly functorial under composition, and ϵ_I is a natural transformation. Disjoint union tensors the tree complexes and preserves terminal corollas, so these transitions are compatible with the factorization coproduct. They therefore assemble the labelled resolutions over the colimit presentation of $\mathrm{Ran}(X)$. $\square$

17.3 Primitives

Let C be a coaugmented conilpotent commutative coalgebra in the chiral monoidal category. The primitive functor is defined invariantly as the right adjoint to the chiral Chevalley functor. With the Chevalley convention of (120), its value is shifted by $[-1]$ to produce the corresponding chiral Lie object.

In a strict conilpotent dg-coalgebra model, let

$$\overline{C} = \text{Cofib}(\mathbf{1} \rightarrow C)$$

be the coaugmentation coideal and let $\overline{\Delta} : \overline{C} \rightarrow \overline{C} \otimes^{\text{ch}} \overline{C}$ be the reduced coproduct. When strict primitives compute the derived right adjoint—in particular for cofree conilpotent models and their fibrant replacements—one may represent the primitive complex by

$$\text{Prim}(C) \simeq \text{Fib}\left(\overline{C} \xrightarrow{\overline{\Delta}} \overline{C} \otimes^{\text{ch}} \overline{C}\right). \quad (121)$$

For a cofree conilpotent coalgebra this complex is the cogenerator complex.

The Chevalley and primitive constructions form an adjunction

$$\text{coChev}^{\text{ch}} : \text{Lie}_{\text{ch}}(X) \rightleftarrows \text{Coalg}_{\text{Com}}^{\text{ch, conil}}(\text{Ran } X) : \text{Prim}[-1]. \quad (122)$$

The shift depends on the Chevalley convention. The substantive point is that the bracket and the reduced coproduct are Koszul-dual operations.

17.4 Francis–Gaitsgory duality

The chiral monoidal category on the Ran space is pro-nilpotent with respect to cardinality. This property removes the convergence obstruction that ordinary Lie–commutative Koszul duality encounters in an arbitrary monoidal category.

Theorem 17.2 (Francis–Gaitsgory chiral Koszul duality). *In the factorization and conilpotent subcategories of the pro-nilpotent chiral monoidal category of D -modules on $\text{Ran}(X)$ considered by Francis and Gaitsgory, the chiral Chevalley functor and the primitive functor are inverse equivalences between chiral Lie algebras and conilpotent factorization coalgebras:*

$$\text{Lie}_{\text{ch}}(X) \simeq \text{CoFact}_{\text{conil}}(\text{Ran } X). \quad (123)$$

The unit and counit of the adjunction are equivalences.

This theorem is the invariant core of chiral Koszul duality [38]. It is stronger than the statement that a quadratic algebra is Koszul. The equivalence is intrinsic and presentation-free.

17.5 Chiral homology

The chiral homology of $\mathfrak{g}$ on a complete curve X is the de Rham cohomology of its factorization coalgebra:

$$\int_X^{\text{ch}} \mathfrak{g} := R\Gamma_{\text{dR}}(\text{Ran } X, \text{coChev}^{\text{ch}}(\mathfrak{g})). \quad (124)$$

The cardinality filtration produces a spectral sequence whose first page is a sum of cohomologies of configuration spaces with coefficients in external tensor powers of $\mathfrak{g}$. The collision differential is induced by the chiral bracket.

For an ordinary Lie algebra $\mathfrak{g}_0$ placed constantly on a point, this reduces to the usual Chevalley complex. For a current chiral algebra on a curve, it becomes a global version of Lie algebra homology with local singularities. For modules inserted at marked points, the pointed collision complex computes derived coinvariants.

17.6 The factorization envelope

Beilinson and Drinfeld associate to a chiral Lie algebra its chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$. Its factorization structure is generated by the same local operation μ , but the output is associative rather than coalgebraic. The Chevalley coalgebra and the enveloping factorization algebra play different roles:

$$\begin{aligned} \text{coChev}^{\text{ch}}(\mathfrak{g}) & \text{ resolves primitives and homology,} \\ U^{\text{ch}}(\mathfrak{g}) & \text{ acts on modules and defines centers.} \end{aligned}$$

The Poincaré–Birkhoff–Witt theorem identifies the associated graded of $U^{\text{ch}}(\mathfrak{g})$ with the symmetric chiral algebra under the standard filtration. This is a theorem about the enveloping construction, a statement about the chiral enveloping functor and its PBW filtration.

17.7 Analytic realization

Over $\mathbb{C}$, regular holonomic D -modules correspond under Riemann–Hilbert to constructible sheaves. The chiral operation becomes a morphism of constructible objects with singular support along diagonals. On the open configuration space, horizontal sections form a local system. The difference between the algebraic and analytic descriptions is then the difference between a regular singular connection and its monodromy representation.

This passage explains the ordered theory. The Ran coalgebra retains symmetric factorization. Pulling its analytic realization to labelled configurations and taking horizontal sections retains paths and braid monodromy. The two constructions are related and preserve distinct information: de Rham cohomology forgets a chosen basis of horizontal solutions, while monodromy acts on it.

17.8 Completion and cardinality

The Ran category is pro-nilpotent by cardinality, but additional completions may enter from conformal weight, pole order, or infinite-dimensional state spaces. These are independent filtrations. If C_N is an inverse system of finite-window complexes with surjective transition maps on cohomology, then the Milnor sequence

$$0 \longrightarrow \varprojlim^1 H^{m-1}(C_N) \longrightarrow H^m(\varprojlim C_N) \longrightarrow \varprojlim H^m(C_N) \longrightarrow 0 \quad (125)$$

shows that completion is exact when the derived limit vanishes. This is the precise content of a Mittag–Leffler argument.

The abstract equivalence of Theorem 17.2 is formulated directly in the pro-nilpotent Ran category. Completion enters when one replaces abstract Ran objects by explicit infinite products of modes or conformal weights.

17.9 The Beilinson hierarchy

The objects generated by one chiral operation form a hierarchy:

$$\mathfrak{g} \longmapsto \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \int_X^{\mathrm{ch}} \mathfrak{g}, \quad \mathfrak{g} \longmapsto U^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \mathrm{Mod}(U^{\mathrm{ch}}(\mathfrak{g})). \quad (126)$$

The first branch is coalgebraic and homological. The second is algebraic and representation-theoretic. Centers, traces, and conformal blocks are built from the second branch; primitives and factorization homology are built from the first. Chiral Koszul duality connects the primitive object to its factorization coalgebra while preserving the algebraic and coalgebraic branches.

18 Chiral Koszul duality, bar–cobar, and presentations

18.1 Three meanings of Koszul duality

The phrase “Koszul duality” is used for three related constructions.

The first is the Lie–commutative equivalence of [Theorem 17.2](#). This is chiral Koszul duality in the strict sense of this paper.

The second is bar–cobar reconstruction for augmented associative or operadic algebras. An augmented algebra A has a conilpotent bar coalgebra $B(A)$, and a conilpotent coalgebra C has a cobar algebra $\Omega(C)$. Under the usual conilpotence and completeness conditions, the unit and counit are equivalences.

The third is quadratic Koszul duality. A quadratic presentation $A = T(V)/(R)$ determines a smaller coalgebra A^i generated by sV with corelations s^2R . The algebra is Koszul when the canonical map $A^i \rightarrow B(A)$ is a quasi-isomorphism.

The first construction is presentation-free. The second is a universal resolution. The third is a property of a presentation. They coincide on the locus where the canonical comparison maps are equivalences.

18.2 Twisting morphisms

Let C be a conilpotent dg coalgebra and A an augmented dg algebra in a complete symmetric monoidal category. A twisting morphism is a degree-one map $\tau : C \rightarrow A$ satisfying

$$d\tau + \mu_A(\tau \otimes \tau)\Delta_C = 0. \quad (127)$$

This is a Maurer–Cartan equation in the convolution algebra $\mathrm{Hom}(C, A)$. It determines a coalgebra map $C \rightarrow B(A)$ and an algebra map $\Omega(C) \rightarrow A$.

Theorem 18.1 (Fundamental theorem of twisting morphisms). *Assume that the bar and cobar constructions are defined and that the coradical filtrations converge. For a twisting morphism $\tau : C \rightarrow A$, the following are equivalent:*

- (i) *the map $C \rightarrow B(A)$ is a quasi-isomorphism;*
- (ii) *the map $\Omega(C) \rightarrow A$ is a quasi-isomorphism;*
- (iii) *the left twisted tensor product $A \otimes_\tau C$ is acyclic;*

(iv) *the right twisted tensor product $C \otimes_{\tau} A$ is acyclic.*

The theorem is classical over a field and extends to filtered chiral settings when the monoidal products preserve the relevant limits and the comparison spectral sequences converge [55]. The formal equation is universal; the chiral hypotheses specify the ambient limits, tensor products, and convergence required to interpret it (127).

18.3 Quadratic recognition

Let $A = T(V)/(R)$ be a connected weight-graded quadratic algebra. Its quadratic coalgebra A^i comes with a canonical twisting morphism. The usual Koszul criterion is diagonal concentration of bar homology.

Theorem 18.2 (Quadratic Koszul criterion). *For a connected weight-graded quadratic algebra A , the following are equivalent:*

- (i) $A^i \rightarrow B(A)$ *is a quasi-isomorphism;*
- (ii) $\Omega(A^i) \rightarrow A$ *is a quasi-isomorphism;*
- (iii) $\mathrm{Tor}_{p,q}^A(\mathbb{C}, \mathbb{C}) = 0$ *for $p \neq q$;*
- (iv) *the quadratic twisted tensor complexes are acyclic.*

Proof. The canonical quadratic twisting morphism $A^i \rightarrow A$ gives comparison maps from the quadratic coalgebra to the bar and from its cobar to A . The fundamental theorem of twisting morphisms identifies either comparison with acyclicity of the left and right twisted tensor products. Filtering the bar by homological and internal weight identifies the homology of those complexes with the off-diagonal groups $\mathrm{Tor}_{p,q}^A(\mathbb{C}, \mathbb{C})$. Thus the four conditions are equivalent [55]. $\square$

For a chiral algebra presented by generators and relations, the same criterion applies in each finite window once the free chiral algebra, the quadratic relation object, and the filtered comparison have been defined. Each universal affine, Virasoro, lattice, or W -algebra becomes Koszul after a presentation and spectral-sequence argument verifies the criterion in its finite windows and completion.

18.4 PBW deformations

A PBW theorem identifies the associated graded of a filtered algebra with a prescribed graded algebra. The bar comparison additionally incorporates the lower-order linear and constant terms: lower-order linear and constant terms alter the Koszul differential and may introduce curvature. The invariant comparison statement is the following filtered lemma.

Criterion 18.3 (Filtered Koszul comparison). *Let $q : C \rightarrow B(A)$ be a morphism from a filtered conilpotent coalgebra to the bar coalgebra of a filtered augmented algebra. Assume that*

- (a) *the filtrations on C and $B(A)$ are exhaustive, separated, complete, and preserved by q ;*
- (b) *the induced map $\mathrm{gr} q : \mathrm{gr} C \rightarrow B(\mathrm{gr} A)$ is a quasi-isomorphism;*
- (c) *the two comparison spectral sequences converge strongly.*

Then q is a quasi-isomorphism.

Proof. The cone of q inherits a complete exhaustive filtration. Its associated graded is the cone of $\text{gr } q$ and is acyclic by hypothesis. Strong convergence therefore gives zero cohomology for the filtered cone, so q is a quasi-isomorphism. $\square$

For a quadratic-linear-constant PBW deformation of a quadratic Koszul algebra, the appropriate C is the nonhomogeneous or curved Koszul coalgebra. The PBW compatibility identities ensure that its coderivation squares to the prescribed curvature, and the associated-graded comparison is the ordinary quadratic Koszul resolution. Thus PBW, classical Koszulness, and the curved compatibility identities together imply the filtered comparison. The three ingredients together give the filtered quasi-isomorphism.

In a vertex algebra, an application must additionally control derivatives, infinitely many modes, and the completion in conformal weight or pole order. A family is declared Koszul after its PBW spanning set has been combined with the filtered coalgebra, curved compatibility identities, and strong convergence.

18.5 Verdier duality and a dual algebra

The chiral Chevalley coalgebra is a factorization coalgebra. When its labelled components are Verdier dualizable, duality exchanges coalgebra maps with algebra maps. Let $\mathbb{D}_{\text{Ran}}$ denote Verdier duality on an appropriate constructible or holonomic subcategory.

Proposition 18.4 (Verdier-dual algebra). *Let C be a factorization coalgebra whose finite labelled components are Verdier dualizable, and suppose Verdier duality commutes with the factorization restriction maps. Then*

$$C^\vee := \mathbb{D}_{\text{Ran}} C$$

is a factorization algebra. If biduality holds componentwise and is compatible with factorization, then $C \rightarrow \mathbb{D}_{\text{Ran}} \mathbb{D}_{\text{Ran}} C$ is an equivalence of factorization coalgebras.

Applied to $C = \text{coChev}^{\text{ch}}(\mathfrak{g})$, this produces a Verdier-dual factorization algebra. Identifying it with a familiar vertex algebra, a reflected level, or a quantum group is a separate recognition theorem.

18.6 Completion

Let $F^1 A$ be the augmentation ideal. The completed bar and cobar constructions use products over tensor length rather than direct sums. Their differentials converge when they raise the filtration. The completed counit

$$\hat{\Omega}\hat{B}(A) \longrightarrow A$$

is an equivalence for complete augmented algebras in the standard pronilpotent setting.

Curved algebras require another distinction. If a curvature element m_0 is present, the unary operation satisfies $m_1^2 = [m_0, -]$ as the inner derivation $[m_0, -]$. A central scalar produces the zero inner derivation, whereas a noncentral curvature element produces the inner derivation $[m_0, -]$. Curved bar–cobar duality belongs to the coderived and contraderived categories of Positselski [60]. The vertex-algebra chiral differential remains square-zero and the curved Koszul differential belongs to the separate curved comparison object.

18.7 Koszul dual modules

A twisting morphism $\tau : C \rightarrow A$ defines adjoint functors between A -modules and C -comodules. Under the hypotheses of Theorem 18.1, these functors give an equivalence on the appropriate derived categories. For a quadratic Koszul algebra, the resolution is linear and yields the familiar exchange of Ext and Tor.

In the chiral setting, modules are factorization modules at marked points. The pointed collision complex is the bar resolution. A module-level Koszul duality therefore requires the algebra comparison and the compatibility of marked-point factorization. Fusion products and conformal blocks are preserved only when a braided tensor equivalence of module categories has been established.

18.8 Maurer–Cartan moduli

Twisting morphisms form a Maurer–Cartan space in a convolution dg Lie or L_∞ -algebra. Gauge equivalence is homotopy of twisting data. The tangent complex at τ is the twisted convolution complex

$$(\mathrm{Hom}(C, A), d + [\tau, -]).$$

Its first cohomology contains infinitesimal deformations and its second contains obstructions.

This is the natural moduli object associated with Koszul presentations. Koszulness itself is the open condition that the comparison cone be acyclic in a finite perfect family; globally it is a derived condition on the vanishing of that cone. The bar coalgebra is classified by its structured comparison cone and Maurer–Cartan datum; scalar invariants arise by applying specified traces.

18.9 Recognition data for the principal implications

The following structured data turn the corresponding coarse properties into valid recognition theorems:

$$\begin{array}{ll} (\text{PBW presentation, curved comparison, convergence}) & \implies \text{Koszul,} \\ (C_2\text{-cofiniteness, linear bar resolution}) & \implies \text{Koszul,} \\ (\text{Verdier dualizability, biduality datum}) & \implies \text{self-duality,} \\ (\text{braided monodromy, fibre functor, presentation recognition}) & \implies \text{specified Yangian,} \\ (\text{central-charge equality, OPE-compatible tensor equivalence}) & \implies \text{equivalent chiral algebras.} \end{array} \tag{128}$$

Each line names the additional structure that upgrades a numerical or filtered shadow to a theorem about the full object.

19 Duality, Hochschild theory, and the derived center

19.1 The associative envelope of the centre

A chiral Lie algebra $\mathfrak{g}$ supplies a bracket. Its chiral enveloping factorisation algebra $U^{\mathrm{ch}}(\mathfrak{g})$ supplies the associative multiplication and bimodule category used by Hochschild theory. The derived centre is formed in this associative envelope.

Definition 19.1 (Derived chiral center). For a chiral Lie algebra $\mathfrak{g}$ for which the chiral enveloping factorization algebra is defined, set

$$Z_{\text{ch}}^{\text{der}}(\mathfrak{g}) := \text{HH}_{\text{Fact}}^*(U^{\text{ch}}(\mathfrak{g}), U^{\text{ch}}(\mathfrak{g})). \quad (129)$$

For an associative factorization algebra A , write $Z_{\text{ch}}^{\text{der}}(A) = \text{HH}_{\text{Fact}}^*(A, A)$.

Hochschild cochains are derived endomorphisms of the diagonal bimodule. The bar resolution supplies a concrete model. In the chiral setting the resolution is supported on configurations with an incoming and an outgoing boundary component, and insertion of configurations produces brace operations.

19.2 The E_2 structure

The higher Deligne theorem gives Hochschild cochains of an E_1 algebra an E_2 structure. The same statement holds for associative factorization algebras in a suitable symmetric monoidal category. The product is cup product; the shifted Lie bracket is the Gerstenhaber bracket; higher braces encode homotopies.

Theorem 19.2 (Derived center structure). *Let A be an associative factorization algebra whose bimodule category admits the diagonal bar resolution. Then $Z_{\text{ch}}^{\text{der}}(A)$ is an E_2 algebra. Its action on A is universal among homotopy-coherent central actions.*

Proof. The endomorphism operad of the diagonal bar resolution carries brace operations obtained by inserting one bar configuration into another. The brace identities are the boundary identities of nested insertions, and the Hochschild differential is the brace commutator with the multiplication of A . The resulting brace algebra is an algebra over a chain model of E_2 , giving the first assertion. The two-coloured Swiss-cheese operad acts on the pair $(Z_{\text{ch}}^{\text{der}}(A), A)$. Its universal property identifies a homotopy-coherent central action of an E_2 algebra B on A with a morphism $B \rightarrow Z_{\text{ch}}^{\text{der}}(A)$, proving universality. $\square$

The universal property is algebraic. An object B acting on A through the Swiss-cheese operad determines a morphism of E_2 algebras

$$B \longrightarrow Z_{\text{ch}}^{\text{der}}(A). \quad (130)$$

The map sends a closed operation to the family of open operations obtained by inserting it next to boundary inputs.

19.3 The center and a physical bulk

Let Obs_{bulk} be the factorization algebra of local observables of a field theory with boundary algebra A . The bulk-to-boundary operator product defines a map

$$\text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ch}}^{\text{der}}(A). \quad (131)$$

A quasi-isomorphism in (131) reconstructs the bulk local observables from the boundary algebra. A product bulk $\mathcal{T} \otimes \mathcal{D}$ with trivial boundary action from $\mathcal{D}$ has the same boundary algebra and centre, while $\text{Obs}_{\mathcal{D}}$ appears in the homotopy fibre of the comparison map.

Separation theorem 19.3 (Open–closed reconstruction criterion). *Tensoring a bulk theory with a factor $\mathcal{D}$ whose boundary action is the unit preserves the boundary ordered chiral algebra and its derived centre. The bulk observable algebra becomes*

$$\mathrm{Obs}_{\mathrm{bulk}} \otimes \mathrm{Obs}_{\mathcal{D}},$$

and the additional factor maps to the unit central action. Hence open–closed reconstruction is equivalent to the assertion that (131) is a quasi-isomorphism, or to an explicit generation theorem that implies this assertion.

Proof. Factorization observables of a product theory are the tensor product of the two observable algebras. When the second factor acts through the boundary unit, its bulk-to-boundary map is the unit map followed by the central action of $\mathbf{1}$. Thus the boundary algebra and the universal centre are those of the first factor, while the reduced observables of $\mathcal{D}$ survive in the homotopy fibre of the product comparison. Therefore the boundary reconstructs all local bulk observables precisely when the comparison map is a quasi-isomorphism; a generation theorem proves the same conclusion by showing that its cone vanishes. $\square$

This criterion states the exact hypothesis carried by holographic reconstruction from local boundary operations.

19.4 The annulus and Hochschild homology

Factorization homology of the circle with coefficients in an E_1 algebra is Hochschild homology:

$$\int_{S^1} A \simeq A \otimes_{A \otimes A^{\mathrm{op}}}^L A. \quad (132)$$

For a boundary theory, the annulus is obtained by thickening the circle. Its state space is therefore a trace of the boundary algebra. The pairing between Hochschild cochains and Hochschild chains is the algebraic form of inserting a bulk local operator into an annulus amplitude.

A cyclic trace on A thickens planar trees to ribbon graphs and supplies the loop contraction. The derived centre already carries its E_2 structure; the trace adds the operation that forms ribbon genus.

19.5 Morita invariance

Hochschild cochains and chains are Morita invariant. If A and B are derived Morita equivalent associative factorization algebras, then

$$\mathrm{HH}_{\mathrm{Fact}}^*(A, A) \simeq \mathrm{HH}_{\mathrm{Fact}}^*(B, B)$$

as E_2 algebras, and their Hochschild homologies agree.

A Morita equivalence induced by Koszul duality therefore produces an equivalence of centres. The proof passes through the equivalence of module categories and preserves the full E_2 structure.

19.6 A small convolution model

Assume A is augmented, complete, and Koszul with a perfect bar coalgebra $B(A)$. The diagonal bar resolution identifies Hochschild cochains with a twisted convolution complex

$$Z_{\text{ch}}^{\text{der}}(A) \simeq \text{Hom}^\tau(B(A), A), \quad (133)$$

where the differential is

$$df = d_A f - (-1)^{|f|} f d_B + [\tau, f].$$

The braces insert one coalgebra input into another before applying f . This model is finite and explicit when the bar coalgebra is locally finite.

The formula (133) explains the relation between deformation theory and the center. A first-order associative deformation is a degree-two Hochschild cocycle; its obstruction is the Gerstenhaber square. The chiral deformation complex of Equation (119) and the factorization Hochschild complex may be compared, but they govern different structures: one deforms the chiral operation, the other the associative factorization envelope.

19.7 Verdier duality and centers

Verdier duality exchanges $!$ - and $*$ -extensions across diagonals. For a perfect factorization algebra A , duality produces an opposite or dual factorization algebra. The induced map on bimodule categories reverses left and right actions. Consequently the Hochschild complex of the dual is the dual of the Hochschild chain complex of A under a Calabi–Yau pairing.

A degree- d Calabi–Yau structure on A gives a nondegenerate pairing

$$\text{HH}^i(A, A) \otimes \text{HH}_{d-i}(A) \longrightarrow \mathbb{C}.$$

The degree is determined by the cyclic structure, by the cyclic structure before any moduli-space realization is chosen. This pairing is the algebraic origin of BV operators on Hochschild cohomology.

19.8 Topological enhancement

The derived centre is E_2 in the transverse directions. A coherent null-homotopy of one further real translation gives an E_3 enhancement. A coherent topologization of the whole complex curve supplies two additional real topological directions and gives an E_4 algebra, as in Proposition 9.8. A holomorphic centre becomes topological after the required translation homotopies have been supplied.

In three-dimensional Chern–Simons theory, local bulk observables are E_3 because the bulk has three topological directions. Comparison with a boundary centre therefore uses the appropriate intermediate topologization of the boundary object. In a four-real-dimensional holomorphic–topological theory whose holomorphic direction is also topologized, the corresponding local structure is E_4 . In either dimension, proving that the bulk-to-centre map is an equivalence is a field-theoretic calculation.

19.9 Centers at critical level

For the universal affine vertex algebra at the critical level, the ordinary vertex-algebra center is large: the Feigin–Frenkel theorem identifies it with functions on opers for the Langlands dual group on the formal disc [34]. This centre is a commutative chiral algebra inside the affine algebra. Its fields

act on the factorisation Hochschild complex, which additionally contains higher diagonal-bimodule self-extensions.

The distinction is visible in degrees. The vertex center consists of fields commuting with all fields under nonnegative modes. The derived center includes higher self-extensions of the diagonal bimodule. At noncritical level the ordinary center may be scalar while the derived center remains nontrivial.

Part III

Exchange, reduction, and quantum symmetry

20 Labelled chiral configurations and monodromy

20.1 Exchange geometry and the ordered line

The order in an E_1 algebra comes from chambers of configurations on a real line. The monodromy in this section comes instead from moving labelled chiral insertion points in the complex curve, or from an E_2 transverse enhancement. The exchange structures below enter through labelled motion in the curve or through an E_2 transverse enhancement.

20.2 Labels retain paths

The labelled configuration space of n points is

$$\mathrm{Conf}_n(X) = \{(x_1, \dots, x_n) \in X^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group acts freely, and the quotient is the unordered configuration space

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/\Sigma_n.$$

For $X = \mathbb{C}$,

$$\pi_1(\mathrm{Conf}_n(\mathbb{C})) = P_n, \quad \pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n,$$

the pure braid and braid groups. A local system on the labelled space therefore retains pure braid monodromy; an equivariant local system descends to the unordered space and retains full braid monodromy.

A symmetric factorization coalgebra carries coherent descent data under all relabelings and diagonals. Labelled analytic continuation is extra information: it is a representation of the fundamental groupoid of the configuration space.

20.3 Derived descent

Let M be a chain complex with an action of a finite group G . The homotopy coinvariants are

$$M_{hG} = M \otimes_{\mathbb{C}[G]}^L \mathbb{C}.$$

The augmentation ideal $I_G = \ker(\mathbb{C}[G] \rightarrow \mathbb{C})$ fits into a distinguished triangle

$$M \otimes_{\mathbb{C}[G]}^L I_G \longrightarrow M \longrightarrow M_{hG}. \quad (134)$$

Thus the homotopy fiber of descent is canonical. In characteristic zero and for finite G , the group algebra is semisimple, so derived coinvariants agree with ordinary coinvariants. The fiber still records all nontrivial isotypic components.

For a braid-group local system, the relevant descent uses the full braid-group algebra before symmetric descent. One must retain the monodromy representation and its equivariance under the extension

$$1 \longrightarrow P_n \longrightarrow B_n \longrightarrow \Sigma_n \longrightarrow 1.$$

Discarding the pure braid action loses analytic continuation even when the underlying vector space is unchanged.

20.4 The descent fiber and collision filtrations

For a coefficient system M_n carrying a Σ_n -action, define the descent fiber

$$\mathcal{C}_n(M) = \text{Fib}(M_n \rightarrow M_{n,h\Sigma_n}). \quad (135)$$

By (134), it is canonically $M_n \otimes_{\mathbb{C}[\Sigma_n]}^L I_{\Sigma_n}$. A universal chain model is obtained from the ordinary group bar resolution of the trivial $\mathbb{C}[\Sigma_n]$ -module. In characteristic zero the group algebra is semisimple, so this derived fiber is represented by the direct sum of the nontrivial isotypic components.

The proper partition complex is a different universal object. It is the incidence and Koszul complex of collision partitions, and its top homology is the Lie representation. For an arbitrary Σ_n -module, the group-bar construction resolves the descent fibre. When M_n carries a functorial filtration indexed by collision partitions, an explicit comparison morphism from the partition-incidence complex to the group-bar model may filter $\mathcal{C}_n(M)$ by binary collisions, ternary collision coherence, and higher strata. Such a filtration belongs to the coefficient system and must be constructed; it is additional filtered coefficient data carried by the local system.

The descent fibre contains the nontrivial isotypic and braid data carried by the coefficient system. An r -matrix appears when the first-order monodromy of a flat connection has the form required by the classical Yang–Baxter equation.

20.5 Dual local systems

Let M be an oriented manifold and $\mathcal{L}$ a finite-rank local system with monodromy $\rho : \pi_1(M) \rightarrow GL(V)$. Its Verdier dual is

$$\mathbb{D}\mathcal{L} \cong \mathcal{L}^\vee[\dim M],$$

up to the orientation line. The monodromy of the dual local system is

$$\rho^\vee(\gamma) = \rho(\gamma^{-1})^\dagger. \quad (136)$$

The inverse occurs because parallel transport on the dual is contragredient; the transpose is the linear dual.

Corollary 20.1. *Verdier duality sends a braid operator R to the inverse transpose R^{-t} . If the representation is identified with its dual by a nondegenerate invariant pairing, this becomes inverse braiding.*

The further notation $q \mapsto q^{-1}$ is justified only when a family of braid operators has been identified with a quantum-group family parametrized by q .

20.6 Associators

The monodromy of a flat connection on the configuration space of three points depends on comparing asymptotic zones. A Drinfeld associator is a group-like formal series satisfying the pentagon and hexagon equations. Associators form a torsor under the Grothendieck–Teichmüller group GRT_1 after the usual normalization [30, 77].

The pentagon is a nonlinear arity-four equation for a group-like formal series; the hexagon relates that series to the braiding. Their normalized solutions form the associator torsor.

In the de Rham realization, the Knizhnik–Zamolodchikov associator is the regularized parallel transport between tangential base points at 0 and 1 on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its coefficients are multiple zeta values. This arithmetic belongs to chiral configuration-space monodromy, to the regularized monodromy of the chiral configuration-space connection.

20.7 Tannaka reconstruction

Let $\mathcal{C}$ be a rigid braided tensor category and let $F : \mathcal{C} \rightarrow \text{Vect}$ be a faithful exact fibre functor. The coend

$$H = \int^{M \in \mathcal{C}} F(M)^\vee \otimes F(M) \quad (137)$$

reconstructs a bialgebra object when F is strong monoidal; rigidity supplies an antipode when the coend belongs to the finite or continuous-dual class. If F is only quasi-monoidal, the associativity constraint produces a coassociator and H is a coquasi-bialgebra. The braiding produces a coquasitriangular structure. Under the relevant finite-dimensional or continuous duality, one obtains a quasitriangular Hopf or quasi-Hopf algebra, respectively.

Theorem 20.2 (Ordered Tannaka principle). *A rigid tensor-compatible monodromy category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra whenever the antipode is defined in the chosen finite or continuous-dual class. With a faithful quasi-monoidal fibre functor, it reconstructs a coquasitriangular coquasi-bialgebra. A quasi-Hopf presentation becomes a strict Hopf presentation after a twist whose coboundary trivializes the coassociator.*

Proof. Form the coend (137). Tensor compatibility defines multiplication of matrix coefficients, while the monoidal constraint of F defines the coproduct and counit. Rigidity transports evaluation and coevaluation to the antipode. The categorical braiding defines the universal R -form; the hexagon identities are the coquasitriangular identities. For a quasi-monoidal functor, the pentagon transports to the coassociator equation. A twist changes the coassociator by its standard coboundary, so strict coassociativity is equivalent to a twist trivialization. This is the Tannaka reconstruction theorem for braided tensor categories. $\square$

The theorem is universal. Identifying the reconstructed object with $U_q(\mathfrak{g})$, a Yangian, an elliptic quantum group, or a Hall algebra is a recognition theorem using the specific connection and

representation category.

20.8 Surface braid groups

For a curve of genus $g > 0$, $\pi_1(\text{Conf}_n(X))$ is a surface pure braid group. Its generators include collisions and transport around the A - and B -cycles of the curve. Flat connections such as the KZB connection represent these generators by operators satisfying elliptic braid relations.

The elliptic theory consists of the genus-zero associator together with quasi-periodicity, the choice of a dynamical parameter, and the mapping-class-group action supply additional data. At higher genus, conformal-block monodromy is a representation of the mapping class group of the punctured surface, and refines the surface braid-group representations.

20.9 What symmetrization preserves

Symmetrization preserves invariant traces, dimensions of invariant subspaces, and any operation that factors through the quotient. It may preserve nontrivial periods when those periods lie in the invariant part of the local system. Symmetric descent retains the invariant arithmetic and monodromy and removes the non-invariant quotient measured by (135).

For conformal blocks, passing from vectors to their dimension preserves the Verlinde number and loses the mapping-class-group representation. Passing from a braided category to its Grothendieck ring preserves fusion coefficients and loses the associator and twists. These are concrete instances of the same descent.

21 KZ geometry, quantum groups, Gaudin systems, and lines

21.1 The logarithmic connection

Let $\mathfrak{g}$ be a finite-dimensional complex simple Lie algebra with invariant tensor $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$. On $V_1 \otimes \cdots \otimes V_n$, define

$$\nabla_{\text{KZ}} = d - \hbar \sum_{i < j} \Omega_{ij} d \log(z_i - z_j), \quad \hbar = \frac{1}{k + h^\vee}. \quad (138)$$

The parameter depends on the normalization of the invariant form; the shifted level is invariantly determined by the Sugawara construction.

The infinitesimal braid relations are

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{if } \{i, j\} \cap \{k, l\} = \emptyset, \quad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0 \quad (139)$$

for distinct i, j, k .

Theorem 21.1 (Flatness of the KZ connection). *The connection (138) is flat.*

Proof. The logarithmic forms are closed, so for the convention $\nabla = d - A$ the curvature is $A \wedge A$. Terms with disjoint index pairs vanish by the first relation in (139). For every triple i, j, k , the remaining terms factor as the Arnold relation

$$d \log z_{ij} \wedge d \log z_{jk} + d \log z_{jk} \wedge d \log z_{ki} + d \log z_{ki} \wedge d \log z_{ij} = 0$$

times the second infinitesimal braid relation. Hence the curvature vanishes. $\square$

This proof displays the exact division of labor. The differential forms supply the Arnold identity. The coefficient tensor supplies the infinitesimal braid identity. Their tensor-product cancellation is the flatness equation.

21.2 From affine currents to KZ

For the affine vertex algebra $V_k(\mathfrak{g})$, current fields satisfy

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \quad (140)$$

Ward identities for primary insertions convert the simple-pole action into the differential equation (138); the double pole and the Sugawara stress tensor produce the shift by $\hbar^\vee$. The KZ connection is therefore the representation-theoretic projection of the affine chiral operation to the bundle of conformal blocks.

21.3 Drinfeld–Kohno

Let $q = \exp(\pi i \hbar)$ in the standard convention. The Drinfeld–Kohno theorem identifies the braided tensor category defined by KZ monodromy with the category of finite-dimensional modules over the Drinfeld–Jimbo quantum group $U_q(\mathfrak{g})$, with an associator determined by KZ parallel transport [51, 30, 50].

Theorem 21.2 (Drinfeld–Kohno). *For generic $\hbar$, the braid-group representations obtained from the monodromy of the KZ connection are naturally equivalent, as braided tensor data, to those obtained from the universal R -matrix of $U_q(\mathfrak{g})$ with $q = e^{\pi i \hbar}$.*

Source. Drinfeld constructs the quasi-Hopf algebra whose associator is the regularized KZ transport and proves its twist equivalence with the Drinfeld–Jimbo quantization; Kohno identifies the resulting braid representations with KZ monodromy. The Kazhdan–Lusztig tensor equivalence supplies the corresponding affine representation-theoretic form. With the invariant-form normalization used in (138), the parameter relation is $q = e^{\pi i \hbar}$ [30, 51, 50]. $\square$

The theorem is the rigorous statement that a quantum group is present in the ordered affine theory. It includes the associator. The Hopf or quasi-Hopf algebra is reconstructed from the braided category and fibre functor.

21.4 The classical r -matrix

The one-form in (138) has coefficient

$$r_{ij}(z_i - z_j) = \hbar \frac{\Omega_{ij}}{z_i - z_j}. \quad (141)$$

The flatness equation at first order in $\hbar$ is the classical Yang–Baxter equation. The monodromy is a path-ordered exponential of the connection; its semiclassical expansion begins with the classical r -matrix.

The same rational kernel appears in several constructions once their normalizations are matched: the KZ connection, the rational Gaudin model, and the line-operator OPE of four-dimensional Chern–Simons theory. The equality is a theorem only after the coefficient spaces and coupling constants have been identified.

21.5 Gaudin Hamiltonians

Fix distinct points $z_1, \dots, z_n$. The Gaudin Hamiltonians are

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}. \quad (142)$$

The infinitesimal braid relations imply

$$[H_i, H_j] = 0.$$

They are the residues of the KZ connection in the z_i variable. Their commutativity is therefore another coefficient of flatness.

For highest-weight modules, the Bethe ansatz constructs simultaneous eigenvectors from critical points of a master function. In the simplest $\mathfrak{sl}_2$ case, Bethe roots t_a satisfy

$$\sum_{i=1}^n \frac{\lambda_i}{t_a - z_i} - \sum_{b \neq a} \frac{2}{t_a - t_b} = 0. \quad (143)$$

The Schechtman–Varchenko hypergeometric solutions of the KZ equations relate these critical points to horizontal sections. Thus KZ monodromy and the Gaudin spectrum are two readings of the same flat connection.

21.6 Four-dimensional Chern–Simons theory

Costello–Witten–Yamazaki construct a four-dimensional gauge theory that is topological in two directions and holomorphic in one complex direction. Wilson lines at distinct holomorphic positions have an operator product governed by a spectral-parameter R -matrix. In the rational case, the algebra of line operators is a Yangian; trigonometric and elliptic choices produce quantum loop and elliptic structures [19, 20].

The spectral parameter is geometric: it is the difference of holomorphic positions of the lines. The Yang–Baxter equation follows from isotopy of three lines. The one-loop computation fixes the first correction to the R -matrix. This is a physical construction of ordered monodromy, the physical realization whose comparison with chiral Koszul duality is supplied by the boundary functor.

21.7 Yangians and recognition

A Yangian $Y_h(\mathfrak{g})$ is a filtered deformation of $U(\mathfrak{g}[t])$ with a rational R -matrix on suitable representations. Recognition of a reconstructed quasi-bialgebra as a Yangian uses a PBW theorem, evaluation representations, the rational difference property, and a presentation theorem showing that the listed relations generate the complete kernel.

Criterion 21.3 (Yangian recognition). *Let H be a filtered quasi-bialgebra reconstructed from an ordered tensor category. If*

- (i) $\mathrm{gr} H \cong U(\mathfrak{g}[t])$ as a bialgebra;
- (ii) the fundamental evaluation modules are faithful on the completed algebra;
- (iii) their R -matrices agree with the rational Yangian R -matrices;
- (iv) the defining RTT or Drinfeld relations generate the full kernel of the representation map;

then H is isomorphic, up to the associator twist already present in the tensor functor, to the corresponding completed Yangian.

The criterion states exactly what an ordered chiral construction must prove before the word “Yangian” is justified.

21.8 Elliptic transport

On an elliptic curve, the KZB connection replaces $d\log(z_i - z_j)$ by elliptic kernels with prescribed quasi-periodicity. Its flatness uses Fay identities and dynamical correction terms. The monodromy represents the elliptic braid group and the modular group. Depending on the setting, reconstruction leads to elliptic or dynamical quantum groups.

Rational, trigonometric, and elliptic quantum groups arise from the corresponding rational, cylindrical, and elliptic connection geometries and are related by the degeneration maps of those curves and coefficient categories. Their relation is controlled by limits of the curve and of the coefficient category.

21.9 Knots and ribbon traces

A ribbon category assigns invariants to framed links by composing braidings, duality maps, and twists. The Reshetikhin–Turaev construction applied to $U_q(\mathfrak{g})$ produces quantum link invariants [63]. Chern–Simons theory supplies the physical path-integral interpretation. The KZ connection supplies a monodromy realization of the same braided tensor category.

The closure of a braid uses a categorical trace. This is the precise point at which ordered monodromy becomes a scalar knot invariant. The scalar is the categorical trace of the individual braid representation, while the full representation retains every braid operator.

21.10 Duality of monodromy

By (136), dualizing the KZ local system sends the monodromy to its inverse transpose. Under the invariant form on representations and the Drinfeld–Kohno equivalence, this agrees with the dual quantum-group representation and inverse braiding. A level reflection such as $k \mapsto -k - 2h^\vee$ has the same effect on the parameter $\hbar = (k + h^\vee)^{-1}$, namely $\hbar \mapsto -\hbar$, hence $q \mapsto q^{-1}$. This is a statement about the KZ family. For a general chiral Koszul pair, the analogous inversion is established by constructing the dual local system and comparing its fibre functor.

22 Quantum line defects and meromorphic tensor categories

22.1 Fusion and planar exchange

Let $A \in \text{Alg}_{E_1}(\text{FactD}(X))$. Its modules can be placed in a fixed order along the transverse line. Relative tensor product gives fusion:

$$M \star_A N := M \otimes_A^L N.$$

Associativity follows from excision for three consecutive intervals. The configuration space of two subintervals has two order chambers. A path between the chambers appears after the transverse line is thickened to a plane, and its parallel transport gives a map $M \star_A N \rightarrow N \star_A M$.

Separation theorem 22.1 (Planar enhancement criterion). *Let A be an associative algebra concentrated in cohomological degree zero. An E_2 refinement whose induced degree-zero product equals the product of A forces that product to be commutative. Hence matrix algebras and free associative algebras acquire exchange through an enlarged derived, categorical, or meromorphic structure rather than through a discrete E_2 refinement of the same multiplication.*

Proof. The homology multiplication of an E_2 algebra is graded commutative. Concentration in degree zero turns graded commutativity into ordinary commutativity. The planar configuration space supplies the geometric refinement: its exchange paths act on a category of modules or on a meromorphic local system while preserving the original ordered fusion inside each chamber. $\square$

Fusion belongs to the ordered chiral algebra. Planar transport, the Hochschild centre, or a meromorphic local system supplies exchange.

22.2 The meromorphic line category

Let $\mathcal{L}$ be a category of line defects varying over the spectral curve X . For distinct spectral points, fusion is defined and is compatible with factorization. An exchange structure consists of isomorphisms

$$R_{L,M}(x, y) : L_x \star M_y \longrightarrow M_y \star L_x, \quad x \neq y,$$

which are horizontal for the D -module connection and satisfy the braid relation on $\text{Conf}_3(X)$.

Definition 22.2 (Meromorphic tensor category). A meromorphic tensor category on X consists of a factorization category $\mathcal{L}$ on the ordered configuration spaces of X , an associative fusion product on disjoint ordered blocks, and a braid-group descent datum on the locus of distinct points. The descent datum is meromorphic along the collision divisors and satisfies the pentagon and hexagon identities.

The associator compares the three asymptotic regions of four ordered insertions. The braiding compares two chambers after a transverse exchange. The pole of the exchange operator is a chiral singularity; its homotopy class is topological.

Theorem 22.3 (Line-category recognition). *Let $\mathcal{T}$ be a renormalized quantum field theory on $X \times \mathbb{R}^2$ that is holomorphic on X and topological on $\mathbb{R}^2$. Assume that the fusion and exchange Ward identities are anomaly-free on a chosen class of line defects, that their morphism complexes are perfect over the meromorphic functions on X , that the line OPE extends meromorphically across the collision*

divisors, and that the class is closed under fusion and duality. Then these line defects form a rigid meromorphic tensor category in the sense of Theorem 22.2.

Proof. Fusion is factorization in one transverse direction. Associativity is isotopy of three parallel lines. The exchange of two lines is a path in the configuration space of two points in the transverse plane. Isotopy of three exchange paths gives the braid relation. Isotopy of decompositions of four lines gives the pentagon; compatibility of exchange with fusion gives the hexagons. Holomorphic dependence on spectral position and locality of counterterms make the structure meromorphic along diagonals. Dual line orientations give categorical duals under the perfectness hypothesis. $\square$

22.3 The centre and categorical exchange

The internal Hochschild centre $Z_{\text{ord, ch}}(A)$ is an E_2 algebra in $\text{FactD}(X)$. This means that its local multiplication carries the homotopy-coherent exchange of little two-disks. Its ordinary module category is naturally E_1 -monoidal. The Drinfeld centre of that monoidal category, or modules over an E_3 enhancement, carries the braided line category.

Corollary 22.4 (Exchange through a complete centre). *Suppose the bulk-to-boundary map*

$$\beta_{\mathcal{T}} : \text{Obs}_{\text{bulk}} \longrightarrow Z_{\text{ord, ch}}(A)$$

is an equivalence of local E_2 algebras. Suppose in addition that a monoidal-affineness theorem identifies the selected bulk line category with the Drinfeld centre $Z(\text{Bimod}_A)$, and identifies this categorical centre with modules over an E_3 topological enhancement of $Z_{\text{ord, ch}}(A)$. Then the corresponding perfect line categories are equivalent as braided monoidal categories.

Proof. The bulk-to-boundary equivalence identifies the local E_2 observable algebra with the internal centre. The monoidal-affineness equivalences identify, respectively, bulk lines with $Z(\text{Bimod}_A)$ and that categorical centre with modules over the E_3 enhancement. Composition of these braided monoidal equivalences gives the asserted equivalence of perfect line categories. $\square$

The hypotheses identify local bulk observables, extended line defects, and the categorical centre in one comparison diagram. Together they yield the braided equivalence stated in the corollary.

22.4 Tannaka reconstruction and its exact output

Choose a faithful exact meromorphic fibre functor

$$F : \mathcal{L} \longrightarrow \text{Vect}_K, \quad K = \mathbb{C}(X).$$

The coend

$$H_{\mathcal{L}} = \int^{L \in \mathcal{L}} F(L)^{\vee} \otimes_K F(L)$$

represents tensor automorphisms of F . The tensor product, associator, and braiding produce respectively a coproduct, coassociator, and universal R -form.

Theorem 22.5 (Quantum symmetry of line defects). *If F is strong monoidal, $H_{\mathcal{L}}$ is a coquasitriangular bialgebra over K , and rigidity supplies a Hopf antipode in the finite or continuous-dual class. If F is only quasi-monoidal, $H_{\mathcal{L}}$ is a coquasitriangular coquasi-bialgebra whose coassociator is the transported*

tensor constraint. When the relevant dual exists, the dual objects are quasitriangular Hopf or quasi-Hopf algebras, respectively. The construction is invariant under braided tensor equivalence compatible with the chosen fibre functor.

Proof. The coend represents natural matrix coefficients of the fibre functor. A strong tensor constraint defines a coassociative coproduct and counit; rigidity defines the antipode in the finite or continuous-dual class. The braiding defines the universal R -form, and the hexagon identities give coquasitriangularity. A quasi-monoidal constraint transports the associator to a coassociator satisfying the pentagon. Braided tensor equivalences compatible with fibre functors induce isomorphisms of the representing coends and preserve all these maps. $\square$

The theorem reconstructs quantum symmetry intrinsically. RTT, current, elliptic, or Hall relations together with a PBW theorem recognize the resulting object as a Yangian, quantum loop algebra, elliptic quantum group, or Hall algebra.

22.5 Rational gauge theory and the Yangian

In four-dimensional Chern–Simons theory on $\Sigma \times X$, where the theory is topological on Σ and holomorphic on X , parallel Wilson lines form a meromorphic tensor category. In the rational theory $X = \mathbb{C}$, perturbation theory produces

$$R_{V,W}(z) = 1 + \hbar \frac{\Omega_{V,W}}{z} + O(\hbar^2). \quad (144)$$

The one-loop anomaly of a line is cancelled by deforming the classical current algebra to the Yangian. The resulting line OPE is the RTT or Drinfeld coproduct of $Y_{\hbar}(\mathfrak{g})$.

Theorem 22.6 (Rational Wilson-line algebra). *For the perturbative Wilson-line categories constructed in anomaly-free rational four-dimensional Chern–Simons theory by Costello–Witten–Yamazaki, the algebra of line operators is the completed Yangian deformation of $U(\mathfrak{g}[z])$ on the stated class of representations; exchange is given by the corresponding rational universal R -matrix. The spectral parameter is the difference of the holomorphic positions. The assertion belongs to the completed formal $\hbar$ -adic category in which the line anomaly and its cancellation are defined [19, 20].*

The theorem is a field-theoretic recognition theorem. It uses the gauge-theory Ward identities and anomaly calculation, together with the ordered bar construction and its boundary realization.

22.6 Trigonometric and elliptic geometries

Changing the holomorphic curve changes the meromorphic tensor category. For $X = \mathbb{C}^\times$, the trigonometric kernel and its periodicity lead to a quantum loop algebra. For an elliptic curve, quasi-periodicity and a dynamical parameter lead to elliptic or dynamical quantum groups. These structures are reconstructed from different meromorphic connections.

Principle 22.7 (The spectral curve selects the quantum group). *For the standard rational, trigonometric, and elliptic meromorphic line categories, Tannaka reconstruction produces the corresponding rational, quantum-loop, and elliptic or dynamical quantum symmetries after their presentation-recognition theorems are applied. A degeneration of the spectral curve induces a degeneration of the exchange kernel only when the line category, fibre functor, and meromorphic connection extend flatly over the degenerating family.*

Proof. On the rational curve, the standard meromorphic connection and its representation category are recognized by the rational RTT or Drinfeld presentation. Periodicity on $\mathbb{C}^\times$ replaces rational currents by loop currents and yields the quantum-loop presentation. Quasi-periodicity on an elliptic curve introduces the elliptic or dynamical parameter and yields the elliptic presentation. Tannaka reconstruction converts each tensor category into its coordinate bialgebra, and the corresponding presentation theorem identifies its continuous dual. Flat extension of the category, fibre functor, and connection transports this identification through a degeneration. $\square$

22.7 Shifted Yangians and boundary conditions

A line defect may end on a boundary or meet an 't Hooft defect. Magnetic charge changes the allowed pole orders of the Drinfeld currents. Algebraically this produces shifted Yangians. Geometrically the shift is a modification of the factorization module at a marked transverse boundary condition.

Let μ be a coweight. A shifted object is represented by a pointed module bar

$$\text{Bar}_X^{\text{ord}}(M_-, A, M_+) = M_- \otimes_A^L M_+,$$

where the endpoint modules encode the two boundary charges. The unshifted algebra is recovered when both modules are the augmentation boundary.

Criterion 22.8 (Shifted recognition). *A line algebra is identified with a shifted Yangian by matching the following data: the coweight filtration of currents, the shifted coproduct, the PBW associated graded, the fundamental line modules, and the rational exchange kernel. The coweight filtration and shifted coproduct determine the shift beyond the unshifted R -matrix.*

Proof. The matched current filtration and coproduct define a homomorphism from the shifted Drinfeld presentation to the line algebra. The fundamental line modules and rational exchange kernel identify the images of the generators and the shift. The PBW comparison identifies the associated graded map with an isomorphism; completeness then makes the filtered homomorphism an isomorphism. The filtration and coproduct distinguish different coweight shifts that share the same unshifted binary kernel. $\square$

22.8 KZ and gauge-theory exchange

For affine conformal blocks, the KZ connection gives braid monodromy with coupling $(k + h^\vee)^{-1}$. For four-dimensional Chern–Simons lines, the gauge coupling fixes the rational R -matrix. These two constructions are related through the Drinfeld–Kohno theorem after the representation categories and normalizations have been identified. The equality of kernels is established by the tensor equivalence and normalization comparison: the affine theory is a two-dimensional chiral theory, whereas the gauge theory is a four-dimensional holomorphic–topological theory.

Theorem 22.9 (Comparison through the line category). *Let $\mathcal{L}_{\text{KZ}}$ be a rigid braided tensor subcategory of the Drinfeld–KZ category at a nonresonant shifted level, and let $\mathcal{L}_{\text{gauge}}$ be a rigid Wilson-line subcategory of a rational gauge theory. A braided tensor equivalence matching tensor generators and the first-order Casimir kernel identifies their braid monodromies. Under compatible Tannaka fibre functors, both reconstruct the same quasitriangular Hopf or quasi-Hopf object up to the transported associator twist.*

Proof. A tensor equivalence preserves the coend reconstruction. Matching the Casimir coefficient fixes the formal deformation parameter. The braid operator on tensor generators then agrees, and rigidity extends the agreement to the generated category. The associator records the difference between strict and KZ tensor structures. $\square$

The theorem isolates the actual comparison datum. The tensor equivalence upgrades similarity of rational kernels to an identity of the corresponding line categories.

22.9 Koszul duality of line couplings

Let A be the algebra of local observables near a line and let $\varepsilon : A \rightarrow \mathbf{1}$ describe the distinguished trivial line. Formal deformations supported on that line are governed by the derived deformation theory of the augmentation module. Its endomorphism algebra is the Koszul dual

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

Theorem 22.10 (Koszul duality for the augmentation-generated line sector). *Let A be a complete augmented E_1 algebra of local observables in the formal neighbourhood of a line defect, and put $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$. The adjunction of Theorem 7.5 identifies the thick, or presentably completed, sector of A -modules generated by the augmentation with the corresponding sector of $A^!$ -modules. The bar complex $\mathbf{1} \otimes_A^L \mathbf{1}$ computes the derived matrix coefficients of the augmentation line.*

Proof. The augmentation module is the distinguished line. Its derived endomorphisms form $A^!$. The endomorphism adjunction is an equivalence on the generator and therefore on its thick closure; compact generation extends it to the stated presentable completion. The two-sided bar computes the derived tensor product of the two augmentation modules. $\square$

This theorem is the precise common content of Koszul duality in defect physics. Identifying $A^!$ with a named Yangian is a further calculation.

22.10 Transfer matrices and scalar observables

Let V be an auxiliary object of a rigid line category and let $M_1, \dots, M_n$ be quantum spaces. The monodromy operator

$$T_V(z) = R_{V, M_n}(z - z_n) \cdots R_{V, M_1}(z - z_1)$$

obeys the RTT relation. Taking the categorical trace over V gives commuting transfer matrices:

$$[t_V(z), t_W(w)] = 0.$$

The proof is the braid relation followed by cyclicity of the trace.

This passage loses information. The transfer matrix is a scalar or endomorphism-valued trace of ordered exchange. Bethe equations and Baxter relations describe its spectrum in selected representations. They follow from the particular reconstructed quantum group and its representation theory.

22.11 Knots and framing

Closing a braid requires duals and a ribbon twist. The Reshetikhin–Turaev invariant is the categorical trace of the closed braid. In Chern–Simons theory the twist is the framing anomaly. Thus three structures enter in order:

$$\text{fusion} \longrightarrow \text{braiding} \longrightarrow \text{ribbon trace}.$$

They are governed respectively by E_1 , E_2 , and cyclic or framed data. The distinction is the categorical form of the distinction between an ordered word, an exchange path, and a closed world-line.

23 Higher transverse dimensions and iterated chiral Koszul duality

A holomorphic–topological theory on

$$X_{\text{hol}} \times \mathbb{R}_{\text{top}}^n$$

assigns two independent kinds of local operation. Collision in the curve produces the chiral D -module operation. Motion and little-disk insertion in the transverse space produce an E_n multiplication. Their common algebraic carrier is an E_n -algebra internal to factorisable D -modules on X .

23.1 The higher noncommutative chiral object

Let

$$\mathcal{C}_X = (\text{FactD}(X), \otimes^{\text{pt}}, \mathbf{1})$$

be the symmetric monoidal category fixed in Part I. For $n \geq 1$, the category

$$\text{Alg}_{E_n}^{\text{aug}}(\mathcal{C}_X) \tag{145}$$

consists of augmented E_n -algebras whose structure maps are morphisms of factorisation objects on the curve. The multiplication therefore preserves every chiral factorisation map.

Dunn additivity supplies a canonical equivalence

$$\text{Alg}_{E_m}(\text{Alg}_{E_n}(\mathcal{C}_X)) \simeq \text{Alg}_{E_{m+n}}(\mathcal{C}_X). \tag{146}$$

The equivalence identifies a configuration of little $(m+n)$ -disks with an m -dimensional configuration of n -dimensional configurations [28, 54].

23.2 The iterated bar

Let $A \in \text{Alg}_{E_n}^{\text{aug}}(\mathcal{C}_X)$. Dunn additivity presents A as an augmented E_1 -algebra in $\text{Alg}_{E_{n-1}}(\mathcal{C}_X)$. Applying the reduced associative bar in that category and repeating gives

$$\text{Bar}_X^{(n)}(A) = \text{Bar}_{E_1} \text{Bar}_{E_1} \cdots \text{Bar}_{E_1}(A). \tag{147}$$

After the j -th bar, the remaining multiplication has type E_{n-j} and the first j directions carry commuting conilpotent coalgebra structures.

Theorem 23.1 (Iterated bar–cobar reconstruction). *Assume that $\mathcal{C}_X$ is stable and presentable, that $\otimes^{\text{pt}}$ preserves geometric realisations separately, and that the augmentation ideal of A has a complete positive filtration with dualisable finite quotients. Assume further that the augmented associative bar–cobar counit is an equivalence in every monoidal category $\text{Alg}_{E_j}(\mathcal{C}_X)$ occurring in the iteration. Then $\widehat{\text{Bar}}_X^{(n)}(A)$ is a conilpotent E_n -coalgebra in the completed category and the composite counit is an equivalence*

$$\widehat{\text{Cobar}}_X^{(n)} \widehat{\text{Bar}}_X^{(n)}(A) \xrightarrow{\simeq} A. \quad (148)$$

Proof. Apply the associative bar–cobar equivalence to A inside $\text{Alg}_{E_{n-1}}(\mathcal{C}_X)$. The resulting bar is an E_{n-1} -algebra in conilpotent coassociative coalgebras. Apply the same argument to the residual E_{n-1} multiplication. Dunn additivity identifies the n commuting coassociative coalgebra directions with an E_n -coalgebra structure. At every finite filtration level, conilpotence makes the bar and cobar sums finite in each total weight. The transition maps between filtration quotients are surjective, so the Milnor exact sequence identifies cohomology of the completed inverse limit with the inverse limit of the finite-stage cohomologies. Composition of the n associative counits gives (148). $\square$

23.3 Factorisation homology of the pointed disk

Equip the boundary sphere of D^n with the augmentation boundary condition. Excision and Fubini for factorisation homology identify the iterated bar with the relative integral

$$\widehat{\text{Bar}}_X^{(n)}(A) \simeq \int_{(D^n, \partial D^n)} A. \quad (149)$$

For $n = 1$, a subdivision of the interval gives the two-sided simplicial bar. Writing $D^n = D^{n-1} \times [0, 1]$ and applying factorisation Fubini proves the general case inductively [1].

The linking manifold of a transverse point collision is S^{n-1} . Its fundamental class extracts the universal degree- $(1 - n)$ operation on homology.

23.4 Configuration-space coefficients and the Poisson shadow

For $d \geq 2$, the rational cohomology of $\text{Conf}_r(\mathbb{R}^d)$ is generated by classes

$$\omega_{ij} \in H^{d-1}(\text{Conf}_r(\mathbb{R}^d); \mathbb{Q}), \quad \omega_{ji} = (-1)^d \omega_{ij},$$

obtained by pulling back the fundamental cohomology class of S^{d-1} along

$$(x_1, \dots, x_r) \mapsto \frac{x_i - x_j}{\|x_i - x_j\|}.$$

The generators satisfy $\omega_{ij}^2 = 0$ and the three-term relation

$$\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0, \quad (150)$$

with the Koszul signs determined by $|\omega_{ij}| = d - 1$. These classes form the screen-cohomology coefficient system of the Fulton–MacPherson compactification of transverse configurations.

Proposition 23.2 (Linking operation). *Let A be an E_d -algebra in cochain complexes. Evaluation of the binary little-disks operation on the fundamental cycle of S^{d-1} defines*

$$\{-, -\} : H^p(A) \otimes H^q(A) \longrightarrow H^{p+q+1-d}(A). \quad (151)$$

Together with the product induced by a point of the binary little-disks space, this operation gives the canonical P_d -algebra structure on $H^\bullet(A)$.

Proof. The space of binary little d -disks is homotopy equivalent to S^{d-1} . Evaluation on its fundamental class has cohomological degree $1 - d$. The codimension-one faces in the compactification of the three-disk configuration give the biderivation and shifted Jacobi identities; their determinant orientations give the standard P_d signs. $\square$

For an E_d -chiral algebra, (151) belongs to the transverse topological structure. The diagonal-supported chiral operation on X simultaneously carries the holomorphic OPE. The interchange law follows from the defining requirement that every transverse multiplication map be a morphism of factorisation objects.

23.5 Higher centres

For $A \in \text{Alg}_{E_n}(\mathcal{C}_X)$, define the higher centre $Z_{E_n}(A)$ by its universal property: it is the final E_{n+1} -algebra equipped with a central action on A . Under the presentability hypotheses above, the centre exists and agrees with the E_n Hochschild cochain object

$$Z_{E_n}(A) = HH_{E_n}^\bullet(A). \quad (152)$$

For $n = 1$, this is the Hochschild cochain object $\text{RHom}_{A \otimes A^{\text{op}}}(A, A)$ with its E_2 structure. The higher Deligne theorem equips (152) with an E_{n+1} multiplication.

Theorem 23.3 (Higher open-closed recognition). *Let B be an E_{n+1} -algebra and let A be an E_n -algebra. Every central action of B on A determines a unique E_{n+1} morphism*

$$B \longrightarrow Z_{E_n}(A) \quad (153)$$

compatible with the action. A protected boundary condition is complete precisely when this morphism is an equivalence for its protected local bulk algebra B .

Proof. The first assertion is the representing property that defines $Z_{E_n}(A)$. Completeness means that a protected local bulk operation is determined uniquely by its action on the boundary. The representing morphism is therefore fully faithful and essentially surjective exactly in the complete case. $\square$

23.6 The dimensional ladder

The three operations associated with one additional transverse direction are now explicit:

integration over a pointed disk	$\longmapsto$	iterated bar,
passage from boundary to universal bulk action	$\longmapsto$	higher centre,
integration over a linking sphere	$\longmapsto$	shifted Poisson bracket.

The chiral coefficient on X remains fixed throughout. Higher transverse dimension acts on the E_n structure, while diagonal collision continues to be expressed by the same factorisable D -module geometry.

24 BRST descent and W -algebras

24.1 A differential compatible with chirality

Let (C, Q) be a cochain complex equipped with a chiral operation X . The statement that Q is a derivation of the chiral structure is

$$[Q, X] = 0 \tag{154}$$

in the chiral deformation Lie algebra. If $Q^2 = 0$ and $[X, X] = 0$, then

$$[Q + X, Q + X] = 0.$$

Thus the BRST differential and the chiral operation form one Maurer–Cartan element with unary and binary parts.

Proposition 24.1 (BRST descent of chiral operations). *If Q is a square-zero derivation of a chiral algebra C , then $H_Q(C)$ inherits a chiral algebra structure. The induced operation is independent of the choice of cocycle representatives.*

Proof. Equation (154) says that the operation sends cocycles to cocycles and that changing one input by a boundary changes the output by a boundary. The chiral Jacobi identity descends because it holds before taking cohomology. $\square$

The proposition applies equally to vertex algebras in a formal coordinate. It is the algebraic core of Hamiltonian reduction.

24.2 The BRST–collision bicomplex

The collision complex of C carries the internal differential induced by Q and the collision differential induced by X . They anticommute by (154). Hence

$$(C_I^{\text{coll}}(C), Q + d_{\text{coll}}) \tag{155}$$

is a bicomplex.

Filter by BRST degree. The first spectral sequence has

$$E_1 \cong C_I^{\text{coll}}(H_Q(C)).$$

Filter by collision degree. The other spectral sequence computes BRST cohomology of the collision resolution.

Theorem 24.2 (Bar–BRST comparison). *Assume that the tensor products and sums forming the coefficient objects are Q -acyclic outside degree zero, and that the total bicomplex (155) is bounded below in one grading and complete with respect to an exhaustive filtration in the other. Then the*

canonical map of collision complexes

$$C_I^{\text{coll}}(H_Q^0(C)) \longrightarrow H_Q^0(C_I^{\text{coll}}(C), Q) \quad (156)$$

is an isomorphism, while $H_Q^j(C_I^{\text{coll}}(C), Q) = 0$ for $j \neq 0$. Consequently the total complex is quasi-isomorphic to $C_I^{\text{coll}}(H_Q^0(C))$ with its induced collision differential.

Proof. Taking vertical Q -cohomology commutes with the coefficient tensor products under the stated acyclicity hypothesis, which proves (156) and the higher vanishing. The BRST-first spectral sequence therefore has one nonzero row. Boundedness and completeness identify its abutment with the total cohomology, giving the final quasi-isomorphism. $\square$

The theorem is exact and strong, but its concentration hypothesis is substantive. In the general case the result is the full derived BRST bar object; concentration identifies it with the ordinary bar complex of degree-zero cohomology.

24.3 Quantum Drinfeld–Sokolov reduction

Let $\mathfrak{g}$ be a simple Lie algebra and $f \in \mathfrak{g}$ nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine vertex algebra $V_k(\mathfrak{g})$, charged ghosts, neutral fields when required, and a differential determined by f . The affine W -algebra is

$$\mathcal{W}^k(\mathfrak{g}, f) = H_{\text{BRST}}^0(C_{\text{DS}}^k(\mathfrak{g}, f)). \quad (157)$$

For the universal W -algebra, the BRST cohomology is concentrated in degree zero in the standard construction; the resulting vertex algebra has an associated graded described by functions on an arc space of the Slodowy slice [2, 3].

The chiral operation on the BRST complex descends by Theorem 24.1. The complicated nonlinear OPEs of W -generators are therefore transferred collision operations. Their composite fields are chiral-geometric operations; they are the output of homological reduction.

24.4 Homotopy transfer

Choose a contraction

$$(H, 0) \begin{matrix} \xrightarrow{i} \\ \xleftarrow{p} \end{matrix} (C, Q), \quad Qi = 0, \quad pQ = 0, \quad Qh + hQ = \text{id} - ip.$$

Homotopy transfer expresses the induced higher operations on H as sums over rooted trees. Vertices carry the original chiral operation, internal edges carry h , leaves carry i , and the root carries p .

If the transferred structure is strict, the tree sums beyond the binary operation cancel. In the general transferred model, H carries a homotopy chiral structure. The cohomology classes of the higher operations measure the obstruction to strictification; their raw coefficients depend on the contraction.

In Drinfeld–Sokolov reduction, the cohomology is known independently to carry a strict vertex-algebra structure. A $\mathcal{P}_\infty$ quasi-isomorphism from the transferred model to the strict BRST-cohomology model identifies the two Maurer–Cartan elements by gauge equivalence. If the contraction extends

to a $\mathcal{P}_\infty$ quasi-isomorphism from the transferred homotopy chiral structure to the strict BRST-cohomology model, then the two Maurer–Cartan elements are gauge equivalent and the transferred higher operations are removable in that model. Before choosing that comparison, the higher operations form the canonical transfer data attached to the contraction.

24.5 Feigin–Frenkel duality for principal W -algebras

For principal nilpotent f , the Feigin–Frenkel duality relates the principal W -algebras of Langlands dual Lie algebras. With lacing number $r^\vee$, the levels k and ℓ satisfy

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1. \quad (158)$$

Under this relation there is an isomorphism

$$\mathcal{W}^k(\mathfrak{g}) \cong \mathcal{W}^\ell({}^L \mathfrak{g}). \quad (159)$$

This is the standard W -algebra duality. It is the multiplicative shifted-level relation of principal W -algebra duality, distinct from the additive affine reflection $k \mapsto -k - 2h^\vee$.

The duality may be constructed through screening operators or free-field realizations. Its relation to a bar coalgebra requires a separate comparison between the BRST resolution and the chiral Chevalley construction.

24.6 Screening operators

A free-field realization embeds a W -algebra into an intersection of kernels of screening operators. The screenings are contour integrals of vertex operators and commute with the W -algebra up to total derivatives. Their kernels form a complex whose cohomology realizes the reduced algebra.

Geometrically, a screening contour is a cycle in a configuration space. Its monodromy and intersection pairing depend on the charges of the free fields. Multiple screenings produce hypergeometric local systems and quantum-group actions. This is an ordered configuration-space realization of BRST reduction.

24.7 Critical level

At $k = -h^\vee$, the Sugawara stress tensor is undefined because its denominator vanishes. On the formal disc D , the center of the critical vacuum vertex algebra is

$$Z(V_{-h^\vee}(\mathfrak{g})) \cong \mathcal{O}(\mathrm{Op}_G(D)). \quad (160)$$

The center of the completed critical enveloping algebra over the punctured disc is described instead by functions on opers on $D^\times$, with the appropriate completion and central-character formulation [34, 36]. These are related but distinct statements. The vertex center is a commutative chiral algebra and underlies the local geometric Langlands construction.

The critical level is singular for KZ because $\hbar = (k + h^\vee)^{-1}$ diverges. The parameter $q = \exp(\pi i / (k + h^\vee))$ becomes singular in this normalization. The affine chiral algebra remains well defined, but its conformal structure and monodromy description change character.

24.8 What BRST reduction preserves

BRST reduction preserves any structure represented by a chain map commuting with Q . It preserves the chiral operation, module actions, and pairings that satisfy this condition. The following structures are preserved after the corresponding chain maps commuting with Q have been supplied:

- a chosen quadratic presentation;
- Koszulness of that presentation;
- a particular braid local system;
- a Verdier-dual identification;
- positive-genus sewing before compatibility with Q is proved.

The precise theorem is [Theorem 24.2](#). Every stronger statement is a comparison theorem for the corresponding additional structure.

24.9 The physical meaning

Classical Hamiltonian reduction imposes constraints and divides by gauge transformations. BRST replaces this quotient by a cochain complex whose differential combines constraints and ghosts. Quantum Drinfeld–Sokolov reduction is the chiral version. The Virasoro or W -algebra is the algebra of gauge-invariant boundary observables after reduction.

From the field-theoretic viewpoint, the ghost central charge is part of the anomaly calculation. BRST nilpotence requires cancellation of the total anomaly in a coupled theory. This is an equality of central extensions in the BRST complex, a tensor-product cancellation of the anomaly classes in the BRST complex.

25 Critical level, opers, and the derived Langlands fibre

At critical level the affine current algebra acquires a large commutative chiral centre. Feigin and Frenkel identify this centre with functions on opers for the Langlands dual group. A central character is therefore a geometric point, and its derived fibre is calculated by the Koszul complex of the oper conormal directions.

25.1 The Feigin–Frenkel centre

Let $\mathfrak{g}$ be a complex simple Lie algebra, let $h^\vee$ be its dual Coxeter number, and set

$$V_{\text{crit}}(\mathfrak{g}) = V_{-h^\vee}(\mathfrak{g}).$$

Let ${}^L G$ be the Langlands dual group and let $D = \text{Spec } \mathbb{C}[[t]]$.

Theorem 25.1 (Feigin–Frenkel). *There is a canonical isomorphism of commutative chiral algebras*

$$Z(V_{\text{crit}}(\mathfrak{g})) \cong \mathcal{O}(\text{Op}_{{}^L G}(D)), \quad (161)$$

where $\text{Op}_{{}^L G}(D)$ is the ind-scheme of ${}^L G$ -opers on the formal disc [\[34, 36\]](#).

If $d_1, \dots, d_r$ are the exponents of $\mathfrak{g}$, the centre has topological generators of conformal weights $d_i + 1$. In a formal coordinate, their Fourier modes identify the centre with the classical W -algebra attached to ${}^L\mathfrak{g}$. Coordinate independence is built into the oper description: an oper is a connection together with a Borel reduction satisfying the oper transversality condition.

25.2 A regular central fibre

The local derived calculation is the Koszul resolution of a regular character.

Theorem 25.2 (Koszul resolution of a regular central fibre). *Let $Z = \text{Sym}(P)$ be a graded polynomial algebra on a finite-dimensional positively graded vector space P . Let A be a flat Z -algebra, with the image of Z central in A , and let $\chi : Z \rightarrow \mathbb{C}$ be a character. Choose polynomial generators $z_1, \dots, z_r$. The complex*

$$K_\chi(A) = A \otimes \Lambda(e_1, \dots, e_r), \quad |e_i| = -1, \quad (162)$$

with differential

$$d(ae_{i_1} \wedge \dots \wedge e_{i_q}) = \sum_{s=1}^q (-1)^{s-1} a(z_{i_s} - \chi(z_{i_s})) e_{i_1} \wedge \dots \wedge \widehat{e}_{i_s} \wedge \dots \wedge e_{i_q} \quad (163)$$

is a free left A -resolution of

$$A_\chi = A \otimes_Z \mathbb{C}_\chi.$$

The derived self-intersection of the character is

$$\mathbb{C}_\chi \otimes_Z^{\mathbb{L}} \mathbb{C}_\chi \simeq \text{Sym}\left(T_\chi^* \text{Spec } Z[1]\right), \quad (164)$$

with zero differential.

Proof. The elements $z_i - \chi(z_i)$ form a regular sequence in Z . The polynomial Koszul complex resolves $\mathbb{C}_\chi$, and flat base change along $Z \rightarrow A$ gives (162). Tensoring this resolution with $\mathbb{C}_\chi$ makes its differential zero. The suspended conormal space is $T_\chi^* \text{Spec } Z[1]$; its graded-symmetric algebra is the exterior algebra on the conormal generators in cohomological degree -1 . $\square$

Choose a finite oper-jet truncation

$$\text{Op}_{LG}^{\leq N}(D) = \text{Spec } Z_N.$$

Each Z_N is a polynomial algebra. Theorem 25.2 therefore calculates every finite jet fibre. The transition maps $Z_{N+1} \rightarrow Z_N$ are surjective, and the completed derived fibre at a smooth oper χ is

$$\mathcal{O}_\chi \widehat{\otimes}_{\mathcal{O}(\text{Op})}^{\mathbb{L}} \mathcal{O}_\chi \simeq \widehat{\text{Sym}}\left(T_\chi^* \text{Op}_{LG}(D)[1]\right). \quad (165)$$

The completion is the inverse limit of the finite exterior factors.

Corollary 25.3 (Oper-cotangent classes in critical reduction). *Let $\chi \in \text{Op}_{LG}(D)$ be a smooth oper and define the derived central reduction*

$$V_{\text{crit}, \chi}^{\mathbb{L}} = V_{\text{crit}}(\mathfrak{g}) \widehat{\otimes}_{Z(V_{\text{crit}})}^{\mathbb{L}} \mathbb{C}_\chi. \quad (166)$$

In every finite oper-jet window, the conormal contribution is represented by (163); in the completed limit it is the factor (165). A flat finite window identifies the derived reduction with the ordinary central quotient in cohomological degree zero.

Proof. Resolve the residue field $\mathbb{C}_\chi$ over each polynomial jet algebra Z_N by the Koszul complex on a regular system of parameters. Tensoring this resolution with the critical vacuum module gives the stated conormal factor and differential. Surjectivity of the jet transition maps removes the derived inverse-limit correction, so the inverse limit is the completed symmetric algebra in (165). Flatness of a finite window makes higher Tor vanish and leaves the ordinary quotient in degree zero. $\square$

25.3 Change of rings

The oper directions enter Ext through a standard change-of-rings spectral sequence.

Theorem 25.4 (Central change of rings). *In the finite-dimensional setting of Theorem 25.2, put $B = A_\chi$. For left B -modules M and N there is a natural convergent spectral sequence*

$$E_2^{p,q} = \text{Ext}_B^p(M, N \otimes \Lambda^q T_\chi \text{Spec } Z) \implies \text{Ext}_A^{p+q}(M, N). \quad (167)$$

The exterior factor is the tangent space dual to the conormal generators of (164).

Proof. Apply the Cartan–Eilenberg change-of-rings spectral sequence for $A \rightarrow B$:

$$\text{Ext}_B^p(M, \text{Ext}_A^q(B, N)) \implies \text{Ext}_A^{p+q}(M, N).$$

Compute $\text{Ext}_A^q(B, N)$ by applying $\text{Hom}_A(-, N)$ to the Koszul resolution (162). The central sequence acts by zero on the B -module N , so the dual Koszul differential vanishes and

$$\text{Ext}_A^q(B, N) \cong N \otimes \Lambda^q T_\chi \text{Spec } Z.$$

Substitution gives (167). Finite length of the Koszul complex gives convergence. $\square$

Applying the theorem to each oper-jet truncation gives a tower of spectral sequences. Surjective transition maps and finite-dimensional weight windows identify the completed spectral sequence with the inverse limit of the finite ones.

25.4 Gaudin Hamiltonians from the centre

Choose pairwise distinct points $z_1, \dots, z_n$ and finite-dimensional $\mathfrak{g}$ -modules $V_1, \dots, V_n$. Evaluation of the critical centre on $V_1 \otimes \dots \otimes V_n$ produces the Gaudin commutative algebra. The quadratic Segal–Sugawara generator maps to

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}, \quad [H_i, H_j] = 0. \quad (168)$$

Higher central generators produce the higher Gaudin Hamiltonians. The spectrum of this commutative algebra is described by ${}^L G$ -opers on $\mathbb{P}^1$ with the corresponding singularity data, and Bethe eigenvectors arise from Miura opers [83].

The two proofs of commutativity expose the same geometry. The critical centre is commutative before evaluation. Equivalently, the infinitesimal braid relations make the Gaudin connection flat, and residues of a flat commuting family commute.

25.5 Local and global geometric Langlands

The critical centre supplies the local spectral base. Frenkel and Gaitsgory identify the spherical critical-level category with quasi-coherent sheaves on the ind-scheme of unramified opers:

$$\widehat{\mathfrak{g}}_{\text{crit}}\text{-Mod}^{G[[t]]} \simeq \text{QCoh}(\text{Op}_{L_G}^{\text{unr}}(D^\times)). \quad (169)$$

The equivalence is the spherical local geometric Langlands theorem [84].

For a smooth projective curve C , a global ${}^L G$ -oper ρ defines a critically twisted D -module on $\text{Bun}_G(C)$ by Beilinson–Drinfeld localization. The resulting object is a Hecke eigensheaf with eigenvalue equal to the ${}^L G$ -local system underlying ρ [81]. The local centre (161) and the global localization functor therefore form the geometric chain

$$\begin{aligned} \text{critical central character } \rho &\longmapsto \text{localized twisted } D\text{-module on } \text{Bun}_G, \\ \text{localized twisted } D\text{-module} &\longmapsto \text{Hecke eigenvalue underlying } \rho. \end{aligned} \quad (170)$$

25.6 The three compatible derived directions

A critical ordered chiral theory can carry three commuting differentials:

- (i) the chiral Chevalley differential of affine-current collision;
- (ii) the interval-bar differential of a specified transverse E_1 multiplication;
- (iii) the oper-Koszul differential of a selected central character.

Linearity of the transverse multiplication over the Feigin–Frenkel centre makes the three structure maps commute. The total complex is then trigraded by chiral collision degree, interval length, and oper-conormal degree. Filtering in the three possible directions produces spectral sequences for one derived object.

In the Gaudin realization, the oper is the joint spectral datum, the evaluated central generators are commuting quantum Hamiltonians, and the Miura-oper fibre encodes Bethe eigenvectors. The evaluation homomorphism from the critical centre to the Gaudin algebra is the exact mathematical map relating the chiral and integrable descriptions.

Part IV

Two genera and two master equations

26 Two genera and the two-edge master equation

26.1 Two loop-forming operations

Let A be an ordered chiral algebra. A cyclic trace on its transverse E_1 multiplication closes an interval into a circle and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues marked points of algebraic curves by plumbing and produces conformal blocks. Both operations contract pairs of flags, but their moduli spaces and anomaly lines are different.

Definition 26.1 (Transverse Calabi–Yau structure). A transverse Calabi–Yau structure of degree $d_\perp$ is a trace

$$\mathrm{tr}_\perp : \mathrm{HH}_0^{\mathrm{FactD}(X)}(A) \longrightarrow \mathbf{1}[-d_\perp]$$

whose pairing $\langle a, b \rangle_\perp = \mathrm{tr}_\perp(ab)$ is perfect and cyclic.

Definition 26.2 (Chiral modular sewing structure). A chiral modular sewing structure of degree d_X consists of perfect pairings on the relevant insertion or module objects and operations assigned to stable pointed complex curves, compatible with separating and nonseparating clutching, the factorisation D -module maps, and the determinant twists.

The first definition is an open two-dimensional topological structure internal to the chiral coefficient. The second varies the complex curve. Their simultaneous presence is additional bicoloured structure.

26.2 Bicoloured stable graphs

A bicoloured stable graph Γ has two sets of internal edges,

$$E(\Gamma) = E_X(\Gamma) \sqcup E_\perp(\Gamma).$$

An X -edge represents a chiral plumbing contraction. A $\perp$ -edge represents a transverse cyclic contraction. Vertices carry the operation appropriate to the incident flags; transverse flags retain their cyclic order. The orientation line is

$$\mathrm{or}(\Gamma) = \det(E_X(\Gamma)) \otimes \det(E_\perp(\Gamma)). \quad (171)$$

Contracting a loop of colour X raises the chiral genus g by one; contracting a loop of colour $\perp$ raises the ribbon genus h by one.

Let $\mathbf{G}^{\mathrm{bi}}$ be the groupoid of these graphs with isomorphisms preserving the edge colour and cyclic orders. The free completed graph object generated by the two sewing systems is

$$\mathbb{F}^{\mathrm{bi}}(A) = \prod_{[\Gamma] \in \mathbf{G}^{\mathrm{bi}}} (\mathrm{or}(\Gamma) \otimes A(\Gamma))_{\mathrm{Aut}(\Gamma)}. \quad (172)$$

This definition directly records the two edge colours and their common vertex operations. A tensor-product presentation is a further comparison model.

26.3 The two boundary operators

Let ∂_X be the signed sum of contractions of one X -edge and $\partial_\perp$ the signed sum of contractions of one $\perp$ -edge.

Proposition 26.3 (Independent graph boundaries). *With the orientation convention (171),*

$$\partial_X^2 = 0, \quad \partial_\perp^2 = 0, \quad \partial_X \partial_\perp + \partial_\perp \partial_X = 0.$$

Hence $\partial_X + \partial_\perp$ is square-zero.

Proof. For two edges of the same colour, the two contraction orders induce opposite orientations in the determinant of that edge set. For edges of different colours, the final graph is the same in either order, while exchanging the two odd determinant factors in (171) changes the sign. $\square$

The proposition is a graph-theoretic identity. A geometric realization must additionally show that the chosen ribbon and Deligne–Mumford boundary maps represent these edge contractions.

26.4 The bicoloured Feynman transform

Let $\mathcal{P}_X$ be the modular or cyclic object governing chiral sewing, and let $\mathcal{P}_\perp$ govern transverse cyclic sewing. Assume that their endomorphism actions on A obey the interchange identities on common flags. Define the *two-edge Feynman transform* to be the free completed bicoloured graph object with differential

$$d_{\mathbb{F}} = d_{\text{int}} + \partial_X + \partial_\perp.$$

By Proposition 26.3, this is a differential.

The corresponding completed convolution complex has components

$$\mathfrak{g}_A^{\text{bi}} = \prod_{g,h,n} \text{Hom}_{\Sigma_n} \left(\mathcal{P}^{\text{bi}}(g, h, n), \text{End}_A(g, h, n) \right), \quad (173)$$

completed by the number of internal edges and vertices. There are brackets $[-, -]_X$ and $[-, -]_\perp$ from joining two vertices with an edge of the indicated colour, and loop operators Δ_X and $\Delta_\perp$ from joining two flags of one vertex.

Theorem 26.4 (Two-edge master equation). *A morphism from the two-edge Feynman transform to the cyclic endomorphism object of A is equivalent to an element*

$$\Theta_A^{\text{bi}} \in \mathfrak{g}_A^{\text{bi}}[[\hbar_X, \hbar_\perp]]$$

satisfying

$$d\Theta_A^{\text{bi}} + \frac{1}{2}[\Theta_A^{\text{bi}}, \Theta_A^{\text{bi}}]_X + \frac{1}{2}[\Theta_A^{\text{bi}}, \Theta_A^{\text{bi}}]_\perp + \hbar_X \Delta_X \Theta_A^{\text{bi}} + \hbar_\perp \Delta_\perp \Theta_A^{\text{bi}} = 0. \quad (174)$$

Proof. The free transform is generated by corollas, and its differential is the sum of the internal differential and all one-edge contractions. Evaluating a morphism on the generators gives Θ_A^{bi} . Compatibility with $d_{\mathbb{F}}$ is exactly (174). Conversely, the equation extends the assignment on corollas uniquely to a chain morphism on the free completed graph object. $\square$

The hypotheses supply both sewing systems, and the theorem identifies their common master equation.

26.5 The two loop operators

On the suspended convolution complex, both loop contractions are odd.

Proposition 26.5 (Supercommutation of loop contractions). *With the determinant convention (171),*

$$\Delta_X \Delta_{\perp} + \Delta_{\perp} \Delta_X = 0.$$

After passing to unsuspended scalar amplitudes, the induced even contractions commute.

Proof. Applying the two contractions in either order produces the same bicoloured graph and the same coefficient contraction. The two new edge-orientation factors occur in opposite orders in the determinant line. Since each loop operator is odd on the suspended convolution complex, interchanging those factors contributes a minus sign. Desuspension removes the two odd shifts and yields commuting even scalar contractions. $\square$

At bidegree $(g, h) = (0, 1)$, one sees the annulus and the Hochschild trace. At $(1, 0)$, one sees genus-one chiral sewing and its projective anomaly. At $(1, 1)$, both loop operations occur; the mixed codimension-two identity is their supercommutation.

26.6 Geometric realization

A geometric realization of the $\perp$ -edges uses ribbon surfaces or cyclically ordered boundary configurations. A realization of the X -edges uses stable curves and clutching morphisms. Compatibility is expressed by a Cartesian or correspondential model in which the two degenerations commute through the prescribed interchange map.

Criterion 26.6 (Geometric two-edge realization). *A pair of geometric sewing theories realizes the two-edge master equation when:*

- (i) *every codimension-one boundary of either type has a coefficient contraction;*
- (ii) *same-colour codimension-two boundaries satisfy the usual modular or ribbon boundary identity;*
- (iii) *mixed codimension-two boundaries form Cartesian comparison squares and carry the product orientation (171);*
- (iv) *the coefficient contractions commute over those squares;*
- (v) *the graph filtration is complete.*

Proof. Conditions (i) and (ii) identify each same-colour geometric face with the corresponding one-edge contraction and make its square vanish. Conditions (iii) and (iv) identify the two orders of a mixed degeneration with one Cartesian boundary stratum and the product-orientation sign, which gives anticommutation. Completeness in (v) makes the graph sum convergent in the edge filtration. Hence the geometric boundary differential maps to $d_{\mathbb{F}}$ and Theorem 26.4 applies. $\square$

26.7 Open topological field theory and chiral conformal theory

A smooth proper Calabi–Yau E_1 algebra defines an oriented open topological field theory; its value on the circle is Hochschild homology,

$$\int_{S^1} A \simeq \mathrm{HH}_\bullet(A).$$

Ribbon graphs compute its amplitudes. This is the $g = 0$ row with arbitrary h .

A chiral conformal theory assigns blocks to complex curves and varies with complex structure through a projectively flat connection. This is the $h = 0$ column with arbitrary g .

Separation theorem 26.7 (Independent genus gradings). *The universal parameter space of the combined theory is bigraded by (g, h) . The coordinate g counts nodes and handles of algebraic curves; the coordinate h counts handles of transverse ribbon surfaces. A string duality relating the two expansions is represented by a specified map of the corresponding modular sewing theories.*

Proof. The two graph genera are changed by contractions of different edge colours and are preserved separately by every remaining operation. They therefore define a genuine $\mathbb{Z}_{\geq 0}^2$ grading on (172). Identifying the two expansions requires a morphism that sends the X -coloured sewing maps to combinations of the transverse ribbon sewing maps and respects their master equations. $\square$

26.8 The double expansion

For a matrix realization, a connected ribbon graph is weighted by a power of N determined by h and the number of boundaries. Chiral sewing is weighted by a string coupling determined by g . When both exist, the connected potential has the form

$$\mathcal{F}_A(N, g_s) = \sum_{g, h, b} N^{2-2h-b} g_s^{2g-2} \mathcal{F}_{g, h, b}(A). \quad (175)$$

The coefficients are traces of the bidegree components of Θ_A^{bi} . Keeping the two parameters separate is the exact algebraic expression of the distinction between matrix-planar and worldsheet-genus expansions.

26.9 Duality of the two-edge theory

Theorem 26.8 (Duality of bicoloured amplitudes). *Suppose:*

- (i) *A and $A^!$ are joined by an invertible bimodule Morita context on the selected complete perfect sector;*
- (ii) *their transverse Calabi–Yau traces correspond;*
- (iii) *their chiral sewing systems are identified by a tensor equivalence preserving projective connections, pairings, and determinant twists.*

Then the equivalence induces an isomorphism of two-edge convolution complexes carrying Θ_A^{bi} to $\Theta_{A^!}^{\mathrm{bi}}$.

Proof. Morita invariance identifies Hochschild chains, ribbon contractions, and $\Delta_\perp$. The sewing equivalence identifies the chiral contractions and Δ_X . Preservation of determinant lines identifies the orientation signs. Hence every term of (174) is intertwined. $\square$

A numerical complementarity statement follows only after a trace has been applied to this structured equivalence.

26.10 Physical meaning

The operator $\Delta_\perp$ contracts two open or line-defect states and creates a loop of the transverse topological theory. The operator Δ_X contracts two chiral insertions through a plumbing fixture and creates a handle on the complex curve. Their supercommutation is locality for two independent degenerations. The universal object is the bicoloured graph operation recording the contracted edge and the moduli space that carries it.

27 Cyclicity, stable graphs, and the master equation

27.1 Why genus requires a pairing

A genus-zero operation has outputs and inputs. To create a loop, one must identify two flags. Such an identification requires a pairing. Let V be a complex with a nondegenerate graded-symmetric pairing

$$\langle -, - \rangle : V \otimes V \longrightarrow \mathbb{C}[d] \quad (176)$$

that is invariant under the genus-zero operations. The pairing converts an output into an input and permits contraction along internal edges.

A cyclic pairing as in (176) contracts two flags and defines nonseparating sewing. A cyclic chiral algebra, a conformal-block pairing, or a field-theoretic propagator supplies this datum and thereby produces the modular object.

27.2 Modular operads

A modular operad has operations indexed by genus g and a finite set of legs. It allows two kinds of composition: joining a leg of one operation to a leg of another, and joining two legs of the same operation. The first adds genera; the second raises genus by one. Stable graphs encode iterated compositions [43].

Let $\mathcal{P}$ be a dg modular cooperad and let $\text{End}_V^{\text{mod}}$ be the endomorphism modular operad determined by the pairing (176). The completed convolution space

$$\mathfrak{g}_{\mathcal{P}, V} = \prod_{g, n} \text{Hom}_{\Sigma_n}(\mathcal{P}(g, n), \text{End}_V^{\text{mod}}(g, n)) \quad (177)$$

contains a differential, a bracket joining two vertices, and a loop operator Δ joining two legs of one vertex.

27.3 The Feynman transform

The Feynman transform $F\mathcal{P}$ is the free modular operad on the shifted dual of $\mathcal{P}$, with the standard determinant twists, and with differential given by the internal differential and the signed sum over edge contractions. A morphism $F\mathcal{P} \rightarrow \text{End}_V^{\text{mod}}$ is determined by its values on generators. Compatibility with the differential is the master equation.

Theorem 27.1 (Feynman transform and master equation). *Giving V the structure of an algebra over the Feynman transform $F\mathcal{P}$ is equivalent to giving an element $\Theta \in \mathfrak{g}_{\mathcal{P},V}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0. \quad (178)$$

Proof. The free modular operad is generated by corollas. The internal part of its differential gives $d\Theta$. Contracting an edge between two vertices gives the bracket term, with the factor $1/2$ accounting for interchange of the vertices. Contracting a loop at one vertex gives $\Delta\Theta$. Equality with the endomorphism differential is exactly (178). $\square$

The theorem is formal and unconditional. Its hypothesis is the existence of a morphism from the Feynman transform, equivalently the operations that Θ is supposed to encode. The constructed sewing maps assign operations to the boundary strata, and boundary-of-boundary proves their master equation.

27.4 Stable curves

The boundary strata of $\overline{\mathcal{M}}_{g,n}$ are indexed by stable graphs. Separating nodes correspond to edges joining distinct vertices. Nonseparating nodes correspond to loops. Clutching morphisms realize graph contraction geometrically.

Borel–Moore chains on the clutching spaces, with the usual determinant twists, form a modular operad under proper pushforward along gluing. After aritywise finite duality, or in a specified completed continuous dual, the corresponding cochains form a modular cooperad. If a system of multilinear maps on V is compatible with every clutching morphism, it defines an algebra over the appropriate Feynman transform. The master equation then states that the two ways of reaching every codimension-two boundary stratum cancel with orientation signs.

Principle 27.2 (Sewing principle). *The topology of $\overline{\mathcal{M}}_{g,n}$ supplies the relations among sewing maps. The coefficient theory supplies the multilinear maps assigned to the clutching strata.*

Proof. The cellular or Borel–Moore boundary of the stable-curve stratification is the signed sum of one-edge graph contractions, and its square vanishes. A coefficient system becomes an algebra over the Feynman transform by assigning a compatible contraction to every such edge. Applying the coefficient assignment to the boundary identity gives the relations among these maps. Thus topology supplies the relations and the coefficient theory supplies their values. $\square$

27.5 Graph filtration and obstruction theory

Filter $\mathfrak{g}_{\mathcal{P},V}$ by the number of internal edges, or more generally by stable-graph complexity. The filtration is complete and compatible with the bracket. Suppose a genus-zero Maurer–Cartan element Θ_0 is fixed. A partial lift

$$\Theta_{\leq r} = \Theta_0 + \Theta_1 + \cdots + \Theta_r$$

has curvature in filtration $r+1$. Its leading term is a cocycle for the twisted differential $d_{\Theta_0} = d + [\Theta_0, -]$.

Theorem 27.3 (Filtered obstruction theorem). *The obstruction to extending $\Theta_{\leq r}$ to filtration $r+1$ is a canonical class*

$$\mathfrak{o}_{r+1}(\Theta_{\leq r}) \in H^2\left(\mathrm{gr}_F^{r+1} \mathfrak{g}_{\mathcal{P},V}, d_{\Theta_0}\right). \quad (179)$$

It vanishes exactly when an extension exists. When it vanishes, the set of extensions modulo gauge transformations trivial in lower filtration is a torsor for the corresponding H^1 .

Proof. This is the standard obstruction theory of a complete filtered Maurer–Cartan equation. The Bianchi identity implies that the leading curvature is closed. Changing a lift alters it by a coboundary. Solving the next equation is equivalent to choosing a primitive; two primitives differ by a closed degree-one element. $\square$

This graph obstruction tower is invariant. A morphism from the modular deformation complex to a primary-line deformation complex transports the bracket and produces a functorial scalar tower.

27.6 Connected graph cumulants

Suppose a cyclic trace assigns a scalar weight $W(\Gamma)$ to every stable graph and the graph sum converges formally in $\hbar$:

$$Z = 1 + \sum_{\Gamma} \frac{\hbar^{g(\Gamma)}}{|\text{Aut } \Gamma|} W(\Gamma). \quad (180)$$

The exponential formula for combinatorial species gives

$$\log Z = \sum_{\Gamma \text{ connected}} \frac{\hbar^{g(\Gamma)}}{|\text{Aut } \Gamma|} W(\Gamma). \quad (181)$$

Thus connected amplitudes are cumulants of the full partition function. This is a canonical scalar projection once the trace is fixed. The complete chiral algebra is retained by the untraced graph-valued master element.

27.7 Graphwise duality

Let V and $V^\vee$ be paired cyclic objects. Transposing every vertex operation and using the inverse pairing on every internal edge sends a graph weight for V to a graph weight for $V^\vee$. Orientation and determinant lines contribute the standard signs.

Proposition 27.4 (Graphwise transpose). *Assume that the coefficient systems are aritywise perfect, that continuous duality commutes with the stable-graph completion, and that the cyclic pairing identifies their endomorphism modular operads. Then transposition at every vertex and contraction with the inverse pairing on every edge induce a filtered isomorphism between the two modular convolution complexes. It intertwines their master equations and sends connected graph weights to their transposes.*

Proof. Aritywise perfection identifies every vertex-operation space with the dual operation space. The inverse cyclic pairing identifies the contraction assigned to each internal edge, and the determinant line transports its sign. Tensoring these identifications over the vertices and edges of a stable graph gives an isomorphism on each graph summand. Compatibility with completion assembles the graphwise maps into a filtered isomorphism. Edge contraction and transposition commute, so the differential, bracket, and loop operator are intertwined; the master equation and connected traces follow. $\square$

Any scalar complementarity formula is a trace of this graphwise statement and depends on the chosen duality and trace. Levels, central charges, and Hodge classes acquire scalar relations only after a specified graphwise duality and trace map sends them to a common target.

27.8 The Batalin–Vilkovisky reading

In the BV formalism, fields form a shifted symplectic space, the bracket is induced by the symplectic pairing, and Δ is the second-order operator corresponding to contraction of a loop. The quantum action S satisfies

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \quad (182)$$

which expands to

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0.$$

This is the same algebraic shape as (178). Equality of the two equations for a particular theory requires a map from the renormalized BV complex to the modular endomorphism complex. Compactified configuration spaces supply that map in perturbative models where the Feynman weights extend to the boundary.

27.9 Anomalies

The degree-one quantum-master defect is the local anomaly cocycle. Its cohomology class is independent of the choice of counterterm. An anomaly can be cancelled by adding a local counterterm exactly when the class vanishes. A projective connection on conformal blocks is a different but related manifestation: the curvature is central rather than zero. Tensoring by a line with opposite curvature converts projective flatness into flatness.

The next section treats this anomaly line geometrically on the moduli space of curves.

28 Conformal blocks, sewing, and the anomaly line

28.1 Coinvariants and blocks

Let V be a conformal vertex algebra and let $M_1, \dots, M_n$ be modules inserted at marked points of a smooth curve C . The global chiral algebra acts on $M_1 \otimes \dots \otimes M_n$. The coinvariants are the quotient by this action; conformal blocks are their duals. Varying the pointed curve produces sheaves on moduli.

Damiolini–Gibney–Tarasca construct sheaves of coinvariants for vertex algebras on moduli of pointed stable curves and prove factorization and vector-bundle results under finiteness and semisimplicity hypotheses [22, 24]. For strongly rational vertex operator algebras, Damiolini–Woike prove that the conformal blocks define a modular functor [25].

The modular functor is the positive-genus coefficient system required by Theorem 27.2. Its genus-zero restriction is built from intertwining operators and the vertex tensor category. Its sewing maps are convergent analytic operations and satisfy the mapping-class-group relations.

28.2 Projective flatness

The sheaf of conformal blocks carries a projectively flat connection. Infinitesimal changes of complex structure act through the stress tensor. The central term in the Virasoro algebra makes the action projective. Algebraically, the sheaf is a module for an Atiyah algebra rather than for the untwisted tangent sheaf.

Let $\lambda = c_1(\mathbb{E})$ be the Hodge class and ψ_i the cotangent-line classes at the markings. If a_i is the

conformal weight of M_i , the interior projective class has the form

$$\mathrm{rk}(\mathbb{V}) \left(\frac{c}{2} \lambda + \sum_i a_i \psi_i \right). \quad (183)$$

On the compactification, boundary terms are determined by factorization through intermediate modules. Their coefficients depend on fusion rules and conformal weights [22, 23].

Theorem 28.1 (Chern-class form of the anomaly). *For a conformal-block bundle satisfying the factorization hypotheses, its first Chern class is the sum of the interior class (183) and explicit boundary divisor contributions obtained by sewing through simple modules. In particular, the Hodge coefficient is $\mathrm{rk}(\mathbb{V})c/2$, and the boundary sewing coefficients supply the remaining divisor classes.*

Proof. The Atiyah algebra acting on conformal blocks is the sum of the Hodge Atiyah algebra with coefficient $c/2$ and the cotangent-line Atiyah algebras with coefficients a_i . Taking the trace of projective curvature gives the interior term (183). Along a boundary divisor, factorization decomposes the bundle into summands indexed by the simple intermediate module; the residue of the logarithmic connection on each summand is its conformal-weight sewing exponent. Summing the traces of these residues gives the boundary divisor coefficients. This is the Chern-class calculation of Damilini–Gibney–Tarasca [22, 23]. $\square$

This theorem separates the local central charge from the global geometry. Two theories with the same central charge may have different ranks, conformal weights, fusion rules, and boundary classes.

28.3 The anomaly line

A projectively flat bundle E has curvature

$$\nabla_E^2 = \Omega \mathrm{id}_E$$

for a scalar two-form Ω . Let L be a line bundle with connection satisfying $\nabla_L^2 = -\Omega$. Then

$$\nabla_{E \otimes L}^2 = 0.$$

If only a rational multiple of the required line exists, one works with a twisted D -module or on the corresponding gerbe.

Proposition 28.2 (Anomaly cancellation by a line). *Tensoring a projectively flat conformal-block system by a line or twisted line of opposite Atiyah class produces a flat system. The construction is compatible with sewing when the anomaly lines satisfy the clutching multiplicativity dictated by the Hodge and cotangent-line bundles.*

The vertex-algebra differential remains fixed, while the proposition twists the connection on the family of blocks over moduli.

28.4 Sewing and the master equation

Choose a perfect pairing between a module and its contragredient. Sewing two marked points inserts the inverse pairing and sums over a homogeneous basis. In a local sewing parameter q , the operation

contains the propagator

$$q^{L_0 - c/24}.$$

Convergence is analytic; factorization identifies the limit at $q = 0$ with the direct sum over intermediate modules.

The modular-functor sewing maps define an algebra over the chains of the surface operad. After the anomaly twist of [Theorem 28.2](#), they determine a solution of the master equation in the modular convolution algebra. This conclusion follows from [Theorem 27.1](#); the coefficients are exactly those carried by the sewing action and anomaly line.

28.5 Genus one

At genus one, Zhu's theorem gives modular transformation laws for trace functions of rational vertex operator algebras [\[78\]](#). For a holomorphic VOA,

$$Z_V(\tau) = \text{Tr}_V q^{L_0 - c/24}$$

is a modular function with the appropriate multiplier. For a rational VOA, the vector of irreducible characters transforms under a finite-dimensional representation of $SL_2(\mathbb{Z})$.

The projective anomaly appears as the factor $q^{-c/24}$ and the multiplier system. The modular S -matrix and fusion coefficients are related by the Verlinde formula when the representation category is modular [\[47, 73\]](#). The dimension of a conformal-block space is a trace of the modular functor; the $SL_2(\mathbb{Z})$ representation is strictly richer.

28.6 Prime forms and analytic kernels

On a compact Riemann surface, the prime form $E(z, w)$ is a canonical section, after the choice of a theta characteristic, that vanishes simply on the diagonal. The differential $d_z \log E(z, w)$ has the correct local simple pole but is quasi-periodic. Adding a harmonic correction produces a single-valued Green kernel. Fay's identities control products of prime forms and Szego kernels [\[32\]](#).

These kernels are analytic representatives of propagators and connections. Their quasi-periodicity produces monodromy and projective curvature. They define the analytic connection in the base directions; the internal block differential occupies the fibre degree, and the total superconnection preserves both gradings.

28.7 Free bosons and determinant lines

For a rank- r free boson, the oscillator partition function is

$$\eta(\tau)^{-r}.$$

The factor η^{-r} is the regularized determinant of the chiral Laplacian, while the nonchiral partition function includes $(\text{Im } \tau)^{-r/2} |\eta(\tau)|^{-2r}$. On higher-genus surfaces, the determinant line of the $\bar{\partial}$ operator carries the Quillen metric and curvature given by the family index theorem.

This is the precise meeting point of Polyakov's conformal anomaly and the geometry of determinant lines. The coefficient is the central charge of the free fields. Interacting theories add non-Gaussian conformal blocks; their higher correlators are determined by the full OPE, fusion rules, and sewing

maps.

28.8 Lattice theories and Siegel theta series

For an even positive-definite lattice L of rank r , the genus-one character of the lattice VOA is

$$Z_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}. \quad (184)$$

At higher genus the zero modes contribute the Siegel theta series $\Theta_L^{(g)}(\Omega)$, while oscillators contribute determinant factors. Equal genus-one theta series refine to lattice or VOA equivalence only after the full higher-genus theta and OPE data agree equal lattices or equal VOAs.

For the two even unimodular lattices of rank sixteen, the degree- g Siegel theta series agree for $g \leq 3$. Their degree-four difference is proportional to the Schottky form and vanishes on the Jacobian locus. Hence their genus-two theta amplitudes coincide. The first ambient Siegel distinction occurs in degree four, and every arithmetic comparison therefore specifies both genus and moduli locus.

28.9 Genus two and beyond

Genus two introduces a three-dimensional Siegel upper half-space, separating and nonseparating boundary divisors, and stable graphs with loops. The period matrix and the off-diagonal sewing parameter couple channels that are independent at genus one. For rational VOAs, the modular functor gives finite-dimensional blocks with mapping-class-group monodromy. For lattice VOAs, Siegel theta series provide explicit sections.

The stable-graph genus expansion is built from the complete vertex tensors and propagators. The central charge controls the anomaly line; fusion and OPE coefficients control the non-scalar graph weights.

29 Anomalies, curvature, and invertible field theories

29.1 An anomaly has a domain

The word *anomaly* denotes an obstruction to making a classical symmetry act strictly after quantization. Its mathematical type depends on the symmetry and on the object on which the symmetry acts. The following classes occur in the present subject:

central extension of a local operator algebra,	$[\omega_{\text{loc}}] \in H_{\text{Lie}}^2(\mathfrak{g}; \mathbb{C}),$
projective curvature of a family of states,	$[\Omega_{\text{fam}}] \in H^2(B; \mathbb{C}),$
BV quantization obstruction,	$[\mathfrak{A}_{\text{BV}}] \in H_{\text{loc}}^1(\text{Obs}^{\text{cl}}),$
BRST obstruction,	$[Q^2] \in H^2(\text{End } C),$
framing anomaly,	$\alpha_{\text{fr}} \in \pi_0 \text{InvFT}_{d+1}^{\text{fr}},$
curvature of a curved Koszul object,	$m_0 \in C^2.$

The symbols lie in different complexes. A theorem can identify their images under transgression or a comparison map. A scalar comparison arises after a specified transgression or trace maps these classes to a common one-dimensional target.

29.2 Local central extensions

The affine current algebra has central term

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}.$$

The Virasoro algebra has central term

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The parameters k and c classify different central extensions. The Sugawara construction relates them for a chosen affine theory:

$$c_{\text{Sug}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}.$$

This rational function is the Sugawara central charge and has a pole at the critical level; the affine level remains the coefficient of the current central extension.

A central extension is part of the chiral operation. Invariance of its cocycle makes the central term compatible with composition. The simplicial identities of the ordered bar and the Jacobi identities of the chiral coefficient then place the central term inside a square-zero total differential.

Proposition 29.1 (Central terms and the ordered bar). *Let A be an augmented dg E_1 algebra in $\text{FactD}(X)$ whose chiral coefficient carries a central extension. The ordered bar coderivation satisfies $b^2 = 0$. The central term contributes to b through the multiplication of A or through its coefficient differential; its contribution is part of the square-zero simplicial bar coderivation.*

Proof. The simplicial identities of the bar construction use only the associativity and chain-map property of multiplication. A central extension satisfying the algebra identities is already included in that multiplication. Hence the standard proof of $b^2 = 0$ is unchanged. $\square$

29.3 Projective curvature over moduli

Let $\mathbb{V} \rightarrow B$ be a bundle of conformal blocks. A projectively flat connection satisfies

$$\nabla_{\mathbb{V}}^2 = \Omega \text{id}_{\mathbb{V}}.$$

The class $[\Omega]$ is the curvature of a line-valued central extension of the tangent action on B . For pointed stable curves, the interior contribution to the first Chern class is

$$\text{rk}(\mathbb{V}) \left(\frac{c}{2} \lambda + \sum_i a_i \psi_i \right),$$

and factorization supplies boundary terms. The central charge controls one coefficient of this class; the conformal weights and fusion channels control the rest.

Theorem 29.2 (Type of the conformal anomaly). *The conformal anomaly of a family of chiral blocks is the central extension encoded by the Atiyah algebra of the block bundle. After a projective connection is chosen, the same class is represented by its scalar curvature. It acts in the moduli direction while the internal differential on each block complex remains square-zero.*

Proof. The internal differential is defined before the base curve varies and has cohomological degree one in the fibre. The projective curvature is a two-form on the base acting by a scalar endomorphism of the fibre. They belong to the vertical and horizontal parts of the superconnection. The family Ward identity identifies the horizontal curvature with the central extension of vector fields represented by the stress tensor. $\square$

29.4 The superconnection equation

The correct object combining fibre and base is a superconnection

$$\mathbb{D} = d_{\text{fib}} + \nabla + \mathbb{A}_2 + \mathbb{A}_3 + \cdots.$$

Its square has total degree two. Flatness means $\mathbb{D}^2 = 0$; projective flatness means $\mathbb{D}^2 = \Omega \text{id}$.

Proposition 29.3 (Cancellation by an anomaly line). *Let $\mathbb{D}_{\mathbb{V}}$ be a projectively flat superconnection with $\mathbb{D}_{\mathbb{V}}^2 = \Omega \text{id}_{\mathbb{V}}$. If a line bundle L with connection satisfies $\nabla_L^2 = -\Omega$, then the tensor-product superconnection on $\mathbb{V} \otimes L$ is flat. If $[\Omega]$ is rationally integral, the same construction takes place in the gerbe of twisted bundles determined by the required denominator.*

Proof. The tensor-product superconnection is

$$\mathbb{D}_{\mathbb{V} \otimes L} = \mathbb{D}_{\mathbb{V}} \otimes 1 + 1 \otimes \nabla_L,$$

where the second summand acts with the Koszul sign. The two odd summands supercommute, so

$$\mathbb{D}_{\mathbb{V} \otimes L}^2 = \Omega \text{id}_{\mathbb{V} \otimes L} - \Omega \text{id}_{\mathbb{V} \otimes L} = 0.$$

For a rationally integral class, local root lines and their descent isomorphisms define the corresponding twisted tensor product; the same curvature calculation holds on every chart and is preserved by descent. $\square$

The cancellation datum consists of the line or gerbe, its connection, and the curvature identification used in the proof.

29.5 Polyakov's Weyl anomaly

Let $g = e^{2\sigma} \hat{g}$ be two metrics in the same conformal class on a closed surface. Fix the convention $W[g] = -\log Z[g]$ and

$$\delta W = \frac{1}{2} \int_{\Sigma} \sqrt{g} \langle T^{\mu\nu} \rangle \delta g_{\mu\nu}.$$

For a conformal field theory of central charge c , the Weyl anomaly integrates to

$$W[e^{2\sigma} \hat{g}] - W[\hat{g}] = -\frac{c}{24\pi} \int_{\Sigma} \left(|\nabla_{\hat{g}} \sigma|^2 + R_{\hat{g}} \sigma \right) d\text{vol}_{\hat{g}}, \quad (185)$$

up to a metric-independent topological constant. Differentiation at $\sigma = 0$ gives

$$\langle T^{\mu}_{\mu} \rangle = -\frac{c}{24\pi} R.$$

The family index theorem identifies the curvature of the corresponding determinant line over moduli. Its holomorphic component supplies the coefficient $c\lambda/2$ in the Atiyah class of the conformal-block bundle [65].

The stress tensor couples the Weyl response to the variation of complex structure. The family index map carries this local response to the determinant line over moduli, while the transverse E_1 multiplication continues to govern ordered fusion.

29.6 BRST anomaly

A BRST operator Q is required to satisfy $Q^2 = 0$. In the bosonic string, the total BRST charge squares to a multiple of the total central-charge defect and the intercept defect. Matter and ghosts cancel the central extension when

$$c_{\text{matter}} + c_{bc} = 0, \quad c_{bc} = -26.$$

This is additivity under tensor product of chiral theories.

Theorem 29.4 (BRST cancellation is tensorial). *Let (V, Q_V) and (G, Q_G) be graded conformal complexes whose odd BRST endomorphisms act on separate tensor factors. On homogeneous tensors define*

$$Q(v \otimes g) = Q_V v \otimes g + (-1)^{|v|} v \otimes Q_G g.$$

Then

$$Q^2 = Q_V^2 \otimes 1 + 1 \otimes Q_G^2.$$

Consequently, central anomaly endomorphisms add under tensor product, and the tensor-product BRST differential is square-zero precisely when their sum vanishes. A Koszul-duality comparison is a further equivalence between the corresponding augmentation-generated sectors.

Proof. The two tensor-factor operators are odd and supercommute. Their two mixed compositions therefore cancel. The remaining diagonal compositions are $Q_V^2 \otimes 1$ and $1 \otimes Q_G^2$, which proves the formula and its cancellation statement. $\square$

29.7 BV anomaly

For a classical interaction I , define the renormalized quantum master defect

$$\mathfrak{A}(I_h) = QI_h + \frac{1}{2}\{I_h, I_h\} + \hbar\Delta I_h.$$

At the first unsolved order in $\hbar$, $\mathfrak{A}$ is closed for the classical BV differential. Its cohomology class is independent of counterterm representatives. A quantization exists to the next order precisely when this class vanishes.

Proposition 29.5 (Locality of the BV anomaly). *The BV anomaly is represented by a local functional. In a configuration-space renormalisation scheme, graph weights extend to Fulton–MacPherson or Axelrod–Singer compactifications. The codimension-one boundary factorisation formula expresses the anomaly as the sum of local counterterm faces and the boundary terms that survive the quantum master cancellation.*

Proof. The renormalised interaction is a local functional, and the BV bracket and BV Laplacian preserve the local deformation complex. Hence the first nonzero quantum-master defect defines a cocycle in local cohomology. In configuration-space renormalisation, Stokes' theorem writes the defect as a sum over compactification faces. Faces describing collapsed subgraphs give local counterterms; the remaining faces give the factorisation terms of the quantum master equation. This is the compactified form of the standard locality theorem for perturbative BV quantisation [9, 14]. $\square$

The quantum coefficient system consists of propagators and renormalised graph weights on the compactified configuration spaces. Its boundary factorisation is the geometric realization of the BV master equation.

29.8 Framing anomaly

A framing, or equivalently a compensating gravitational Chern–Simons term, lifts the Chern–Simons partition function to an invariant of the framed three-manifold. The framing change multiplies the partition function and Wilson-line observables by phases determined by the central charge of the associated modular tensor category. This anomaly lives in three-dimensional geometry.

The ribbon twist of the line category records the same phase on framed knots. Its relation to the two-dimensional WZW central charge follows from the Chern–Simons/WZW correspondence. The Chern–Simons/WZW comparison map relates this ribbon phase to the two-dimensional central charge in the corresponding boundary theory.

29.9 Curved Koszul duality

A curved dg algebra or coalgebra has a curvature element m_0 satisfying

$$d^2(x) = [m_0, x]$$

in the associative case, with the corresponding A_∞ identities in the homotopy case. For central scalar m_0 , the commutator is zero, while m_0 remains as the constant term of the curved Maurer–Cartan equation.

Curvature arises naturally when an inhomogeneous relation has a constant term. For example, quadratic-linear-constant Koszul duality places the constant term in m_0 . A family comparison or index map may send this algebraic curvature to a class such as $c_1(\mathbb{E})$.

Separation theorem 29.6 (Native anomaly tuple). *The anomaly data of an ordered chiral theory form a tuple*

$$(k, c, [\Omega_{\det}], [\mathfrak{A}_{\text{BRST}}], [\mathfrak{A}_{\text{fr}}], m_0)$$

in the product of the current-extension line, Virasoro-extension line, determinant-line cohomology, BRST local cohomology, framing-anomaly group, and curved Koszul algebra. A scalar relation among components is the value of a specified comparison morphism or trace on this tuple.

Proof. Each component is defined by its native deformation problem: current extensions, stress-tensor extensions, projective connections, ghost differentials, framed three-dimensional amplitudes, and inhomogeneous quadratic relations. Tensor products and family-index maps act componentwise. Therefore every equality between components is carried by an explicit morphism between their deformation complexes. $\square$

29.10 Anomalies as invertible theories

For local anomalies admitting the standard anomaly-field-theory description, the anomaly is represented by an invertible field theory one dimension higher. The anomalous theory is relative to its state lines, and cancellation is a trivialisation of the tensor product anomaly.

For chiral conformal blocks, the invertible theory is represented by a determinant or Hodge line. For fermions it is the determinant line of the Dirac operator. For Chern–Simons theory it includes the framing line. This viewpoint explains additivity: tensoring theories tensors anomaly lines and adds their characteristic classes.

Principle 29.7 (Anomaly comparison). *A morphism of invertible theories identifies two anomalies by matching their state lines, connections, and gluing laws. Numerical equalities are traces of this identification.*

Proof. An invertible theory assigns a line to every closed codimension-one manifold and an isomorphism of tensor products of lines to every bordism. A morphism is a compatible family of line isomorphisms, so it identifies the corresponding connections and their curvature classes and intertwines every gluing map. Evaluation on a closed bordism, or application of a character of the coefficient line, takes this structured identification to the associated numerical equality. $\square$

29.11 The anomaly square

The complete theory contains four compatible arrows:

$$\begin{array}{ccc}
 \text{local central extension} & \xrightarrow{\text{Ward identity}} & \text{Atiyah algebra of blocks} \\
 \text{BV descent} \downarrow & & \downarrow c_1 \\
 \text{local BV anomaly} & \xrightarrow{\text{family index}} & \text{curvature of determinant line.}
 \end{array} \tag{186}$$

Each arrow is constructed in a specified field theory. For free chiral fields and WZW models, the standard Ward-identity and family-index calculations construct the arrows of this square. For a general ordered chiral algebra, the upper-left corner exists while the other corners require a stress tensor, a family of states, and a BV realization.

The square is the precise form of the unity sought by the subject. It gives one language through the typed comparison maps joining its distinct objects.

Part V

Calculated noncommutative chiral algebras

30 A calculus of strictly noncommutative chiral algebras

The transverse multiplication is genuine structure. A factorisable D -module M may support several associative products, and each product has its own interval bar. The calculations below begin with

generators and relations in the symmetric monoidal category

$$\mathcal{C}_X := (\text{FactD}(X), \otimes^{\text{pt}}, \mathbf{1}).$$

Every tensor product, kernel, cokernel, and dual in this section is formed in $\mathcal{C}_X$. Exactness statements are read after a conservative family of component functors to complexes of vector spaces. This convention gives concrete proofs while retaining the full chiral coefficient system on X .

30.1 The multiplication-dependence theorem

Let $M \in \mathcal{C}_X$ be connective and tensor-flat. Two augmented associative algebras arise from the same generator object:

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \cdots, \quad Q_X(M) = \mathbf{1} \oplus M, \quad M \cdot M = 0.$$

The first uses concatenation; the second uses the square-zero product.

Theorem 30.1 (Multiplication determines the ordered bar). *There are natural equivalences*

$$\overline{\text{Bar}}_X^{\text{ord}}(T_X(M)) \simeq M[1], \quad \overline{\text{Bar}}_X^{\text{ord}}(Q_X(M)) \simeq \bigoplus_{n \geq 1} M[1]^{\otimes n}.$$

The first reduced coproduct vanishes. The second is deconcatenation.

Proof. The augmentation module of $T_X(M)$ has the two-term free resolution

$$0 \longrightarrow T_X(M) \otimes M \xrightarrow{\text{conc}} T_X(M) \longrightarrow \mathbf{1} \longrightarrow 0.$$

Tensoring over $T_X(M)$ with $\mathbf{1}$ produces $M[1]$. The simplicial bar gives the same equivalence through its extra degeneracy: a bar word with an internal cut contracts, and a one-letter word survives.

For $Q_X(M)$, every inner face of the normalized bar multiplies two elements of M and therefore vanishes. The normalized complex is the tensor coalgebra on $M[1]$ with its internal differential and deconcatenation coproduct. $\square$

Corollary 30.2 (Presentation is part of the invariant). *The ordered bar is the invariant of the pair $(M, m_{\perp})$, where M carries the chiral coefficient and $m_{\perp}$ carries the transverse associative multiplication. Distinct products on the same coefficient object produce distinct bar coalgebras.*

Proof. The normalized bar differential is the coderivation induced by $m_{\perp}$. For the free tensor algebra the extra-degeneracy contraction leaves one copy of $M[1]$; for the square-zero product the differential vanishes and the whole tensor coalgebra survives. These two products on the same coefficient object therefore give inequivalent bar coalgebras, and the general bar is functorial in the pair $(M, m_{\perp})$. $\square$

The corollary supplies the first exact test for every proposed numerical bar invariant: the multiplication must appear in the chain differential that produces the number.

30.2 Tensor products and independent sectors

Theorem 30.3 (Eilenberg–Zilber for internal bars). *Let A and B be augmented tensor-flat associative algebras in $\mathcal{C}_X$. The shuffle and Alexander–Whitney maps give a natural equivalence of conilpotent*

coalgebras

$$\mathrm{Bar}_X^{\mathrm{ord}}(A \otimes B) \simeq \mathrm{Bar}_X^{\mathrm{ord}}(A) \otimes \mathrm{Bar}_X^{\mathrm{ord}}(B).$$

If the Kunnetth spectral sequence converges, then

$$\mathrm{Tor}_*^{A \otimes B}(\mathbf{1}, \mathbf{1}) \cong \mathrm{Tor}_*^A(\mathbf{1}, \mathbf{1}) \otimes \mathrm{Tor}_*^B(\mathbf{1}, \mathbf{1}).$$

Proof. The bisimplicial object in bidegree (p, q) is

$$\mathbf{1} \otimes \bar{A}^{\otimes p} \otimes \bar{B}^{\otimes q} \otimes \mathbf{1}.$$

Its diagonal is the normalized bar of $A \otimes B$. The Eilenberg–Zilber theorem identifies the realization of the diagonal with the iterated realization. The shuffle map respects deconcatenation because a cut of a shuffle is the sum of shuffles of the two induced cuts. $\square$

This theorem constructs doubly structured examples directly. One factor may carry a complicated chiral coefficient and a simple transverse product; the second may carry a highly noncommutative transverse product. Their bar invariants multiply while their chiral factorization maps remain visible in each tensor factor.

30.3 Quadratic algebras in a chiral coefficient category

Let $K \in \mathcal{C}_X$ be tensor-flat and let V be a finite-dimensional vector space. Set $M = K \otimes V$. A subobject

$$R = K^{\otimes 2} \otimes R_0 \subset M^{\otimes 2}$$

for $R_0 \subset V^{\otimes 2}$ defines

$$A_X(V, R_0; K) := T_X(M)/(R).$$

The chiral geometry lives in the powers of K ; the transverse relation lives in R_0 .

Proposition 30.4 (Base change of a quadratic resolution). *Suppose the ordinary quadratic algebra $A_0 = T(V)/(R_0)$ has a linear free resolution whose matrices have scalar entries in the generators. Tensoring every weight- n term with $K^{\otimes n}$ produces an internal free resolution of the augmentation of $A_X(V, R_0; K)$. Consequently the ordinary Koszul homology of A_0 lifts coefficientwise to $\mathcal{C}_X$.*

Proof. Every identity among the matrices of the ordinary resolution is an identity in the free tensor category generated by V . The symmetric monoidal functor $V^{\otimes n} \mapsto K^{\otimes n} \otimes V^{\otimes n}$ preserves these identities. Tensor-flatness preserves exactness after component evaluation, and conservativity returns exactness in $\mathcal{C}_X$. $\square$

30.4 The quantum plane

Fix $q \in \mathbb{C}^\times$. The quantum-plane chiral algebra is

$$A_q(K) = T_X(Kx \oplus Ky) / (yx - qxy). \quad (187)$$

The relation orients the transverse product. The powers of K retain their chiral collision data.

Theorem 30.5 (Quantum-plane resolution and dual). *The augmentation of $A_q(K)$ has the length-two free resolution*

$$0 \longrightarrow A_q(K) \otimes K^{\otimes 2}[-2] \xrightarrow{d_2} A_q(K) \otimes (Kx \oplus Ky)[-1] \xrightarrow{d_1} A_q(K) \longrightarrow \mathbf{1} \longrightarrow 0, \quad (188)$$

where

$$d_1(e_x) = x, \quad d_1(e_y) = y, \quad d_2(1) = ye_x - qxe_y.$$

Hence

$$\mathrm{Tor}_0^{A_q(K)}(\mathbf{1}, \mathbf{1}) = \mathbf{1}, \quad \mathrm{Tor}_1^{A_q(K)}(\mathbf{1}, \mathbf{1}) = Kx[1] \oplus Ky[1], \quad \mathrm{Tor}_2^{A_q(K)}(\mathbf{1}, \mathbf{1}) = K^{\otimes 2}\rho[2],$$

and the higher groups vanish.

For dualizable K , the quadratic dual is

$$A_q(K)^\dagger = T_X(K^\vee \xi \oplus K^\vee \eta) / (\xi^2, \eta^2, \xi\eta + q\eta\xi). \quad (189)$$

Proof. The single reduction rule $yx \mapsto qxy$ gives the unique normal form $x^i y^j$, so these monomials form a PBW basis. Formula (188) is a complex because

$$d_1 d_2(1) = yx - qxy = 0.$$

Write a syzygy as $ae_x + be_y$ with $ax + by = 0$. Comparison of PBW normal forms shows that the terms ending in x and y cancel precisely in pairs generated by $ye_x - qxe_y$. The Ore-extension realization of A_q makes it a domain, and therefore d_2 is injective. This proves exactness.

Let ξ, η be dual to x, y . A tensor $a\xi^2 + b\xi\eta + c\eta\xi + d\eta^2$ is orthogonal to $yx - qxy$ exactly when $c - qb = 0$. The orthogonal relation space is generated by

$$\xi^2, \quad \eta^2, \quad \xi\eta + q\eta\xi,$$

which proves (189). □

The reduced coproduct of the degree-two bar class records the relation:

$$\overline{\Delta}(\rho) = sy \otimes sx - qsx \otimes sy.$$

Thus the coalgebra remembers the ordered commutation parameter before any trace is taken.

30.5 The Jordan plane

The Jordan-plane chiral algebra is

$$A_J(K) = T_X(Kx \oplus Ky) / (yx - xy - x^2). \quad (190)$$

Theorem 30.6 (Jordan-plane resolution and dual). *The augmentation has the free resolution*

$$0 \longrightarrow A_J(K) \otimes K^{\otimes 2}[-2] \longrightarrow A_J(K) \otimes (Kx \oplus Ky)[-1] \longrightarrow A_J(K) \longrightarrow \mathbf{1} \longrightarrow 0$$

with

$$d_2(1) = (y - x)e_x - xe_y.$$

The quadratic dual is

$$A_J(K)^\dagger = T_X(K^\vee \xi \oplus K^\vee \eta) / (\eta^2, \xi\eta + \eta\xi, \xi^2 + \eta\xi). \quad (191)$$

Proof. The relation writes A_J as the Ore extension $\mathbb{C}[x][y; \delta]$ with $\delta(x) = x^2$. Hence $x^i y^j$ is a PBW basis and A_J is a domain. The displayed differential is a complex because

$$(y - x)x - xy = yx - x^2 - xy = 0.$$

PBW comparison gives the same one-generator syzygy argument as in the quantum plane. Orthogonality to $yx - xy - x^2$ gives the linear condition $c - b - a = 0$ on the coefficients of $a\xi^2 + b\xi\eta + c\eta\xi + d\eta^2$. The three displayed relations form a basis of this orthogonal subspace. $\square$

The quantum and Jordan planes have the same Hilbert series $(1 - t)^{-2}$ and the same bar dimensions $(1, 2, 1)$. Their reduced coproducts and dual multiplication distinguish them. Dimension is therefore a shadow of the ordered coalgebra rather than its complete invariant.

30.6 The Weyl deformation and curved duality

Let $\hbar$ be central. The Rees Weyl algebra is

$$\mathcal{W}_\hbar(K) = T_X(Kx \oplus Ky)[\hbar] / (yx - xy - \hbar). \quad (192)$$

The filtration by total degree in x, y has associated graded the commutative polynomial algebra.

Theorem 30.7 (Curved dual of the Weyl relation). *The nonhomogeneous quadratic dual of (192) is the curved algebra*

$$(\Lambda_X(K^\vee \xi \oplus K^\vee \eta), d = 0, h = \hbar \xi \eta),$$

in the suspension convention of Appendix A. Its curved cobar construction recovers $\mathcal{W}_\hbar(K)$.

Proof. The quadratic part $yx - xy$ has orthogonal relations ξ^2, η^2 , and $\xi\eta + \eta\xi$, which form the exterior algebra. The constant component of the relation is the functional sending the quadratic relation to $\hbar$. Under quadratic duality this functional is the curvature element $\hbar \xi \eta$. The curved cobar differential has a constant term dual to h ; its defining relation is therefore $yx - xy = \hbar$. $\square$

Theorem 30.7 places $\hbar \xi \eta$ inside the curved Koszul dual algebra. A family over moduli carries a second geometric object: the curvature two-form of its determinant-line connection. The algebraic element and the geometric two-form are related only through a constructed family index or anomaly map.

30.7 Relative quiver bars

Let Q be a finite quiver and

$$S = \bigoplus_{i \in Q_0} \mathbf{1} e_i.$$

Assign a factorisation object M_a to each arrow $a : i \rightarrow j$ and put

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i.$$

The path algebra is $T_S(M)$. Its augmentation takes a path to its length-zero part.

Theorem 30.8 (Relative bar of a path algebra). *The relative bar of $T_S(M)$ contracts to*

$$S \oplus M[1].$$

The relative bar of the radical-square-zero algebra $S \oplus M$ is

$$T_S^c(M[1]) = S \oplus M[1] \oplus (M \otimes_S M)[2] \oplus \cdots$$

with zero multiplication differential.

Proof. The first assertion is the relative form of Theorem 30.1. The second follows because every product of two arrows vanishes. $\square$

Example 30.9 (The doubled A_2 quiver). Let $a : 1 \rightarrow 2$ and $a^* : 2 \rightarrow 1$. The preprojective algebra of type A_2 is

$$\Pi_{A_2} = \mathbb{C}\langle e_1, e_2, a, a^* \rangle / (aa^*, a^*a),$$

with the standard idempotent relations. Relative to $S = \mathbb{C}e_1 \oplus \mathbb{C}e_2$, its radical is square-zero. For every $n \geq 1$, the tensor power $M^{\otimes n}$ has two basis paths, the alternating path starting at vertex 1 and the alternating path starting at vertex 2. Hence

$$\dim_S \operatorname{Tor}_n^{\Pi_{A_2}}(S, S) = 2 \quad (n \geq 1),$$

and the completed dual is the completed path algebra of the doubled quiver. Decorating a and a^* by factorisation D -modules produces boundary-changing chiral fields with the same exact relative bar.

The example exhibits infinite homological depth in a four-dimensional algebra. The depth comes from transverse relations; the chiral coefficient may be varied independently.

30.8 Finite-group and orbifold extensions

Let a finite group G act by augmented algebra automorphisms on $A \in \operatorname{Alg}_{E_1}(\mathcal{C}_X)$, and form the smash product $A \rtimes G$. Give $\mathbf{1}$ the trivial G -action.

Theorem 30.10 (Characteristic-zero smash-product bar). *Over $\mathbb{C}$ there is a natural isomorphism*

$$\operatorname{Tor}_*^{A \rtimes G}(\mathbf{1}, \mathbf{1}) \cong \operatorname{Tor}_*^A(\mathbf{1}, \mathbf{1})^G.$$

The coalgebra structure is the invariant part of the bar coalgebra of A .

Proof. The extension $A \subset A \rtimes G$ gives the Cartan–Eilenberg spectral sequence

$$H_p(G, \operatorname{Tor}_q^A(\mathbf{1}, \mathbf{1})) \implies \operatorname{Tor}_{p+q}^{A \rtimes G}(\mathbf{1}, \mathbf{1}).$$

Maschke semisimplicity gives $H_p(G, -) = 0$ for $p > 0$ and identifies H_0 with invariants. Multiplicativity and compatibility with the bar diagonal identify the coalgebra structure. $\square$

Example 30.11 (Cyclic orbifold of the quantum plane). Let $G = \mu_m$, $m > 1$, and let a generator act by

$$x \mapsto \zeta x, \quad y \mapsto \zeta^{-1} y.$$

On the degree-one dual generators, $\xi \mapsto \zeta^{-1} \xi$ and $\eta \mapsto \zeta \eta$. The degree-two class ρ is invariant. Therefore

$$\mathrm{Tor}_*^{A_q \rtimes \mu_m}(\mathbf{1}, \mathbf{1}) \cong \mathbf{1} \oplus \mathbf{1}\rho[2].$$

The orbifold removes the charged one-particle boundary states and retains the neutral two-particle relation.

30.9 Universal enveloping algebras

Let $\mathfrak{g}$ be a tensor-flat Lie algebra object in $\mathcal{C}_X$. Its enveloping algebra $U_X(\mathfrak{g})$ is the quotient of $T_X(\mathfrak{g})$ by

$$x \otimes y - (-1)^{|x||y|} y \otimes x - [x, y].$$

Theorem 30.12 (Internal PBW–Chevalley comparison). *The antisymmetrization twisting morphism gives a filtered quasi-isomorphism*

$$C_*^{\mathrm{CE}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{Bar}_X^{\mathrm{ord}}(U_X(\mathfrak{g})).$$

When $\mathfrak{g}$ is an ordinary complex semisimple Lie algebra with exponents $m_1, \dots, m_r$, viewed as a constant coefficient object, its Poincaré polynomial is

$$\sum_{n \geq 0} \dim \mathrm{Tor}_n^{U(\mathfrak{g})}(\mathbb{C}, \mathbb{C}) t^n = \prod_{i=1}^r (1 + t^{2m_i+1}). \quad (193)$$

Proof. PBW identifies $\mathrm{gr} U_X(\mathfrak{g})$ with $\mathrm{Sym}_X(\mathfrak{g})$. The associated graded of the Chevalley complex is the Koszul resolution of the symmetric algebra. Strong convergence of the PBW filtration gives the first statement. For a complex semisimple Lie algebra, Hopf–Koszul–Samelson theory identifies Lie algebra homology with the exterior algebra on primitive generators in degrees $2m_i + 1$. $\square$

Example 30.13 (The complete $\mathfrak{sl}_2$ calculation). Choose $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. In the homological Chevalley complex

$$0 \longrightarrow \Lambda^3 \mathfrak{sl}_2 \xrightarrow{d_3} \Lambda^2 \mathfrak{sl}_2 \xrightarrow{d_2} \mathfrak{sl}_2 \longrightarrow \mathbb{C} \longrightarrow 0,$$

the map d_2 is the bracket. It is an isomorphism because

$$e \wedge f \mapsto h, \quad h \wedge e \mapsto 2e, \quad h \wedge f \mapsto -2f.$$

The top element $e \wedge f \wedge h$ is a cycle. Hence

$$H_0 \cong \mathbb{C}, \quad H_3 \cong \mathbb{C}, \quad H_1 = H_2 = 0.$$

Thus

$$\mathrm{Tor}_*^{U(\mathfrak{sl}_2)}(\mathbb{C}, \mathbb{C}) \cong \Lambda(\zeta_3).$$

For $\mathfrak{sl}_3$, whose exponents are 1, 2, the polynomial is

$$(1 + t^3)(1 + t^5) = 1 + t^3 + t^5 + t^8.$$

For G_2 , whose exponents are 1, 5, it is

$$(1 + t^3)(1 + t^{11}) = 1 + t^3 + t^{11} + t^{14}.$$

These formulas give exact ordered bars for the specified enveloping multiplications.

30.10 The first structural conclusion

The examples produce three independent axes:

chiral coefficient	transverse multiplication	exchange enhancement
factorisable D -module	E_1 algebra law	E_2 transport or flat connection

A change along the middle axis changes the interval bar. A change along the first axis changes collision singularities and conformal blocks. A change along the third axis changes braid monodromy and the Tannaka reconstruction. The full theory records all three axes and the comparison maps among them.

31 Exact enumeration and the meaning of the legacy tables

The original calculations contain genuine combinatorics. Their mathematical meaning becomes precise after the transverse bar and the chiral collision complex are placed in separate degrees. This section performs that separation and identifies every sequence that survives as an exact invariant.

31.1 Three chain models

Let $A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X))$. Three complexes occur naturally.

The interval bar is

$$B_p^\perp(A) = \bar{A}[1]^{\otimes p}, \quad d = b_A + d_A.$$

Its differential uses the transverse multiplication.

For a chiral Lie coefficient $\mathfrak{g}$, the collision Chevalley complex is

$$B_I^X(\mathfrak{g}) = \bigoplus_{\pi \in \mathrm{Part}(I)} \Delta_{\pi, !}^{\mathrm{unord}} \bigotimes_{B \in \pi} \mathfrak{g}[1],$$

with differential induced by the diagonal-supported chiral bracket.

A screen-enhanced calculation uses a bicomplex

$$B_{\mathrm{scr}}^{p,q}(A) = \bar{A}[1]^{\otimes p} \otimes C_{\mathrm{scr}}^q, \tag{194}$$

where C_{scr}^q is a chosen chain or cochain model of compactified relative configurations. Its total differential has typed components

$$D = b_A + d_A + d_{\text{scr}} + \delta_{\text{int}},$$

and δ_{int} is a supplied interaction map satisfying the mixed square-zero identities.

Theorem 31.1 (Separation spectral sequence). *Assume (194) is first-quadrant and the four differential components form a square-zero total differential. Filtration by screen degree gives*

$$E_{p,q}^1 = H_q(C_{\text{scr}}^*) \otimes B_p^\perp(A),$$

with the induced interaction differential. Filtration by bar length gives

$${}^{\prime}E_{p,q}^1 = \text{Tor}_p^A(\mathbf{1}, \mathbf{1}) \otimes C_{\text{scr}}^q,$$

again with the induced interaction differential.

Proof. Each filtration is exhaustive and bounded in every finite total degree. The first differential on the associated graded is respectively d_{scr} and $b_A + d_A$. Standard comparison of filtered complexes gives the two spectral sequences. $\square$

The theorem gives a precise home to configuration-space numbers. A dimension from C_{scr}^* is a geometric coefficient. A dimension from Tor^A is an interval-bar invariant. Their product is a chain-space dimension on the first page; later differentials encode the supplied interaction.

31.2 The Orlik–Solomon factor

For the configuration space of n ordered points on the affine line,

$$\text{Conf}_n(\mathbb{C}) = \{(z_1, \dots, z_n) : z_i \neq z_j\},$$

Arnold’s presentation gives

$$H^*(\text{Conf}_n(\mathbb{C}); \mathbb{C}) = \Lambda(\omega_{ij}) / (\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij}),$$

with $\omega_{ij} = d \log(z_i - z_j)$.

Proposition 31.2 (Poincare polynomial of ordered configurations). *The Poincare polynomial is*

$$P_n(t) = \prod_{j=1}^{n-1} (1 + jt).$$

In particular,

$$\dim H^{n-1}(\text{Conf}_n(\mathbb{C})) = (n-1)!.$$

Proof. The forgetful map $\text{Conf}_n(\mathbb{C}) \rightarrow \text{Conf}_{n-1}(\mathbb{C})$ has fibre $\mathbb{C}$ with $n-1$ points removed. The Fadell–Neuwirth fibration and the trivial action on cohomology give $P_n(t) = (1 + (n-1)t)P_{n-1}(t)$, with $P_1(t) = 1$. $\square$

Consequently, for a d -dimensional coefficient space V , the vector space

$$V^{\otimes n} \otimes H^{n-1}(\text{Conf}_n(\mathbb{C}))$$

has dimension

$$d^n(n-1)!. \quad (195)$$

Formula (195) is an exact chain-space count. The interval bar of an associative multiplication on V has chain dimension d^n before normalization. A homology dimension requires the differential.

31.3 Two planar generating functions

Let M_n be the Motzkin numbers, characterized by

$$M(z) = \sum_{n \geq 0} M_n z^n = 1 + zM(z) + z^2 M(z)^2.$$

Set $a_n = M_{n+1} - M_n$ for $n \geq 1$.

Proposition 31.3 (Motzkin-difference series). *The generating function of $(a_n)_{n \geq 1}$ is*

$$A(z) = \sum_{n \geq 1} a_n z^n = \frac{4z}{(1 - z + \sqrt{1 - 2z - 3z^2})^2}. \quad (196)$$

Its first values are

$$1, 2, 5, 12, 30, 76, 196, 512, 1353, 3610.$$

Proof. Solving the quadratic equation for the branch with $M(0) = 1$ gives

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}.$$

Since

$$A(z) = \frac{M(z) - 1}{z} - (M(z) - 1),$$

algebraic simplification gives (196). □

Let R_n be the Riordan numbers, with generating function

$$R(z) = \sum_{n \geq 0} R_n z^n = \frac{1 + z - \sqrt{1 - 2z - 3z^2}}{2z(1 + z)}.$$

They count Motzkin paths whose horizontal steps occur strictly above the horizontal axis.

Proposition 31.4 (Riordan series). *The shifted sequence R_{n+3} begins*

$$3, 6, 15, 36, 91, 232, 603, 1585.$$

Both (196) and $R(z)$ have the discriminant

$$1 - 2z - 3z^2 = (1 + z)(1 - 3z),$$

so both sequences have exponential growth rate 3.

Proof. Coefficient extraction from the displayed algebraic function gives the values. The dominant singularity is $z = 1/3$, and standard singularity analysis gives growth proportional to $3^n n^{-3/2}$. $\square$

These propositions retain the arithmetic of the tables exactly. A comparison with an ordered bar becomes a theorem after a chain complex and a quasi-isomorphism are supplied.

31.4 The corrected bar values for standard transverse products

Let the same three-dimensional generator space underlie several associative algebras. The exact ordered-bar homologies are:

transverse algebra	Poincare series of Tor	mechanism
$T(V)$	$1 + 3t$	free-algebra contraction
$\mathbf{1} \oplus V, V^2 = 0$	$\frac{1}{1 - 3t}$	zero bar multiplication
$U(\mathfrak{sl}_2)$	$1 + t^3$	Chevalley homology
A_q	$1 + 2t + t^2$	quadratic Koszul resolution

The final row is the rank-two quantum-plane calculation of Theorem 30.5.

Theorem 31.5 (Exact audit of the affine sequences). *For the transverse multiplication $U(\mathfrak{g})$, the corrected ordered bar is the Chevalley complex and has Poincare polynomial (193). Hence*

$\mathfrak{g}$	$\sum_n \dim \operatorname{Tor}_n^{U(\mathfrak{g})}(\mathbb{C}, \mathbb{C}) t^n$
$\mathfrak{sl}_2$	$1 + t^3$
$\mathfrak{sl}_3$	$1 + t^3 + t^5 + t^8$
G_2	$1 + t^3 + t^{11} + t^{14}$

The Riordan sequence records a different planar enumeration. An identification with an affine ordered bar requires a transverse multiplication and a chain equivalence whose induced Poincare series equals $R(z)$.

Proof. The first assertion is Theorem 30.12. Substituting the exponent sets $\{1\}$, $\{1, 2\}$, and $\{1, 5\}$ gives the table. The final sentence is the universal property of homology: an equality of dimensions follows from a quasi-isomorphism or an independently proved equality of homology groups. $\square$

31.5 Virasoro coefficients and transverse presentations

The Virasoro OPE specifies a chiral coefficient. A transverse E_1 product arises from additional data, such as an endomorphism algebra of boundary conditions, a tensor algebra generated by the Virasoro coefficient, or an enveloping algebra of a chosen mode Lie algebra.

Theorem 31.6 (Exact alternatives for a Virasoro coefficient). *Let M_{Vir} denote a factorisation object carrying the universal Virasoro chiral operation.*

- (i) *The free transverse algebra $T_X(M_{\text{Vir}})$ has reduced ordered-bar homology $M_{\text{Vir}}[1]$.*
- (ii) *The square-zero transverse algebra $\mathbf{1} \oplus M_{\text{Vir}}$ has ordered bar $T^c(M_{\text{Vir}}[1])$.*

- (iii) For a Lie subalgebra $\mathfrak{l}$ of Virasoro modes and the transverse algebra $U(\mathfrak{l})$, the ordered bar is $C_*^{\text{CE}}(\mathfrak{l})$.

A fourth transverse presentation realizes the Motzkin-difference series as a Virasoro bar invariant when its bar complex is connected to the Motzkin complex by a proved quasi-isomorphism.

Proof. The three calculations are Theorem 30.1 and Theorem 30.12. The final assertion states the data needed to define an invariant of an associative algebra. $\square$

The theorem refines the theory: the Virasoro singularity and the transverse multiplication are coordinates on different axes. A physical boundary condition selects a transverse endomorphism algebra, and its bar can then be calculated.

31.6 A reproducible finite-window test

Let $A = T(V)/(R)$ be a homogeneous algebra with finite-dimensional graded pieces. For weights at most N and bar length at most L , form the finite complex

$$B_{\leq L, \leq N}(A) = \bigoplus_{p \leq L} \bigoplus_{w \leq N} (\bar{A}[1]^{\otimes p})_w.$$

Proposition 31.7 (Finite-window certification). *A matrix computation of the bar differential on $B_{\leq L, \leq N}(A)$ certifies the homology in every bidegree (p, w) satisfying*

$$p \leq L, \quad w \leq N, \quad p + 1 \leq L \text{ for incoming boundaries.}$$

For a quadratic Koszul algebra generated in weight one, the length-two Koszul resolution certifies every weight for the quantum and Jordan planes.

Proof. The differential preserves total weight and lowers bar length by one. Homology at length p uses only the maps from lengths $p + 1$ and p . The stated window contains both. For a length-two free resolution, the resolution itself computes all derived tensors, so every weight is included simultaneously. $\square$

This finite-window principle supplies arithmetic checks while preserving the mathematical definition of the complex. The matrix is a verification of a specified differential; the theorem identifies the range certified by the matrix.

32 Modules, boundary conditions, and orbifold operations

A noncommutative chiral algebra acts on boundary conditions. The correct derived object for two boundaries is the two-sided interval amplitude.

32.1 The pointed ordered bar

Let $A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X))$, let M_R be a right A -module, and let M_L be a left A -module. Define

$$B_A(M_R, M_L) = \left| [n] \longmapsto M_R \otimes \bar{A}^{\otimes n} \otimes M_L \right|. \quad (197)$$

Theorem 32.1 (Boundary-to-boundary amplitude). *There is a natural equivalence*

$$B_A(M_R, M_L) \simeq M_R \otimes_A^L M_L.$$

If the modules arise from boundary conditions in a rectangular holomorphic-topological factorisation algebra on $X \times \mathbb{R}$, the same object is the factorisation homology of an interval labelled by M_R and M_L at its endpoints.

Proof. The simplicial object (197) is the standard derived coequalizer of the right and left actions. Geometric realization computes the relative tensor product. Excision for the interval identifies the same realization with the labelled factorisation homology. $\square$

A marked chiral insertion produces a comodule over $\text{Bar}_X^{\text{ord}}(A)$ by cutting the interval between the marked point and either endpoint. The two cuts are the left and right coactions.

32.2 Quiver boundary conditions

For a finite quiver Q , let $A_Q = T_S(M)$ be the internal path algebra of Section 30. The vertex idempotent e_i defines a boundary condition and a simple relative module S_i .

Proposition 32.2 (Derived boundary-changing fields). *For the free path algebra A_Q , the relative derived morphisms satisfy*

$$\text{Tor}_0^{A_Q}(S_j, S_i) \cong \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad \text{Tor}_1^{A_Q}(S_j, S_i) \cong \bigoplus_{a:i \rightarrow j} M_a[1],$$

and the higher groups vanish.

Proof. The relative bar of a tensor algebra contracts to $S \oplus M[1]$. Multiplication by e_j on the left and e_i on the right selects the summands with target j and source i . $\square$

For the radical-square-zero doubled A_2 algebra of Example 30.9, the corresponding derived morphism has one alternating path in every admissible length. The complete bar therefore records an infinite tower of boundary-changing composites even though the algebra itself has radical square zero.

32.3 Koszul modules for the quantum plane

Let $A_q = A_q(\mathbf{1})$ and let $\mathbb{C}$ be the augmentation module. The resolution (188) gives

$$\text{Ext}_{A_q}^*(\mathbb{C}, \mathbb{C}) \cong A_q^! = \mathbb{C}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

The Yoneda product is the multiplication of the quadratic dual. The two degree-one extension classes compose to the degree-two relation class, and their ordered composition carries the parameter q .

Let $L_x = A_q/(A_q x)$ and $L_y = A_q/(A_q y)$. Their defining one-step resolutions give

$$\text{Tor}_1^{A_q}(\mathbb{C}, L_x) \cong \mathbb{C}x[1], \quad \text{Tor}_1^{A_q}(\mathbb{C}, L_y) \cong \mathbb{C}y[1].$$

The modules distinguish the two ordered generators even when the ungraded Poincare series of A_q agrees with that of the Jordan plane.

32.4 Equivariant modules and crossed products

Let a finite group G act on A by augmented automorphisms. The crossed product $A \rtimes G$ contains both the algebra and the symmetry operators.

Theorem 32.3 (Equivariantization). *There is an equivalence of module categories*

$$\text{Mod}_{A \rtimes G} \simeq \text{Mod}_A^G,$$

where the right side is the category of G -equivariant A -modules. Under this equivalence, the two-sided bar of crossed-product modules is the total complex of the group bar and the A -bar.

Proof. An $A \rtimes G$ action restricts to an A action and a G action satisfying $g(am) = g(a)g(m)$. Conversely, this compatibility defines the crossed-product action. The standard bar resolution of the semidirect product is the diagonal of the resulting bisimplicial resolution. $\square$

In characteristic zero, group averaging contracts the positive group-homology rows. The invariant formula of Theorem 30.10 follows.

32.5 Invariant algebras and crossed products

The fixed algebra A^G and the crossed product $A \rtimes G$ encode complementary operations. The fixed algebra contains invariant local operators. The crossed product contains equivariant boundary conditions and symmetry defects. The canonical functor

$$\text{Mod}_{A \rtimes G} \longrightarrow \text{Mod}_{A^G}$$

forgets the equivariant structure and restricts scalars. A Morita equivalence between these algebras is supplied by a Galois or freeness theorem for the action.

This distinction becomes geometric for orbifolds. Untwisted invariant fields belong to A^G ; twisted sectors arise from defect boundary conditions in the equivariant category. Sewing the twisted sectors produces the orbifold conformal blocks.

32.6 Symmetric powers and plethystic exponentials

Let $V = \bigoplus_{n \geq 0} V_n$ be a graded vector space with finite-dimensional pieces and character

$$f(q) = \sum_{n \geq 0} (\dim V_n) q^n.$$

The symmetric algebra has the second-quantized character

$$\sum_{N \geq 0} p^N \text{ch}(\text{Sym}^N V) = \exp \left(\sum_{r \geq 1} \frac{p^r}{r} f(q^r) \right). \quad (198)$$

Proof. Choose homogeneous bases. Each even basis element of weight n contributes $(1 - pq^n)^{-1}$. Multiplication over the basis and logarithmic expansion of the Euler factors gives (198). The super version replaces dimensions by superdimensions and includes the fermionic Euler factors. $\square$

The formula is a trace identity. The orbifold operator algebra additionally contains twisted sectors and their OPEs.

32.7 The DMVV formula

Let Y be a compact complex manifold for which the elliptic genus

$$\text{Ell}(Y; q, y) = \sum_{m \geq 0, \ell \in \mathbb{Z}} c(m, \ell) q^m y^\ell$$

is defined. The symmetric-product orbifold elliptic genera satisfy the Dijkgraaf–Moore–Verlinde–Verlinde product formula [26]:

$$\sum_{N \geq 0} p^N \text{Ell}_{\text{orb}}(\text{Sym}^N Y; q, y) = \prod_{n \geq 1} \prod_{m \geq 0} \prod_{\ell \in \mathbb{Z}} \left(1 - p^n q^m y^\ell\right)^{-c(nm, \ell)}. \quad (199)$$

Theorem 32.4 (Cycle decomposition behind DMVV). *Formula (199) is the exponential of connected cycle contributions. A length- n cycle contributes the seed coefficient with energy multiplied by n and with the standard $1/n$ automorphism factor. Exponentiation assembles disjoint cycles.*

Proof. Conjugacy classes in the symmetric group are indexed by partitions, equivalently by multiplicities of cycles of each length. The orbifold trace factorizes over cycles. Summing the multiplicity of length- n cycles exponentiates its connected contribution. Expanding the logarithm of the product in (199) reproduces the cycle sum. $\square$

The same combinatorics appears in a cyclic bar: connected annular contributions exponentiate to disconnected traces. The coefficient object and the orbifold twisted sectors determine the actual amplitudes.

32.8 Boundary categories as the physical carrier

A physical boundary condition b in a holomorphic–topological theory supplies

$$A_b = \text{RHom}(b, b) \in \text{Alg}_{E_1}(\text{FactD}(X)).$$

Two boundaries b_0, b_1 supply the bimodule

$$M_{01} = \text{RHom}(b_0, b_1).$$

Composition of boundary-changing operators gives the module structure; collision on X gives the chiral coefficient. The interval amplitude

$$M_{10} \otimes_{A_{b_0}}^L M_{01}$$

counts derived excursions from b_0 to b_1 and back. A quiver of boundary conditions is therefore a literal source of the quiver chiral algebras above.

This construction realizes strict noncommutative chirality in field theory. The associative order is composition in the topological direction, and the chiral singularity is the operator product on the curve.

33 Rational exchange, RTT algebras, and Yangian recognition

The interval supplies associative order. A transverse plane supplies paths that exchange ordered lines. This section calculates the resulting rational exchange algebra and states the exact recognition theorem that produces a Yangian.

33.1 The Yang rational matrix

Let $V = \mathbb{C}^N$ and let $P \in \text{End}(V \otimes V)$ be the flip, $P(v \otimes w) = w \otimes v$. Set

$$R(u) = 1 - \frac{\hbar P}{u}. \quad (200)$$

Theorem 33.1 (Rational Yang–Baxter equation). *The meromorphic operator (200) satisfies*

$$R_{12}(u)R_{13}(u+v)R_{23}(v) = R_{23}(v)R_{13}(u+v)R_{12}(u) \quad (201)$$

on $V^{\otimes 3}$. It also satisfies

$$R_{12}(u)R_{21}(-u) = \left(1 - \frac{\hbar^2}{u^2}\right) 1. \quad (202)$$

Proof. Expand both sides of (201) in the group algebra of S_3 . The linear terms agree. The quadratic terms agree by

$$P_{12}P_{13} = P_{23}P_{12}, \quad P_{13}P_{23} = P_{23}P_{12}, \quad P_{12}P_{23} = P_{13}P_{12},$$

together with

$$\frac{1}{u(u+v)} + \frac{1}{(u+v)v} = \frac{1}{uv}.$$

The cubic terms agree because $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$. Formula (202) follows from $P_{21} = P_{12}$ and $P^2 = 1$. $\square$

The decomposition

$$V \otimes V = \text{Sym}^2 V \oplus \Lambda^2 V$$

diagonalizes P . Hence

$$R(u)|_{\text{Sym}^2 V} = 1 - \frac{\hbar}{u}, \quad R(u)|_{\Lambda^2 V} = 1 + \frac{\hbar}{u}. \quad (203)$$

The two eigenvalues are the exchange amplitudes in the symmetric and antisymmetric channels.

33.2 The RTT algebra

Write

$$T(u) = \sum_{i,j=1}^N E_{ij} \otimes t_{ij}(u), \quad t_{ij}(u) = \delta_{ij} + \hbar \sum_{r \geq 1} t_{ij}^{(r)} u^{-r}.$$

The RTT relation is

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v). \quad (204)$$

Proposition 33.2 (Coefficient relations). *With $t_{ij}^{(0)} = \delta_{ij}$, equation (204) is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar(t_{kj}^{(r)}t_{il}^{(s)} - t_{kl}^{(s)}t_{ij}^{(r)}) \quad (205)$$

for $r, s \geq 0$.

Proof. Multiply (204) by $u - v$ and use $P(E_{ij} \otimes E_{kl}) = (E_{kj} \otimes E_{il})P$. The matrix coefficient in $E_{ij} \otimes E_{kl}$ gives

$$(u - v)[t_{ij}(u), t_{kl}(v)] = \hbar(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)).$$

Expansion in u^{-1} and v^{-1} gives (205). $\square$

The coproduct

$$\Delta T(u) = T_{13}(u)T_{23}(u), \quad \Delta t_{ij}(u) = \sum_a t_{ia}(u) \otimes t_{aj}(u), \quad (206)$$

is coassociative and preserves the RTT relation. The resulting bialgebra is the RTT Yangian $Y_h(\mathfrak{gl}_N)$.

Proposition 33.3 (Evaluation representation). *For $a \in \mathbb{C}$, the assignment*

$$T(u) \mapsto 1 + \frac{\hbar E}{u - a}, \quad E = \sum_{i,j} E_{ij} \otimes E_{ji}, \quad (207)$$

defines a homomorphism $Y_h(\mathfrak{gl}_N) \rightarrow U(\mathfrak{gl}_N)[[\hbar]]$.

Proof. The operator E obeys the same permutation identities as P under the defining representation. Substitution of (207) into the RTT equation reduces to the rational Yang–Baxter equation of Theorem 33.1 and the commutation relations of matrix units. $\square$

Two evaluation modules at parameters a, b are intertwined by $R(a - b)$. The pole at $a = b$ is the collision locus of the two spectral lines.

33.3 The KZ connection

Let $\mathfrak{g}$ be simple with invariant tensor $\Omega = \sum_a x_a \otimes x^a$. On a tensor product of $\mathfrak{g}$ -modules define

$$\nabla_{\text{KZ}} = d - \frac{1}{\kappa} \sum_{1 \leq i < j \leq n} \Omega_{ij} d \log(z_i - z_j). \quad (208)$$

Theorem 33.4 (Flatness of KZ). *The connection (208) is flat. Its local monodromy around $z_i = z_j$ is conjugate to*

$$\exp\left(\frac{2\pi i}{\kappa} \Omega_{ij}\right).$$

Proof. The exterior derivative of each logarithmic form vanishes on configuration space. The curvature is therefore the commutator term. Operators with disjoint supports commute, and invariance of Ω gives the infinitesimal braid relations

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad (\{i, j\} \cap \{k, l\} = \emptyset), \quad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0.$$

Combining these relations with Arnold's identity

$$d \log z_{ij} \wedge d \log z_{jk} + d \log z_{jk} \wedge d \log z_{ki} + d \log z_{ki} \wedge d \log z_{ij} = 0$$

gives zero curvature. Integration around a small positive loop about $z_i = z_j$ gives the residue $2\pi i \Omega_{ij} / \kappa$. $\square$

For two fundamental $\mathfrak{sl}_2$ modules, write $V_{1/2} \otimes V_{1/2} = V_1 \oplus V_0$. With the Casimir normalized by $C_j = j(j+1)$,

$$\Omega_{12}|_{V_J} = \frac{1}{2} (C_J - 2C_{1/2}),$$

so

$$\Omega_{12}|_{V_1} = \frac{1}{4}, \quad \Omega_{12}|_{V_0} = -\frac{3}{4}. \quad (209)$$

The two local monodromy eigenvalues are therefore

$$\exp\left(\frac{\pi i}{2\kappa}\right), \quad \exp\left(-\frac{3\pi i}{2\kappa}\right).$$

33.4 Drinfeld–Kohno recognition

The KZ connection produces a braided tensor category of suitable affine conformal blocks. The Drinfeld–Kohno theorem identifies this braided tensor category with the corresponding quantum-group category, with

$$q = \exp\left(\frac{\pi i}{\kappa}\right)$$

under the chosen normalization [51, 50].

Theorem 33.5 (Recognition from exchange data). *Let $\mathcal{L}$ be a rigid meromorphic tensor category generated by evaluation objects $V(a)$, and let $F : \mathcal{L} \rightarrow \text{Vect}_{\mathbb{C}((\hbar))}$ be a faithful exact tensor functor. Assume:*

- (i) *the exchange on $F(V(a)) \otimes F(V(b))$ is the rational matrix $R(a-b)$ of (200);*
- (ii) *fusion is compatible with the RTT coproduct (206);*
- (iii) *tensor products of evaluation objects generate $\mathcal{L}$ under finite colimits, subquotients, and duals;*
- (iv) *their matrix coefficients separate the spectral completion of $Y_h(\mathfrak{gl}_N)$.*

Then the Tannaka coend

$$\mathcal{O}_F = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L)$$

is the spectral finite-dual bialgebra of the RTT Yangian. When the spectral graded pieces are finite-dimensional, the continuous dual satisfies

$$\mathcal{O}_F^\vee \cong \widehat{Y_h(\mathfrak{gl}_N)}.$$

Proof. The coend is generated by matrix coefficients of the evaluation objects. Naturality of exchange gives the RTT relation on matrix coefficients, while tensor compatibility gives the dual of the coproduct (206). The generation hypothesis makes the resulting morphism from the spectral finite dual onto $\mathcal{O}_F$ surjective. Separation makes it injective. Finite-dimensionality of the spectral graded pieces identifies the continuous dual of the finite dual with the completed RTT algebra. $\square$

This theorem locates the Yangian precisely. The interval E_1 product supplies fusion. The meromorphic exchange supplies $R(u)$. The fibre functor reconstructs the bialgebra.

33.5 Four-dimensional Chern–Simons theory

Costello–Witten–Yamazaki study a gauge theory on $\Sigma \times C$, topological along the real surface Σ and holomorphic along the complex curve C [19, 20]. Wilson lines supported on curves in Σ carry a spectral parameter in C . Perturbative exchange of two lines gives the classical rational kernel

$$r(z) = \frac{\hbar\Omega}{z},$$

and isotopy of three lines gives the Yang–Baxter equation. Their all-orders analysis produces the RTT presentations of the rational, trigonometric, and elliptic quantum groups in the corresponding geometries.

Proposition 33.6 (Ordered chiral structure of Wilson lines). *Fix an oriented line in the topological surface Σ . The fusion of Wilson lines along that line is an E_1 multiplication. Holomorphic dependence on their positions in C is a factorisable coefficient. Allowing the lines to move in the full surface supplies exchange paths and promotes the line category to a braided meromorphic tensor category.*

Proof. Disjoint intervals on the oriented line compose by topological gluing, which is the little-interval operation. Holomorphic translation in C preserves the protected sector and produces meromorphic collision kernels. In the full surface, the configuration space of two lines has a nontrivial exchange path; topological invariance identifies isotopic paths and yields braid relations. $\square$

The proposition is the physical realization of an E_1 algebra in a chiral coefficient category. The Yangian is the Tannaka reconstruction of its planar exchange enhancement.

33.6 A complete rational example for $N = 2$

For $V = \mathbb{C}^2$, choose the ordered basis

$$e_1 \otimes e_1, \quad e_1 \otimes e_2, \quad e_2 \otimes e_1, \quad e_2 \otimes e_2.$$

Then

$$R(u) = \begin{pmatrix} 1 - \hbar/u & 0 & 0 & 0 \\ 0 & 1 & -\hbar/u & 0 \\ 0 & -\hbar/u & 1 & 0 \\ 0 & 0 & 0 & 1 - \hbar/u \end{pmatrix}. \quad (210)$$

The symmetric vectors $e_1 \otimes e_1$, $e_2 \otimes e_2$, and $e_1 \otimes e_2 + e_2 \otimes e_1$ have eigenvalue $1 - \hbar/u$; the antisymmetric vector $e_1 \otimes e_2 - e_2 \otimes e_1$ has eigenvalue $1 + \hbar/u$.

The normalized operator

$$\tilde{R}(u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1/2} R(u)$$

satisfies $\tilde{R}(u)\tilde{R}_{21}(-u) = 1$ after a choice of formal square root. The scalar normalization changes the framing convention while preserving the RTT relation.

34 Prime forms, sewing, and exact higher-genus amplitudes

The complex curve contributes a geometric genus through plumbing and conformal blocks. A cyclic transverse algebra contributes a ribbon genus through Hochschild and open topological sewing. The constructions below calculate the chiral-curve side and display its exact relation to the local collision operation.

34.1 The prime form and the normalized bidifferential

Let C be a compact Riemann surface of genus g , choose a symplectic basis of $H_1(C, \mathbb{Z})$, and let $\omega = (\omega_1, \dots, \omega_g)^T$ be the normalized holomorphic one-forms with period matrix Ω . Choose an odd theta characteristic δ . The prime form is

$$E(p, q) = \frac{\theta[\delta]\left(\int_q^p \omega \mid \Omega\right)}{h_\delta(p)h_\delta(q)}, \quad (211)$$

where h_δ^2 is the holomorphic one-form obtained from the gradient of the odd theta function at the origin [32].

Proposition 34.1 (Local form of the prime form). *The section $E(p, q)$ is antisymmetric, has a simple zero on the diagonal, and in a local coordinate z satisfies*

$$E(p, q) = \frac{z(p) - z(q)}{\sqrt{dz(p)}\sqrt{dz(q)}} \left(1 + O((z(p) - z(q))^2)\right). \quad (212)$$

The bidifferential

$$B(p, q) = d_p d_q \log E(p, q) \quad (213)$$

has a double pole of biresidue 1 on the diagonal, vanishing A -periods, and B -periods

$$\oint_{B_i} B(p, q) = 2\pi i \omega_i(q).$$

Proof. Oddness of the theta characteristic gives a first-order zero at the origin of the Abel–Jacobi variable. The denominator in (211) converts the numerator into a $(-1/2, -1/2)$ -form. Taylor expansion gives (212). Two derivatives produce the double pole. The transformation law of the theta function gives zero A -periods and the stated B -periods. $\square$

The normal symbols of Section 15 are the algebraic local coefficients of (212). The prime form supplies their global analytic transport.

34.2 The genus-one single-valued kernel

On $C_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, set

$$S_\tau(z) = \frac{\theta'_1(z|\tau)}{\theta_1(z|\tau)} + 2\pi i \frac{\Im z}{\Im \tau}. \quad (214)$$

Proposition 34.2 (Torus kernel). *The one-form $S_\tau(z) dz$ is doubly periodic and has residue 1 at $z = 0$. Its antiholomorphic derivative is*

$$\bar{\partial}(S_\tau(z)dz) = -\frac{\pi}{\Im \tau} d\bar{z} \wedge dz. \quad (215)$$

Proof. The theta transformation laws give

$$\frac{\theta'_1(z+1)}{\theta_1(z+1)} = \frac{\theta'_1(z)}{\theta_1(z)}, \quad \frac{\theta'_1(z+\tau)}{\theta_1(z+\tau)} = \frac{\theta'_1(z)}{\theta_1(z)} - 2\pi i.$$

The second term of (214) changes by $2\pi i$ under $z \mapsto z + \tau$, so the sum is periodic. The local expansion $\theta_1(z) = \theta'_1(0)z + O(z^3)$ gives residue 1. Since $\partial_{\bar{z}}\Im z = i/2$, differentiation gives (215). $\square$

The equation is a curvature identity for a global analytic kernel. The local residue remains the coefficient of the simple collision symbol. Higher OPE modes continue to live in the higher normal symbols of the chiral coefficient.

The scalar Green function

$$G_\tau(z) = -\log \left| \frac{\theta_1(z|\tau)}{\eta(\tau)} \right|^2 + \frac{2\pi(\Im z)^2}{\Im \tau} \quad (216)$$

is doubly periodic. For the Euclidean Laplacian $\Delta = \partial_x^2 + \partial_y^2 = 4\partial_z\partial_{\bar{z}}$ and the flat area $\text{Area}(C_\tau) = \Im \tau$, it satisfies

$$\Delta G_\tau = -4\pi \left(\delta_0 - \frac{1}{\Im \tau} \right).$$

The distributional identity $\Delta \log |z|^2 = 4\pi\delta_0$ and the equality $\Delta(\Im z)^2 = 2$ prove the formula. Differentiating G_τ recovers the kernel (214) in the stated complex-coordinate convention.

34.3 Formal sewing of chiral states

Let $V = \bigoplus_{n \geq 0} V_n$ carry a nondegenerate invariant pairing, and choose dual homogeneous bases $\{u_\alpha\}$ and $\{u^\alpha\}$. The sewing tensor is

$$\mathcal{S}(q) = \sum_{\alpha} q^{\text{wt}(u_\alpha)} u_\alpha \otimes u^\alpha. \quad (217)$$

Theorem 34.3 (Algebraic sewing tensor). *In the q -adic completion, contraction with (217) defines the clutching operation that identifies the two punctures. Associativity of successive sewing is the completeness identity*

$$\sum_{\alpha} \langle v, u_\alpha \rangle u^\alpha = v$$

together with associativity of the underlying correlation maps. The anomaly line supplies the coordinate-change factor associated with the chosen local parameters.

Proof. Insert the identity operator on the completed state space, decomposed into L_0 eigenspaces. The operator q^{L_0} gives the weight factor in (217). Two sewings compose by two insertions of the identity, and completeness contracts the intermediate basis. Coordinate changes act through the Virasoro representation and therefore through the determinant or anomaly line. $\square$

For strongly rational vertex operator algebras, conformal blocks form vector bundles and the sewing maps define a modular functor [22, 24, 25]. The theorem supplies the exact analytic realization of the chiral modular operad in that class.

34.4 The Heisenberg determinant

For the rank- r Heisenberg vertex algebra, the oscillator Fock space is

$$\mathrm{Sym} \left(\bigoplus_{n \geq 1} \mathbb{C}^r q^n \right).$$

Its graded trace is

$$\mathrm{Tr} \, q^{L_0 - r/24} = q^{-r/24} \prod_{n \geq 1} (1 - q^n)^{-r} = \eta(\tau)^{-r}. \quad (218)$$

Proof. Each bosonic oscillator of frequency n contributes the geometric series $(1 - q^n)^{-1}$. There are r independent oscillators at every positive frequency. The zeta-regularized vacuum energy is $-r/24$. $\square$

The same product is the inverse Fredholm determinant of the one-particle sewing operator with eigenvalues q^n , repeated r times:

$$\det(1 - T_q)^{-r} = \prod_{n \geq 1} (1 - q^n)^{-r}.$$

The operator trace and the chiral character are two realizations of the same sewing contraction.

34.5 The Heisenberg–Eisenstein calculation

Set $q = e^{2\pi i \tau}$ and normalize the Eisenstein series by

$$E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n, \quad E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, \quad E_6(\tau) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n.$$

Proposition 34.4 (Stress-tensor trace and E_2). *For the rank- r Heisenberg character $Z_r(\tau) = \eta(\tau)^{-r}$,*

$$q \frac{d}{dq} \log \eta(\tau) = \frac{E_2(\tau)}{24}, \quad q \frac{d}{dq} \log Z_r(\tau) = -\frac{r}{24} E_2(\tau). \quad (219)$$

Equivalently,

$$\frac{\mathrm{Tr}_{\mathcal{H}_r} \left((L_0 - r/24) q^{L_0 - r/24} \right)}{\mathrm{Tr}_{\mathcal{H}_r} q^{L_0 - r/24}} = -\frac{r}{24} E_2(\tau). \quad (220)$$

Proof. The Euler product gives

$$q \frac{d}{dq} \log \eta = \frac{1}{24} - \sum_{n \geq 1} \frac{n q^n}{1 - q^n} = \frac{1}{24} - \sum_{m \geq 1} \sigma_1(m) q^m = \frac{E_2}{24}.$$

The second identity follows from $Z_r = \eta^{-r}$. Differentiating the graded trace inserts $L_0 - r/24$, which gives (220). $\square$

Let $J^a(z)$, $1 \leq a \leq r$, be Heisenberg currents with $J^a(z) J^b(w) \sim k \delta^{ab} (z - w)^{-2}$. Let

$$B_\tau(z, w) = d_z d_w \log E(z, w)$$

be the normalized torus bidifferential.

Proposition 34.5 (Elliptic Wick theorem). *The normalized vacuum-module traces satisfy*

$$\langle J^a(z)J^b(w) \rangle_\tau = k\delta^{ab}B_\tau(z, w). \quad (221)$$

Every odd current correlator vanishes, and every even correlator is

$$\langle J^{a_1}(z_1) \cdots J^{a_{2m}}(z_{2m}) \rangle_\tau = \sum_{\text{pairings } P} \prod_{\{i,j\} \in P} k\delta^{a_i a_j} B_\tau(z_i, z_j). \quad (222)$$

Proof. The Heisenberg Fock representation is Gaussian. The oscillator trace contracts one positive mode with one negative mode, and the resulting Fourier series is the normalized bidifferential B_τ . Commutation moves every positive mode to the right and every negative mode to the left. Iterating this identity gives the sum over pairings; a word of odd length leaves one unpaired oscillator and has zero trace. $\square$

The modular derivatives close on the Eisenstein ring through Ramanujan's identities

$$q \frac{dE_2}{dq} = \frac{E_2^2 - E_4}{12}, \quad q \frac{dE_4}{dq} = \frac{E_2 E_4 - E_6}{3}, \quad q \frac{dE_6}{dq} = \frac{E_2 E_6 - E_4^2}{2}. \quad (223)$$

Consequently, repeated stress-tensor insertions in the Heisenberg trace are explicit quasimodular polynomials. Equations (219), (222), and (223) give three faithful realizations of the same genus-one operation: oscillator trace, elliptic propagator, and modular differential equation.

34.6 Eisenstein series from even unimodular lattices

Let L be an even positive-definite lattice. Its genus-one theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{(x,x)/2}.$$

For an even unimodular lattice of rank r , Θ_L is a modular form of weight $r/2$.

Proposition 34.6 (The E_8 identity). *For the E_8 lattice,*

$$\Theta_{E_8}(\tau) = E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n. \quad (224)$$

Proof. The theta series is a weight-four modular form for $SL_2(\mathbb{Z})$ with constant term 1. The space $M_4(SL_2(\mathbb{Z}))$ is one-dimensional and generated by E_4 . $\square$

The lattice character is

$$Z_L(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}. \quad (225)$$

Its stress-tensor trace separates oscillator and zero-mode contributions:

$$q \frac{d}{dq} \log Z_L = q \frac{d}{dq} \log \Theta_L - \frac{r}{24} E_2. \quad (226)$$

For $L = E_8$, Ramanujan's identity gives the closed expression

$$q \frac{d}{dq} \log Z_{E_8} = \frac{E_2 - E_6/E_4}{3} - \frac{E_2}{3} = -\frac{E_6}{3E_4}. \quad (227)$$

Thus the oscillator quasimodularity cancels against the E_2 term in the weight-four theta derivative, leaving a meromorphic modular function of weight two. The cancellation is the explicit genus-one manifestation of modular covariance of the self-dual lattice theory.

For a Niemeier lattice L with Coxeter number h of its root system,

$$\Theta_L = E_4^3 + (24h - 720)\Delta. \quad (228)$$

The coefficient of q counts the $24h$ roots. For the Leech lattice, $h = 0$ and

$$\Theta_{\text{Leech}} = E_4^3 - 720\Delta. \quad (229)$$

These formulas are complete genus-one arithmetic data for the lattice zero modes.

34.7 Siegel theta series and the rank-sixteen test

For a lattice L , the degree- g Siegel theta series is

$$\Theta_L^{(g)}(\Omega) = \sum_{x_1, \dots, x_g \in L} \exp \left(\pi i \sum_{a,b=1}^g (x_a, x_b) \Omega_{ab} \right).$$

Theorem 34.7 (Rank-sixteen theta comparison). *Let $L_1 = E_8 \oplus E_8$ and $L_2 = D_{16}^+$. Then*

$$\Theta_{L_1}^{(g)} = \Theta_{L_2}^{(g)} \quad (g = 1, 2, 3).$$

For $g = 4$, their difference is a nonzero scalar multiple of the Schottky form on Siegel space, and its restriction to the Jacobian locus of smooth genus-four curves vanishes [59, 66].

The theorem gives a precise genus threshold. Degree-four Siegel modular data distinguishes the lattices on the ambient Siegel space, while smooth genus-four partition functions remain equal on the Jacobian locus.

34.8 Separating and nonseparating plumbing

A separating plumbing identifies local coordinates $xy = q$ at punctures of two curves. In the state-space realization, the amplitude is

$$\mathcal{Z}_{C_1 \#_q C_2} = \sum_{\alpha} q^{\text{wt}(u_{\alpha})} \mathcal{Z}_{C_1}(\dots, u_{\alpha}) \mathcal{Z}_{C_2}(u^{\alpha}, \dots), \quad (230)$$

with the anomaly line carrying the vacuum-energy and coordinate factors. A nonseparating plumbing contracts two insertions on one curve with the same tensor (217).

Proposition 34.8 (Graph expansion of sewing). *Iterated plumbing expands a conformal block as a sum over stable graphs. Vertices carry lower-genus blocks, edges carry (217), and automorphism*

groups supply the standard symmetry factors. Absolute convergence follows in every region where the q^{L_0} sewing operators are trace class.

Proof. Apply (230) at every node. Choices of intermediate states label the half-edges. Completeness contracts the labels along each edge. Quotienting labelled gluings by graph automorphisms gives the symmetry factor. Trace-class bounds permit rearrangement of the sums. $\square$

34.9 The two genus variables

Let Θ carry a chiral modular sewing structure and a transverse cyclic E_1 structure. Denote the corresponding edge contractions by

$$\Delta_X, \quad \Delta_\perp.$$

The first forms nodes in the algebraic curve. The second closes open topological boundaries into ribbon handles.

Theorem 34.9 (Bigraded sewing equation). *Assume the two cyclic structures act on commuting colors of a two-colored modular operad and the determinant orientation lines are ordered as $\text{or}_X \otimes \text{or}_\perp$. Then*

$$\Delta_X^2 = 0, \quad \Delta_\perp^2 = 0, \quad \Delta_X \Delta_\perp + \Delta_\perp \Delta_X = 0,$$

and the combined master equation is

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_X + \frac{1}{2}[\Theta, \Theta]_\perp + \hbar_X \Delta_X \Theta + \hbar_\perp \Delta_\perp \Theta = 0.$$

Proof. Each square is the oriented codimension-two boundary identity for two nodes of the same color. A mixed pair of nodes has product normal line $N_X \oplus N_\perp$. Interchanging the two odd normal generators reverses the determinant orientation, so the mixed contractions anticommute. The master equation is the image of the total boundary differential under the supplied coefficient operations. $\square$

The two formal variables retain the physical distinction between complex-curve sewing and open-string ribbon topology. A duality relating them is represented by a map between their two modular operads.

Part VI

Exact mathematics and exact physics

35 Exact examples

35.1 Heisenberg

Let $\mathcal{H}_k$ be generated by a current J with

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \tag{231}$$

The chiral operation is concentrated in the second normal symbol of the diagonal. The corresponding ordinary Lie conformal bracket is

$$[J_\lambda J] = k\lambda.$$

The Jacobi identity is immediate because the bracket is central.

The chiral Chevalley coalgebra is cofree on the abelian chiral Lie algebra with the central extension built into the operation. Ordered monodromy is scalar. For vacuum and Fock-module insertions on a compact curve, the conformal blocks admit Fock-space descriptions and their partition functions are determinant expressions. The rank-one conformal vector has central charge one, independent of the nonzero normalization k after rescaling the current. This already separates level from central charge.

35.2 Affine Kac–Moody

For $V_k(\mathfrak{g})$ the current OPE is (140). The simple pole is the Lie bracket; the double pole is the invariant bilinear form. The chiral Jacobi identity splits into the ordinary Jacobi identity and invariance of the form:

$$([a, b], c) = (a, [b, c]).$$

The Chevalley differential formed from the simple-pole bracket squares to zero by itself. The level term supplies a central extension and enters conformal and representation-theoretic constructions.

At noncritical level the Sugawara vector has central charge

$$c_{\text{aff}} = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \quad (232)$$

The KZ connection and its quantum-group monodromy are described in Section 21. At positive integral level, the integrable modules of the simple affine VOA $L_k(\mathfrak{g})$ form a modular tensor category, and their conformal blocks satisfy the Verlinde formula.

At level one for simply laced $\mathfrak{g}$, the Frenkel–Kac lattice realization expresses the affine currents inside a lattice VOA [35]. The Frenkel–Kac realization preserves the nonabelian simple-pole bracket inside the lattice VOA.

35.3 Virasoro

The Virasoro vertex algebra is generated by T with

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (233)$$

The three singular terms lie in the fourth, second, and first normal symbols. Their relation is fixed by conformal covariance. The central charge c is the coefficient of the fourth-order symbol and the central extension of the vector-field algebra.

The vacuum Verma module has a level-four quasi-primary

$$\Lambda =: TT : - \frac{3}{10} \partial^2 T$$

whose norm is proportional to $c(5c + 22)$. The zeros at $c = 0$ and $c = -22/5$ signal null vectors. This Gram determinant is representation-theoretic data controlling the level-four null-vector locus; any

quartic deformation invariant requires a separately defined deformation complex and projector.

Conformal blocks carry a projective connection with Hodge coefficient $c/2$. At $c = 26$, coupling to the (b, c) ghost system of weights $(2, -1)$ cancels the total central charge. The value $c = 13$ is the fixed point of the numerical reflection $c \mapsto 26 - c$ used in bosonic string bookkeeping; a Virasoro self-duality theorem requires an OPE-compatible equivalence at that point.

35.4 Free fermions

A complex free fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The chiral operation lies in the first normal symbol. Wick's theorem determines all higher correlators as determinants. The representation category, spin structures, and modular transformation law depend on whether one uses a real, complex, or lattice extension.

Boson–fermion correspondence identifies the charged free fermion VOA with a rank-one lattice VOA. This is an isomorphism of vertex algebras after the lattice extension. A linear Koszul comparison between the Heisenberg and fermion chiral Lie algebras is a different construction.

35.5 The $\beta\gamma$ and bc systems

A first-order bosonic system has

$$\beta(z)\gamma(w) \sim \frac{1}{z-w},$$

while a fermionic bc system has the same singular kernel with odd parity. Their oscillator algebras are Weyl and Clifford algebras. For fields of weights $(\lambda, 1 - \lambda)$, the central charges are

$$c_{\beta\gamma} = 2(6\lambda^2 - 6\lambda + 1), \quad c_{bc} = -2(6\lambda^2 - 6\lambda + 1). \quad (234)$$

The opposite signs reflect statistics. The determinant lines cancel when the systems are paired with identical weights.

The full homotopical structure also contains the operations generated by composite fields and modules. Pole order and deformation cohomology are independent.

35.6 Lattice vertex algebras

Let L be an even positive-definite lattice. The lattice VOA V_L has central charge $\text{rk } L$. Its fields include Heisenberg currents and exponentials e^α . Their operator products contain powers

$$(z-w)^{(\alpha, \beta)}$$

and a cocycle $\epsilon(\alpha, \beta)$. Locality requires the commutator of the cocycle to match the parity determined by the lattice form.

If L is even unimodular, V_L is holomorphic. The character is (184). Inverting the cocycle gives the contragredient braiding; a symmetric cocycle may make the theory self-contragredient. This statement belongs to the lattice extension and its cocycle-reversal involution.

35.7 Leech and Moonshine

The Leech lattice VOA V_Λ and the Moonshine VOA $V^\natural$ both have central charge 24. Their weight-one spaces differ:

$$(V_\Lambda)_1 \cong \mathbb{C}^{24}, \quad (V^\natural)_1 = 0.$$

Their characters also differ in the constant term after the vacuum normalization. Thus the weight-one algebra, graded character, OPE, and factorization coalgebra refine central charge in the classification of holomorphic VOAs.

The Monster action on $V^\natural$ produces McKay–Thompson series. These are traces of automorphisms on the state space. They are arithmetic traces of Monster automorphisms on the VOA and become coefficients of any modular invariant only through the corresponding twining trace.

35.8 Principal W -algebras

The principal W -algebra $\mathcal{W}^k(\mathfrak{g})$ is obtained by BRST reduction. At generic level its standard strong generators have conformal weights equal to the exponents of $\mathfrak{g}$ plus one; special levels may impose decoupling relations or collapse generators. Their OPEs are nonlinear and contain normally ordered composites. The full chiral operation nevertheless satisfies the same Maurer–Cartan equation (116).

For $\mathfrak{sl}_2$, the principal W -algebra is Virasoro. For $\mathfrak{sl}_3$, it is generated by T and a weight-three field W ; the WW OPE contains T , its derivatives, and the quasi-primary Λ . These terms are fixed by Jacobi and conformal covariance. The WW OPE is determined by the complete family of structure coefficients and Jacobi identities.

Feigin–Frenkel duality uses (158). The central charges of dual pairs agree because the vertex algebras are isomorphic. This duality is distinct from KZ inverse braiding.

35.9 Critical affine algebras

At $k = -h^\vee$, the affine OPE remains (140), but the Sugawara conformal vector ceases to exist. The Feigin–Frenkel center becomes large and is identified with opers. KZ equations in their usual WZW normalization become singular. The critical level is therefore the structural locus where Sugawara conformal data degenerates and the oper centre emerges.

35.10 A comparison table

family	highest singular symbol	ordered structure	positive-genus structure
Heisenberg	second normal symbol k	scalar monodromy	determinant line and Fock sewing
affine $V_k(\mathfrak{g})$	bracket plus second symbol	KZ braid monodromy	WZW conformal blocks
Virasoro	fourth symbol $c/2$	conformal-block monodromy	projective anomaly $c\lambda/2$ plus boundary data
free first-order	first symbol	Weyl/Clifford statistics	determinant lines
lattice V_L	lattice powers and cocycle	lattice braiding	Siegel theta series
principal W	nonlinear finite OPEs	screening monodromy	W -conformal blocks when finite
critical affine	bracket and critical central extension	oper geometry replaces KZ	Feigin–Frenkel center

The table lists the structures and the comparison maps that distinguish the examples. Each row realizes the diagonal principle in a different representation category.

36 Strictly noncommutative chiral algebras

36.1 Free ordered chiral algebras

Let $M \in \text{FactD}(X)$. The free augmented ordered chiral algebra is

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus M^{\otimes 3} \oplus \dots, \quad (235)$$

with concatenation. The tensor products are taken in $\text{FactD}(X)$, so each word retains the full chiral coefficient system of its letters.

Proposition 36.1 (Bar homology of the free algebra). *Assume the tensor products in (235) are exact. The reduced bar of $T_X(M)$ contracts onto $M[1]$. Equivalently,*

$$\mathbf{1} \otimes_{T_X(M)}^L \mathbf{1} \simeq \mathbf{1} \oplus M[1]$$

as a conilpotent coalgebra, with zero reduced coproduct on $M[1]$.

Proof. The word-length decomposition gives a fibre sequence of left $T_X(M)$ -modules

$$T_X(M) \otimes M \xrightarrow{\text{mult}} T_X(M) \longrightarrow \mathbf{1}.$$

On the summand of words of positive length, multiplication is the canonical identification obtained by separating the last letter; the cofiber is the empty word. Applying $\mathbf{1} \otimes_{T_X(M)}^L -$ leaves the two-term complex $M \rightarrow \mathbf{1}$ with zero reduced differential, hence the class $M[1]$. Equivalently, the standard extra degeneracy of the normalized bar contracts every bar word having an internal cut and leaves the one-letter cogenerator. $\square$

Thus the Koszul dual algebra of a free associative object is the square-zero extension

$$T_X(M)^! \simeq \mathbf{1} \oplus M^\vee[-1]$$

when M is dualizable. Noncommutative freedom on one side becomes a first-order infinitesimal object on the other.

36.2 Square-zero chiral algebras

Conversely, let

$$A = \mathbf{1} \oplus M, \quad M^2 = 0.$$

The bar multiplication differential vanishes, and

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = \bigoplus_{r \geq 0} M[1]^{\otimes r}$$

with deconcatenation. Its continuous dual is the completed tensor algebra

$$A^! \simeq \widehat{T}_X(M^\vee[-1]).$$

This pair is the local model of an infinitesimal boundary coupling and its algebra of iterated defect insertions.

36.3 Quiver chiral algebras

Let Q be a finite quiver, let

$$S = \bigoplus_{i \in Q_0} \mathbf{1}e_i,$$

and assign an object $M_a \in \mathrm{FactD}(X)$ to every arrow $a : i \rightarrow j$. Put

$$M = \bigoplus_{a \in Q_1} e_{t(a)} M_a e_{s(a)}.$$

The relative tensor algebra

$$A_Q = T_S(M) \tag{236}$$

is an E_1 algebra in factorization D -modules. Multiplication concatenates composable arrows and is zero otherwise. The augmentation $A_Q \rightarrow S$ forgets nontrivial paths.

The relative bar

$$B_S(A_Q) = S \otimes_{A_Q}^L S$$

retains the ordered path structure. If Q is free over S , it contracts to $S \oplus M[1]$. If relations $R \subset M \otimes_S M$ are imposed, the quadratic coalgebra begins

$$S \oplus M[1] \oplus R[2] \oplus \cdots.$$

Example 36.2 (Two boundary conditions). For the quiver $1 \rightleftarrows_u^v 2$, the objects M_u and M_v are boundary-changing fields. The composites uv and vu are endomorphisms of different boundaries and belong to different vertex endomorphism spaces and are independently specified. Their chiral insertion

points may collide on X , while the order of composition is fixed by the transverse interval. This is strict noncommutative chirality in its smallest form.

A cyclic potential W determines a cyclic functional on the completed path algebra. After choosing the standard Ginzburg or cyclic A_∞ construction, its cyclic derivatives enter the differential and are the classical equations of motion of the corresponding open-string field theory. The ribbon genus of Section 26 is then the genus of the resulting open-string Feynman surface.

36.4 Matrix and Chan–Paton extensions

Let V be a commutative algebra object in $\text{FactD}(X)$ and let E be a finite-dimensional vector space. The matrix extension

$$A = \text{End}(E) \otimes V \quad (237)$$

is an E_1 algebra in $\text{FactD}(X)$. Its multiplication is matrix multiplication times the chiral factorization product. The OPE coefficients are $\text{End}(E)$ -valued and preserve their order.

The algebra (237) is Morita equivalent to V . Hence

$$Z_{\text{ord, ch}}(A) \simeq Z_{\text{ord, ch}}(V)$$

and their perfect module categories have equivalent centres. The open amplitudes differ by Chan–Paton traces, producing powers of $\dim E$ on ribbon graphs.

Proposition 36.3 (Chan–Paton weight). *For a connected ribbon graph Γ contributing to a cyclic amplitude of (237), contraction of matrix indices contributes*

$$(\dim E)^{F(\Gamma)},$$

where $F(\Gamma)$ is the number of index faces. For a closed ribbon surface of genus h with b boundary components, this is $(\dim E)^{2-2h-b}$ after the standard normalization of vertices and propagators.

Proof. Write every propagator and vertex in matrix indices. Along each ribbon edge the two Kronecker symbols identify adjacent indices. Following an index strand around a closed face leaves one freely summed index and contributes $\dim E$. Independent faces give the product $(\dim E)^{F(\Gamma)}$. With the conventional factor $(\dim E)^{-1}$ for each normalized propagator and the matching vertex normalization, Euler’s identity $V - E + F = 2 - 2h - b$ gives $(\dim E)^{2-2h-b}$. $\square$

This is the exact algebraic origin of the ’t Hooft expansion. It belongs to the transverse ribbon genus and is independent of the genus of the chiral curve.

36.5 Universal enveloping ordered chiral algebras

Let $\mathfrak{g}$ be a Lie algebra object in $\text{FactD}(X)$. Its universal enveloping algebra

$$U_X(\mathfrak{g}) \in \text{Alg}_{E_1}(\text{FactD}(X))$$

is defined by the usual coequalizer of the tensor algebra by

$$x \otimes y - (-1)^{|x||y|} y \otimes x - [x, y].$$

The Poincare–Birkhoff–Witt filtration has

$$\mathrm{gr} U_X(\mathfrak{g}) \simeq \mathrm{Sym}_X(\mathfrak{g})$$

under the standard flatness hypotheses.

Theorem 36.4 (Associative bar and internal Chevalley chains). *For a flat Lie object $\mathfrak{g}$ in $\mathrm{FactD}(X)$, the canonical twisting morphism induces a quasi-isomorphism*

$$\mathrm{Chev}_X(\mathfrak{g}) \xrightarrow{\sim} \mathrm{Bar}_X^{\mathrm{ord}}(U_X(\mathfrak{g})).$$

Proof. Filter both sides by symmetric degree. The associated graded map is the ordinary Koszul resolution of $\mathrm{Sym}_X(\mathfrak{g})$ by the exterior coalgebra on $\mathfrak{g}[1]$. PBW identifies the associated graded of the enveloping algebra. Completeness and exactness of the filtration lift the associated-graded quasi-isomorphism. $\square$

This theorem is the internal Lie–associative Koszul comparison in the pointwise coefficient category. It is distinct from the Beilinson–Drinfeld chiral Chevalley construction on the Ran space. The internal Lie bracket maps to the commutator in its enveloping algebra, and internal Lie chains compute the ordered bar of that enveloping algebra.

36.6 Affine currents as chiral coefficients

The affine current algebra is a Beilinson–Drinfeld chiral Lie algebra: its simple pole is the Lie bracket and its double pole is the invariant cocycle. Its chiral enveloping construction produces the universal affine chiral coefficient. This construction takes place in the chiral pseudo-tensor category and is distinct from the internal enveloping algebra of Theorem 36.4, which begins with a Lie object for the pointwise tensor product.

A strictly noncommutative ordered chiral algebra from affine data consists of a transverse associative object: for example an endomorphism algebra of affine boundary conditions, a quiver of affine modules, or a free tensor algebra generated by the affine factorisation coefficient. The resulting internal bar resolves that transverse multiplication. The chiral Chevalley complex resolves the affine chiral Lie operation. A comparison between them is an enveloping or Morita theorem and is provided by an enveloping or Morita theorem. In either construction the simple-pole bracket satisfies Jacobi and invariance of the level cocycle makes the central extension chiral-Lie.

36.7 Weyl and Clifford chiral algebras

Let M carry a perfect pairing $\omega : M \otimes M \rightarrow \mathbf{1}[d]$. The tensor algebra modulo

$$xy - (-1)^{|x||y|}yx = \hbar \omega(x, y) \tag{238}$$

produces a Weyl or Clifford object according to parity. In the chiral category, M may be a factorization D -module and ω may contain a higher normal symbol along the diagonal.

The relation is quadratic-linear-constant. Its Koszul dual is curved: the constant term becomes the curvature m_0 of the dual coalgebra. The curvature belongs to the algebraic presentation. A Hodge class on moduli enters through a separate family-index or sewing construction.

Example 36.5 (The $\beta\gamma$ system). The chiral Weyl algebra generated by β and γ has singular relation

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}.$$

With matrix Chan–Paton factors, the ordered product is matrix multiplication while the singular coefficient is the Weyl pairing. The two operations are visibly independent.

Example 36.6 (The bc system). The odd analogue has Clifford relation. Its determinant line contributes the ghost conformal anomaly. The Weyl–Clifford Koszul exchange is a statement about quadratic relations; the cancellation of their conformal anomalies is a separate determinant-line calculation.

36.8 Chiral differential operators

For a smooth variety Y , a sheaf of chiral differential operators exists after its obstruction gerbe has been trivialized. It supplies a chiral factorisation coefficient whose local fields include functions and vector fields on Y . The vertex or chiral operation of that coefficient becomes the transverse E_1 multiplication after a factorization-compatible associative product has been specified.

Strict ordered examples are obtained from associative endomorphism objects in a category of chiral-differential-operator modules, from quiver endomorphisms of boundary conditions, or from matrix and tensor-algebra extensions of a dualizable coefficient. When Y carries a group action, BRST or quantum Hamiltonian reduction acts on such an ordered endomorphism algebra provided the BRST operator is a derivation of the transverse product. The resulting comparison is governed by Section 24.

36.9 Boundary endomorphism algebras

Let $\mathcal{B}$ be a factorization category of boundary conditions on X . For objects b_i , set

$$A_{ij} = \mathrm{RHom}_{\mathcal{B}}(b_i, b_j).$$

Composition gives

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The direct sum

$$A_{\mathcal{B}} = \bigoplus_{i,j} A_{ij}$$

is an algebra over the semisimple idempotent object $S = \oplus_i \mathbf{1}e_i$. Its relative bar computes chains of boundary changes. This is the categorical form of the quiver example and the native algebra of open strings.

Theorem 36.7 (Morita reconstruction from a boundary generator). *Let $\mathcal{B}$ be a stable presentable $\mathrm{FactD}(X)$ -module category whose action preserves colimits separately, and let $b = \oplus_i b_i$ be a compact generator. Then*

$$A_b = \mathrm{RHom}_{\mathcal{B}}(b, b)$$

is an E_1 algebra in $\mathrm{FactD}(X)$ and

$$\mathcal{B} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

by the enriched Hom functor.

Proof. Composition makes A_b an E_1 algebra in the enriched coefficient category. The adjunction

$$- \otimes_{A_b} b \dashv \mathrm{RHom}_{\mathcal{B}}(b, -)$$

has equivalence unit on the free module A_b and equivalence counit on the generator b . Both functors are exact and preserve the colimits used to form the presentable categories. Since b and A_b compactly generate their respective sides, the unit and counit are equivalences on every object. $\square$

The theorem explains why noncommutative chiral algebras are primary in boundary physics. The category of boundary conditions comes first; its endomorphism algebra is a coordinate chart on that category.

36.10 Factorization quantum groups

A factorization quantum group combines an algebra object, a compatible coalgebra or tensor-category structure, and meromorphic exchange. In the ordered language these pieces occupy different levels:

E_1 multiplication	: fusion,
factorization coproduct	: splitting of lines or representations,
E_2 enhancement	: exchange,
ribbon trace	: closure and link invariants.

A Yangian possesses all these structures in a compatible completed sense. A generic ordered chiral algebra possesses only the first.

36.11 A noncommutative Koszul pair

Let V be dualizable and let $R \subset V \otimes V$ be a factorization subobject. Define

$$A = T_X(V)/(R), \quad A^{\mathrm{l}, \mathrm{quad}} = T_X(V^\vee[-1])/(R^\perp).$$

The canonical twisting morphism $A^i \rightarrow A$ gives the Koszul complex

$$K(A) = A \otimes A^i.$$

Theorem 36.8 (Quadratic recognition in the ordered chiral category). *Assume positive weight, degreewise dualizability, and strong convergence of the weight filtration. The following are equivalent:*

- (i) $A^i \rightarrow \mathrm{Bar}_X^{\mathrm{ord}}(A)$ is a quasi-isomorphism;
- (ii) $\mathrm{Cobar}_X(A^i) \rightarrow A$ is a quasi-isomorphism;
- (iii) the left and right quadratic Koszul complexes are acyclic off weight zero;
- (iv) $\mathrm{Tor}^A(\mathbf{1}, \mathbf{1})$ is concentrated on the quadratic diagonal.

Proof. The quadratic twisting morphism defines the left and right twisted tensor products and the comparison maps in (i) and (ii). Component evaluation in $\mathrm{FactD}(X)$ preserves tensor products, the

quadratic subcoalgebra, and the weight filtration. In each finite weight the fundamental theorem of Koszul twisting morphisms identifies (i), (ii), and acyclicity of both twisted tensor products. The bar spectral sequence identifies their homology with $\mathrm{Tor}^A(\mathbf{1}, \mathbf{1})$, so diagonal concentration is equivalent to the same acyclicity. Strong convergence and conservativity of component evaluation lift these equivalences to the ordered chiral category. $\square$

When these conditions hold and finite duals exist, dualizing the quadratic coalgebra identifies $A^{!,\mathrm{quad}}$ with the boundary endomorphism algebra $\mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$.

36.12 What these examples establish

The examples exhibit three independent kinds of noncommutativity.

Quiver and matrix algebras have noncommutative transverse multiplication even when their chiral coefficient is free or commutative.

Internal enveloping algebras have noncommutative transverse multiplication generated by an internal Lie object. Affine current algebras supply chiral coefficients; a strictly ordered affine example requires the additional boundary, quiver, tensor, or endomorphism multiplication described above.

Yangians and line categories add meromorphic braid exchange to noncommutative fusion.

The ordered chiral language accommodates all three through their typed comparison maps. That separation is the condition under which their Koszul dualities can be stated correctly.

37 Arithmetic traces and automorphic forms

37.1 Arithmetic enters through periods and traces

The chiral operation is defined over an algebraic or analytic coefficient field. Arithmetic appears when one takes periods of configuration spaces, traces of monodromy, dimensions of conformal blocks, theta sums over lattices, or regularized determinants. These operations are functorial shadows. Their arithmetic is controlled by the geometry of the chosen trace.

There are two basic sources. Ordered iterated integrals produce multiple zeta values and their elliptic analogues. Symmetric and modular traces produce modular forms, theta series, and characteristic numbers. The two sources can interact, and interact through explicit period and trace maps.

37.2 The KZ associator

The regularized holonomy of the KZ connection on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ is the Drinfeld associator. Its coefficients are multiple zeta values. The weight filtration is the length of iterated integration. The pentagon equation expresses compatibility around the five degenerations forming the boundary of a real associahedral pentagon inside $\overline{\mathcal{M}}_{0,5}$; the complex Deligne–Mumford boundary itself has ten irreducible divisors.

This is a direct arithmetic consequence of ordered configuration geometry. Passing to the Grothendieck ring of the representation category discards the associator but retains fusion multiplicities. Thus ordered and symmetric traces have visibly different arithmetic content.

37.3 Verlinde numbers

For a modular tensor category with simple objects indexed by λ and modular S -matrix, the dimension of the genus- g conformal-block space is

$$\dim \mathbb{V}_g(\lambda_1, \dots, \lambda_n) = \sum_{\mu} (S_{0\mu})^{2-2g-n} \prod_{i=1}^n S_{\lambda_i \mu}. \quad (239)$$

For $\widehat{\mathfrak{sl}}_2$ at level k with the vacuum insertion,

$$\dim \mathbb{V}_g = \left(\frac{k+2}{2} \right)^{g-1} \sum_{j=1}^{k+1} \sin \left(\frac{\pi j}{k+2} \right)^{2-2g}. \quad (240)$$

The cosecant power sums are polynomials in $k+2$ with Bernoulli-number coefficients. This arithmetic follows from the modular tensor category and the geometry of flat connections. A bar-cohomology dimension formula requires a separate identification of the relevant bar complex with this conformal-block trace.

37.4 Hodge integrals

Faber and Pandharipande evaluate the one-point Hodge integrals

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g. \quad (241)$$

Their generating function is

$$1 + \sum_{g \geq 1} \left(\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g \right) t^{2g} = \frac{t/2}{\sin(t/2)}. \quad (242)$$

This identity is a theorem of the tautological geometry of moduli [33]. It contributes to any field theory whose stable-graph trace reduces to these Hodge integrals. A chiral algebra acquires this genus expansion when its stable-graph trace factors through these Hodge integrals.

37.5 Theta series

For an even lattice L , the degree- g Siegel theta series is

$$\Theta_L^{(g)}(\Omega) = \sum_{x_1, \dots, x_g \in L} \exp \left(\pi i \sum_{a,b} (x_a, x_b) \Omega_{ab} \right). \quad (243)$$

It is a Siegel modular form when L is even unimodular. The genus- g partition function of a lattice theory contains (243) multiplied by an oscillator determinant.

Theta series are traces over zero modes. They forget the cocycle and many OPE coefficients. Nonisometric lattices may have equal theta series in low genus. The Schottky form illustrates the first place at which the two rank-sixteen even-unimodular lattice theta series differ on the full Siegel space, while their restrictions to the genus-four Jacobian locus agree [59, 66].

37.6 Symmetric products and the DMVV formula

Let $\phi_M(\tau, z)$ be the elliptic genus of a compact Calabi–Yau manifold M . Dijkgraaf–Moore–Verlinde–Verlinde prove the product formula

$$\sum_{N \geq 0} p^N \phi_{\text{Sym}^N M}(\tau, z) = \prod_{n > 0, m \geq 0, \ell \in \mathbb{Z}} \left(1 - p^n q^m y^\ell\right)^{-c(nm, \ell)}, \quad (244)$$

where $c(m, \ell)$ are the Fourier coefficients of ϕ_M [26]. The logarithm of the product is a sum of Hecke transforms. This is the plethystic exponential of connected single-particle data.

The formula has the same combinatorial exponential structure as a cofree commutative coalgebra and as the connected-graph identity (181). The equality of combinatorial forms becomes an identification of the DMVV state space with a chiral Chevalley coalgebra. after constructing a functor from the sigma-model or BPS category to chiral factorization data.

37.7 Borchers products

A weak Jacobi form with integral Fourier coefficients can produce an automorphic product whose exponents are those coefficients. Gritsenko and Nikulin use such products to construct automorphic corrections of Lorentzian Kac–Moody algebras [44]. The Igusa cusp form Φ_{10} is the multiplicative lift relevant to the K3 elliptic genus, and the DMVV formula is naturally expressed using $1/\Phi_{10}$.

The weight of a Borchers product is half the constant coefficient of its input in the standard normalization. This weight is an automorphic invariant. The central charge, lattice rank, and Koszul data retain their own definitions and can be compared to the automorphic weight through explicit index formulas.

37.8 K3 and geometric quantum groups

Maulik and Okounkov construct Yangians from the geometry of Nakajima quiver varieties using stable envelopes and R -matrices [56]. The cohomology of Hilbert schemes of points on a surface carries Heisenberg and, in suitable settings, larger quantum-group actions. Braverman–Finkelberg–Nakajima construct Coulomb-branch algebras that include shifted Yangians [10].

These are exact geometric constructions of quantum groups. A K3 surface also supplies a lattice of signature $(4, 20)$, elliptic genera, and Hilbert-scheme Fock spaces. An identification of a K3 Hall algebra with the ordered Koszul dual of a particular chiral algebra. is established by a functor preserving products, coproducts, pairings, completions, and R -matrices.

37.9 Moonshine

Moonshine associates modular functions to traces of Monster elements on $V^\natural$. Borchers’ proof uses a generalized Kac–Moody algebra and an automorphic denominator identity. The chiral algebra supplies the graded representation and its vertex operators; the denominator identity supplies an automorphic trace.

Chiral Koszul duality applies to the local chiral operation of $V^\natural$. Moonshine is additional equivariant and arithmetic structure. A trace remembers the conjugacy class and graded multiplicities while the complete OPE is recovered from all vertex-operator products.

37.10 Genus one and genus two

Genus one detects characters, modular transformations, and one-variable theta series. Genus two detects interactions between two cycles and introduces Siegel modular forms. Some distinctions first appear only on the full Siegel space and disappear on the Jacobian locus. Therefore the phrase “genus two detects” has mathematical content only after the function, the ambient moduli space, and the restriction locus have been named.

The same precision applies to L -functions. A modular form determines an L -function; a graph amplitude equals a special value after an integral representation or Rankin–Selberg unfolding has identified the graph amplitude with that special value. An operator-product coefficient becomes an L -value through this explicit period map.

37.11 Arithmetic duality

Dualizing a local system sends its period matrix to the contragredient one. Poincaré duality pairs homology and cohomology periods. Functional equations of L -functions reflect duality of motives. These are distinct dualities, connected only when the relevant realization functors commute.

The common language is a diagram of realizations:

$$\text{algebraic coefficient object} \longrightarrow \text{de Rham local system} \longrightarrow \text{Betti monodromy} \longrightarrow \text{period or trace}.$$

Chiral Koszul duality acts at the first stage. Arithmetic enters at the final stages.

38 Quantum field theory in the same language

38.1 Observables as a factorization algebra

A perturbative quantum field theory assigns a cochain complex of observables to each open set. Observables with disjoint supports multiply. Locality and descent make these complexes a factorization algebra [14, 15]. The differential combines the equations of motion, gauge symmetry, and quantum corrections.

A holomorphic theory on a complex curve produces a holomorphic factorization algebra. Under suitable regularity, its local observables determine a vertex algebra or chiral algebra. A topological–holomorphic theory on $M \times X$ may produce a chiral algebra on a boundary parallel to X . The OPE is the short-distance expansion of the factorization product.

38.2 Renormalization and compactified configurations

A Feynman graph with vertices at points of a manifold defines a differential form or distribution on the configuration space of distinct vertices. Divergences occur at collision diagonals. Compactifying the configuration space resolves these diagonals into boundary faces indexed by forests. Renormalization extends the weight to the boundary by adding local counterterms.

The forest formula and the Fulton–MacPherson forest resolution have the same incidence category. The coefficient systems differ: a Feynman weight contains propagators, gauge-fixing data, and densities; the chiral Chevalley complex contains the diagonal operation. A comparison theorem identifies them when the boundary asymptotics of the propagator reproduce the chiral operation.

38.3 Classical and quantum master equations

Let $\mathcal{E}$ be the BV complex of fields with odd Poisson bracket $\{-, -\}$. A classical action satisfies

$$QS + \frac{1}{2}\{S, S\} = 0.$$

A quantization satisfies

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0$$

after renormalization. The anomaly is the cohomology class of the quantum-master defect.

The modular equation (178) has the same form because both are generated by graph contractions. The identification is exact when the Feynman rules define an algebra over the same modular cooperad. The boundary strata of compactified configurations then prove the master equation by Stokes' theorem.

38.4 Chern–Simons theory and affine currents

Chern–Simons theory on a three-manifold with the standard chiral boundary condition induces a chiral Wess–Zumino–Witten boundary theory. The boundary currents form an affine Kac–Moody algebra. In the abelian theory, the boundary current OPE is proportional to the level and is computed directly from the Green kernel. In the nonabelian theory, one-loop effects produce the familiar shifted coupling in KZ and surgery formulas.

Gwilliam and Williams construct a one-loop exact quantization of a mixed holomorphic–topological presentation of Chern–Simons theory and analyze its boundary behavior [45]. The factorization algebra of boundary observables is the analytic counterpart of the affine chiral algebra.

Wilson lines form a braided tensor category. Their exchange is described by quantum-group R -matrices; their closure produces knot invariants. This is the physical realization of Theorems 20.2 and 21.2.

38.5 Four-dimensional Chern–Simons and integrability

In four-dimensional Chern–Simons theory, line operators are topological along a surface and holomorphic in a spectral parameter. Their OPE produces rational, trigonometric, or elliptic R -matrices depending on the curve. The Yang–Baxter equation follows from moving three lines through an isotopy in the regular configuration space. Transfer matrices commute because separated line networks are isotopic.

The Gaudin model arises from the semiclassical rational connection. Spin chains arise from choosing representations on parallel lines. Baxter relations are exact sequences or functional relations in the representation category. These structures are ordered because the sequence and positions of lines matter.

38.6 BRST, ghosts, and strings

Gauge fixing introduces ghosts and a BRST differential. The total central charge of matter and ghosts controls the conformal anomaly of the worldsheet theory. For the bosonic string, the (b, c) ghost system has central charge -26 ; BRST nilpotence requires matter central charge 26 together with the usual normal-ordering or intercept condition in the state-space formulation.

Polyakov’s path integral produces determinant lines over moduli. The Weyl anomaly is the curvature of these determinant lines. The Belavin–Knizhnik theorem and the theory of Quillen metrics make this relation geometric. The cancellation of matter and ghost determinant lines is the global form of the central-charge condition.

This anomaly cancellation is a tensor-product cancellation in the conformal-block and determinant-line theory.

38.7 Brown–Henneaux and Cardy

Three-dimensional gravity with negative cosmological constant has asymptotic Virasoro symmetry with

$$c = \frac{3\ell}{2G} \quad (245)$$

by the Brown–Henneaux theorem [12]. In a unitary modular-invariant two-dimensional CFT with a sparse light spectrum and vacuum dominance, the Cardy formula gives the high-energy density of states. Substituting (245) reproduces the Bekenstein–Hawking entropy of the BTZ black hole in the semiclassical regime.

The rigorous algebraic input is the boundary Virasoro algebra and modular covariance of its partition function. A nonperturbative theory of three-dimensional gravity additionally requires a spectrum, integration cycle, modular completion, and a bulk–boundary comparison with the Virasoro centre.

38.8 Entanglement in two-dimensional conformal field theory

For the vacuum of a unitary CFT on the line, the entanglement entropy of an interval of length L with ultraviolet cutoff ϵ is

$$S_A = \frac{c + \bar{c}}{6} \log \frac{L}{\epsilon} + \text{const.} \quad (246)$$

The replica derivation uses twist fields and conformal maps. The coefficient is fixed by the left- and right-moving stress-tensor anomalies; in a parity-invariant theory with $c = \bar{c}$ it reduces to $c/3$.

Equation (246) is a trace statement. The OPE, braid monodromy, and representation category provide the complementary operator-algebraic data. Two CFTs with the same c may have different entanglement spectra and operator algebras.

38.9 Bulk–boundary maps

A boundary condition for a factorization algebra of bulk observables defines an action on boundary observables. By the universal property of the derived center, there is a map

$$\text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ch}}^{\text{der}}(A).$$

Costello and Gaiotto use this structure in twisted holography, where protected bulk sectors and boundary vertex algebras can sometimes be compared exactly [16]. The comparison is a theorem of the particular twisted theory, a theorem of the specified twisted bulk–boundary system.

A decoupled bulk factor lies in the homotopy fibre of boundary restriction. A fixed boundary algebra may therefore admit several bulk completions with different global observables. The centre classifies their central boundary actions, while the homotopy fibre records the additional bulk sectors.

38.10 Celestial chiral algebras

In celestial and twistor constructions, scattering amplitudes can be encoded by correlators of chiral algebras. Costello–Paquette and collaborators construct examples in which gauge-theory amplitudes arise from current-algebra correlators and their deformations [17]. Associativity of the celestial OPE is then a nontrivial constraint on loop corrections.

The precise statement depends on the background geometry, asymptotic states, and the map from scattering observables to chiral fields. Soft theorems provide currents in the chiral algebra, and a theory-specific comparison map identifies the resulting current insertions with the gravitational soft expansion.

38.11 Calabi–Yau categories and topological strings

A Calabi–Yau A_∞ category has a cyclic pairing. Costello’s construction associates to suitable Calabi–Yau categories a two-dimensional topological conformal field theory whose closed states are Hochschild chains. The Connes operator and the cyclic pairing supply the BV structure.

A holomorphic field theory on a Calabi–Yau variety may produce factorization algebras and BPS operator algebras. Cohomological Hall algebras, shifted Yangians, and Borchers products arise in specific geometries. Passing from such a category to a chiral algebra on a curve requires an explicit compactification, reduction, or boundary functor. The dimension shift and cyclic pairing supply inputs to the functor once the compactification, reduction, and boundary morphisms are specified.

38.12 The physical comparison diagram

The exact relation between the mathematical and physical objects is a commutative diagram of maps, a commutative diagram in the relevant categories:

$$\begin{array}{ccc}
 \text{coChev}^{\text{ch}}(\mathfrak{g}) & \xrightarrow{\quad\quad\quad} & \text{algebraic factorization coalgebra} \\
 \downarrow & & \downarrow \\
 \text{renormalized collision complex} & \xrightarrow{\quad\quad\quad} & \text{Obs}_\partial
 \end{array} \tag{247}$$

The bulk-to-boundary action has the separately typed form

$$\text{Obs}_{\text{bulk}}|_\partial \longrightarrow Z_{\text{ch}}^{\text{der}}(\text{Obs}_\partial). \tag{248}$$

The vertical maps are Riemann–Hilbert, analytification, or explicit field-theoretic comparison maps. When they are quasi-isomorphisms, the mathematical and physical master equations are the same equation in two models.

39 The physical derivation of ordered chirality

39.1 A local experiment

Consider a quantum field theory on

$$X \times \mathbb{R}_t,$$

holomorphic along the complex curve X and topological along the oriented coordinate t . Let $\mathcal{O}_i(z_i, t_i)$ be local observables with

$$t_1 < t_2 < \cdots < t_n.$$

Topological invariance permits the t_i to move inside a fixed order chamber while preserving the correlator. Holomorphy permits the z_i to move on the complement of collision diagonals. Hence a correlator is locally constant in the ordered chamber of the t_i and meromorphic or distributional in the z_i .

The short-distance product has the form

$$\mathcal{O}_1(z_1, t_1) \cdots \mathcal{O}_n(z_n, t_n) \sim \sum_{\alpha} C_{\alpha}(z_1, \dots, z_n) \mathcal{O}_{\alpha}(z, t), \quad (249)$$

where the coefficient C_{α} is a chiral distribution and the multiplication order is the order of the t_i . Formula (249) is the physical content of an E_1 algebra in factorization D -modules.

Principle 39.1 (Ordered OPE principle). *Holomorphic collision determines the coefficient distribution. Topological ordering determines the associative word. The full product is their interchange law.*

Proof. Choose pairwise disjoint product disks $U_i \times I_i \subset X \times \mathbb{R}$ with $I_1 < \cdots < I_n$. The factorisation map first multiplies observables according to the linear order of the intervals and then transports the result along the holomorphic configuration of the U_i . Shrinking the U_i toward a diagonal produces the chiral coefficient distribution, while isotopy of the ordered intervals preserves the word. Associativity of factorisation and compatibility with products of embeddings give the interchange identity. $\square$

This principle is more primitive than a vertex-algebra expansion. A local coordinate on X converts the coefficient distribution into modes. Choosing an ordering prescription on $\mathbb{R}$ converts the topological factor into a time-ordered product.

39.2 Topological descent and the E_1 product

Let Q be the cohomological supercharge of the twist. Topological invariance in the real direction means that the translation generator is Q -exact:

$$P_t = [Q, G_t].$$

For a Q -closed observable $\mathcal{O}$,

$$\partial_t \mathcal{O} = [Q, G_t \mathcal{O}].$$

The difference between insertions at two values of t is therefore Q -exact. Ordered products are independent of the positions on cohomology.

The identity $P_t = [Q, G_t]$ is the first descent equation. A chain-level E_1 structure requires the coherent extension of this identity to families of interval embeddings: the descendant operators must satisfy the Cartan homotopy relations and their higher compatibility on compactified configuration cells. Integrating the coherent descendants gives the A_{∞} operations. The boundary of an associahedron is the sum of its parenthesized degenerations, so Stokes' theorem gives the Stasheff identities.

Theorem 39.2 (Descent produces ordered multiplication). *Let a factorization algebra of observables on $X \times \mathbb{R}$ be holomorphic in X and equipped with coherent topological descent in the real direction, so*

that its restriction to intervals is a locally constant factorization algebra. Then restriction to product disks determines an E_1 algebra in holomorphic factorization observables on X . A local operator G_t with $P_t = [Q, G_t]$ exhibits the first coherence; when it extends to the stated descent datum, the product on Q -cohomology is independent of the transverse positions.

Proof. Locally constant factorisation algebras on an oriented line are algebras over the little-interval operad. For each disk $U \subset X$, restriction to $U \times \mathbb{R}$ therefore gives an E_1 algebra. Holomorphic factorisation in U makes these algebras a factorisation object on X , and factorisation on product embeddings makes the E_1 operations morphisms of that object. The descent identity integrates to

$$\mathcal{O}(t_2) - \mathcal{O}(t_1) = [Q, \int_{t_1}^{t_2} G_t \mathcal{O} dt],$$

so ordered insertions at different points of one chamber define the same cohomology class. Higher Cartan homotopies extend this equality over compactified interval configurations; Stokes' formula on their cells is the collection of Stasheff identities. $\square$

The theorem is the physical proof of Theorem 10.2 for a cohomological field theory.

39.3 Why the OPE is a D -module

The coefficient in (249) is a distribution with a supported singular part at the collision locus. Ward identities imply differential equations in the insertion points. Contact terms are distributions supported on diagonals. A right D -module records both the differential equations away from collisions and the allowed supported terms at collisions.

For a conserved holomorphic current,

$$\bar{\partial}J = 0$$

away from other insertions. In a correlator,

$$\bar{\partial}_z \frac{1}{z - w} = \pi \delta^{(2)}(z - w)$$

in the distributional normalization. Thus the singular OPE and the contact Ward identity are the same distribution seen before and after applying $\bar{\partial}$. The D -module language retains this equality intrinsically, before any choice of Laurent contour.

39.4 Boundary conditions and open strings

Let the theory occupy $X \times \mathbb{R}_{\geq 0}$ and let b be a boundary condition at $t = 0$. Boundary operators form the endomorphism algebra of b . If several boundary conditions b_i are present, a boundary-changing field lies in

$$A_{ij} = \mathrm{RHom}(b_i, b_j).$$

Composition has the order

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

This is the open-string product. The insertion point on X supplies its chiral dependence.

The strip with boundary conditions b_i and b_j has state complex A_{ij} . Gluing strips is composition. Closing a strip into an annulus gives Hochschild homology when a cyclic trace and the required

dualizability are present. Thickening the Feynman graphs of a cyclic open-string interaction then gives ribbon surfaces. These hypotheses are exactly the transverse modular data of Section 26.

39.5 The interval bar as a propagator-independent amplitude

Place the augmentation boundary at the two ends of an interval. Factorization homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes_A^L \mathbf{1}.$$

A cellular decomposition of the interval yields the bar complex. Its differential composes adjacent operator insertions. This construction is the exact topological interval gluing of the E_1 algebra.

A Feynman expansion is another model of the same derived tensor product when the field theory admits an action and a gauge fixing. The comparison map sends a bar word to the corresponding time-ordered amplitude. Its proof is a homological perturbation or factorization-excision theorem, an explicit comparison between time-ordered amplitudes and simplicial interval gluing.

39.6 A second topological direction and exchange

Now place the theory on $X \times \mathbb{R}^2$. Two parallel lines can move around one another. The exchange path defines

$$R_{12}(z_1, z_2) : L_{z_1} \otimes L_{z_2} \longrightarrow L_{z_2} \otimes L_{z_1}.$$

Three-line isotopy gives the Yang–Baxter equation. The exchange depends meromorphically on $z_1 - z_2$ because a singularity occurs when the spectral points collide.

This derivation explains the codimension hierarchy:

$$\begin{aligned} \mathbb{R}^1 & : E_1 \text{ fusion,} \\ \mathbb{R}^2 & : E_2 \text{ braid exchange,} \\ \mathbb{R}^3 & : E_3 \text{ higher linking operations.} \end{aligned}$$

The chiral curve is a coefficient direction and contributes an additional little-disks direction when its holomorphic translations become topological.

39.7 Four-dimensional Chern–Simons theory

The action

$$S[A] = \frac{1}{2\pi\hbar} \int_{\Sigma \times X} \omega \wedge \left(\frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A]) \right)$$

is topological on the oriented surface Σ and holomorphic on the curve X . A Wilson line lies on a curve in Σ and at a point of X . The point of X is its spectral parameter.

At tree level, exchange of two Wilson lines gives the Casimir kernel

$$r(z) = \frac{\Omega}{z}.$$

The configuration of three lines has two isotopic resolutions, which gives the classical Yang–Baxter equation. Quantization replaces r by an R -matrix. The one-loop anomaly of line gauge invariance forces the Yangian deformation in the rational theory. In the formal perturbative rational theory it is

selected by the gauge-theory anomaly and Ward identities [19, 20].

For $X = \mathbb{C}^\times$ and for elliptic X , the propagator has trigonometric and elliptic periodicity. The quantum symmetry changes accordingly. The spectral curve is part of the physics.

39.8 Three-dimensional gauge theory and boundary chiral algebra

A holomorphic twist of a three-dimensional supersymmetric theory on $X \times \mathbb{R}_{\geq 0}$ may admit a boundary condition preserving the holomorphic supercharge. Boundary local operators form a chiral algebra on X . Boundary line junctions carry an ordered product inherited from the normal direction. Bulk lines act by modules.

In a Lagrangian theory, the boundary algebra is computed from boundary fields and BRST reduction. Gauge anomalies must cancel for the boundary condition to exist quantum mechanically. Different boundary conditions can give different E_1 chiral algebras with equivalent central bulk actions. This is the physical origin of Morita and Koszul dual boundary descriptions.

39.9 Drinfeld–Sokolov reduction as gauge fixing

Start from affine currents and couple them to ghosts imposing a moment-map constraint. The BRST charge is

$$Q = \oint \left(\sum_a c^a (J_a - \chi_a) - \frac{1}{2} \sum_{a,b,c} f_{ab}^c : c^a c^b b_c : + \dots \right).$$

The equation $Q^2 = 0$ follows from the current algebra, invariance of the level cocycle, and cancellation of the gauge anomaly. The reduced chiral algebra is $H_Q(A)$.

Because Q is a derivation of the transverse product, it acts on the ordered bar. The total differential is

$$D = b_A + Q_{\text{bar}}.$$

The spectral sequence obtained by first taking BRST cohomology compares the bar of the reduced algebra with the reduction of the bar. Degeneration is a theorem of the particular good grading and filtration. The higher W -operations are the homotopy-transferred remnants of affine multiplication and ghost contraction.

39.10 Polyakov’s determinant and the chiral anomaly

For a free scalar on a Riemann surface,

$$Z[g] = (\det' \Delta_g)^{-1/2}$$

up to zero-mode volume. Under $g \mapsto e^{2\sigma} g$, zeta regularization gives the Polyakov action

$$\log \frac{Z[e^{2\sigma} g]}{Z[g]} = \frac{1}{24\pi} \int (|\nabla \sigma|^2 + R\sigma) d\text{vol}.$$

For a theory of central charge c , the right side is multiplied by c . The same coefficient appears in the fourth-order pole of the stress tensor because both measure the central extension of local conformal transformations.

On a family of curves, the determinant forms a line bundle with Quillen connection. Its curvature is computed by the family index theorem. This is the exact bridge from the local Virasoro cocycle to the Hodge class. The affine level, ribbon genus, and Koszul curvature retain their independent definitions.

39.11 Worldsheet sewing

Choose local coordinates $zw = q$ around two punctures. Sewing inserts

$$q^{L_0 - c/24}$$

between dual states. The factor $q^{-c/24}$ is the vacuum energy on the cylinder. The sum over a homogeneous basis is the inverse two-point pairing. Convergence for $|q| < 1$ is the analytic sewing theorem.

A nonseparating sewing increases the genus of the chiral curve. An annular trace of the transverse E_1 algebra increases ribbon genus. The two operations can occur in the same amplitude, giving the bidegree of Section 26.

39.12 The Chern–Simons/WZW comparison theorem and its hypotheses

For compact gauge group at integral level, and after the standard choice of polarization and framing data, the Chern–Simons/WZW and Reshetikhin–Turaev constructions give the fundamental example in which the quantized surface state spaces, boundary current algebra, Wilson-line labels, conformal blocks, and ribbon-category invariants are connected by comparison theorems.

The boundary current algebra is affine Kac–Moody. Canonical quantization on a surface gives the WZW conformal-block space. Wilson lines are labelled by admissible representations. The Chern–Simons/WZW and Drinfeld–Kohno comparison theorems identify their exchange with the corresponding quantum-group braiding. Closing lines with the ribbon trace gives link invariants. Under these comparison theorems, the framing dependence of the three-dimensional theory matches the ribbon twist and the projective anomaly data of the WZW conformal blocks.

Each statement has a different mathematical carrier:

current OPE	: chiral D -module,
boundary composition	: E_1 algebra,
Wilson exchange	: E_2 line category,
surface states	: conformal blocks,
framing phase	: invertible anomaly theory.

The strength of the example lies in the comparison maps between these carriers, among these distinct carriers.

39.13 Large N and the double genus

For N coincident branes, boundary operators acquire matrix indices. A connected ribbon graph contributes N^{2-2h-b} . Independently, a closed-string worldsheet of genus g contributes g_s^{2g-2} . Open–closed duality can reorganize the first expansion into the second after summing boundaries and applying a dynamical transition.

Before that transition, the amplitude is naturally

$$\sum_{g,h,b} g_s^{2g-2} N^{2-2h-b} \mathcal{F}_{g,h,b}.$$

The bimodular formalism is the algebraic form of this double expansion. The relation between g_s and N is a physical duality parameter, a parameter of the physical duality relating the two expansions.

39.14 Holographic reconstruction

Let A be the ordered chiral algebra of a boundary condition. Bulk observables act centrally on A , giving

$$\beta : \text{Obs}_{\text{bulk}} \longrightarrow Z_{\text{ord, ch}}(A).$$

If the boundary condition is complete, β is a quasi-isomorphism. Then the protected bulk local algebra is reconstructed from the boundary. Two boundary descriptions have the same centre when an explicit Morita context identifies their bimodule categories; a Koszul equivalence of augmentation-generated sectors upgrades to this stronger statement when it extends to an explicit Morita context on the full bimodule categories.

This is the precise algebraic statement behind twisted holography. It reconstructs a protected local sector. A metric, Hilbert space, nonperturbative spectrum, and integration cycle require additional physical structure.

39.15 The field-theoretic meaning of the dual

The augmentation $A \rightarrow \mathbf{1}$ is a distinguished boundary or defect. Its derived endomorphism algebra

$$A^\dagger = \text{RHom}_A(\mathbf{1}, \mathbf{1})$$

acts on deformations of that defect. The bar complex

$$\mathbf{1} \otimes_A^L \mathbf{1}$$

is the complex of interval amplitudes between two copies of the defect. This is why Koszul duality repeatedly appears in line-operator physics: it exchanges local bulk couplings near the defect with operators living on the defect.

The dual is a completion near the chosen augmentation. Sectors connected to the chosen boundary condition form the completed local theory recovered by the double dual. The physical meaning of double duality is therefore completion of the local theory around the chosen vacuum.

39.16 The physical theorem in one sentence

A holomorphic-topological quantum field theory whose protected observables are product-generated, algebraizable as factorisable D -modules, locally constant in the ordered direction, and anomaly-free produces an ordered chiral algebra: collision on the complex curve gives the chiral coefficient and ordered fusion gives the E_1 product. A second topological direction supplies exchange, a cyclic trace supplies ribbon loops, variation of the curve supplies conformal sewing, and each anomaly is the invertible theory obstructing the corresponding gluing.

40 Replica defects, cyclic bars, and entanglement

Entanglement is a property of a state and a factorization of its observable algebra. In two dimensions, the replica construction represents the required traces by branched covers and cyclic permutation defects. The chiral algebra supplies local twist operators and conformal blocks; the full state and the antiholomorphic sector supply the density matrix.

40.1 Twisted Hochschild chains

Let A be a unital E_1 algebra and let $\sigma \in \text{Aut}(A)$. The σ -twisted bimodule ${}_{\sigma}A$ has left action $a \cdot m = \sigma(a)m$ and its usual right action $m \cdot a = ma$. The twisted Hochschild chain complex is

$$C_m(A, {}_{\sigma}A) = A^{\otimes(m+1)} \quad (250)$$

with differential

$$b_{\sigma}(a_0 \otimes \cdots \otimes a_m) = \sum_{i=0}^{m-1} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_m \quad (251)$$

$$+ (-1)^m \sigma(a_m) a_0 \otimes a_1 \otimes \cdots \otimes a_{m-1}. \quad (252)$$

Associativity gives $b_{\sigma}^2 = 0$.

Proposition 40.1 (Cyclic defect trace). *Let $A^{\otimes n}$ carry the cyclic permutation $\sigma = (1\ 2\ \cdots\ n)$. The factorisation homology of a circle with a σ -defect is*

$$\int_{S^1_{\sigma}} A^{\otimes n} \simeq HH_{\bullet}(A^{\otimes n}, {}_{\sigma}(A^{\otimes n})). \quad (253)$$

The degree-zero group is the twisted coinvariant space

$$HH_0(A^{\otimes n}, {}_{\sigma}(A^{\otimes n})) = A^{\otimes n} / \langle xa - \sigma(a)x \rangle.$$

Proof. Cut the defect circle at one point. Excision identifies the circle integral with the two-sided bar of the interval algebra, and gluing the two endpoints applies the automorphism σ to the final face. The resulting simplicial differential is exactly (251). $\square$

The crossed product

$$A^{\otimes n} \rtimes \mathbb{C}[C_n] \quad (254)$$

packages the untwisted and twisted sectors in one associative algebra. In an ordered chiral theory, A is an E_1 algebra in $\text{FactD}(X)$, so every term of (250) remains a factorisable D -module on the curve.

40.2 Replica geometry

Let ρ_A be the reduced density matrix of a region A in a Euclidean quantum field theory. The Renyi entropy is

$$S_n(A) = \frac{1}{1-n} \log \text{Tr } \rho_A^n. \quad (255)$$

The path integral for $\text{Tr } \rho_A^n$ is the partition function on the n -sheeted cover obtained by cyclically gluing the cuts along A . Equivalently, it is a correlation function of cyclic permutation defects in the

n -fold tensor theory [80].

For a two-dimensional CFT, the endpoint of the cut carries a twist field $\mathcal{T}_n$. The covering map $w = z^{1/n}$ and the Schwarzian transformation law of the stress tensor give

$$h_n = \frac{c}{24} \left(n - \frac{1}{n} \right), \quad \bar{h}_n = \frac{\bar{c}}{24} \left(n - \frac{1}{n} \right). \quad (256)$$

Proof. The vacuum expectation of $T(w)$ on the cover is zero. Under $w = z^{1/n}$,

$$\langle T(z) \rangle_{\mathcal{R}_n} = \frac{c}{12} \{z^{1/n}, z\} = \frac{c}{24} \left(1 - \frac{1}{n^2} \right) z^{-2}.$$

The Ward identity for a primary insertion gives $\langle T(z) \mathcal{T}_n(0) \rangle / \langle \mathcal{T}_n(0) \rangle = h_n z^{-2}$ after summing the stress tensors of the n replicas, which multiplies the coefficient by n . This yields (256); the antiholomorphic calculation is identical. $\square$

40.3 The vacuum interval

For an interval of length L in the vacuum on the line, conformal invariance fixes

$$\text{Tr } \rho_A^n = c_n \left(\frac{L}{\epsilon} \right)^{-\frac{c+\bar{c}}{12}(n-1/n)}. \quad (257)$$

Consequently,

$$S_n(A) = \frac{c+\bar{c}}{12} \left(1 + \frac{1}{n} \right) \log \frac{L}{\epsilon} + s_n, \quad (258)$$

and analytic continuation to $n = 1$ gives

$$S(A) = \frac{c+\bar{c}}{6} \log \frac{L}{\epsilon} + s_1. \quad (259)$$

The constants c_n and s_n depend on the ultraviolet normalization of the twist fields.

On a thermal cylinder of circumference β , the exponential map gives

$$S_n(A) = \frac{c+\bar{c}}{12} \left(1 + \frac{1}{n} \right) \log \left[\frac{\beta}{\pi \epsilon} \sinh \left(\frac{\pi L}{\beta} \right) \right] + s_n. \quad (260)$$

40.4 Conformal-block decomposition

For several intervals, the replica surface has positive genus and the twist-field correlation function decomposes into conformal blocks. For two intervals, the four twist insertions depend on the cross-ratio x :

$$\langle \mathcal{T}_n(0) \tilde{\mathcal{T}}_n(x) \mathcal{T}_n(1) \tilde{\mathcal{T}}_n(\infty) \rangle = \sum_{i,j} M_{ij} \mathcal{F}_i^{(n)}(x) \overline{\mathcal{F}_j^{(n)}(x)}. \quad (261)$$

The chiral algebra determines the admissible blocks $\mathcal{F}_i^{(n)}$, their differential equations, and their braid monodromy. The modular invariant matrix M_{ij} and the chosen state determine the full correlator. Sewing the replica surface uses the same state-sum tensor (217) as ordinary conformal blocks, now in the permutation orbifold.

40.5 Topological entanglement and categorical trace

For a unitary modular tensor category with simple objects a , quantum dimensions d_a , and total quantum dimension

$$\mathcal{D} = \sqrt{\sum_a d_a^2},$$

the vacuum entanglement entropy of a simply connected region in the associated $(2+1)$ -dimensional topological phase has the universal constant term

$$S_{\text{top}} = -\log \mathcal{D}. \quad (262)$$

The same category supplies the Reshetikhin–Turaev trace, and the categorical dimension is the trace of the identity line. Equation (262) is therefore the logarithm of a categorical normalization of the line sector.

40.6 Bulk–boundary interpretation

A complete boundary condition gives an open–closed map

$$\text{Obs}_{\text{bulk}} \longrightarrow Z_{E_1}(A).$$

The entangling defect is a module for the replica crossed product (254). A bulk replica path integral and a boundary twisted Hochschild trace agree when the open–closed comparison extends to the replica defect category. This statement gives the exact bridge between bulk entanglement and the cyclic bar of the ordered boundary algebra.

The replica construction therefore adds a concrete physical realization to the chiral Koszul language:

ordered interval product $\longrightarrow$ cyclic bar $\longrightarrow$ permutation defect,

permutation defect $\longrightarrow$ branched-cover amplitude $\longrightarrow$ Renyi and entanglement entropies.

Every arrow is a gluing operation, and every scalar entropy is the final logarithmic trace of a structured defect amplitude.

41 Higher-dimensional sources and reduction to a curve

41.1 Pushforward is the source of chirality

Let Y be a compact framed d -manifold and let $p : Y \times X \times \mathbb{R} \rightarrow X \times \mathbb{R}$. Suppose $\mathcal{F}$ is a factorization algebra that is locally constant in the Y and $\mathbb{R}$ directions and holomorphic in X . Define

$$(p_*^{\text{fact}} \mathcal{F})(U \times I) := \int_{Y \times U \times I} \mathcal{F}. \quad (263)$$

Excision gives factorization maps on the target.

Theorem 41.1 (Topological pushforward to ordered chirality). *Assume the local constancy, dualisability, and excision hypotheses required for factorisation homology, and assume that the pushforward is product-generated and its holomorphic coefficient admits the factorisable D -module realisation of*

Theorem 10.1. Then (263) is a rectangular holomorphic–topological factorisation algebra on $X \times \mathbb{R}$ and determines an ordered chiral algebra in $\text{Alg}_{E_1}(\text{FactD}(X))$.

Proof. Fubini for factorization homology identifies integration over disjoint product opens with iterated integration over Y and over the target open. Local constancy in Y and $\mathbb{R}$ is preserved by the pushforward. Holomorphic differential equations in X commute with integration in the topological directions. Weiss descent follows from excision. Apply Theorem 10.2. $\square$

The theorem gives the invariant shape of the higher-dimensional constructions treated here under these hypotheses. The source is a factorization algebra; the reduction is factorization homology; the output is an E_1 algebra in chiral coefficients.

41.2 Dimensional bookkeeping

If a theory is locally constant in m transverse directions, its local operators carry E_m structure. Integrating over a framed r -manifold with $r < m$ leaves E_{m-r} structure. A holomorphic curve remains a coefficient direction. Thus a theory on

$$Y^r \times X \times \mathbb{R}^{m-r}$$

can reduce to an E_{m-r} algebra in $\text{FactD}(X)$.

This formula replaces a universal “chirality ladder.” The output depends on the actual number of topological directions and on the tangential structure used in the pushforward.

41.3 Calabi–Yau categories

A smooth proper Calabi–Yau A_∞ category $\mathcal{C}$ has a nondegenerate cyclic trace. Its open sector is the family of morphism complexes $\text{RHom}(E, F)$ with A_∞ composition. Its closed state complex is Hochschild homology $\text{HH}_\bullet(\mathcal{C})$. The Connes operator and the cyclic pairing supply the circle and surface operations of the associated topological conformal field theory.

These data are topological. To obtain a chiral algebra on X , one needs a family $\mathcal{C}_x$ varying holomorphically with $x \in X$, or a holomorphic field theory whose boundary category is $\mathcal{C}$. The ordered chiral algebra is then the endomorphism object of a chosen boundary generator, or the pushforward (263).

Theorem 41.2 (Categorical source theorem). *Let $\mathcal{C}$ be a stable presentable $\text{FactD}(X)$ -module category whose action preserves colimits separately, let b be a compact generator, and let the ordered topological direction act by composition of boundary morphisms. Then*

$$A_b = \text{RHom}_{\mathcal{C}}(b, b)$$

is an ordered chiral algebra and the enriched Morita functor gives an equivalence

$$\text{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \xrightarrow{\cong} \text{Mod}_{A_b}(\text{FactD}(X)).$$

A Calabi–Yau trace on $\mathcal{C}$ induces the transverse cyclic structure of Theorem 26.1 on A_b .

Proof. Composition of endomorphisms gives A_b its E_1 multiplication. Factorisation of the enriched Hom objects over disjoint configurations gives the chiral coefficient, and functoriality of composition

makes multiplication a morphism of factorisation objects. The adjunction

$$- \otimes_{A_b} b \dashv \mathrm{RHom}_{\mathcal{C}}(b, -)$$

has unit and counit equivalences on the generators A_b and b . Exactness, preservation of colimits, and compact generation extend those equivalences to the presentable categories. A Calabi–Yau trace pairs $\mathrm{RHom}(b_0, b_1)$ with $\mathrm{RHom}(b_1, b_0)$ cyclically; specializing to $b_0 = b_1 = b$ gives a cyclic trace on A_b , compatible with the factorisation maps by naturality of the categorical trace. $\square$

This theorem is the correct passage from a Calabi–Yau category to noncommutative chirality. It requires a factorization family over the curve; the Calabi–Yau trace supplies cyclic contractions once the factorization family over the curve has been constructed.

41.4 Naturality of obstruction classes

Let $\mathfrak{g}$ and $\mathfrak{h}$ be complete filtered L_∞ algebras controlling two deformation problems, and let

$$F : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a filtered L_∞ morphism. It sends Maurer–Cartan elements to Maurer–Cartan elements and gauge equivalences to gauge equivalences.

Proposition 41.3 (Transgression of obstructions). *Let $\alpha_{\leq r}$ be a Maurer–Cartan solution modulo filtration $F^{r+1}\mathfrak{g}$, with obstruction*

$$o_{r+1}(\alpha) \in H^2(\mathrm{gr}^{r+1}\mathfrak{g}, d_\alpha).$$

Let F_1^α denote the linear term of the L_∞ morphism twisted by $\alpha_{\leq r}$. The obstruction to extending $F_(\alpha_{\leq r})$ is*

$$[\mathrm{gr}^{r+1} F_1^\alpha] o_{r+1}(\alpha) \in H^2(\mathrm{gr}^{r+1}\mathfrak{h}, d_{F_*(\alpha)}).$$

Proof. Write the Maurer–Cartan curvature as

$$\mathcal{R}_{\mathfrak{g}}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n^{\mathfrak{g}}(\alpha^{\otimes n}).$$

The L_∞ identities give the curvature identity

$$\mathcal{R}_{\mathfrak{h}}(F_*(\alpha)) = \sum_{m \geq 0} \frac{1}{m!} F_{m+1}(\mathcal{R}_{\mathfrak{g}}(\alpha), \alpha^{\otimes m}).$$

Modulo F^{r+2} , the right-hand side is precisely F_1^α applied to the filtration- $(r+1)$ curvature. Twisted L_∞ functoriality states that F_1^α is a cochain map from $(\mathfrak{g}, d_\alpha)$ to $(\mathfrak{h}, d_{F_*(\alpha)})$. Passing to the associated graded and then to cohomology proves the formula. $\square$

The proposition gives the rigorous form of a Calabi–Yau, Hall, or gauge-theoretic obstruction becoming a chiral obstruction: the comparison is the twisted linear term of a constructed filtered L_∞ morphism.

41.5 Cohomological Hall algebras

Let $\mathcal{M}$ be a moduli stack of objects in a suitable abelian or derived category. Correspondences of short exact sequences define a convolution product on an appropriate homology theory of $\mathcal{M}$. The resulting cohomological Hall algebra is an E_1 algebra. Its multiplication is noncommutative because extension flags are ordered.

A coproduct requires additional geometry, typically localization, a stability condition, or a choice of chamber. A Drinfeld double requires a nondegenerate Hopf pairing. An R -matrix requires a stable-envelope or line-exchange construction. These are separate structures.

Criterion 41.4 (Hall-to-quantum-group construction). *A completed Hall algebra together with the following data determines a quasitriangular completed Hopf algebra on the chosen representation category:*

- (i) *a compatible completed coproduct;*
- (ii) *a nondegenerate pairing and a double;*
- (iii) *a PBW theorem and a presentation of the relation ideal;*
- (iv) *a universal or categorical R -matrix;*
- (v) *convergence of the completion on the chosen representation category.*

Proof. The completed coproduct and counit give the bialgebra structure. The nondegenerate pairing identifies the negative half with the continuous dual of the positive half and constructs the Drinfeld double; the PBW theorem identifies the completed vector space and proves that the displayed relation ideal presents it. The universal R -matrix furnished in (iv) intertwines the coproduct with its opposite and satisfies the two hexagon identities, hence the quantum Yang–Baxter equation. Rigidity of the representation category supplies the antipode, while (v) makes every structure map continuous and every matrix coefficient convergent. These operations are exactly the data of a quasitriangular completed Hopf algebra. $\square$

The ordered chiral pushforward of a Hall theory carries the E_1 multiplication. Transport of items (i)–(v) through the pushforward identifies its line category with the reconstructed quantum group.

41.6 Coulomb branches and shifted Yangians

For three-dimensional $\mathcal{N} = 4$ gauge theories, the Braverman–Finkelberg–Nakajima construction defines Coulomb-branch algebras by convolution on affine-Grassmannian spaces. In type A quiver examples, quantized Coulomb branches are truncated shifted Yangians. The coweight shift records magnetic or boundary data.

In the holomorphic–topological twist, these algebras act on line defects. A boundary reduction to X gives a factorization family of modules with spectral parameter. The ordered chiral algebra records fusion of these lines. The BFN identification supplies the named shifted Yangian after the geometric convolution algebra and the Drinfeld presentation have been compared.

41.7 K3 geometry

Let S be a K3 surface. Its Mukai lattice

$$\tilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

has signature $(4, 20)$. Nakajima correspondences give a Heisenberg action on

$$\bigoplus_{n \geq 0} H^*(S^{[n]}).$$

The elliptic genera of the symmetric products satisfy the Dijkgraaf–Moore–Verlinde–Verlinde product formula. The reciprocal Igusa cusp form encodes the second-quantized K3 elliptic genus.

These are exact statements about cohomology, modular forms, and second quantization. A specialization functor carrying their products, coproducts, and traces defines the corresponding ordered chiral algebra and Koszul dual.

Principle 41.5 (K3 comparison datum). *A common ordered chiral realization of K3 geometry consists of explicit functors from the Mukai-lattice, Nakajima, Hall, elliptic-genus, Siegel-modular, and line-defect constructions to one factorisation category, together with natural transformations identifying their products, coproducts, pairings, and traces.*

Proof. The six inputs occupy different native categories: an integral quadratic lattice, a representation on Hilbert-scheme cohomology, a convolution algebra, a Jacobi-form index, a section over Siegel moduli, and a braided tensor category. Each structure reaches an ordered chiral algebra through a definite functor: lattice vertex construction, factorisation pushforward, Hall convolution, trace, sewing, or Tannaka reconstruction. A common realization is therefore precisely a cone over this diagram whose comparison transformations preserve the named operations. Such a cone supplies one chiral object; conversely, any chiral object realizing all six structures furnishes these functors and transformations by restriction to the corresponding operations. $\square$

A valid construction may begin with a K3-associated field theory on $S \times X \times \mathbb{R}$, integrate over S or a cycle in S , and apply Theorem 41.1. Its comparison with a Hall algebra or a Borchers product is then a theorem about that pushforward.

41.8 Borchers products and scalar traces

A Borchers lift takes a suitable vector-valued modular or Jacobi form to an automorphic product. The weight is determined by the constant term of the input. The product can be interpreted as a denominator identity of a generalized Kac–Moody algebra in favorable cases.

The automorphic product is a scalar or section of a line bundle. It may arise as a trace, index, or partition function of a categorical or chiral object. The trace has a fibre consisting of the categories with that index. The ordered multiplication, braiding, and extension data distinguish points in this fibre.

41.9 Omega backgrounds and BPS algebras

An Omega background makes rotations equivariantly exact and can reduce a higher-dimensional field theory to a noncommutative algebra of protected operators. In M-theory and gauge-theory examples,

the resulting algebras include Yangians and their deformations. The deformation parameters are equivariant weights; line-defect spectral parameters are positions in a residual holomorphic curve.

The algebraic structure is obtained from the factorization OPE of protected operators. Koszul duality appears when local bulk observables are coupled to a defect and one computes derived endomorphisms of the trivial defect. The named BPS algebra follows from a separate comparison with geometric correspondences or representation-theoretic generators.

41.10 Shifted symplectic geometry

Derived moduli stacks of fields or objects often carry shifted symplectic forms. A boundary condition is represented by a Lagrangian morphism. The derived intersection of two Lagrangians has a shifted Poisson structure and may quantize to the algebra of boundary-changing operators.

Let $L_0, L_1 \rightarrow \mathcal{M}$ be Lagrangians in an n -shifted symplectic stack. Their derived intersection is $(n-1)$ -shifted symplectic. Its $(n-1)$ -shifted symplectic form determines a P_n -structure. Whenever an E_n -quantisation of that P_n -structure has been constructed and varies factorisably over a chiral curve, the resulting observables form an E_n algebra in factorisation D -modules.

This gives a geometric source of ordered chiral algebras when the ambient symplectic shift is $n=1$, so that the Lagrangian intersection is 0-shifted symplectic and its quantisation is associative. The quantization and the factorization family are part of the theorem; the shifted symplectic form alone is classical data.

41.11 A source-to-chiral comparison

Suppose the higher-dimensional factorisation pushforward constructs

$$A_{\text{geom}} \in \text{Alg}_{E_1}(\text{FactD}(X)),$$

and a Hall, BPS, or localization construction independently produces

$$A_{\text{src}} \in \text{Alg}_{E_1}(\text{FactD}(X)).$$

The comparison datum is an E_1 -algebra morphism

$$\phi : A_{\text{src}} \longrightarrow A_{\text{geom}}$$

compatible with the chiral coefficient. Applying the ordered bar gives the commutative square

$$\begin{array}{ccc} A_{\text{src}} & \xrightarrow{\phi} & A_{\text{geom}} \\ \text{Bar}_X^{\text{ord}} \downarrow & & \downarrow \text{Bar}_X^{\text{ord}} \\ \text{Bar}_X^{\text{ord}}(A_{\text{src}}) & \xrightarrow{\text{Bar}(\phi)} & \text{Bar}_X^{\text{ord}}(A_{\text{geom}}). \end{array} \quad (264)$$

The upper objects are associative and the lower objects are coassociative; the functorial bar map is what relates the two types. The construction of A_{src} contains the Hall or BPS correspondences, while the construction of A_{geom} contains factorisation pushforward. Their equality is the theorem that ϕ is an equivalence.

If ϕ is an equivalence on a chosen complete sector, then (264) identifies the corresponding Hall or BPS resolution with the ordered chiral bar. A quantum-group identification still requires the coproduct, pairing, R -matrix, and PBW data of Criterion 41.4.

41.12 The higher-dimensional conclusion

Higher-dimensional geometry enters ordered chiral Koszul duality through factorization pushforward, boundary endomorphisms, and morphisms of deformation complexes. Cyclic traces produce surfaces; Hall correspondences produce associative multiplication; Omega backgrounds produce deformation parameters; shifted symplectic forms produce Poisson brackets. The chiral curve carries the spectral dependence of the resulting operators.

The language remains unchanged: an E_1 algebra in factorization D -modules is the target through which these sources can be compared while retaining their geometric origin.

42 The single language

42.1 The theorem of the construction

Theorem 42.1 (Noncommutative chiral Koszul geometry). *Let X be a smooth complex algebraic curve and let*

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)).$$

Then the following statements hold in the indicated ambients.

- (i) *The operadic Grothendieck construction for $E_1 \otimes \text{Ch}_X$ presents A by ordered sequences of entire finite chiral configurations. Chiral collision occurs inside a block; the associative operadic maps compose adjacent blocks.*
- (ii) *The pointwise tensor and chiral convolution on the Ran category satisfy the duoidal interchange law. The E_1 multiplication of A is a morphism of chiral factorisation coalgebras. Consequently the ordered bar is a coassociative coalgebra object in chiral factorisation coalgebras.*
- (iii) *The internal ordered bar is the derived interval tensor product*

$$\text{Bar}_X^{\text{ord}}(A) \simeq \mathbf{1} \otimes_A^L \mathbf{1}.$$

Its differential contracts adjacent transverse factors, while the complete chiral singularity remains in the coefficient D -module.

- (iv) *On connected positive-weight objects and their complete Mittag-Leffler limits, internal associative bar and cobar reconstruct one another. The boundary Koszul dual $A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$ is the endomorphism algebra of the augmentation boundary, and $A^{!!}$ is the derived augmentation completion.*
- (v) *When A is admissibly positive, the universal twisting morphism identifies augmentation-complete A -modules with conilpotent comodules over $C = \text{Bar}_X^{\text{ord}}(A)$. Its two-sided form sends the diagonal bimodule to C_Δ , carries relative tensor products to derived cotensor products, and identifies Hochschild homology and cohomology with the corresponding coHochschild theories.*

- (vi) *The cardinality pro-nilpotence used by Francis–Gaitsgory is formulated in $D(\text{Ran } X)$ with the chiral convolution product; the pointwise coefficient tensor of $\text{FactD}(X)$ carries the independent transverse E_1 structure. A conservative strong monoidal realization*

$$J : (\text{FactD}(X), \otimes^{\text{pt}}) \longrightarrow (D(\text{Ran } X), \otimes^{\text{ch}})$$

whose image is closed under enhanced bar and cobar transports the internal constructions to the Ran constructions. In the chiral monoidal category, the enhanced theorem gives the reduced associative bar–cobar equivalence and, in the corresponding Lie case, the chiral Lie–commutative-coalgebra equivalence.

- (vii) *A cofibrant resolution of $E_1 \otimes \text{Ch}_X$ is described by two-coloured levelled forests. Transverse edges encode associative homotopy, chiral edges encode collision homotopy, and the two edge-contraction differentials anticommute.*
- (viii) *For a chiral coefficient admitting a meromorphic pole filtration along the collision divisors, the associated graded records every normal symbol of the singular product. On a formal disc, a translation-equivariant chiral coefficient becomes a vertex or nonlocal-vertex operation. In the standard chiral operadic model, its Maurer–Cartan equation is the integral Jacobi identity and hence the Borchers identity.*
- (ix) *An E_2 enhancement or a flat configuration-space connection supplies exchange. A rigid meromorphic line category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra when the antipode exists. A faithful quasi-monoidal fibre functor reconstructs a coquasi-bialgebra, or a quasi-Hopf algebra under the corresponding rigidity hypotheses. A Yangian or quantum loop algebra is obtained after its presentation has been recognized by a separate theorem.*
- (x) *A square-zero BRST derivation extends to the ordered bar and forms a bicomplex with its differential. The comparison with the bar of BRST cohomology is the corresponding convergent spectral sequence; degree-zero Drinfeld–Sokolov reduction is recovered when BRST cohomology is concentrated.*
- (xi) *A transverse Calabi–Yau trace and a chiral modular sewing structure define two edge colours in a completed graph transform. Their structure element satisfies the two-edge master equation (174). Ribbon genus and chiral genus are independent gradings.*
- (xii) *A perturbatively renormalized holomorphic–topological BV theory whose observables satisfy the hypotheses of Theorem 10.5 produces such an A . Its renormalized graph weights give a morphism from the mixed forest operad when they extend to compactified configurations and satisfy the quantum master equation.*

Proof. Statement (i) is the block-ordered recognition theorem, Theorem 4.2. Statements (ii) and (iii) follow from the duoidal Ran theorem and the interval-bar identification, Theorems 3.1, 3.3, 6.2 and 6.3. Statement (iv) is Theorems 7.2, 7.3, 7.5 and 7.6. Statement (v) is the Koszul–Morita square of Theorems 8.1 to 8.3, 8.5 and 8.6. Statement (vi) combines the declared strong monoidal realization with Theorems 7.4 and 17.2. Statement (vii) is the mixed Boardman–Vogt resolution, Theorems 5.3 and 5.4. Statement (viii) follows from the normal-cone collision operad, the chiral-to-classical symbol map, and the formal-disc selection theorem, Theorems 15.1, 15.2 and 16.1. Statement

(ix) is Theorem 20.2 together with the relevant representation-theoretic recognition theorem. Statement (x) is Theorems 24.1 and 24.2. Statement (xi) is Theorems 26.4, 26.6 and 26.7. Statement (xii) is Theorems 5.5 and 10.5. Every implication therefore factors through a construction proved in its native monoidal category. $\square$

The theorem obtains each construction from one object by a functor whose source, target, and hypotheses are visible.

42.2 The Platonic object

The invariant object is

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X), \otimes^{\text{pt}}). \quad (265)$$

It is equivalently an algebra over the tensor-product operad

$$E_1 \otimes \text{Ch}_X,$$

or, on the regular-holonomic product-generated analytic locus, a factorization algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic in X and locally constant in $\mathbb{R}$.

This is the minimal language because it remembers exactly the two primitive experiments:

collision on the complex curve, ordered composition on the real line.

Removing the first loses the operator product. Removing the second loses noncommutative fusion. Adding a second real direction produces exchange rather than altering either primitive operation.

42.3 The functorial enrichments

enrichment or realization	mathematical and physical output
formal coordinate and translation	modes and vertex or nonlocal-vertex operations
pole filtration	normal symbols and the Poisson-vertex associated graded
internal interval bar	boundary amplitudes and a conilpotent coassociative coalgebra
augmentation boundary	the endomorphism algebra $A^!$ and derived completion
E_2 transverse enhancement	exchange paths and the Yang–Baxter equation; braided defect categories after categorical extension
strong or quasi-monoidal fibre functor	Tannaka bialgebra or coquasi-bialgebra, followed by quantum-group recognition
transverse Calabi–Yau trace	Hochschild chains, annuli, and ribbon genus
chiral modular sewing	conformal blocks, plumbing, and chiral genus
conformal vector	projective connection and determinant-line anomaly
BRST derivation	derived reduction and W -algebra cohomology
BV quantization	renormalized graph weights and the quantum master equation
Calabi–Yau or Hall source	factorization pushforward and line algebras after a specified functor
trace or index	characters, Verlinde numbers, theta series, and automorphic functions

A scalar appears only in the last row or as a coefficient of an invertible anomaly theory. Its value is the image of the structured operations under the stated trace or index map.

42.4 The clarity obtained

The full OPE is the diagonal D -module with its complete pole filtration. Arnold's relation is a relation among configuration-space forms that may be paired with a coefficient identity. A Fulton–MacPherson screen carries its intrinsic configuration-space cohomology, while the terminal corolla contracts the separate incidence resolution. The internal E_1 bar uses the pointwise coefficient tensor, and the Francis–Gaitsgory chiral bar uses chiral convolution. Planar exchange or a flat configuration-space connection braids the ordered line. A quantum group is a reconstruction of a tensor-compatible exchange category. A constructed sewing action satisfies the modular master equation through its codimension-two boundary identities. Curved Koszul data and projective conformal curvature occupy their respective algebraic and moduli-theoretic deformation complexes. The derived centre is the universal target of central bulk actions.

Each correction strengthens the original intuition. The subject becomes unified by preserving the outputs in their native categories and by deriving them from the same local object through exact functors.

42.5 Mathematics and physics

Mathematically, the primitive datum (265) is the algebraic form of one holomorphic and one topological direction. The chiral coefficient is controlled by diagonals and the Ran space. The ordered product is controlled by intervals. Their common cofibrant resolution is controlled by two-coloured forests. Koszul duality is the passage from an augmented algebra to the coalgebra of interval states and to the endomorphisms of its augmentation boundary.

Physically, the same datum is the protected local algebra of a holomorphic–topological system. Chiral collisions are short-distance singularities. The E_1 product is time- or normal-ordered composition. The bar is interval evolution between boundary states. The E_2 centre is the universal algebra of central bulk operations. Braiding is exchange of lines in a transverse plane. Ribbon contraction is an open loop. Deligne–Mumford contraction is worldsheet plumbing. The BV master equation asserts compatibility of renormalized amplitudes with these boundary degenerations.

The two descriptions coincide when a factorization-observable comparison has been constructed. Before that comparison, they are parallel realizations of the same operadic geometry.

42.6 Structured invariants and their scalar traces

Corollary 42.2 (Structured completeness precedes scalarization). *The complete invariant considered here is the tuple consisting of the E_1 multiplication, complete chiral coefficient, interval bar coalgebra, exchange local system, and the two sewing theories. Ordinary and supertraces, Euler characteristics, conformal-block ranks, Grothendieck classes, determinants, and numerical indices are functorial images of this tuple. An equality between two scalar expressions is the image of a specified comparison or trace.*

Proof. The commutator records the antisymmetric part of multiplication. A trace records the cyclic central contribution. A Grothendieck ring records decategorified fusion. A conformal-block rank

records the dimension of the state space. A central charge records the Virasoro extension. The complete ordered chiral object is recovered from the full structured tuple rather than from any one of these projections. $\square$

The universal object is the noncommutative chiral algebra (265), together with the functorial bars, centres, exchange structures, and sewing operations that exist for the structures actually supplied.

Coda: the geometry remembers

A quantum field carries its collision law. On a curve, the law is a D -module operation supported on a diagonal.

An ordered product carries the orientation of an interval. Its algebraic form is an E_1 multiplication.

An exchange carries a path in a transverse configuration space. Its algebraic form is monodromy, and braided Tannaka reconstruction gathers all matrix coefficients into a quantum symmetry algebra.

A trace closes an interval into a circle. Its algebraic forms are Hochschild homology, cyclic homology, and the self-cotensor of the diagonal bar state.

Ribbon genus closes open topological boundaries. Chiral genus plumbs complex curves. The bicoloured graph transform records both gradings and their commuting sewing operations.

An anomaly is a class in the deformation theory of an invertible field theory. A determinant line, connection, surface, and normalization evaluate that class numerically.

Koszul duality resolves an operation into coalgebraic boundary states and identifies their endomorphisms at the augmentation boundary. On the Koszul–Morita locus, the boundary category and its universal centre pass through the duality unchanged.

The common syntax is factorisation. Disjoint regions multiply. Colliding regions produce singular coefficients. Cutting an interval produces a derived tensor product. Gluing coalgebraic states produces a derived cotensor product. Moving a defect produces monodromy. Closing a boundary produces a trace. Degenerating a surface produces a master equation.

The equations inherit the locality of their geometry. Associativity is the boundary identity of an interval compactification. Chiral Jacobi is the boundary identity among partial diagonals. Yang–Baxter is flatness on three-point configurations. The Batalin–Vilkovisky equation is Stokes’ theorem for renormalised graph weights. The modular equation is the oriented cancellation of codimension-two clutching strata.

The subject has one invariant sentence and many exact realizations:

$$\boxed{A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X))}. \quad (266)$$

The diagonal supplies chirality. The interval supplies noncommutativity. The plane supplies exchange. The circle supplies trace. The stable curve supplies sewing. The bar supplies resolution. The centre supplies universal bulk action. The determinant line supplies anomaly. The BV complex supplies quantum consistency.

Each construction preserves the category of its input. Their comparison maps form the inner music of chiral Koszul duality.

A Orientation lines and the collision differential

The square-zero identities used throughout the paper are identities of oriented correspondences. This appendix fixes one convention and proves the two cancellations on which the collision complex and the corolla-normalized forest resolution rest. A coherent change of convention conjugates every differential by diagonal signs and preserves the resulting complex.

A.1 Cohomological suspension and unordered tensor products

Let (K, d_K) be a cochain complex. We write $sK = K[1]$ in the cohomological convention

$$(sK)^n = K^{n+1}, \quad |sx| = |x| - 1, \quad d_{sK}(sx) = -s(d_K x).$$

For a finite set S , the unordered tensor product is

$$\bigotimes_{s \in S}^{\text{unord}} K_s := \left(\bigoplus_{\alpha: \{1, \dots, |S|\} \xrightarrow{\sim} S} K_{\alpha(1)} \otimes \cdots \otimes K_{\alpha(|S|)} \right)_{\Sigma_{|S|}}, \quad (267)$$

where a permutation acts by the Koszul symmetry. An ordering of S identifies (267) with an ordinary tensor product, and different orderings are related by the Koszul symmetry.

Let $\mu : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$ be a degree-zero graded Lie bracket. Its suspended form

$$\bar{\mu} : s\mathfrak{g} \otimes s\mathfrak{g} \longrightarrow s\mathfrak{g}$$

has cohomological degree +1 and is defined by

$$\bar{\mu}(sx, sy) = (-1)^{|x|} s[x, y]. \quad (268)$$

With this convention, graded antisymmetry of $[-, -]$ becomes graded symmetry of $\bar{\mu}$ with respect to the shifted degrees:

$$\bar{\mu}(u, v) = (-1)^{|u||v|} \bar{\mu}(v, u).$$

The Jacobi identity becomes

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)} \bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)} \bar{\mu}(\bar{\mu}(w, u), v) = 0. \quad (269)$$

The same formulas hold for a chiral Lie bracket after the tensor products are replaced by chiral products and the outputs are pushed forward to the common small diagonal.

A.2 Determinant lines of collision chains

Let I be finite and let $\Pi(I)$ be its partition lattice, ordered by refinement. A cover $\pi \triangleleft \rho$ merges exactly two blocks. Write $e_{\pi, \rho}$ for the one-dimensional conormal direction of that merger. For a saturated chain

$$\gamma = (\pi_0 \triangleleft \pi_1 \triangleleft \cdots \triangleleft \pi_r)$$

define

$$\text{or}(\gamma) = \bigwedge_{i=0}^{r-1} (e_{\pi_i, \pi_{i+1}} \oplus \cdots \oplus e_{\pi_{r-1}, \pi_r}).$$

Deleting the j -th edge gives the simplicial boundary sign $(-1)^{j-1}$. Interchanging two independent collision directions reverses the determinant orientation.

For a partition π , let

$$C_\pi(\mathfrak{g}) = \Delta_{\pi, !} \left(\bigotimes_{B \in \pi}^{\text{unord}} s\mathfrak{g} \right).$$

If $\pi \prec \rho$ merges blocks B and C , the map $d_{\pi,\rho} : C_\pi(\mathfrak{g}) \rightarrow C_\rho(\mathfrak{g})$ is the identity on the other blocks and $\bar{\mu}$ on the B, C factors, tensored with the oriented conormal line. The total collision differential is

$$d_{\text{coll}} = d_{\mathfrak{g}} + \sum_{\pi \prec \rho} d_{\pi,\rho}. \quad (270)$$

Lemma A.1 (Internal-collision cancellation). *If $d_{\mathfrak{g}}$ is a derivation of the chiral bracket, then*

$$d_{\mathfrak{g}} d_{\text{coll}}^{(1)} + d_{\text{coll}}^{(1)} d_{\mathfrak{g}} = 0,$$

where $d_{\text{coll}}^{(1)}$ is the sum over covers.

Proof. On the two merged factors this is the suspended form of

$$d[x, y] = [dx, y] + (-1)^{|x|} [x, dy].$$

The identity $d(sx) = -s(dx)$ gives the sign required when the degree-+1 map $\bar{\mu}$ is interchanged with the unary differential. Untouched factors contribute the usual tensor-product signs. Because the operation is defined on the unordered tensor product, the calculation is independent of the temporary ordering used to write it. $\square$

A.3 Rank-two intervals in the partition lattice

Every coefficient of $(d_{\text{coll}}^{(1)})^2$ is supported on a rank-two interval $[\pi, \sigma]$ of $\Pi(I)$. Such an interval has exactly one of two forms: two disjoint pairs of blocks are merged, or three blocks are merged into one.

Proposition A.2 (Disjoint-pair cancellation). *Let B, C, D, E be four distinct blocks of π . The two composites that merge B with C and D with E in the two possible orders sum to zero.*

Proof. The two coefficient operations act on disjoint factors. The transposition of the two odd collision directions can be recorded either as the interchange sign of the two degree-one maps or as the reversal

$$e_{BC} \wedge e_{DE} = -e_{DE} \wedge e_{BC}$$

of the determinant line. These are two presentations of the same transposition, two presentations of one orientation transposition. In the total oriented convention the two paths induce opposite maps to the common codimension-two component. $\square$

Proposition A.3 (Triple-collision coefficient). *Let B, C, D be three blocks of π , and let σ be obtained by replacing them with $B \cup C \cup D$. The component $C_\pi(\mathfrak{g}) \rightarrow C_\sigma(\mathfrak{g})$ of $(d_{\text{coll}}^{(1)})^2$ is one half of the suspended chiral Jacobiator.*

Proof. The three intermediate partitions are obtained by first merging B, C , first merging C, D , or first merging D, B . For a temporary ordering u, v, w of the suspended factors, the oriented sum is

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)} \bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)} \bar{\mu}(\bar{\mu}(w, u), v),$$

which is (269). The factor 1/2 is the conventional conversion between the square of an odd coderivation and one half of its graded commutator with itself. $\square$

Theorem A.4 (Square of the collision differential). *For a graded chiral antisymmetric operation μ compatible with $d_{\mathfrak{g}}$,*

$$d_{\text{coll}}^2 = \frac{1}{2}d_{[\mu, \mu]}.$$

Consequently $d_{\text{coll}}^2 = 0$ if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal square is zero and the mixed terms vanish by Theorem A.1. Disjoint mergers cancel by Theorem A.2; triple mergers give the Jacobiator by Theorem A.3. Conversely, the component from the discrete three-block partition to the one-block partition is precisely the Jacobiator. $\square$

A.4 Signs in the corolla-normalized forest resolution

For a nonempty block B , let $K(B) = N_*(N \text{Tree}(B))$ be the normalized nerve complex from Section 14. Its vertical differential ∂_B is the alternating simplicial boundary and satisfies $\partial_B^2 = 0$. The augmentation and terminal-corolla section

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C} \xrightarrow{\iota_B} K(B)$$

are chain maps. For disjoint blocks B, C , set

$$m_{B,C} = \iota_{B \cup C}(\epsilon_B \otimes \epsilon_C) : K(B) \otimes K(C) \longrightarrow K(B \cup C). \quad (271)$$

The maps $m_{B,C}$ are symmetric and strictly associative, and commute with the vertical differentials.

The total complex

$$W_I^{\text{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \Pi(I)} C_{\pi}(\mathfrak{g}) \otimes \bigotimes_{B \in \pi}^{\text{unord}} K(B)$$

has vertical differential $\partial = \sum_B \partial_B$ and horizontal differential obtained by tensoring $d_{\pi, \rho}$ with $m_{B,C}$. The mixed commutator vanishes because (271) is a chain map. Strict associativity of the $m_{B,C}$ reduces the horizontal square to the partition-lattice calculation above. Hence the total square is zero exactly when the chiral Jacobi identity holds.

This construction is intentionally different from an assertion that a single internal-edge contraction merges disconnected roots. Vertical contractions resolve a rooted tree inside a fixed block. Horizontal collision maps merge blocks through the partition differential and insert the terminal corolla of the new block.

A.5 Residues and coorientations

Let Y be smooth and $D_1, \dots, D_r$ a normal-crossing divisor. A local logarithmic form can be written

$$\alpha = \sum_{J \subseteq \{1, \dots, r\}} \left(\bigwedge_{j \in J} \frac{dt_j}{t_j} \right) \wedge \alpha_J.$$

The iterated residue along an ordered tuple is the corresponding coefficient. Therefore

$$\text{Res}_{D_i} \text{Res}_{D_j} = -\text{Res}_{D_j} \text{Res}_{D_i} \quad (272)$$

when the residue maps include their conormal orientation lines. This concerns the logarithmic conormal factors. Higher OPE poles are first paired with normal jets, as in Section 15, and then subjected to

the same residue orientation law.

A.6 Stable graphs

For a stable graph Γ , set $\text{or}(\Gamma) = \det \mathbb{R}^{E(\Gamma)}$. A separating clutching inserts an edge joining two vertices; a nonseparating clutching inserts a loop. Ordering the newly inserted edges gives the graph differential. The determinant-line calculation proves that two successive node-forming operations anticommute. The automorphism group of Γ acts on $\text{or}(\Gamma)$, and the modular convolution complex takes coinvariants with this twist.

In a BV presentation, joining different vertices gives the bracket and joining two flags of one vertex gives the loop operator Δ . With the standard modular-operad determinant convention, the codimension-two relations give

$$d^2 = 0, \quad d\Delta + \Delta d = 0, \quad \Delta^2 = 0,$$

together with the second-order identity expressing the bracket as the deviation of Δ from being a derivation. These equations are orientation identities of two node-forming parameters; their proof is entirely the orientation calculus of the two node-forming parameters.

B Low-arity calculations

Low arity is where every abstract object can be seen in its complete elementary form. This appendix computes the collision complex in arities two, three, and four; identifies the Arnold and infinitesimal-braid cancellations; and records the first examples of pole symbols, BRST descent, and modular sewing.

B.1 Two labels

Let $I = \{1, 2\}$. The partition lattice has two elements,

$$1|2 < 12.$$

For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is

$$\Delta_{1|2,!}(s\mathfrak{g} \boxtimes s\mathfrak{g}) \xrightarrow{\bar{\mu}} \Delta_{12,!}(s\mathfrak{g}). \quad (273)$$

Since $\Delta_{1|2}$ is the identity of X^2 and Δ_{12} is the diagonal, the map is exactly the chiral bracket

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

written after suspension. The first quadratic compatibility condition appears in arity three. Antisymmetry is the compatibility with the transposition of labels.

On a formal disc, a translation-invariant binary chiral operation is represented by

$$\{a_\lambda b\} + :ab:, \quad \{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b.$$

The summand $a_{(n)}b$ is the coefficient of the normal principal part of order $n + 1$. Thus the arity-two chain already contains the full singular OPE and the regular product.

Example B.1 (Heisenberg). For the rank-one Heisenberg chiral algebra with field J and

$$J(z)J(w) \sim \frac{k}{(z-w)^2},$$

the binary operation is concentrated in the second normal symbol. Its singular part is the second normal symbol $k N_{\Delta/X^2}^{\otimes 2}$. The coefficient k is supplied by the order-two principal-part component of the chiral operation.

Example B.2 (Affine currents). For a finite-dimensional Lie algebra $\mathfrak{g}$ with invariant form $(\cdot, \cdot)$,

$$J^a(z)J^b(w) \sim \frac{k(a, b)}{(z-w)^2} + \frac{J^{[a, b]}(w)}{z-w}.$$

The first normal symbol is the Lie bracket and the second is the invariant form multiplied by the level. They have different tensorial weights and satisfy different identities: Jacobi for the first, invariance for the second.

B.2 Three labels and the Jacobiator

For $I = \{1, 2, 3\}$, the partition lattice is

$$1|2|3 < \{12|3, 13|2, 23|1\} < 123.$$

Suppressing the diagonal pushforwards, the collision complex has the form

$$(s\mathfrak{g})_{\Sigma_0}^{\otimes 3} \xrightarrow{d_1} ((s\mathfrak{g})^{\otimes 2})^{\oplus 3} \xrightarrow{d_2} s\mathfrak{g}. \quad (274)$$

Here the notation Σ_0 only reminds us that the labels are fixed; the calculation is performed before symmetric descent. For homogeneous $u, v, w \in s\mathfrak{g}$,

$$d_1(u \otimes v \otimes w) = \begin{pmatrix} \bar{\mu}(u, v) \otimes w \\ (-1)^{|v||w|} \bar{\mu}(u, w) \otimes v \\ (-1)^{|u|(|v|+|w|)} \bar{\mu}(v, w) \otimes u \end{pmatrix},$$

where the three rows correspond to $12|3$, $13|2$, and $23|1$. With boundary orientations chosen cyclically,

$$d_2(x_{12|3}, x_{13|2}, x_{23|1}) = \bar{\mu}(x_{12}, x_3) - \bar{\mu}(x_{13}, x_2) + \bar{\mu}(x_{23}, x_1),$$

with the Koszul permutations implicit. Multiplication gives the suspended Jacobiator. This is the smallest calculation in which the square of the collision differential contains information.

B.3 Arnold's relation

On $\text{Conf}_3(\mathbb{C})$ put

$$\omega_{ij} = d \log(z_i - z_j).$$

Set $x = z_1 - z_2$ and $y = z_2 - z_3$, so that $z_1 - z_3 = x + y$. Then

$$\omega_{12} = \frac{dx}{x}, \quad \omega_{23} = \frac{dy}{y}, \quad \omega_{13} = \frac{dx + dy}{x + y}.$$

A direct calculation gives

$$\begin{aligned} \omega_{12} \wedge \omega_{23} - \omega_{12} \wedge \omega_{13} - \omega_{13} \wedge \omega_{23} &= \left(\frac{1}{xy} - \frac{1}{x(x+y)} - \frac{1}{y(x+y)} \right) dx \wedge dy \\ &= 0. \end{aligned}$$

Equivalently,

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0. \quad (275)$$

This relation lies in the coefficient algebra of configuration-space forms. The Jacobi identity enters after this form relation is tensored with the coefficient endomorphisms. A flat connection combines the two by tensoring ω_{ij} with endomorphisms that satisfy the infinitesimal braid relations.

B.4 The KZ curvature in arity three

Let V_1, V_2, V_3 be $\mathfrak{g}$ -modules and let Ω_{ij} denote the invariant Casimir acting in the i -th and j -th factors. Put

$$A = \hbar(\Omega_{12}\omega_{12} + \Omega_{13}\omega_{13} + \Omega_{23}\omega_{23}), \quad \nabla = d - A.$$

Since $d\omega_{ij} = 0$, the curvature is $F_\nabla = A \wedge A$. The terms with identical pairs vanish. After collecting coefficients,

$$\begin{aligned} \frac{1}{\hbar^2} F_\nabla &= [\Omega_{12}, \Omega_{23}] \omega_{12} \wedge \omega_{23} + [\Omega_{12}, \Omega_{13}] \omega_{12} \wedge \omega_{13} \\ &\quad + [\Omega_{13}, \Omega_{23}] \omega_{13} \wedge \omega_{23}. \end{aligned} \quad (276)$$

Invariance of the Casimir gives

$$[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0, \quad [\Omega_{13}, \Omega_{12} + \Omega_{23}] = 0.$$

Hence the three commutators in (276) are the same up to signs, and the curvature is that common commutator multiplied by (275). Therefore $F_\nabla = 0$.

This calculation displays the division of labor:

$$\text{Arnold relation} \quad \times \quad \text{invariance and Jacobi of } \mathfrak{g} \quad = \quad \text{flatness of KZ.}$$

The geometric and algebraic factors are individually meaningful and their tensor product supplies the flatness calculation.

B.5 The full vertex identity in arity three

For a vertex algebra, the coefficient operation contains the full system of modes in addition to Ω_{ij} . Write

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}.$$

The equality between the expansions in the regions $|z_1 - z_2| < |z_2 - z_3|$ and $|z_2 - z_3| < |z_1 - z_3|$ is the formal-distribution Jacobi identity. Multiplying by powers of the differences and comparing coefficients gives the Borchers identity

$$\sum_{i \geq 0} \binom{m}{i} (a_{(n+i)} b_{(m+k-i)} c = \sum_{i \geq 0} (-1)^i \binom{n}{i} (a_{(m+n-i)} b_{(k+i)} c - (-1)^n b_{(n+k-i)} a_{(m+i)} c), \quad (277)$$

with the usual parity modification in the super case. The binomial coefficients come from the expansion of translated formal variables. The chiral operad encodes (277) as a single Maurer–Cartan equation. Configuration-space forms can realize the domains and their boundary combinatorics, and the chiral coefficient supplies the mode operations and their binomial translation factors.

B.6 Four labels

The partition lattice of four labels has ranks 0, 1, 2, 3. The rank-one elements are the six partitions with one pair and two singletons. Rank two has two species:

$$12|34 \quad \text{and its two analogues,} \quad 123|4 \quad \text{and its three analogues.}$$

Rank three is 1234. The two species of rank-two partitions separate the two mechanisms in the square of the differential:

- The interval from $1|2|3|4$ to $12|34$ contains two paths. Their cancellation is the disjoint-pair sign.
- The interval from $1|2|3|4$ to $123|4$ contains three paths. Their sum is the Jacobiator on the first three factors, tensored with the identity on the fourth.

The next differential, from rank two to rank three, tests compatibility of Jacobi with a fourth insertion. The higher relation follows because the collision complex is the Chevalley complex of the chiral Lie algebra. In the Fulton–MacPherson resolution, however, the rank-three geometry has screen spaces with their own cohomology. Their cohomology contributes additional form classes while the algebraic Chevalley differential remains the algebraic Chevalley differential.

B.7 The screen of three points

Consider three points on a complex tangent line modulo translation and dilation. Fix the first two at 0 and 1; the third gives a coordinate

$$u \in \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

A compactification is $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, with boundary at $0, 1, \infty$. Thus the screen carries the cohomology of a thrice-punctured projective line. Its open cohomology is

$$H^0 \cong \mathbb{C}, \quad H^1 \cong \mathbb{C} d \log u \oplus \mathbb{C} d \log(1 - u).$$

The relation among the three boundary residues is the global residue theorem. The combinatorial forest category over the triple collision does have a terminal corolla and therefore contractible nerve.

This explicit example proves why the incidence resolution and the geometric screen complex must be distinguished.

B.8 Virasoro pole symbols

The Virasoro OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (278)$$

Let D be the pair diagonal. The three singular terms of (278) lie in

$$N_{D/X^2}^{\otimes 4}, \quad N_{D/X^2}^{\otimes 2} \otimes \langle T \rangle, \quad N_{D/X^2} \otimes \langle \partial T \rangle.$$

The regular term lies in filtration degree zero. A change of coordinate acts triangularly on this filtered object; the Schwarzian term records the extension by the fourth-order central symbol. The complete filtered operation consists of all these graded terms together with their extension data.

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity for this polynomial is equivalent to the Virasoro Lie conformal algebra identity. The normally ordered product is then part of the corresponding vertex algebra operation. This gives a direct formal-disc test of the chiral selection theorem.

B.9 A BRST bicomplex

Let (A, d_A) be a dg chiral algebra and let $Q : A \rightarrow A$ be a degree-one square-zero derivation anticommuting with d_A :

$$Q^2 = 0, \quad d_A Q + Q d_A = 0, \quad Q\mu = \mu(Q \otimes 1 + 1 \otimes Q).$$

On the collision complex, Q acts on every coefficient factor. Denote this extension by Q_{coll} . The derivation identity gives

$$Q_{\text{coll}} d_{\text{coll}} + d_{\text{coll}} Q_{\text{coll}} = 0.$$

Hence

$$D = d_A + d_{\text{coll}} + Q_{\text{coll}}$$

satisfies $D^2 = 0$. Filter by ghost number. If the spectral sequence is bounded below and its first page is concentrated in ghost degree zero, then

$$H_Q^0(C^{\text{coll}}(A)) \cong C^{\text{coll}}(H_Q^0(A)).$$

This is the low-arity mechanism behind bar-BRST interchange. In Drinfeld–Sokolov reduction the complex is infinite and filtered, so convergence must be proved from the good grading and PBW filtration; the algebraic identity above remains exact at every finite stage.

B.10 The first modular sewings

The stable pairs with smallest complexity are $(0, 3)$ and $(1, 1)$. The space $\overline{\mathcal{M}}_{0,3}$ is a point, so a cyclic three-point operation is a tensor

$$\theta_{0,3} \in (V^{\otimes 3})^{\Sigma_3}$$

with the appropriate degree and determinant twist. Contracting two legs with the inverse pairing produces a genus-one one-point tensor

$$\Delta\theta_{0,3} \in V.$$

The complex compactification $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ has three boundary divisors, and the global residue theorem relates their logarithmic residues. A real associahedral interval inside $\overline{\mathcal{M}}_{0,4}(\mathbb{R})$ has two endpoints; Stokes' theorem on that interval gives equality between the two adjacent parenthesizations. Analytic continuation, or the symmetric operadic action, relates the three complex channels. With coefficient tensors inserted, these statements give the associativity or Jacobi relation appropriate to the chosen operad.

A basic codimension-two modular relation occurs when a loop contraction and a separating contraction meet. Its algebraic image is the BV second-order identity. The relation is unconditional for an algebra over the modular operad: all contractions are already defined. A given chiral algebra enters this modular construction after a nondegenerate cyclic pairing and convergent sewing amplitudes have been supplied.

B.11 The free boson determinant

Let q satisfy $|q| < 1$. On the one-particle space of a chiral boson, sewing a cylinder acts diagonally with eigenvalues q^n , $n \geq 1$. Its Fredholm determinant is

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

Second quantization gives the oscillator partition function

$$\mathrm{Tr}_{\mathrm{Fock}} q^{L_0 - 1/24} = q^{-1/24} \prod_{n \geq 1} (1 - q^n)^{-1} = \eta(\tau)^{-1}, \quad q = e^{2\pi i \tau}.$$

This equality has three exact readings: a trace in the representation, a determinant of the sewing operator, and a Gaussian functional integral. The equality identifies three computations of the same free-field amplitude, whose anomaly coefficient is the central charge.

B.12 A diagnostic table

Low-arity datum	Mathematical identity	Physical reading
pair collision	chiral operation supported on Δ	local OPE/contact term
triple collision	chiral Jacobi or Borchers identity	independence of nested short-distance expansion
Arnold form	relation in $H^2(\text{Conf}_3)$	cancellation of configuration-space propagators
Casimir coefficient	infinitesimal braid relation	flat KZ transport
three-point screen	$\mathcal{M}_{0,4}$ geometry	relative configuration in a collision
loop contraction	BV operator Δ	one-loop sewing
anomaly cocycle	cohomology class of the quantum-master defect	obstruction class for quantization
Fredholm determinant	trace-class sewing identity	Gaussian partition function

Every row compares objects only after the indicated coefficient map or realization has been constructed. This is the low-arity form of the paper's governing rule: geometry supplies the carrier, algebra supplies the local operation, and physics supplies a representation by observables and amplitudes.

C Low-arity calculations for the ordered theory

C.1 Arity zero and one

Let $A \rightarrow \mathbf{1}$ be an augmented ordered chiral algebra and write $\bar{A}$ for the augmentation ideal. The reduced bar begins

$$B_0 = \mathbf{1}, \quad B_1 = \bar{A}[1].$$

The internal differential on B_1 is the suspension of the differential of A ; multiplication first appears in arity two.

The coaugmentation $\mathbf{1} \rightarrow B$ represents the empty ordered configuration. The projection $B \rightarrow \bar{A}[1]$ is the primitive quotient. The reduced coproduct vanishes on B_1 .

C.2 Arity two

In arity two,

$$B_2 = \bar{A}[1] \otimes \bar{A}[1].$$

For homogeneous $a, b \in \bar{A}$, the multiplication face is

$$b_2[a|b] = (-1)^{|a|}[ab]$$

with the suspension convention of Section A. The internal differential is

$$b_1[a|b] = [da|b] + (-1)^{|a|-1}[a|db].$$

The identity $b_1b_2 + b_2b_1 = 0$ is the Leibniz rule for multiplication.

Geometrically, the configuration of two ordered intervals has one collision face. The chiral coefficient of ab may have a pole along $z_1 = z_2$. The bar face composes the two letters, while the chiral coefficient transports the pole along $z_1 = z_2$.

C.3 Arity three and associativity

There are two multiplication faces:

$$[a|b|c] \mapsto [ab|c], \quad [a|b|c] \mapsto [a|bc].$$

Applying the multiplication differential twice gives

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

The two boundary points of the compactified one-dimensional moduli of three ordered intervals have opposite orientations. Associativity makes the coefficient operations equal. Hence $b_2^2 = 0$.

For an A_∞ algebra, the compactification is the interval K_4 . Its interior operation is m_3 , and the boundary identity is

$$dm_3 + m_2(m_2 \otimes 1) - m_2(1 \otimes m_2) + \text{terms containing } m_1 = 0.$$

The strict ordered chiral algebra is the case $m_{r \geq 3} = 0$.

C.4 The first mixed square

Take four product disks

$$U_i \times I_i \subset X \times \mathbb{R}, \quad I_1 < I_2 < I_3 < I_4.$$

Suppose the first two chiral points collide while the last two transverse intervals become adjacent. There are two codimension-two paths:

$$\text{chiral collision then interval fusion,} \quad \text{interval fusion then chiral collision.}$$

The coefficient maps agree by the interchange law (75). The conormal orientation lines occur in opposite orders, so the two boundary contributions cancel in the total differential.

This is the smallest calculation of the total complex with its chiral-resolution and ordered-bar differentials kept in separate bidegrees.

C.5 Planar exchange in arity two

The ordered chamber

$$\{t_1 < t_2\} \subset \text{Conf}_2(\mathbb{R})$$

is contractible. Parallel transport inside the chamber is trivial. Thickening the line to a plane connects the two order chambers by exchange paths.

In two transverse dimensions,

$$\text{Conf}_2(\mathbb{R}^2) \simeq S^1.$$

A half-rotation exchanges the two points; its square is the generator of the pure braid group $P_2 \cong \mathbb{Z}$.

Thus the R -matrix first appears after the E_2 thickening.

C.6 The free algebra

Let $A = T_X(M)$. In bar degree two the map

$$M^{\otimes p} \mid M^{\otimes q} \longmapsto M^{\otimes(p+q)}$$

concatenates words. The contracting homotopy splits off the first letter of every word of total tensor length at least two. Consequently only the one-letter class survives. This gives the contraction of Theorem 36.1 explicitly.

C.7 The square-zero algebra

For $A = \mathbf{1} \oplus M$ with $M^2 = 0$, all multiplication faces vanish. Hence

$$H_\bullet \text{Bar}_X^{\text{ord}}(A) \cong \bigoplus_{r \geq 0} H_\bullet(M[1]^{\otimes r}).$$

The deconcatenation coproduct cuts a word at every position. Continuous duality gives the completed free algebra on $M^\vee[-1]$.

C.8 A quiver calculation

For the quiver $1 \xrightarrow{u} 2 \xrightarrow{v} 3$, the relative bar over $S = \mathbf{1}e_1 \oplus \mathbf{1}e_2 \oplus \mathbf{1}e_3$ has

$$B_1 = M_u \oplus M_v, \quad B_2 = M_v \otimes M_u.$$

The multiplication face sends $[v|u]$ to the path vu . For the free path algebra, the path belongs to the algebra and the two-term resolution is exact. If the relation $vu = 0$ is imposed, $[v|u]$ becomes the quadratic Koszul cogenerator. The calculation is internal to $\text{FactD}(X)$, so M_u and M_v may carry singular chiral coefficients.

C.9 The enveloping algebra

Let $A = U_X(\mathfrak{g})$. Antisymmetrization sends

$$x_1 \wedge \cdots \wedge x_r \longmapsto \sum_{\sigma \in \Sigma_r} \text{sgn}(\sigma) [x_{\sigma(1)} \mid \cdots \mid x_{\sigma(r)}].$$

The bar multiplication differential becomes the Chevalley differential

$$\sum_{i < j} (-1)^{\epsilon_{ij}} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_r.$$

In arity three its square is the Jacobiator. PBW proves that antisymmetrization is a quasi-isomorphism.

C.10 BRST in arity two

Let Q be an odd derivation of A with $Q^2 = 0$. On $[a|b]$,

$$Q_{\text{bar}}[a|b] = [Qa|b] + (-1)^{|a|-1}[a|Qb].$$

The mixed commutator with the bar face is

$$(Q_{\text{bar}}b + bQ_{\text{bar}})[a|b] = \pm(Q(ab) - (Qa)b - (-1)^{|a|}a(Qb)) = 0.$$

Thus $b + Q_{\text{bar}}$ is square-zero. A convergent spectral sequence compares its cohomology with the bar of $H_Q(A)$.

C.11 The annulus and the torus

The transverse annulus amplitude is

$$\text{HH}_\bullet(A) = A \otimes_{A \otimes A^{\text{op}}}^L A.$$

It closes an ordered interval and raises the ribbon genus. The genus-one chiral block is obtained by sewing two marked points of a chiral sphere and raises the curve genus. Their lowest chain models both contain a trace, but they act on different moduli spaces. In the bimodular complex they occupy bidegrees $(0, 1)$ and $(1, 0)$.

C.12 The first Tannaka coefficient

Let a meromorphic line category have exchange

$$R(z) = 1 + \hbar r(z) + O(\hbar^2).$$

The quantum Yang–Baxter equation at order $\hbar^2$ gives

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0.$$

For $r(z) = \Omega/z$, the coefficient of each commutator is paired with the Arnold three-term relation. This is the first nontrivial compatibility between chiral collision geometry and transverse exchange. It is carried by the E_2 line category that enhances the E_1 bar.

D The stratified Fulton–MacPherson spectral sequence

The Fulton–MacPherson space carries two different complexes that are easily conflated. The incidence complex remembers how boundary strata meet. The geometric complex also remembers the cohomology of the screens. A spectral sequence relates them and states exactly when the incidence complex is sufficient.

D.1 A filtered space with normal-crossing boundary

Let Y be a smooth complex variety and

$$D = \bigcup_{a \in A} D_a$$

a simple normal-crossing divisor. For $J \subseteq A$, put

$$D_J = \bigcap_{a \in J} D_a, \quad D_J^\circ = D_J \setminus \bigcup_{b \notin J} D_b.$$

Let $j : Y \setminus D \hookrightarrow Y$. The increasing filtration by depth of support on the boundary gives a Cousin-type spectral sequence for $Rj_*\mathbb{C}$ or, algebraically, for the logarithmic de Rham complex. In one standard indexing,

$$E_1^{-p,q} = \bigoplus_{|J|=p} H^{q-2p}(D_J)(-p) \implies H^{q-p}(Y \setminus D). \quad (279)$$

The first differential is the alternating sum of Gysin or residue maps. Its signs are the determinant signs of Section A. Deligne's mixed Hodge theory identifies (279) with the weight spectral sequence when Y is proper.

For coefficients in a complex $\mathcal{L}$ constructible along the boundary stratification, the same filtration gives

$$E_1^{-p,q}(\mathcal{L}) = \bigoplus_{|J|=p} \mathbb{H}^{q-2p}(D_J, \mathcal{L}|_{D_J})(-p) \implies \mathbb{H}^{q-p}(Y \setminus D, \mathcal{L}). \quad (280)$$

The construction accepts an arbitrary constructible coefficient system $\mathcal{L}$. In chiral applications it is assembled from external tensor powers of the coefficient D -module and from principal-part lines along the normal directions.

D.2 Boundary strata of Fulton–MacPherson space

Let X be a smooth curve and $Y_I = \text{FM}_I(X)$. Boundary divisors are indexed by subsets $S \subseteq I$, $|S| \geq 2$, subject to the convention determined by the chosen compactification. A collection $\mathcal{N}$ of subsets labels a nonempty stratum exactly when it is nested: any two members are disjoint or one contains the other. The nested set is equivalently a rooted collision forest F .

The open stratum associated with F has the form of an iterated bundle whose base records the root configurations on X and whose fibers are screen configuration spaces. On a curve, the screen attached to a vertex with r incoming branches is the compactification of r points on an affine line modulo translations and dilations. Its interior is

$$\text{Conf}_r(\mathbb{A}^1)/(\mathbb{G}_a \rtimes \mathbb{G}_m).$$

For $r = 2$ the screen is a point. For $r = 3$ its interior is $\mathcal{M}_{0,4}$ and its compactification is $\overline{\mathcal{M}}_{0,4}$. For larger r it has nontrivial cohomology in several degrees.

Consequently, a stratum is determined by its forest incidence type together with the internal topology of its screens.

D.3 The forest resolution as the row of constants

Let $p : Y_I \rightarrow X^I$ be the blowdown. Consider the filtration of a geometric coefficient complex $\mathcal{K}_I$ by boundary depth. The E_1 page is a sum over forests,

$$E_1^{p,q} = \bigoplus_{\substack{F \in \text{For}(I) \\ |E(F)|=p}} R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) \otimes \text{or}(F), \quad (281)$$

where p_F is the projection of the stratum to the corresponding partial diagonal. The differential contracts one edge and applies the induced restriction or residue map.

The row $q = 0$ contains the locally constant functions on every screen. If the coefficient system is concentrated in screen degree zero and carries trivial local monodromy, then after barycentric or nerve replacement of the face incidence category and blockwise terminal-corolla normalization, this row maps to the combinatorial forest resolution $W_I^{\text{FM}}(\mathfrak{g})$ of Section 14. The augmentation to the partition collision complex is a quasi-isomorphism because every rooted-tree category has a terminal corolla.

Criterion D.1 (Incidence sufficiency). *The incidence forest complex computes the proper pushforward of $\mathcal{K}_I$ if the following two conditions hold:*

- (i) $R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) = 0$ for every $q > 0$ and every forest F ;
- (ii) *the degree-zero pushforwards identify with the coefficient system used in the forest complex, compatibly with the vertical incidence differential and the corolla-normalized horizontal collision merge.*

Proof. Condition (i) concentrates the E_1 page (281) in the row $q = 0$. Condition (ii) identifies that row, including its first differential, with the incidence forest complex. Every higher differential has source or target in a positive q -row and therefore vanishes. The spectral sequence consequently collapses at E_2 , and its edge morphism identifies the forest complex with the proper pushforward of $\mathcal{K}_I$. $\square$

For constant coefficients on complex screens, the positive screen rows carry the additional Orlik–Solomon cohomology.

D.4 The first nonconstant row

For a triple-collision vertex, the screen interior is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore the $q = 1$ row of (281) already appears in arity three. Its logarithmic generators may be taken to be

$$d \log u, \quad d \log(1 - u).$$

Their residues at $0, 1, \infty$ satisfy one linear relation. This row carries the first period data of ordered configurations. Integrating its forms along paths gives logarithms; iterating at higher arity gives multiple polylogarithms and, for KZ transport, multiple zeta values.

The chiral Chevalley differential uses the diagonal coefficient operation and the incidence of partitions. Configuration-space integral realizations additionally use the screen periods. The spectral sequence locates the exact extra data.

D.5 Orlik–Solomon cohomology

For $X = \mathbb{A}^1$, the cohomology algebra of $\text{Conf}_n(X)$ is the Orlik–Solomon algebra of the braid arrangement. It is generated in degree one by ω_{ij} with relations

$$\omega_{ij} = \omega_{ji}, \quad \omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

A broken-circuit-free basis gives the Poincare polynomial

$$P_{\text{Conf}_n(\mathbb{C})}(t) = \prod_{j=1}^{n-1} (1 + jt). \quad (282)$$

The coefficient $(n-1)!$ in top degree is geometric cohomology, the geometric top cohomology rank of the configuration space. To form a geometric bar model one tensors or convolves (282) with the algebraic coefficient complex and includes all differentials. Its homology is computed from the total differential on the tensor product.

D.6 Nested residues

On a normal-crossing chart with local boundary parameters t_e indexed by the edges of a collision forest, a logarithmic form has an expansion

$$\alpha = \sum_{J \subseteq E(F)} \left(\bigwedge_{e \in J} \frac{dt_e}{t_e} \right) \wedge \alpha_J.$$

The depth filtration counts $|J|$. The first differential of the residue spectral sequence extracts $\alpha_{\{e\}}$ and restricts it to $t_e = 0$. A second residue extracts $\alpha_{\{e,f\}}$. Because

$$\frac{dt_e}{t_e} \wedge \frac{dt_f}{t_f} = -\frac{dt_f}{t_f} \wedge \frac{dt_e}{t_e},$$

the codimension-two contribution cancels between the two orders. When the coefficient system itself changes under contraction, the remaining coefficient is the associator or Jacobiator of that system.

For higher poles, ordinary Poincare residue still acts only on the logarithmic conormal factor. The order- m_e coefficient is first paired with the dual m_e -jet of the normal line, producing a logarithmic coefficient; ordinary residue then removes dt_e/t_e . Invariantly this is the pairing

$$N_e^{\otimes m_e} \otimes N_e^{\vee \otimes m_e} \longrightarrow \mathcal{O}$$

followed by Poincare residue. Different pole orders remain in different associated-graded normal lines.

D.7 The Rees spectral sequence

Let $F_{\mathbf{m}}$ be the multi-pole filtration along the collision divisors. The multi-Rees object

$$\mathcal{R}(\mathcal{K}_I) = \bigoplus_{\mathbf{m} \geq 0} F_{\mathbf{m}} \mathcal{K}_I \mathbf{t}^{\mathbf{m}}$$

interpolates between the filtered chiral coefficient system and its normal-cone associated graded. Filtering the geometric complex simultaneously by boundary depth and pole order gives a bicomplex. One spectral sequence first resolves collision incidence and then pole order; the other first passes to normal symbols and then resolves the forest geometry. Under boundedness in each finite arity and pole window, the two converge to the same total cohomology.

This interchange explains the relation between chiral and Poisson vertex structures. The pole associated graded is classical; the extension data between pole orders is quantum. Fulton–MacPherson depth is independent of pole degree; it records how many nested collisions occur. The two filtrations form a genuine bifiltration.

D.8 A comparison theorem for finite windows

Fix a finite set I , a maximum pole order M , and a bounded cohomological range. Let $\mathcal{K}_I^{\leq M}$ be the truncated principal-part coefficient system. Suppose there is a morphism

$$\Phi_I : W_I^{\text{FM}}(\mathfrak{g}; \leq M) \longrightarrow Rp_*\mathcal{K}_I^{\leq M} \quad (283)$$

that is compatible with the vertical incidence differential and the corolla-normalized collision merge, and induces an isomorphism on every stratumwise cohomology group in the chosen range.

Theorem D.2 (Finite-window comparison). *Under the preceding hypotheses, (283) is a quasi-isomorphism in the chosen cohomological range.*

Proof. Filter source and target by boundary depth. The map induces a morphism of the corresponding spectral sequences. By hypothesis it is an isomorphism on E_1 in the range. Since the filtration is finite for fixed I , the comparison theorem for spectral sequences gives an isomorphism on the associated graded of cohomology, hence on cohomology itself. $\square$

The theorem is deliberately local in arity, pole order, and cohomological degree. Passage to an unbounded completed object requires control of inverse limits. Surjective transition maps imply the Mittag–Leffler condition and kill $\varprojlim^1$; the derived inverse limit records any remaining $\varprojlim^1$ contribution.

D.9 Renormalization and compactified configurations

Let M be a real manifold and consider a perturbative quantum field theory. A Feynman amplitude associated with a graph Γ is initially a differential form or distribution on the configuration space of its vertices away from the diagonals. Pulling it to a compactification separates the asymptotic regimes of every collision forest. Renormalization is an extension across the boundary satisfying locality and compatibility with disjoint unions.

The forest formula of perturbative renormalization has the same incidence category as the Fulton–MacPherson boundary. A divergent subgraph corresponds to a cluster of vertices; nested and disjoint divergent subgraphs correspond to nested and disjoint boundary divisors. Counterterms live on the associated diagonals. Stokes’ theorem on the compactification converts a Ward-identity defect into boundary integrals. The quantum master equation holds when the resulting local cocycle is exact or vanishes.

This physical realization uses more than the incidence complex:

- (i) a propagator or parametrix;
- (ii) a gauge-fixing operator;
- (iii) an extension of singular distributions;
- (iv) a renormalization prescription;
- (v) a proof that the counterterms preserve the required symmetries.

When these data are fixed, the geometric boundary complex computes the anomaly. The combinatorial identity $\partial^2 = 0$ becomes an anomaly-cancellation theorem after the coefficient cocycle is shown to be exact.

D.10 Degeneration of curves

For a family of curves approaching a node, the relevant compactification contains two kinds of boundary simultaneously: collisions of marked points in the fibers and degeneration of the curve. Logarithmic Fulton–MacPherson spaces resolve the first relative to the second. A local model of the node is

$$xy = q,$$

where q is the smoothing parameter. Collision parameters in the two branches and the nodal parameter q define independent conormal directions. The total residue complex therefore has a multigrading.

Separating degeneration corresponds to two components joined at the node. Nonseparating degeneration identifies two points on one component. The associated maps on conformal blocks are the two sewing operations of a modular operad. The screen cohomology of point collisions and the vanishing-cycle or nearby-cycle data of the node are distinct. A correct global comparison must include both.

D.11 Consequences

The spectral sequence proves four statements that are used repeatedly in the body of the paper.

- (1) The partition collision complex is canonical and computes chiral Chevalley chains.
- (2) The forest incidence complex is a canonical resolution of the partition complex.
- (3) The geometric Fulton–MacPherson complex contains additional screen cohomology.
- (4) An equality between the geometric and algebraic complexes is a comparison theorem, proved by an explicit comparison from the geometric coefficient complex to the algebraic collision complex.

From the physical side, the same four statements say that locality determines the pattern of counterterms, while the propagator and its periods determine their numerical amplitudes. The first is combinatorial and functorial. The second is analytic and theory-dependent. Chiral Koszul duality is the algebraic local-to-global theorem; configuration integrals are geometric realizations of its coefficient maps.

E Formal neighborhoods, normal jets, and degeneration coordinates

The local geometry of a chiral operation is the geometry of a divisor and its formal neighborhood. The formulas in this appendix make explicit how pole coefficients, delta derivatives, coordinate changes, and sewing parameters are different expressions of normal geometry.

E.1 The formal completion of a diagonal

Let X be a smooth curve and let $\Delta \subset X \times X$ be the diagonal. Its ideal sheaf $\mathcal{I}_\Delta$ satisfies

$$\mathcal{I}_\Delta/\mathcal{I}_\Delta^2 \cong \Omega_X^1, \quad N_{\Delta/X^2} \cong T_X.$$

The m -th infinitesimal neighborhood is

$$\Delta^{(m)} = \text{Spec}(\mathcal{O}_{X^2}/\mathcal{I}_\Delta^{m+1}).$$

The associated graded algebra of the formal completion is

$$\text{gr}_{\mathcal{I}} \widehat{\mathcal{O}_{X^2}_\Delta} \cong \text{Sym}_{\mathcal{O}_X}(\mathcal{I}_\Delta/\mathcal{I}_\Delta^2) \cong \text{Sym}_{\mathcal{O}_X}(\Omega_X^1). \quad (284)$$

Dualizing gives the algebra of normal differential operators

$$\text{Sym}_{\mathcal{O}_X}(T_X).$$

A coordinate difference $t = z_1 - z_2$ trivializes the conormal line and identifies the formal completion with $\mathcal{O}_X[[t]]$. A different coordinate changes t by a formal power series with invertible linear term. The formal neighborhood is the coordinate-invariant object, and each chosen series is one of its trivializations.

E.2 Principal parts

Let $D \subset Y$ be a smooth divisor with local equation $t = 0$. For $m \geq 0$, the sheaf $\mathcal{O}_Y(mD)$ consists locally of functions with poles of order at most m . There is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \xrightarrow{\sigma_{m+1}} N_{D/Y}^{\otimes(m+1)} \longrightarrow 0. \quad (285)$$

To verify the target, write a section as

$$f = t^{-(m+1)}a_{-(m+1)} + t^{-m}a_{-m} + \cdots.$$

Modulo $\mathcal{O}_Y(mD)$ only the leading coefficient remains. If $t' = ut + O(t^2)$, then

$$(t')^{-(m+1)} = u^{-(m+1)}t^{-(m+1)} + O(t^{-m}),$$

so $a_{-(m+1)}$ transforms as the $(m+1)$ -st power of the normal line. This proves (285) intrinsically from the divisor and its normal line.

The pole filtration

$$F_m \mathcal{O}_Y(*D) = \mathcal{O}_Y(mD)$$

has associated graded

$$\mathrm{gr}_m^F \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes m}, \quad m \geq 1. \quad (286)$$

A complete OPE is a filtered morphism with coefficients in $\mathcal{O}_Y(*D)$. Its mode of order $m - 1$ is the image under σ_m .

E.3 Differential operators and delta derivatives

Let $i : D \hookrightarrow Y$ be a smooth divisor. Distributions supported on D are generated by normal derivatives of the delta distribution. Algebraically, the Kashiwara equivalence identifies right D_Y -modules supported on D with right D_D -modules. Locally, a supported distribution has the form

$$T = \sum_{m=0}^M T_m \partial_t^m \delta(t). \quad (287)$$

The coefficient pairing of principal parts with normal jets is the formal residue identity

$$\mathrm{Res}_{t=0}(t^n t^{-m-1} dt) = \delta_{mn}. \quad (288)$$

The dual jet functional may be represented distributionally by a normalized derivative of $\delta(t)$. In complex analysis, (288) is Cauchy's coefficient formula; in D -module theory it is the pairing between local cohomology along the diagonal and normal differential operators.

For a vertex algebra, the singular commutator is

$$[Y(a, z), Y(b, w)] = \sum_{m \geq 0} Y(a_{(m)} b, w) \frac{1}{m!} \partial_w^m \delta(z - w). \quad (289)$$

Equation (289) is the distributional form of the chiral operation. It contains every singular mode. The normally ordered product is recovered from the regular extension, from the complementary regular extension.

E.4 The Rees deformation

The Rees algebra of the ideal of a divisor is

$$\mathcal{R}(\mathcal{I}_D) = \bigoplus_{m \geq 0} \mathcal{I}_D^m u^{-m}.$$

Its relative spectrum over $\mathbb{A}_u^1$ is the deformation to the normal cone. In a local chart one may write

$$t = ux. \quad (290)$$

For $u \neq 0$, the fiber is isomorphic to the original neighborhood. For $u = 0$, the coordinate x is the normal coordinate. The corresponding construction for the pole filtration uses inverse powers and interpolates between meromorphic kernels and normal symbols.

For several normal-crossing divisors D_e with local equations t_e , the multi-Rees space has equations

$$t_e = u_e x_e.$$

Its deepest special fiber is the direct sum of the normal lines. On a Fulton–MacPherson stratum indexed by a forest F , the indices e are the internal edges of F . This proves the normal-bundle splitting

$$N_{D_F/\text{FM}_I(X)} \cong \bigoplus_{e \in E(F)} N_e. \quad (291)$$

Operadic substitution on the associated graded is substitution of the normal coordinates x_e along contracted subtrees.

E.5 Coordinate changes and triangularity

Let $w = f(z)$ with $f'(z) \neq 0$. Near $z_1 = z_2 = z$, put $t = z_1 - z_2$. Taylor expansion gives

$$f(z+t) - f(z) = f'(z)t + \frac{1}{2}f''(z)t^2 + \frac{1}{6}f'''(z)t^3 + O(t^4), \quad (292)$$

$$\frac{1}{f(z+t) - f(z)} = \frac{1}{f'(z)t} - \frac{f''(z)}{2f'(z)^2} + O(t), \quad (293)$$

$$\frac{1}{(f(z+t) - f(z))^2} = \frac{1}{f'(z)^2 t^2} - \frac{f''(z)}{f'(z)^3 t} + O(1). \quad (294)$$

A pole of order m therefore transforms into a linear combination of poles of orders at most m , with leading factor $f'(z)^{-m}$. The associated graded is tensorial, while the filtered object is triangular.

This is exactly the transformation law of vertex modes. The top pole of an OPE between primary fields transforms by the tensor power prescribed by its conformal weights. Lower poles acquire corrections from derivatives of f . The complete chiral operation is coordinate-free because these corrections are the transition functions of the filtered principal-parts sheaf.

E.6 The Schwarzian derivative

The Schwarzian derivative is

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)} \right)^2.$$

It obeys the cocycle identity

$$\{f \circ g, z\} = \{f, g(z)\}g'(z)^2 + \{g, z\}. \quad (295)$$

A stress tensor of central charge c transforms as

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12} \{f, z\}. \quad (296)$$

The first term is the tensorial transformation of a quadratic differential. The second is an affine cocycle. Thus stress tensors form a torsor for projective connections rather than the vector space of quadratic differentials.

Equation (296) follows from the fourth-order central term of the Virasoro OPE. The central term is a fourth normal symbol, while the Schwarzian is the extension class that mixes it into filtration degree zero under coordinate change. This is a precise example of why the associated graded and the filtered chiral operation contain different information.

E.7 The affine connection and the shifted level

For affine currents, the invariant form gives the second normal symbol. The KZ connection contains instead the Casimir divided by $k + h^\vee$:

$$\nabla = d - \frac{1}{k + h^\vee} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

The level k and the shifted denominator $k + h^\vee$ arise in different calculations. The first is the coefficient of the central extension in the current OPE. The second is the coefficient of the Sugawara stress tensor and the KZ transport. Their relation is a theorem of the current-algebra representation, a representation-theoretic theorem relating two independently defined coefficients.

At $k = -h^\vee$, the Sugawara construction is singular. The current OPE and its level remain defined. The KZ connection in this normalization is singular. The Feigin–Frenkel centre appears. These facts are compatible because they concern different pieces of structure.

E.8 Prime form and normalized bidifferential

Let X be a compact Riemann surface of genus g , choose a symplectic basis of homology, and let Ω be the period matrix. The prime form $E(p, q)$ is a $(-1/2, -1/2)$ -form with a simple zero on the diagonal. Locally,

$$E(p, q) = \frac{z(p) - z(q)}{\sqrt{dz(p)} \sqrt{dz(q)}} (1 + O((z(p) - z(q))^2)).$$

The normalized fundamental bidifferential is

$$B(p, q) = d_p d_q \log E(p, q) + \sum_{i,j=1}^g c_{ij} \omega_i(p) \omega_j(q), \quad (297)$$

where the coefficients c_{ij} are chosen, according to convention, so that the A -periods vanish. It has a double pole with biresidue one on the diagonal. Changing the symplectic basis acts by a modular transformation and changes the decomposition into prime-form and holomorphic pieces, while the normalized bidifferential transforms covariantly.

The prime form is a section of a line bundle with quasi-periodicity. A propagator constructed from it requires a choice of normalization, spin structure, or Green function appropriate to the theory. Its local singular symbol is universal; its harmonic completion is global.

E.9 Plumbing coordinates

A nodal curve has local equation $xy = 0$. A smoothing is

$$xy = q, \quad |q| < 1. \quad (298)$$

Removing discs around $x = 0$ and $y = 0$ and identifying the annuli by $y = q/x$ produces the smooth fiber. The parameter q is a normal coordinate to the Deligne–Mumford boundary divisor.

For a conformal field theory, sewing inserts a complete set of states:

$$\mathcal{A}_q = \sum_{\alpha} q^{L_0 - c/24} |\alpha\rangle \langle \alpha|, \quad (299)$$

with the dual basis defined by the two-point pairing. In a rational theory the sum is organized by finitely many simple modules and convergent graded traces. In a general vertex algebra the expression may be formal. The algebraic clutching map and the analytic convergence of (299) are separate assertions.

The factor $q^{-c/24}$ is the vacuum energy on the cylinder. Under a modular change of plumbing coordinate it contributes to the projective anomaly. The same central charge appears in the Virasoro OPE because both are representations of the same central extension of vector fields. The equality is specific and exact; the independently defined levels and arithmetic weights remain separate invariants.

E.10 Separating and nonseparating nodes

A separating node joins curves of genera g_1 and g_2 . Sewing gives a bilinear composition

$$\mathbb{V}_{g_1, n_1+1} \otimes \mathbb{V}_{g_2, n_2+1} \longrightarrow \mathbb{V}_{g_1+g_2, n_1+n_2}.$$

A nonseparating node identifies two marked points of one curve and contracts them with the inverse pairing:

$$\mathbb{V}_{g-1, n+2} \longrightarrow \mathbb{V}_{g, n}.$$

These are the two modular-operad compositions. Their codimension-two compatibility is the geometric source of the bracket and loop terms in the quantum master equation.

The normal parameter q belongs to the base of the family of curves. Collision parameters t_e belong to relative configuration spaces in the fibers. In a degeneration with colliding insertions, both occur. A superconnection may mix their differentials, but an equation that equates a fiberwise endomorphism with dq/q or with a Hodge class becomes a typed equation in the total superconnection complex after its base degrees have been defined.

E.11 Tangential base points

The spaces $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and $\mathcal{M}_{0, n}$ carry nontrivial fundamental groups. Iterated integrals between boundary points require tangential base points. A tangent vector at 0 specifies the asymptotic direction of approach and regularizes logarithmic divergences. The KZ associator is the regularized parallel transport from the tangent vector at 0 to the tangent vector at 1.

The choice is geometric: it is a nonzero vector in the normal line to a boundary divisor. Changing it conjugates the regularized monodromy by an exponential of the residue. This is the local origin of gauge dependence of associators. The braid category is invariant up to braided equivalence; a particular power-series associator is a representative.

E.12 Reality and positivity

Algebraic chiral operations are defined over a field and form the algebraic sector; a Hilbert-space structure is added by. A unitary conformal field theory adds an antilinear involution and a positive Hermitian form compatible with the modes. The adjoint relation is

$$L_n^\dagger = L_{-n}$$

for the Virasoro modes in radial quantization, with analogous formulas for primary fields. Reflection positivity is an analytic inequality, an additional analytic axiom compatible with the chiral Jacobi identity.

Similarly, a Euclidean Green function depends on a real structure and metric, while the singular chiral kernel depends only on complex geometry. The chiral algebra determines protected holomorphic quantities. Reconstruction of a unitary full conformal field theory requires left–right pairing, modular invariance, and positivity. These are additional structures.

E.13 Local geometry summarized

The formal neighborhood of a diagonal contains the jets. The pole filtration contains the singular modes. The Rees deformation contains the classical limit. The Fulton–MacPherson normal bundle contains nested collision scales. The plumbing parameter contains nodal sewing. A projective connection contains the stress-tensor anomaly. Each object is a line or filtered sheaf attached to a geometric normal direction. This is the local geometry in which the coordinate formulas of chiral algebra and the short-distance formulas of quantum field theory become intrinsic.

F Field-theoretic derivations

This appendix derives the physical structures used in the paper from actions and Ward identities. Every identification is stated after the boundary observables, pairings, and comparison maps have been specified.

F.1 Classical BV theory

A classical gauge theory in the Batalin–Vilkovisky formalism consists of a graded space of fields $\mathcal{E}$, an odd symplectic pairing of cohomological degree -1 , and an action $S \in \mathcal{O}(\mathcal{E})$ of degree zero. The odd symplectic form induces a Poisson bracket $\{-, -\}$ of degree one. The classical master equation is

$$QS + \frac{1}{2}\{S, S\} = 0, \quad (300)$$

where Q is the linearized differential. Equivalently, the Hamiltonian vector field

$$d_S = Q + \{S, -\}$$

satisfies $d_S^2 = 0$.

The Taylor coefficients of S define cyclic multilinear operations. Equation (300) is their L_∞ relation. If the fields are local sections of bundles on a manifold, the operations are supported on the small diagonal. Thus the local L_∞ algebra of fields is the field-theoretic analogue of a diagonal-supported chiral operation.

Gauge fixing chooses a Lagrangian complement and a parametrix for the kinetic operator. This choice is needed to define Feynman amplitudes. Different gauge-fixing choices produce canonically homotopy-equivalent local operations and master-equation solutions.

F.2 Quantum master equation and anomaly

A regularized BV Laplacian Δ is a second-order operator of degree one. The quantum master equation is

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \quad (301)$$

or equivalently

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0. \quad (302)$$

In an interacting field theory the action is replaced by a scale-dependent effective interaction and Δ by a scale-dependent regularized operator. Renormalization constructs a compatible family of solutions when the obstruction vanishes [14, 15].

Suppose a solution has been constructed modulo $\hbar^N$. The coefficient of $\hbar^N$ in the quantum-master defect of (302) is closed for the classical differential d_S . Changes of counterterm preserve its cohomology class. This anomaly class is zero exactly when the solution extends to order $\hbar^N$. The resulting obstruction theory is the physical counterpart of the filtered Maurer–Cartan obstruction theory of Section 27.

The equality between the stable-graph master equation and the BV quantum master equation is exact for a cyclic local field theory: vertices are Taylor coefficients of S , internal edges are contractions by the propagator, and loops are contractions of two flags of one connected graph. The equality is an equality of graph expansions. A boundary realization additionally consists of a factorization-compatible comparison from the field theory to the chiral algebra.

F.3 Factorization observables

For an open set U , let $\text{Obs}^{\text{cl}}(U)$ be the Chevalley–Eilenberg cochains of compactly supported fields in U . Extension by zero and disjoint union define structure maps

$$\text{Obs}^{\text{cl}}(U_1) \otimes \cdots \otimes \text{Obs}^{\text{cl}}(U_n) \longrightarrow \text{Obs}^{\text{cl}}(V)$$

for disjoint $U_i \subset V$. The cosheaf and descent properties make Obs^{cl} a factorization algebra. A perturbative quantization deforms it to Obs^q over $\mathbb{C}[[\hbar]]$.

If translations in some directions are homotopically trivial, the factorization algebra is locally constant in those directions and determines an E_n algebra. If dependence on one complex direction is holomorphic, restriction to a holomorphic curve yields a chiral factorization algebra. Boundary conditions replace the bulk field complex by a relative complex and produce a boundary factorization algebra.

A comparison with an algebraic chiral algebra is a quasi-isomorphism

$$\Phi : U^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{Obs}_{\partial}^q$$

compatible with factorization. Once Φ is given, the algebraic chiral bracket equals the renormalized boundary OPE in cohomology.

F.4 Chern–Simons theory and affine currents

Let M be an oriented three-manifold, G a compact Lie group with invariant form $(\cdot, \cdot)$, and $k \in \mathbb{Z}$. The Chern–Simons action is

$$S_{\text{CS}}(A) = \frac{k}{4\pi} \int_M \left((A, dA) + \frac{2}{3} (A, A \wedge A) \right). \quad (303)$$

On a manifold with boundary, variation gives

$$\delta S_{\text{CS}} = \frac{k}{2\pi} \int_M (\delta A, F_A) + \frac{k}{4\pi} \int_{\partial M} (A, \delta A).$$

A boundary condition chooses a Lagrangian subspace for the boundary symplectic form. In complex coordinates on the boundary, fixing one component produces a chiral boundary theory. The residual gauge symmetry has current algebra

$$J^a(z)J^b(w) \sim \frac{k(a, b)}{(z - w)^2} + \frac{f^{ab}_c J^c(w)}{z - w}. \quad (304)$$

The double pole is the boundary central extension determined by the coefficient of (303). The simple pole is the Lie bracket. The one-loop shift by $\hbar^\vee$ enters the effective coupling and the Sugawara construction; the classical central extension retains its original level coefficient.

Wilson lines are labelled by representations of G . Their expectation values depend on framing. Canonical quantization on a surface gives the space of conformal blocks of the affine algebra. Witten’s path-integral derivation of knot invariants identifies the braiding and ribbon trace with quantum-group data [74, 63]. The Drinfeld–Kohno theorem supplies the algebraic comparison of braid categories.

F.5 One-loop exactness in a holomorphic gauge

A partially holomorphic gauge can make Chern–Simons theory a deformation of a topological–holomorphic BF theory. In the gauge constructed by Gwilliam and Williams, the perturbative quantization is one-loop exact and the bulk–boundary relation with chiral WZW theory is explicit [45]. One-loop exactness means that higher-loop counterterms vanish in that gauge after the relevant local cohomology calculation. Boundary correlators continue to contain tree-level nonlinear vertices and representation-theoretic sums.

The exactness result supplies a rigorous model in which the boundary current OPE, the anomaly shift, and the factorization algebra of observables can be compared at chain level. It is one of the clearest physical realizations of chiral Koszul duality: the local Lie algebra of gauge transformations passes to Chevalley cochains, while boundary factorization organizes their states and operators.

F.6 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface and C a complex curve with meromorphic one-form ω . Four-dimensional Chern–Simons theory has action

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma \times C} \omega \wedge \left((A, dA) + \frac{2}{3} (A, A \wedge A) \right). \quad (305)$$

The theory is topological along Σ and holomorphic along C . Wilson lines supported on curves in Σ and located at points of C possess a spectral parameter. Crossing two lines gives an R -matrix.

The quantum Yang–Baxter equation follows from isotopy invariance of three line crossings. Rational, trigonometric, and elliptic kernels arise from the geometry of (C, ω) [19, 20].

The line-operator OPE gives a tensor category with spectral parameters. A Yangian or quantum loop algebra is recognized when the category is generated by evaluation representations and the R -matrix and relations agree with the corresponding presentation. The field theory provides the monodromy and operator products; the algebraic recognition identifies the named Hopf algebra.

F.7 The Polyakov anomaly

For a two-dimensional conformal field theory on a Riemann surface with metric g , a Weyl rescaling of the metric changes the partition function by the local conformal anomaly $g \mapsto e^{2\sigma}g$. The infinitesimal anomaly is

$$\delta_\sigma \log Z[g] = \frac{c}{24\pi} \int_X \sigma R_g d\text{vol}_g + \text{boundary terms.} \quad (306)$$

Integrating gives the Polyakov action, up to local counterterms [65]. The coefficient c is the Virasoro central charge. On the moduli space of complex structures, the same anomaly is the curvature of a determinant or Hodge line.

For a free scalar, Gaussian integration gives

$$Z[g] \propto (\det' \Delta_g)^{-1/2}$$

with zero-mode factors. The variation of the zeta-regularized determinant reproduces (306) with $c = 1$. For free fermions, ghosts, and systems of several fields, determinants and Pfaffians give the corresponding central charges. The equality between OPE central charge and determinant-line curvature is a local index theorem for that field theory.

F.8 BRST quantization of the bosonic string

In conformal gauge, the bosonic string contains a matter CFT of central charge c_m and a (b, c) ghost system of central charge -26 . The BRST current has the form

$$j_{\text{BRST}} = c \left(T_m + \frac{1}{2} T_{gh} \right) + \text{derivative term.}$$

The BRST charge $Q = \oint j_{\text{BRST}}$ satisfies

$$Q^2 \propto (c_m - 26) \sum_{n \in \mathbb{Z}} (n^3 - n) c_{-n} c_n. \quad (307)$$

Therefore $Q^2 = 0$ precisely when the total Virasoro central charge vanishes, apart from the intercept condition in the state-space formulation. This is anomaly cancellation. It is an exact equality of central extensions in the coupled matter–ghost chiral algebra.

The number 26 is the magnitude of the central charge of the reparametrization ghost system of weights $(2, -1)$. Similar BRST systems in other reductions have different ghost contents and different anomaly polynomials.

F.9 Conformal blocks as states of quantization

Canonical quantization of Chern–Simons theory on a surface Σ produces a Hilbert space depending on the complex structure used in geometric quantization. Projective flatness identifies these spaces as the complex structure varies. The resulting projective bundle is identified with affine conformal blocks. Sewing surfaces corresponds to gluing three-manifolds and summing over intermediate representations.

For a modular tensor category with simple objects i , the dimension of the genus- g state space is

$$\dim \mathcal{H}_g = \sum_i S_{0i}^{2-2g}. \quad (308)$$

This is the Verlinde formula. The full modular functor consists of the state spaces, their mapping-class-group representations, and their gluing maps; equation (308) gives their dimensions [73, 47].

F.10 Cardy asymptotics

Let

$$Z(\tau, \bar{\tau}) = \text{Tr}_{\mathcal{H}} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}$$

be a modular-invariant torus partition function with a discrete spectrum, nonnegative degeneracies, and a unique vacuum that dominates the low-temperature limit. Applying the modular transformation $\tau \mapsto -1/\tau$ and a saddle-point estimate gives the asymptotic density

$$\log \rho(h, \bar{h}) \sim 2\pi \sqrt{\frac{c_{\text{eff}} h}{6}} + 2\pi \sqrt{\frac{\bar{c}_{\text{eff}} \bar{h}}{6}}. \quad (309)$$

The stated hypotheses determine the Cardy saddle; logarithmic theories, continuous spectra, noncompact targets, and additional light states require their corresponding asymptotic analysis.

For asymptotically AdS_3 gravity with Brown–Henneaux boundary conditions,

$$c = \frac{3\ell}{2G}.$$

When the gravitational spectrum satisfies the hypotheses leading to (309), the Cardy entropy agrees with the Bekenstein–Hawking entropy of the BTZ black hole. The equality identifies a regime of a gravitational theory with a regime of a boundary CFT; it uses the full modular invariant, spectrum, and vacuum-dominance hypotheses in addition to the Virasoro symmetry [12].

F.11 Entanglement entropy

For the vacuum of a unitary two-dimensional CFT on the line, the replica method computes the entropy of an interval of length L with ultraviolet cutoff ϵ :

$$S_A = \frac{c + \bar{c}}{6} \log \frac{L}{\epsilon} + s_0. \quad (310)$$

The proof uses twist fields of dimensions

$$h_n = \frac{c}{24} \left(n - \frac{1}{n} \right), \quad \bar{h}_n = \frac{\bar{c}}{24} \left(n - \frac{1}{n} \right),$$

and analytic continuation in the replica number. Equation (310) depends on the state, geometry, and full left–right CFT. A holomorphic chiral algebra supplies the central charge and twist-field constraints while the state and the full left–right theory determine the density matrix.

Relative entropy, modular Hamiltonians, and topological entanglement contain further information. A classification by chiral obstruction data is obtained from a theorem identifying those data with the relevant replica correlators.

F.12 Boundary algebras in supersymmetric gauge theory

Topological or holomorphic twists of supersymmetric gauge theories often produce protected boundary operator algebras. Costello and Gaiotto construct vertex algebras associated with boundary conditions in three-dimensional $\mathcal{N} = 4$ gauge theories and relate their modules to bulk line defects [16]. Costello, Dimofte, and Gaiotto construct boundary chiral algebras for holomorphic twists of three-dimensional $\mathcal{N} = 2$ theories.

These constructions supply examples of the comparison map Φ . Mirror symmetry or duality of the bulk can induce equivalences of boundary chiral algebras or their module categories. The precise statement depends on the chosen twist and boundary condition. A boundary chiral algebra is the protected holomorphic sector selected by the chosen twist and boundary condition.

F.13 Celestial and twistor chiral algebras

In suitable four-dimensional gauge theories, scattering amplitudes can be organized by a chiral algebra on the celestial sphere or on twistor space. Costello and Paquette derive form-factor integrands from correlators of a chiral algebra obtained from a six-dimensional holomorphic theory [17]. The OPE associativity constrains loop corrections; anomalies of the chiral algebra correspond to obstructions in the bulk theory.

The exact object is a protected holomorphic observable algebra. Mellin transformation converts momentum variables to conformal weights when passing to the celestial basis. Soft theorems provide current insertions. Equality with a complete gravitational S -matrix requires control of infrared divergences, reality conditions, and the reconstruction transform; such an equality is supplied by a reconstruction theorem controlling infrared behavior, reality conditions, and the inverse transform.

F.14 Calabi–Yau categories and closed strings

A smooth and proper Calabi–Yau dg category has a nondegenerate cyclic trace. Its Hochschild chains carry the Connes operator, and its Hochschild cochains carry a Gerstenhaber structure. Costello constructs a topological conformal field theory from Calabi–Yau categorical data [18]. The cyclic trace permits the contraction of graph edges, exactly as in the modular envelope.

To obtain a chiral algebra on a curve from a higher-dimensional Calabi–Yau geometry one needs a field theory or factorization algebra, a compactification or defect, and a restriction map to the curve. The resulting morphism of deformation complexes may carry cyclic classes to chiral operations. The construction is geometry-dependent. Numerical invariants such as the Euler characteristic or a Borchers-product weight determine the chiral operation only after the specialization morphism has been constructed.

F.15 The physical synthesis

The equations derived above form one coherent structure:

$$\begin{aligned} \text{local action} &\longrightarrow \text{BV Maurer–Cartan element} \longrightarrow \text{factorization observables}, \\ &\longrightarrow \text{boundary chiral operation}. \end{aligned}$$

A propagator resolves internal edges. Fulton–MacPherson geometry resolves collisions. The quantum master equation governs boundary strata of Feynman graphs. KZ transport governs exchange of current-algebra insertions. Conformal blocks govern sewing of states. Determinant lines govern projective anomalies. Each passage is exact when its comparison map has been constructed.

The resulting unity is physical in Feynman’s sense and mathematical in Beilinson’s sense. One begins with local interactions, follows their support under collision and gluing, and identifies objects precisely through the comparison maps that preserve their operations.

G Exact models seen from every side

A general language is convincing only when its principal objects agree in examples by independent calculations. This appendix treats five models. In each case the chiral operation, collision coalgebra, ordered transport, modular amplitude, and field-theoretic realization are kept distinct and then compared by explicit formulas.

G.1 The rank-one Heisenberg algebra

Let H_k be generated by a field J with OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \quad (311)$$

Equivalently,

$$[J_\lambda J] = k\lambda.$$

The chiral Jacobi identity is immediate because the lambda bracket is central. The full vertex algebra is the symmetric algebra on the negative modes J_{-n} , $n \geq 1$, applied to the vacuum.

Normal geometry. The singular operation in (311) is the second normal symbol of the diagonal. Under $z \mapsto f(z)$, a weight-one current transforms as a one-form, and the coefficient k pairs the two normal lines. Let $\mathfrak{h}_{\text{ch}}$ denote the abelian current chiral Lie algebra and let

$$0 \longrightarrow \omega_X \mathbf{K} \longrightarrow \widehat{\mathfrak{h}}_{\text{ch}} \longrightarrow \mathfrak{h}_{\text{ch}} \longrightarrow 0$$

be its central extension. The bracket of two current sections lands in the central line through the second normal symbol. Hence the Chevalley differential of $\widehat{\mathfrak{h}}_{\text{ch}}$ contracts two current cogenerators to the central cogenerator; the quotient $\mathfrak{h}_{\text{ch}}$ has the zero bracket.

Fock modules. For a momentum α , the Fock module $\mathcal{F}_\alpha$ is generated by a vector $|\alpha\rangle$ such that

$$J_n |\alpha\rangle = 0 \quad (n > 0), \quad J_0 |\alpha\rangle = \alpha |\alpha\rangle.$$

The oscillator modes satisfy

$$[J_m, J_n] = km \delta_{m+n,0}.$$

After choosing a conformal vector

$$T = \frac{1}{2k} : JJ :$$

for $k \neq 0$, the central charge is one. The character is

$$\text{ch } \mathcal{F}_\alpha(\tau) = \frac{q^{\alpha^2/(2k)-1/24}}{\eta(\tau)}. \quad (312)$$

The level k controls the momentum quadratic form, while the oscillator determinant has central charge one. This is an explicit demonstration that level and Virasoro anomaly are different invariants.

Sewing. Let T_q be the one-particle sewing operator with eigenvalues q^n . Then

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

The bosonic Fock functor turns a one-particle determinant into its inverse, giving (312). On a compact Riemann surface of genus g , the nonzero-mode Gaussian integral gives a zeta-regularized determinant of the scalar Laplacian, while zero modes give a volume factor. A compactified boson with momentum lattice Λ multiplies the oscillator determinant by a Siegel theta series.

Field theory. Abelian Chern–Simons theory with action

$$\frac{k}{4\pi} \int_M A \wedge dA$$

and a chiral boundary condition has boundary current algebra (311). A Wilson line of charge q acquires the braiding phase determined by the bilinear form $q_1 q_2 / k$ after the standard normalization of charges and level. The boundary algebra, line category, and Gaussian determinant therefore arise from different operations in one free field theory.

Koszul reading. The quotient $\mathfrak{h}_{\text{ch}}$ has a cofree commutative Chevalley coalgebra with zero bracket differential. For the central extension $\widehat{\mathfrak{h}}_{\text{ch}}$, the Chevalley coderivation contracts two current cogenerators to the central cogenerator. In the level- k vacuum representation the central element acts as $k\mathbf{1}$; after reducing by the vacuum line, this contraction is represented as curvature in the reduced bar model. The completed cobar reconstructs the corresponding chiral enveloping algebra. Linear duality exchanges symmetric algebras and symmetric coalgebras under finite-weight completion.

G.2 Affine Kac–Moody currents

Let $\mathfrak{g}$ be simple and let $(,)$ be an invariant form. The universal affine vertex algebra at level k has currents with OPE

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \quad (313)$$

The lambda bracket is

$$[J_\lambda^a J^b] = J^{[a,b]} + k\lambda(a, b).$$

Its Jacobi identity has two coefficients. The coefficient independent of λ is Jacobi in $\mathfrak{g}$. The coefficient linear in the formal variables is invariance of $(,)$. The simple-pole bracket satisfies Jacobi, and the central term is the compatible invariant cocycle.

Collision complex. For the centrally extended chiral Lie algebra, the partition collision differential has two components: the simple-pole bracket merges currents to $J^{[a,b]}$, and the central cocycle merges them to the central line. The simple-pole component squares to zero by Jacobi; compatibility with the central component is exactly invariance of $(,)$. A geometric coefficient model that includes all principal parts carries both the first and second normal symbols.

Sugawara tensor. For $k \neq -h^\vee$, the stress tensor is

$$T(z) = \frac{1}{2(k + h^\vee)} \sum_a : J^a J_a : (z), \quad (314)$$

with central charge

$$c = \frac{k \dim \mathfrak{g}}{k + h^\vee}. \quad (315)$$

The shift by $h^\vee$ comes from normal ordering. Equation (315) gives the affine conformal anomaly as the rational function $k \dim \mathfrak{g} / (k + h^\vee)$.

KZ transport. For modules $V_1, \dots, V_n$, conformal blocks satisfy the KZ equation

$$\left(\partial_{z_i} - \frac{1}{k + h^\vee} \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} \right) F = 0.$$

Flatness is the calculation of Section B. The monodromy is a braid-group representation. Under the Drinfeld–Kohno comparison, it agrees with the braiding of the corresponding quantum-group category at a parameter fixed by the normalization of the invariant form and the exponentiation of $(k + h^\vee)^{-1}$.

Chern–Simons theory. Three-dimensional Chern–Simons theory at integral level produces the same conformal blocks after quantization on a surface. Wilson-line braiding gives the same ribbon category. The bulk–boundary theorem identifies the quantized current algebra and its Wilson-line category with these conformal blocks and ribbon operations. Framing dependence corresponds to the ribbon twist and the gravitational Chern–Simons counterterm.

Critical level. At $k = -h^\vee$, (314) is undefined. The affine algebra itself remains defined and develops the Feigin–Frenkel centre. The KZ equation in the displayed normalization is singular. The centre is identified with functions on opers for the Langlands dual group. These statements identify criticality with the singular Sugawara denominator and the simultaneous emergence of the oper centre.

Koszul and representation-theoretic dualities. Chiral bar–cobar reconstruction applies in the pro-nilpotent Ran category. Quadratic PBW recognition can be tested after choosing a presentation.

Kazhdan–Lusztig equivalence, Drinfeld–Kohno monodromy, Feigin–Frenkel duality, and quadratic Koszul duality are different theorems. They can be composed only when their hypotheses and categories match.

G.3 The Virasoro chiral algebra

The Virasoro algebra is the central extension of vector fields on the circle. Its OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (316)$$

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity is a polynomial identity in two formal variables. It expresses the cocycle identity of the Gelfand–Fuchs central extension and the Lie bracket of vector fields.

Normal symbols. The three singular terms of (316) occupy normal degrees four, two, and one. The normally ordered product occupies degree zero. A logarithmic residue detects only the degree-one component after a logarithmic form has been supplied. The chiral operation remembers all four pieces and their extensions.

Projective connections. The transformation law

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12} \{f, z\}$$

shows that the stress tensor is an affine projective connection. The Schwarzian cocycle is the coordinate form of the Virasoro central extension. On moduli, it gives a projective connection on conformal blocks. The Hodge coefficient is proportional to c , while insertions and boundary factorization contribute further classes.

Verma modules and null vectors. A highest-weight module $M(c, h)$ is generated by v with

$$L_0 v = h v, \quad L_n v = 0 \quad (n > 0).$$

The Kac determinant detects singular vectors. Passing from the universal Virasoro algebra to a minimal-model quotient changes the homological algebra because relations generated by null vectors are imposed. Each simple quotient therefore requires its own bar calculation after the null-vector relations have been imposed.

The bosonic string. The (b, c) reparametrization ghosts have central charge -26 . Coupling them to a matter theory gives a nilpotent BRST charge when the total central charge and intercept anomaly vanish. The number 26 is therefore the cancellation value of a specific ghost system. The value 13 is the fixed point of the numerical reflection $c \mapsto 26 - c$; a Virasoro self-duality theorem additionally requires an equivalence of the corresponding chiral algebras.

Polyakov and determinant geometry. The Weyl anomaly has coefficient c . For free fields it is computed from determinants. The curvature of the determinant line agrees with the central extension represented by (316). This is the rigorous bridge between local OPE and global moduli curvature. A full nonchiral CFT or gravitational path integral is obtained after adjoining the antiholomorphic sector, spectrum, modular invariant, and integration cycle.

Three-dimensional gravity. Brown–Henneaux boundary conditions produce two Virasoro algebras with

$$c = \bar{c} = \frac{3\ell}{2G}.$$

Under modular invariance, a sparse light spectrum, and vacuum dominance, Cardy asymptotics reproduce the BTZ entropy. The chiral algebra supplies the asymptotic symmetry and central extension. The spectrum and modular invariant are additional physical data.

G.4 Free fermions and ghost systems

A free complex fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The associated Clifford algebra is quadratic. Wick’s theorem expresses every correlator as a determinant. On a Riemann surface, a choice of spin structure enters the Szego kernel. The partition function is a determinant or Pfaffian of the Dirac operator.

A (b, c) system with conformal weights $(\lambda, 1 - \lambda)$ has stress tensor

$$T = (1 - \lambda):(\partial b)c: - \lambda:b\partial c:$$

and central charge

$$c_{bc} = 1 - 3(2\lambda - 1)^2. \tag{317}$$

A bosonic (β, γ) system with the same weights has the opposite determinant statistics and central charge

$$c_{\beta\gamma} = -c_{bc}.$$

The cancellation in a tensor product is a determinant identity and a Virasoro central-charge identity. The associated graded polynomial and exterior algebras are ordinary Koszul duals. The Weyl and Clifford algebras are PBW deformations of these two algebras and are related by the corresponding curved or inhomogeneous Koszul formalism when their constant quadratic terms are retained. An equivalence of their conformal module categories requires a further tensor-functor construction compatible with conformal weights and sewing.

Ghost number provides a second grading. BRST reduction uses the ghost differential and the vertex differential simultaneously. Spectral-sequence concentration can make the reduction exact. In the general case, every higher BRST cohomology group remains part of the reduced derived object.

G.5 Lattice vertex algebras

Let Λ be an even lattice with bilinear form $(\cdot, \cdot)$ and let ε be a normalized two-cocycle on Λ . The lattice vertex algebra is

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

For $\alpha, \beta \in \Lambda$, the vertex operators have leading product

$$Y(e^\alpha, z)Y(e^\beta, w) = (z - w)^{(\alpha, \beta)} \varepsilon(\alpha, \beta) Y(e^{\alpha+\beta}, w) + \cdots. \quad (318)$$

Evenness gives locality. The cocycle controls the ordered phase. Replacing ε by a cohomologous cocycle gives an isomorphic algebra; replacing it by its inverse reverses the commutator phase.

The genus-one character is

$$\text{ch } V_\Lambda(\tau) = \frac{\Theta_\Lambda(\tau)}{\eta(\tau)^{\text{rk } \Lambda}}. \quad (319)$$

For an even unimodular positive-definite lattice, the chiral central charge is $\text{rk } \Lambda$ and the theory is holomorphic. At genus g , the zero-mode sum is the degree- g Siegel theta series

$$\Theta_\Lambda^{(g)}(\Omega) = \sum_{x_1, \dots, x_g \in \Lambda} \exp \left(\pi i \sum_{i,j} (x_i, x_j) \Omega_{ij} \right).$$

The oscillator and lattice pieces are independent factors.

For the two even unimodular rank-sixteen lattices $E_8 \oplus E_8$ and D_{16}^+ , the Siegel theta series agree in degrees $g = 1, 2, 3$. In degree four their difference is a nonzero scalar multiple of the Schottky form on $\mathfrak{H}_4$; its restriction to the genus-four Jacobian locus vanishes [59, 66]. Thus the first Siegel modular distinction occurs in degree four on $\mathfrak{H}_4$, while restriction to the smooth genus-four Jacobian locus identifies the two theta series again.

The Leech lattice VOA and the Moonshine VOA both have central charge 24. Their weight-one spaces differ: the Leech theory has a rank-twenty-four Heisenberg sector, while the Moonshine weight-one space is zero. The weight-one Lie algebra, OPE coefficients, and automorphism group therefore refine the shared central charge.

G.6 Principal W -algebras

Let $\mathfrak{g}$ be simple and f principal nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine algebra, charged ghosts, and neutral fields when required. Its degree-zero cohomology is the principal W -algebra $\mathcal{W}^k(\mathfrak{g}, f)$ under the standard hypotheses.

For the universal principal W -algebra at generic level, the standard strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$; special levels may impose decoupling relations or collapse generators. Their OPEs contain nonlinear normally ordered polynomials. These terms are transferred operations of the BRST contraction and require the complete chiral operation, including regular normally ordered products.

Feigin–Frenkel duality relates principal W -algebras for Langlands dual Lie algebras at levels satisfying

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1, \quad (320)$$

with the lacing convention included. The relation is multiplicative in the shifted levels and differs

from the additive affine reflection $k \mapsto -k - 2h^\vee$.

If a filtered BRST contraction is compatible with the chiral bar construction and its spectral sequence converges, then bar and BRST cohomology commute. This transports collision operations and Koszul comparisons through the reduction. The conclusion holds for each nilpotent orbit and completion carrying the stated compatible contraction and convergent filtration.

G.7 Five comparisons in one table

Model	highest local pole	ordered datum	modular datum	physical realization
Heisenberg	double pole k	abelian phase	oscillator determinant and zero modes	abelian Chern–Simons boundary
affine	double pole plus Lie pole	KZ braid monodromy	affine conformal blocks	nonabelian Chern–Simons/WZW
Virasoro	fourth-order central pole	monodromy of Virasoro blocks	projective connection and determinant line	conformal anomaly; AdS_3 asymptotic symmetry
free ghosts	simple contraction	fermionic or bosonic exchange	determinant/Pfaffian	gauge fixing and BRST
lattice	integral power and cocycle	cocycle phase	Siegel theta series	compactified free boson
W -algebra	nonlinear higher-spin OPE	module-category braiding when constructed	W -conformal blocks	Hamiltonian reduction

The columns record independent structures linked by the comparison maps constructed for each model. The local pole controls the singular operation. Ordered transport requires modules and a connection. Modular data requires sewing. A physical realization requires an action and boundary condition. The exact models show that the same chiral operation can feed all four constructions while preserving their distinct categories and operations.

G.8 What the examples establish

The examples test the common language by independent constructions whose comparison maps are known.

For Heisenberg theory, the boundary quantization map identifies the current OPE with the abelian Chern–Simons boundary bracket, while Gaussian sewing identifies the oscillator trace with the determinant formula.

For affine theory, the Ward identities produce the KZ connection; Drinfeld–Kohno identifies its braid category with the corresponding quantum-group category, and the Chern–Simons/WZW comparison identifies the same braid data with Wilson-line exchange.

For Virasoro theory, the fourth normal symbol gives the central cocycle, the Ward identity gives the Schwarzian transformation, and the family index theorem identifies its moduli-space image with determinant-line curvature and the Weyl anomaly.

For ghost systems, the Clifford or Weyl quadratic relations determine the parity signs, the corresponding determinant ratio gives the central charge, and the BRST square measures the same total anomaly after the BRST comparison is made.

For lattice theories, the group-algebra cocycle determines ordered phases, Poisson summation gives the modular transformation of the theta series, and the compactified-boson path integral yields the same theta factor after zero-mode and oscillator separation.

Each agreement is mediated by an explicit construction. This is the standard of evidence appropriate to chiral Koszul duality: one operation, calculated in several faithful realizations, with every passage preserving the category of the object.

H Object types, historical lineage, and the common language

A unifying language earns its force by preserving the native category of every object. This appendix records the precise domains of the constructions used in the paper and the comparison maps that connect them.

H.1 The primitive object

Fix a smooth complex curve X . The primitive datum is

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}). \quad (321)$$

Thus A consists of a factorisable D -module, a unital associative multiplication

$$\star : A \otimes^{\mathrm{pt}} A \longrightarrow A,$$

and an augmentation $A \rightarrow \mathbf{1}$. The factorisation structure carries holomorphic collision data; the E_1 multiplication carries transverse ordered composition. The multiplication is a morphism of factorisation objects and therefore obeys the duoidal interchange law.

A diagonal-supported chiral Lie operation

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta! \mathfrak{g}$$

supplies the Francis–Gaitsgory Chevalley branch. The transverse associative multiplication is an additional operation on the coefficient object. Together they form the ordered chiral algebra of (321).

Construction		Output	Input carried by the construction
ordered interval bar		conilpotent coassociative coalgebra in $\text{FactD}(X)$	augmentation and a convergent completion
boundary endomorphisms		$A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1})$	augmentation boundary and compactness for double centralization
quadratic comparison		$A^i \rightarrow \text{Bar}_X^{\text{ord}}(A)$	quadratic or filtered presentation
chiral Chevalley		commutative factorisation coalgebra on $\text{Ran } X$	chiral Lie coefficient and Ran pronilpotence
formal-disc realization		vertex or quantum-vertex operations	coordinate, translation, and exchange data
Verdier dual		dual constructible or holonomic object	dualisability, continuity, and monoidal base change
planar enhancement		exchange paths and Yang–Baxter coherence	a transverse plane or flat exchange connection
Tannaka reconstruction		coordinate bialgebra and its continuous dual	rigid tensor category and tensor fibre functor
two sewing structures		bicoloured stable-graph operations	transverse cyclic trace and chiral modular sewing
conformal blocks		sheaf or complex on moduli	modules, finiteness, projective connection, and sewing
internal Hochschild cochains		E_2 derived centre	diagonal bimodule resolution
physical observables		analytic topological algebra	holomorphic factorisation
			fields, action, gauge fixing, renormalisation, and algebraisation

A comparison morphism between two rows states their relation. A quasi-isomorphism, monoidal equivalence, or index map specifies the strength of that relation.

H.2 Bar, boundary dual, quadratic dual, Verdier dual, and centre

Five neighboring constructions occur.

Bar. For an augmented associative object A , the bar construction is a coalgebraic resolution governed by twisting morphisms. Its definition uses augmentation, associative multiplication, and the chosen completion.

Boundary Koszul dual. The universal dual algebra is

$$A^! = \text{RHom}_A(\mathbf{1}, \mathbf{1}).$$

Its double centralizer is the derived augmentation completion under the compactness and proxy-smallness hypotheses stated in the text.

Quadratic dual. A presentation $A = T(V)/(R)$ defines a quadratic coalgebra A^i . The canonical comparison

$$A^i \longrightarrow \text{Bar}(A)$$

is a quasi-isomorphism on the quadratic Koszul locus. PBW and curved versions replace the homogeneous presentation by filtered and quadratic-linear-constant data.

Verdier dual. Verdier duality is contravariant on sheaves and D -modules. It sends a factorisation coalgebra to a factorisation algebra when duality commutes with the tensor products, diagonals, and limits appearing in the factorisation structure.

Derived centre. For an associative factorisation algebra A ,

$$Z_{\text{ch}}^{\text{der}}(A) = \text{HH}_{\text{Fact}}^{\bullet}(A, A)$$

is the derived endomorphism algebra of the diagonal bimodule. For a chiral Lie algebra $\mathfrak{g}$, the typed construction applies to $U^{\text{ch}}(\mathfrak{g})$. The higher Deligne theorem gives the centre an E_2 structure.

The Koszul–Morita square relates these constructions on the finite, complete, and perfect locus. The comparison maps carry the mathematical content.

H.3 Four squares and four curvatures

Four operators carry square identities in this subject.

- (i) A differential on a dg object satisfies $d^2 = 0$.
- (ii) A curved A_{∞} algebra satisfies the curved Stasheff identities with curvature element m_0 .
- (iii) A connection satisfies $\nabla^2 = F_{\nabla}$, where F_{∇} is a two-form on the base with values in endomorphisms.
- (iv) A BV quantum differential squares to the anomaly operator, and the quantum master equation sets that anomaly class to zero.

For a central scalar curvature element, the commutator $[m_0, -]$ is zero while m_0 remains in the curved Maurer–Cartan equation. An exact BV anomaly is removed by a local counterterm. A projective curvature is cancelled by tensoring with a line of opposite Atiyah class when that line has been constructed. These mechanisms occupy distinct deformation complexes and meet through explicit comparison maps.

H.4 Residue, principal part, and propagator

Residue. Poincaré residue extracts the coefficient of the logarithmic normal form dt/t .

Principal part. The coefficient of a pole of order m is a section of $N_{D/Y}^{\otimes m}$. The pole filtration and its associated graded carry all singular modes of a chiral operation.

Propagator. A propagator is a Green kernel or parametrix selected by a kinetic operator and a gauge-fixing condition. Its singular part represents local principal-part data; its regular and harmonic parts encode global geometry.

The logarithmic form $d \log(z - w)$ generates the first cohomology of a two-point affine configuration screen. The normal-symbol tower carries the full OPE. The propagator provides an analytic realization of these local symbols in a chosen field theory.

H.5 Ordering, descent, and braiding

Labelled configurations carry pure-braid monodromy. Unlabelled configurations arise by descent along the symmetric-group cover. Homotopy invariants and coinvariants agree for finite symmetric groups over characteristic zero after averaging, while the labelled local system retains its braid representation.

A Drinfeld associator is a group-like series satisfying pentagon and hexagon equations. It compares parenthesized monodromies. The Grothendieck–Teichmüller group acts on the torsor of such associators.

Verdier duality sends a local system with monodromy ρ to the dual local system with monodromy ρ^{-T} . A chosen nondegenerate bilinear form may identify this with inverse monodromy. A Tannakian or presentation-theoretic recognition theorem then identifies the resulting symmetry with a named quantum group.

H.6 Modular sewing and the master equation

The compactification $\overline{\mathcal{M}}_{g,n}$ has boundary divisors indexed by one-edge stable graphs and codimension-two strata indexed by two-edge stable graphs. Their oriented incidence relation is the differential of the Feynman transform. A cyclic object equipped with clutching-compatible contractions produces a structure element satisfying

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0.$$

The clutching maps provide the coefficients; the boundary incidence provides the signs.

For conformal blocks, the modular-functor hypotheses provide sewing and a projectively flat connection. The first Chern class contains a Hodge term from the central charge, cotangent-line terms from conformal weights, and boundary terms from factorisation channels. The complete modular datum is therefore the bundle with connection, fusion decomposition, and sewing maps.

H.7 The mathematical history

The language assembled here arose through a sequence of exact advances.

Historical note H.1 (Distributions). Dirac’s delta function made an operation supported on a submanifold calculable. Schwartz’s distribution theory made support and differentiation intrinsic. A chiral bracket is the holomorphic D -module-valued descendant: the product is defined on the complement of the diagonal and its singular commutator is supported on the diagonal.

Historical note H.2 (Operator products). Wilson separated short-distance singularity from long-distance state dependence. The operator product expansion is a local asymptotic expansion whose coefficients are local fields. Vertex algebra theory expresses this expansion by formal identities. Chiral algebra theory replaces the chosen coordinate by a geometric correspondence.

Historical note H.3 (Borcherds and Beilinson–Drinfeld). Borcherds recognized locality, commutator formulas, and associativity as coefficient forms of one formal-distribution identity. Beilinson and Drinfeld placed that identity on a curve: operations are defined on configuration space and land on diagonals. The formal variable becomes the normal coordinate of a collision.

Historical note H.4 (Fulton–MacPherson and Kontsevich). Fulton and MacPherson resolved simultaneous collisions by adding screens that record relative configurations. Kontsevich used compactified configuration spaces to organize deformation quantization. Perturbative field theory uses the same spaces to separate ultraviolet regions and encode the forest structure of counterterms.

Historical note H.5 (Ran space and chiral Koszul duality). The Ran space treats a finite subset of a curve as one variable. Factorisation multiplies over disjoint unions. Francis and Gaitsgory proved the Chevalley–primitive equivalence between chiral Lie algebras and conilpotent commutative factorisation coalgebras in the pro-nilpotent chiral monoidal category.

Historical note H.6 (Braids and quantum groups). Knizhnik and Zamolodchikov constructed a flat connection from current algebra. Kohno and Drinfeld identified its braid monodromy with the braided tensor structure of quantum groups. Witten’s Chern–Simons theory produced the same representations from Wilson-line amplitudes. Costello–Witten–Yamazaki obtained rational, trigonometric, and elliptic integrable systems from four-dimensional gauge theory.

Historical note H.7 (Polyakov, Segal, and modular geometry). Polyakov’s path integral expressed the conformal anomaly as curvature on the space of metrics. Segal formulated conformal field theory through sewing of Riemann surfaces. Tsuchiya–Ueno–Yamada constructed conformal blocks in families. Determinant and Hodge lines record the projective anomaly, while modular operads encode the topology of sewing.

Historical note H.8 (Feynman and BV). Feynman’s graphs expand an action perturbatively. The Batalin–Vilkovisky formalism expresses gauge invariance through classical and quantum master equations. Costello’s renormalisation theory constructs effective interactions on compactified configuration spaces and places anomalies in local cohomology. A comparison of fields, pairings, and vertices identifies a cyclic chiral graph equation with a BV theory.

H.8 The physical dictionary

Let Obs be a factorisation algebra of quantum observables whose protected boundary geometry is $X^{\text{an}} \times \mathbb{R}$. A quasi-isomorphism

$$\Phi : A^{\text{an}} \xrightarrow{\sim} \text{Obs}_{\partial} \tag{322}$$

of $E_1 \otimes \text{Ch}_X$ factorisation objects identifies the following structures.

Chiral geometry	Quantum-field-theoretic realization
D -module of fields	cochain complex of boundary local observables
chiral operation	renormalised boundary OPE
partial diagonal	coincidence locus of insertions
principal-part filtration	scaling hierarchy of contact singularities
partition collision complex	local short-distance factorisation complex
Fulton–MacPherson screen	relative ultraviolet configuration
chiral Chevalley coalgebra	coalgebra of factorised insertions
ordered local system	exchange transport of defects or lines
cyclic pairing	two-point function used to sew legs
modular graph operation	amplitude of fixed topology
BV anomaly class	obstruction class of the quantum master equation
Hochschild centre map	bulk observable acting centrally on the boundary

The map Φ turns the two columns into two models of the same operations. Its homotopy fibre measures the remaining physical data.

H.9 Exact data requirements

Separation theorem H.9 (Quantum-group recognition data). *A quantum-group reconstruction consists of a rigid braided or meromorphic tensor category, a tensor fibre functor, and matrix coefficients satisfying a recognition theorem. The dimension of $\ker(V^{\otimes n} \rightarrow \mathrm{Sym}^n V)$ records a vector-space loss under symmetrisation; the categorical data supply multiplication, coproduct, monodromy, and coherence.*

Proof. The coend of a rigid braided category with a tensor fibre functor carries multiplication from tensor product, coproduct from matrix coefficients, and the coquasitriangular form from braiding. These maps depend on categorical composition and coherence. The displayed kernel is the fibre of a linear projection and carries only its inherited vector-space structure. Thus the recognition data are exactly the structures that construct the quantum group. $\square$

Separation theorem H.10 (Coefficient data for modular sewing). *A modular master element consists of cyclic contraction maps assigned to separating and nonseparating clutching strata, together with their compatibility on codimension-two boundaries. The incidence relation $\partial^2 = 0$ then proves the master equation.*

Proof. A cyclic contraction labels every internal edge of a stable graph. Pullback to a codimension-two boundary composes two such contractions in the two possible orders. Equality of these pullbacks identifies the two orientations of each codimension-two stratum. Their signed sum is zero, so the induced convolution differential satisfies the modular master equation. $\square$

Separation theorem H.11 (Open–closed reconstruction data). *The derived centre is terminal among algebraic central actions on the boundary. A physical bulk is identified with it by a bulk-to-centre quasi-isomorphism or by a generation theorem that produces such a quasi-isomorphism. The homotopy fibre records bulk sectors with unit boundary action.*

Proof. The Swiss-cheese action of bulk observables on boundary observables factors through the universal central action, producing the bulk-to-centre map. A quasi-isomorphism identifies the two

local observable algebras. For a product bulk whose second factor acts through the unit, the reduced observables of that factor map to zero and therefore lie in the homotopy fibre. This proves both reconstruction and the stated residual datum. $\square$

Separation theorem H.12 (Classification data). *Central charge determines the Virasoro central extension and its Hodge coefficient. The complete chiral operation, representation category, braid monodromy, and higher-genus blocks provide additional coordinates. Holomorphic theories at fixed central charge display these independent coordinates concretely.*

Proof. The central charge is the coefficient of the Virasoro central extension. The remaining structures depend on the other OPE coefficients, module intertwiners, flat transport, and sewing maps. Even lattices of a fixed rank give holomorphic chiral theories with the same central charge and different lattice products and higher-genus theta series. Hence the listed coordinates refine the central-charge fibre. $\square$

Separation theorem H.13 (Anomaly comparison data). *Current level, Virasoro central charge, determinant-line curvature, BRST anomaly, framing anomaly, and Borchers-product weight live in their respective deformation theories. A family theorem relates selected components by a morphism of invertible theories, an index map, or a trace.*

Proof. Each entry is a class in the deformation complex native to its geometric carrier: current extensions, Virasoro extensions, determinant lines, BRST local functionals, framed invertible theories, and automorphic line bundles. A morphism of invertible theories, a family index, or a trace induces the corresponding map on these complexes and hence the stated relation among selected components. $\square$

H.10 The common language as a commutative diagram

$$\begin{array}{ccccc}
 & A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X)) & & & \\
 & \swarrow \text{Bar}_X^{\text{ord}} & \downarrow \text{RHom}_A(1, -) & \searrow \text{HH}_{\text{FactD}}^\bullet & \\
 \text{Coalg}_{\text{CoAss}}^{\text{conil}}(\text{FactD}(X)) & \xleftarrow{\text{Cobar}_X^{\text{ord}}} A' = \text{RHom}_A(1, 1) & \xrightarrow{\text{bulk action}} & Z_{\text{ord, ch}}(A) & \xrightarrow{E_2/E_3 \text{ extension}} \mathcal{L}_A \\
 & \downarrow & & & \downarrow \text{Tannaka} \\
 & \text{Obs}_\partial(X \times \mathbb{R}) & & & \mathcal{H}_A
 \end{array} \tag{323}$$

A chiral Lie coefficient $\mathfrak{g}$ supplies the independent branch

$$\mathfrak{g} \xrightarrow{\text{coChev}^{\text{ch}}} \text{CoFact}_{\text{conil}}(\text{Ran } X).$$

A strong monoidal comparison relates this branch to the ordered interval bar. A transverse cyclic trace and a chiral modular sewing system add the bicoloured Feynman transform and its master element. A field-theoretic realization adds propagators, renormalised graph weights, and states.

H.11 The final principle

The mathematical phenomenon is the coexistence of holomorphic collision and transverse associative composition. The physical phenomenon is the short-distance product of observables together with

their ordered fusion along a topological direction. The geometric carrier is the operadic product of chiral collision spaces and oriented intervals. The algebraic laws are the chiral identities of the coefficient, associativity of the E_1 multiplication, and their interchange. Chiral Koszul duality produces factorised coalgebraic states in the Ran direction; ordered bar–cobar duality produces interval states in the transverse direction. Ordering records composition, planar transport records exchange, cyclicity permits loops, moduli record sewing, and the derived centre records universal central action.

Every noun has a category. Every verb is a functor, correspondence, contraction, or trace. This typed syntax is the common language of the subject.

I Historical lineage of the language

I.1 Distributions and local fields

Dirac’s delta distribution made support a mathematical datum rather than a limiting metaphor [27]. Quantum field theory inherited this lesson: products of fields are defined away from coincidence and acquire contact terms on diagonals. The chiral D -module operation is the algebraic form of that distributional fact. It records the singular product and its support before a coordinate expansion is chosen.

Wilson’s operator product expansion separated short-distance coefficients from local operators [75]. Borchers and the vertex-algebra tradition encoded the consistency of those coefficients in formal distributions. Beilinson and Drinfeld replaced the formal coordinate by a curve and the formal delta function by a morphism of D -modules supported on the diagonal. The present paper uses the Beilinson–Drinfeld object as the chiral coefficient of a genuinely associative ordered product.

I.2 Time ordering and the associative direction

Feynman’s time-ordered product is an ordered associative multiplication [39]. It depends on an ordering of insertion times, and its perturbative expansion is indexed by graphs. Stasheff’s associahedra later supplied the homotopy geometry of all parenthesizations [68, 69]. Factorization homology identifies the same geometry with the bar construction on an interval.

The ordered chiral algebra combines these two historical lines. Dirac and Beilinson govern support on the complex diagonal. Feynman and Stasheff govern composition along the ordered real direction.

I.3 Configuration spaces and renormalization

Arnold computed the cohomology of configuration spaces and isolated the three-term relation that underlies the flatness of logarithmic connections [4]. Fulton and MacPherson compactified configurations by replacing collisions with screens [40]. Axelrod–Singer and Kontsevich used related compactifications to make perturbative gauge-theory and deformation-quantization integrals finite and compatible with boundary strata [9, 53].

The compactification has two roles that must remain distinct. Its face category resolves collision incidence. Its screen spaces carry nontrivial cohomology and propagator integrals. The first is combinatorial; the second contains the analytic weight of Feynman graphs.

I.4 Chiral algebras and Koszul duality

Koszul, Priddy, Quillen, and the operadic tradition developed the passage between algebraic operations and coalgebraic resolutions. Beilinson–Drinfeld placed local Lie operations on the Ran space. Francis–Gaitsgory proved the corresponding chiral Koszul duality in a pro-nilpotent monoidal category.

The associative extension considered here uses the same Ran geometry with one additional ordered operadic direction. Its bar is the interval bar internal to factorization D -modules. Its dual is the derived endomorphism algebra of the augmentation boundary. This is the form of Koszul duality that appears in defect physics.

I.5 Anomalies and geometry

Polyakov identified the Weyl anomaly with a local action for the conformal factor [65]. Quillen and the family index theorem identified the same anomaly with curvature of a determinant line [62]. Belavin and Knizhnik related determinant geometry to string measures on moduli [11]. These are different realizations of a central extension of conformal transformations.

The language of the present paper preserves the chain of maps: local stress-tensor cocycle, Ward identity, Atiyah algebra of conformal blocks, determinant-line curvature. Every scalar coefficient appears as the final trace of this chain.

I.6 Gauge theory, knots, and quantum groups

Witten’s Chern–Simons theory connected three-dimensional gauge fields, two-dimensional current algebras, conformal blocks, braid groups, and knot invariants [74]. Drinfeld and Jimbo supplied the quantum groups [29, 48]; Kohnno identified their braid representations with KZ monodromy [51]. The decisive structure is the category of lines and its exchange, and the Tannakian reconstruction from that exchange category.

Costello, Witten, and Yamazaki added a holomorphic spectral curve to topological gauge theory [19, 20]. Wilson-line OPEs then produced rational, trigonometric, and elliptic quantum groups. Gauge-theoretic constructions of Gaiotto, Moore, and Neitzke and of Nekrasov placed line defects, framed BPS states, Omega backgrounds, and BPS/CFT Ward identities in the same protected sector [41, 57]; Coulomb-branch convolution gives a complementary algebraic realization [10]. The spectral curve carries chiral dependence; transverse topology carries fusion and exchange.

I.7 Factorization algebras and BV theory

Costello and Gwilliam formulated perturbative quantum field theory as a factorization algebra of observables [14, 15]. The BV classical and quantum master equations became local cohomological identities. Boundary and defect observables became factorization modules. This made the passage from an action functional to a chiral or topological operator algebra a functor with a defined source and target.

The present construction is compatible with that formalism. An ordered chiral algebra is the algebraic local observable structure on a holomorphic–topological product. A BV theory is one realization of it, together with fields, pairing, integration, and renormalization.

I.8 Open–closed theories and centres

Hochschild cochains have long governed deformations of associative algebras. Deligne’s conjecture and the Swiss-cheese operad identified their E_2 structure and their universal action on an E_1 algebra. In open–closed field theory, the centre is the universal closed algebra acting on the open sector.

This universal property is exact but limited. It reconstructs the protected bulk from a boundary only when the bulk-to-centre map is an equivalence. Witten’s open-string field theory and Costello’s Calabi–Yau categories supply settings in which Hochschild chains and cyclic operations are physical closed states [76, 18]. Twisted holography supplies settings in which a boundary algebra determines a protected bulk sector.

I.9 The resulting synthesis

The historical constructions meet in one typed sentence. A noncommutative chiral algebra is an E_1 algebra in factorization D -modules; its chiral coefficient is a diagonal-supported distribution, its multiplication is ordered fusion, its bar is interval factorization homology, its centre is the universal exchange algebra, its quantum group is Tannaka reconstruction of a braided line category, its ribbon loops come from a cyclic trace, its worldsheet sewing comes from conformal blocks, and its anomalies are invertible theories identified by transgression.

Every clause has an independent historical origin. Their unity is the geometry of support, order, exchange, and sewing.

J Concordance with the extended source corpus

The extended source corpus contains calculations and constructions developed in several earlier languages. The present paper assigns each construction a geometric carrier and records the map that places it in the ordered chiral theory. The concordance below is part of the mathematics: it specifies the object whose invariant is being calculated.

J.1 The carrier map

Every construction in the corpus enters through one of the following carriers:

$$\begin{aligned}
 C_{\text{int}}(A) &= \text{Bar}_X^{\text{ord}}(A) = \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1}, & \text{interval composition,} \\
 C_{\text{ch}}(\mathfrak{g}) &= \text{Chev}_X^{\text{ch}}(\mathfrak{g}), & \text{diagonal collision,} \\
 C_{\text{scr}}(I) &= R\Gamma(\text{Screen}_I, \mathcal{K}_I), & \text{screen cohomology and propagators,} \\
 C_{\text{br}}(A) &= \{\mathcal{L}_n, \nabla_n\}_{n \geq 0}, & \text{exchange local systems,} \\
 C_{\text{cyc}}(A) &= \text{HH}_{\bullet}(A), & \text{annular and ribbon sewing,} \\
 C_{\text{mod}}(A) &= \{\mathbb{V}_{g,n}, \xi_{\Gamma}\}, & \text{stable-curve sewing,} \\
 C_{\text{sc}}(A; \tau) &= \tau(C(A)), & \text{specified scalar trace.}
 \end{aligned} \tag{324}$$

Here $\mathcal{K}_I$ denotes the chosen geometric coefficient system, $\mathcal{L}_n$ a tensor-compatible local system on ordered configurations, ξ_{Γ} the clutching map attached to a stable graph, and τ a trace, index, period, determinant, or character map.

Proposition J.1 (Typed promotion of a calculation). *A chain-level calculation from the source corpus defines an invariant of the present theory when it is accompanied by a quasi-isomorphism from its source complex to one of the carriers in (324), compatible with the relevant product, coproduct, module action, or sewing maps. A numerical calculation defines a scalar invariant after composition with the displayed trace τ .*

Proof. A quasi-isomorphism transports cohomology together with every operation preserved by the map. Compatibility with the structural maps identifies the transported cohomology as an invariant of the target carrier. Composition with a trace produces a functorial scalar. These are the standard invariance properties of derived functors and traces. $\square$

J.2 Ordered associative chiral duality

The source chapter on ordered associative chiral duality contributes the transverse algebra, the two-sided bar, the diagonal bicomodule, and Hochschild–coHochschild comparison. Their intrinsic realization is the Koszul–Morita square of Part I:

$$A \longmapsto C = \mathrm{Bar}_X^{\mathrm{ord}}(A) \longmapsto (C_\Delta, \mathrm{coHH}_\bullet(C), \mathrm{coHH}^\bullet(C)).$$

The chiral coefficient accompanies each term as a factorisable D -module. The multiplication differential is the simplicial interval differential. Collision and screen differentials occupy the independent carriers C_{ch} and C_{scr} .

J.3 Bar tables and named sequences

The tables containing Motzkin and Riordan sequences are exact enumerative outputs of the legacy collision complex. Section 31 gives their generating functions and recurrences. Proposition J.1 supplies their promotion criterion:

$$C_{\mathrm{legacy}}(A) \xrightarrow{\simeq} \mathrm{Bar}_X^{\mathrm{ord}}(A, m_\perp) \tag{325}$$

for a specified transverse multiplication $m_\perp$. The concrete algebras of Section 30 provide comparison targets with independently calculated homology. For enveloping multiplication, the target is the Chevalley complex and its Poincaré polynomial is (193).

J.4 Modules and boundary conditions

The source chapter on chiral modules separates into three exact constructions:

$$\begin{aligned} \text{pointed interval} &\longmapsto M_R \otimes_A^{\mathbb{L}} M_L, \\ \text{marked chiral curve} &\longmapsto \text{pointed conformal block}, \\ \text{boundary category} &\longmapsto \mathrm{RHom}(b_i, b_j). \end{aligned}$$

Section 32 calculates the first and third constructions for quivers, quantum planes, and crossed products. Section 28 constructs the second from module insertions and sewing. A module-level comparison is therefore a map between these displayed objects rather than an identification of their dimensions.

J.5 Higher E_n structures

The higher-dimensional source material enters through

$$A \in \text{Alg}_{E_n}^{\text{aug}}(\text{FactD}(X)), \quad \text{Bar}_X^{(n)}(A) \simeq \int_{(D^n, \partial D^n)} A, \quad Z_{E_n}(A) = HH_{E_n}^\bullet(A).$$

Section 23 proves the iterated reconstruction theorem under explicit completeness hypotheses, derives the P_n bracket from the linking sphere, and states the universal higher open–closed map.

J.6 Yangians and Drinfeld–Kohno transport

The three Yangian source chapters contribute three different layers:

$$R(u) = 1 - \frac{\hbar P}{u} \tag{326}$$

solves the matrix Yang–Baxter equation. The connection

$$\nabla_{\text{KZ}} = d - \hbar \sum_{i < j} \Omega_{ij} d \log(z_i - z_j)$$

is flat. The Tannaka coordinate bialgebra is the coend

$$\int^L F(L)^\vee \otimes F(L).$$

Sections 20, 21, and 33 calculate the three layers and their comparison maps. The coordinate coend reconstructs matrix coefficients; its continuous dual recovers the completed RTT algebra under the finite-grade separation hypotheses of Theorem 33.5.

J.7 Heisenberg and Eisenstein calculations

The Heisenberg source chapter contributes an exact Gaussian sewing theory. Section 34 realizes it in four compatible forms:

$$\eta(\tau)^{-r}, \quad q \frac{d}{dq} \log \eta^{-r} = -\frac{r}{24} E_2, \quad \langle J^a(z) J^b(w) \rangle_\tau = k \delta^{ab} B_\tau(z, w), \quad \text{Wick pairings of } B_\tau.$$

The lattice extension multiplies the oscillator determinant by Θ_L . The E_8 identity $\Theta_{E_8} = E_4$, the Niemeier formula (228), and the rank-sixteen Siegel comparison calculate the arithmetic zero-mode sector.

J.8 Symmetric orbifolds

The symmetric-orbifold source material has two carriers. Equivariant boundary conditions are modules over $A \rtimes G$. Invariant local operators belong to A^G . Their relation is restriction along $A^G \rightarrow A \rtimes G$, with Morita equivalence supplied by a Galois or freeness theorem. The DMVV product (199) is the plethystic trace of cycle sectors. The twisted Hochschild complexes of Section 40 provide the local defect carrier for the same cyclic permutations.

J.9 Critical level and derived Langlands

The derived-Langlands source chapter enters through the exact chain

$$Z(V_{\text{crit}}(\mathfrak{g})) \cong \mathcal{O}(\text{Op}_{LG}(D)), \quad \mathbb{C}_\chi \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{C}_\chi \simeq \widehat{\text{Sym}}(T_\chi^* \text{Op}[1]), \quad \text{critical localization.}$$

Section 25 proves the central Koszul fibre calculation, derives the change-of-rings spectral sequence, and places the Gaudin Hamiltonians and Hecke eigenvalue construction on the oper base.

J.10 Entanglement and replica defects

The entanglement source chapter enters through the cyclic permutation defect. Its algebraic carrier is

$$HH_\bullet(A^{\otimes n}, {}_\sigma(A^{\otimes n})),$$

and its geometric carrier is the n -sheeted branched cover. Section 40 derives the twist weight, interval Renyi entropy, conformal-block decomposition, and topological constant from these carriers.

J.11 Higher genus

The higher-genus source chapter contains prime forms, plumbing expansions, Siegel theta functions, and stable graphs. The present theory distributes them between two loop variables:

$$\hbar_X \quad \text{for algebraic-curve plumbing,} \quad \hbar_\perp \quad \text{for transverse ribbon topology.}$$

Sections 26, 27, and 34 construct the bicoloured Feynman transform, its master equation, and the exact Heisenberg and lattice amplitudes. Each stable-graph scalar is the trace of a graph operation already present in the corresponding modular carrier.

J.12 Arithmetic shadows and scalar projections

The arithmetic source material contributes established modular forms, theta series, Hecke operations, determinant products, and period integrals. Section 37 attaches each scalar to its trace map. The structured objects appear first; their arithmetic values appear after character, determinant, theta, period, or index functors. The relation among two scalar values is therefore the image of a displayed comparison morphism.

J.13 The complete concordance

The constructions from the source corpus now occupy the following mathematical positions:

Source theme	Carrier in this paper	Exact output
ordered associative duality	interval bar and Koszul–Morita square	bar–cobar, module–comodule, Hochschild–coHochschild
bar tables	specified source complex plus comparison map	certified homology and promotion criterion
chiral modules	pointed bars and marked conformal blocks	derived boundary-changing fields and sewing
higher E_n Yangians	iterated pointed-disk integral RTT matrix, KZ local system, Tannaka coend	iterated bar, higher centre, P_n bracket exchange, coordinate bialgebra, completed dual algebra
Heisenberg–Eisenstein	oscillator trace and elliptic bidifferential	η^{-r} , E_2 , Wick correlators
orbifolds	crossed products, invariant algebras, cycle defects	equivariant modules, DMVV trace, twisted sectors
critical Langlands	oper centre and derived central fibre	Koszul conormal classes, Gaudin algebra, Hecke eigenvalue
entanglement	twisted Hochschild circle and branched cover	twist dimensions and entropy traces
higher genus	prime forms, plumbing, bicoloured stable graphs	conformal sewing and ribbon sewing
arithmetic	specified scalar functor	modular forms, theta series, periods, determinants

The concordance realizes one principle throughout: an operation is calculated in several faithful realizations, and every passage preserves the category of the object through an explicit map.

References

- [1] D. Ayala and J. Francis, *Factorization homology of topological manifolds*, J. Topol. **8** (2015), 1045–1084.
- [2] T. Arakawa, *Representation theory of W -algebras*, Invent. Math. **169** (2007), 219–320.
- [3] T. Arakawa, *Introduction to W -algebras and their representation theory*, in *Perspectives in Lie Theory*, Springer INdAM Ser. 19, Springer, 2017, 179–250.
- [4] V. I. Arnold, *The cohomology ring of the colored braid group*, Mat. Zametki **5** (1969), 227–231.
- [5] A. Beilinson and V. Drinfeld, *Chiral Algebras*, Amer. Math. Soc. Colloq. Publ. 51, 2004.
- [6] B. Bakalov, A. De Sole, R. Heluani, and V. G. Kac, *An operadic approach to vertex algebra and Poisson vertex algebra cohomology*, Jpn. J. Math. **14** (2019), 249–342.
- [7] B. Bakalov, A. De Sole, R. Heluani, and V. G. Kac, *Chiral versus classical operad*, Int. Math. Res. Not. IMRN (2020), no. 19, 6463–6488.
- [8] B. Bakalov, A. De Sole, and V. G. Kac, *Computation of cohomology of vertex algebras*, Jpn. J. Math. **16** (2021), 81–154.
- [9] S. Axelrod and I. M. Singer, *Chern–Simons perturbation theory II*, J. Differential Geom. **39** (1994), 173–213.
- [10] A. Braverman, M. Finkelberg, and H. Nakajima, *Coulomb branches of $3d \mathcal{N} = 4$ quiver gauge theories and slices in the affine Grassmannian*, Adv. Theor. Math. Phys. **23** (2019), 75–166.
- [11] A. A. Belavin and V. G. Knizhnik, *Algebraic geometry and the geometry of quantum strings*, Phys. Lett. B **168** (1986), 201–206.
- [12] J. D. Brown and M. Henneaux, *Central charges in the canonical realization of asymptotic symmetries*, Comm. Math. Phys. **104** (1986), 207–226.
- [13] D. Butson and S. Nair, *On the deformation theory of chiral quantizations*, arXiv:2606.27341, 2026.
- [14] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory, Vol. 1*, New Mathematical Monographs 31, Cambridge Univ. Press, 2017.
- [15] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory, Vol. 2*, New Mathematical Monographs 41, Cambridge Univ. Press, 2021.
- [16] K. Costello and D. Gaiotto, *Vertex operator algebras and $3d \mathcal{N} = 4$ gauge theories*, J. High Energy Phys. 05 (2019), 018.
- [17] K. Costello and N. M. Paquette, *Celestial holography meets twisted holography: $4d$ amplitudes from chiral correlators*, J. High Energy Phys. 10 (2022), 193.
- [18] K. Costello, *Topological conformal field theories and Calabi–Yau categories*, Adv. Math. **210** (2007), 165–214.
- [19] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, I*, ICCM Not. **6** (2018), 46–119.
- [20] K. Costello, E. Witten, and M. Yamazaki, *Gauge theory and integrability, II*, ICCM Not. **6** (2018), 120–146.
- [21] P. Deligne, *Équations différentielles à points singuliers réguliers*, Lecture Notes in Math. 163, Springer, 1970.
- [22] C. Damiolini, A. Gibney, and N. Tarasca, *Conformal blocks from vertex algebras and their connections on moduli spaces of curves*, Geom. Topol. **25** (2021), 2235–2286.
- [23] C. Damiolini, A. Gibney, and N. Tarasca, *Vertex algebras of CohFT-type*, in *Facets of Algebraic Geometry, Vol. I*, LMS Lecture Note Ser. 472, Cambridge Univ. Press, 2022, 164–189.

- [24] C. Damiolini, A. Gibney, and N. Tarasca, *On factorization and vector bundles of conformal blocks from vertex algebras*, Ann. Sci. Éc. Norm. Supér. **57** (2024), 241–292.
- [25] C. Damiolini and L. Woike, *Modular functors from conformal blocks of rational vertex operator algebras*, arXiv:2507.05845, 2025.
- [26] R. Dijkgraaf, G. Moore, E. Verlinde, and H. Verlinde, *Elliptic genera of symmetric products and second quantized strings*, Comm. Math. Phys. **185** (1997), 197–209.
- [27] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford Univ. Press, 1958.
- [28] G. Dunn, *Tensor product of operads and iterated loop spaces*, J. Pure Appl. Algebra **50** (1988), 237–258.
- [29] V. G. Drinfeld, *Quantum groups*, in *Proceedings of the ICM, Berkeley 1986*, 798–820.
- [30] V. G. Drinfeld, *Quasi-Hopf algebras*, Leningrad Math. J. **1** (1990), 1419–1457.
- [31] P. Etingof and D. Kazhdan, *Quantization of Lie bialgebras, I*, Selecta Math. **2** (1996), 1–41.
- [32] J. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Math. 352, Springer, 1973.
- [33] C. Faber and R. Pandharipande, *Hodge integrals and Gromov–Witten theory*, Invent. Math. **139** (2000), 173–199.
- [34] B. Feigin and E. Frenkel, *Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras*, Int. J. Mod. Phys. A **7** (1992), Suppl. 1A, 197–215.
- [35] I. B. Frenkel and V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980), 23–66.
- [36] E. Frenkel and D. Ben-Zvi, *Vertex Algebras and Algebraic Curves*, 2nd ed., Math. Surveys Monogr. 88, Amer. Math. Soc., 2004.
- [37] E. Frenkel and D. Hernandez, *Baxter’s relations and spectra of quantum integrable models*, Duke Math. J. **164** (2015), 2407–2460.
- [38] J. Francis and D. Gaitsgory, *Chiral Koszul duality*, Selecta Math. **18** (2012), 27–87.
- [39] R. P. Feynman, *Space-time approach to non-relativistic quantum mechanics*, Rev. Mod. Phys. **20** (1948), 367–387.
- [40] W. Fulton and R. MacPherson, *A compactification of configuration spaces*, Ann. of Math. **139** (1994), 183–225.
- [41] D. Gaiotto, G. W. Moore, and A. Neitzke, *Framed BPS states*, Adv. Theor. Math. Phys. **17** (2013), 241–397.
- [42] S. Gautam and V. Toledano Laredo, *Yangians and quantum loop algebras*, Selecta Math. **19** (2013), 271–336.
- [43] E. Getzler and M. M. Kapranov, *Modular operads*, Compos. Math. **110** (1998), 65–126.
- [44] V. Gritsenko and V. Nikulin, *Siegel automorphic form corrections of some Lorentzian Kac–Moody Lie algebras*, Amer. J. Math. **119** (1997), 181–224.
- [45] O. Gwilliam and B. R. Williams, *A one-loop exact quantization of Chern–Simons theory*, arXiv:1910.05230, 2020.
- [46] D. Hernandez and M. Jimbo, *Asymptotic representations and Drinfeld rational fractions*, Compos. Math. **148** (2012), 1593–1623.
- [47] Y.-Z. Huang, *Vertex operator algebras and the Verlinde conjecture*, Comm. Contemp. Math. **10** (2008), 103–154.

- [48] M. Jimbo, *A q -difference analogue of $U(\mathfrak{g})$ and the Yang–Baxter equation*, Lett. Math. Phys. **10** (1985), 63–69.
- [49] V. G. Kac, *Vertex Algebras for Beginners*, 2nd ed., Univ. Lecture Ser. 10, Amer. Math. Soc., 1998.
- [50] D. Kazhdan and G. Lusztig, *Tensor structures arising from affine Lie algebras, I–IV*, J. Amer. Math. Soc. **6–7** (1993–1994).
- [51] T. Kohno, *Monodromy representations of braid groups and Yang–Baxter equations*, Ann. Inst. Fourier **37** (1987), 139–160.
- [52] V. Knizhnik and A. Zamolodchikov, *Current algebra and Wess–Zumino model in two dimensions*, Nuclear Phys. B **247** (1984), 83–103.
- [53] M. Kontsevich, *Deformation quantization of Poisson manifolds*, Lett. Math. Phys. **66** (2003), 157–216.
- [54] J. Lurie, *Higher Algebra*, available from the author’s webpage, 2017 version.
- [55] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren Math. Wiss. 346, Springer, 2012.
- [56] D. Maulik and A. Okounkov, *Quantum groups and quantum cohomology*, Astérisque 408, 2019.
- [57] N. Nekrasov, *BPS/CFT correspondence: non-perturbative Dyson–Schwinger equations and qq-characters*, J. High Energy Phys. 03 (2016), 181.
- [58] D. Mumford, *Towards an enumerative geometry of the moduli space of curves*, in *Arithmetic and Geometry, Vol. II*, Birkhäuser, 1983, 271–328.
- [59] M. Oura, C. Poor, R. Salvati Manni, and D. Yuen, *Modular forms of weight 8 for $\Gamma_g(1, 2)$* , Math. Ann. **346** (2010), 477–498.
- [60] L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule–contramodule correspondence*, Mem. Amer. Math. Soc. **212** (2011), no. 996.
- [61] S. Priddy, *Koszul resolutions*, Trans. Amer. Math. Soc. **152** (1970), 39–60.
- [62] D. Quillen, *Determinants of Cauchy–Riemann operators over a Riemann surface*, Funct. Anal. Appl. **19** (1985), 31–34.
- [63] N. Reshetikhin and V. Turaev, *Ribbon graphs and their invariants derived from quantum groups*, Comm. Math. Phys. **127** (1990), 1–26.
- [64] A. N. Schellekens, *Meromorphic $c = 24$ conformal field theories*, Comm. Math. Phys. **153** (1993), 159–185.
- [65] A. M. Polyakov, *Quantum geometry of bosonic strings*, Phys. Lett. B **103** (1981), 207–210.
- [66] S. Grushevsky and R. Salvati Manni, *The superstring cosmological constant and the Schottky form in genus 5*, Amer. J. Math. **133** (2011), 1007–1027; erratum incorporated in arXiv:0809.1391v5.
- [67] G. Segal, *The definition of conformal field theory*, in *Topology, Geometry and Quantum Field Theory*, London Math. Soc. Lecture Note Ser. 308, Cambridge Univ. Press, 2004, 421–577.
- [68] J. D. Stasheff, *Homotopy associativity of H -spaces. I*, Trans. Amer. Math. Soc. **108** (1963), 275–292.
- [69] J. D. Stasheff, *Homotopy associativity of H -spaces. II*, Trans. Amer. Math. Soc. **108** (1963), 293–312.
- [70] V. V. Schechtman and A. N. Varchenko, *Arrangements of hyperplanes and Lie algebra homology*, Invent. Math. **106** (1991), 139–194.
- [71] D. Tamarkin, *Deformations of chiral algebras*, in *Proceedings of the ICM, Vol. II*, Beijing, 2002, 105–116.
- [72] A. Tsuchiya, K. Ueno, and Y. Yamada, *Conformal field theory on universal family of stable curves with gauge symmetries*, Adv. Stud. Pure Math. **19** (1989), 459–566.

- [73] E. Verlinde, *Fusion rules and modular transformations in 2D conformal field theory*, Nuclear Phys. B **300** (1988), 360–376.
- [74] E. Witten, *Quantum field theory and the Jones polynomial*, Comm. Math. Phys. **121** (1989), 351–399.
- [75] K. G. Wilson, *Non-Lagrangian models of current algebra*, Phys. Rev. **179** (1969), 1499–1512.
- [76] E. Witten, *Noncommutative geometry and string field theory*, Nuclear Phys. B **268** (1986), 253–294.
- [77] T. Willwacher, *M. Kontsevich’s graph complex and the Grothendieck–Teichmüller Lie algebra*, Invent. Math. **200** (2015), 671–760.
- [78] Y. Zhu, *Modular invariance of characters of vertex operator algebras*, J. Amer. Math. Soc. **9** (1996), 237–302.
- [79] T. Torii, *On duoidal ∞ -categories*, J. Homotopy Relat. Struct. **20** (2025), no. 1, 125–162.
- [80] P. Calabrese and J. Cardy, *Entanglement entropy and quantum field theory*, J. Stat. Mech. Theory Exp. (2004), P06002.
- [81] E. Frenkel, *Langlands Correspondence for Loop Groups*, Cambridge Studies in Advanced Mathematics 103, Cambridge Univ. Press, 2007.
- [82] D. Ayala and J. Francis, *Poincaré/Koszul duality*, arXiv:1409.2478, 2014.
- [83] E. Frenkel, *Gaudin model and opers*, arXiv:math/0407524, 2005.
- [84] E. Frenkel and D. Gaitsgory, *Local geometric Langlands correspondence: the spherical case*, Adv. Math. **223** (2010), 34–102.